Module Introduction:

This module consists of four units. The first unit deals with real sequence and infinite series. In this unit we will look briefly at many terms and concepts related to the real sequences and infinite series. The second unit deals with power series, which is one of the most useful types of infinite series, and their applications. In particular we will also discuss the two special types of power series named Taylor series and Maclaurin series. The third unit discusses on calculus of functions of several variables, specifically focuses on the limit, continuity and partial derivatives of functions of several variables and their applications. The fourth unit deals with multiple integrals particularly, double integrals and triple integrals of functions of two and three variables respectively together with their applications. By doing so students will be able to express terms and concepts related to infinite series, power series, partial derivatives of functions of several variables and multiple integrals.

SOME FEATURES OF THE MODULE

Visualization: This module makes extensive use of modern computer graphics to clarify concepts and to develop the student's ability to visualize mathematical objects, particularly those in 3 dimensional space.

Quick Check Exercises: Each exercise set begins with approximately five practice exercises that are designed to provide students with an immediate assessment of whether they have mastered key ideas from the section.

Applicability of Calculus: One of the good feature, primary goals of this module is to link calculus to the real world and the student's own experience. This theme is carried through in the examples and exercises.

Career Preparation: This module is written at a mathematical level that will prepare students for a wide variety of careers that require a sound mathematics background, including engineering, the various sciences, and business**.**

Historical Notes: Some biographies and historical notes have been included in the module, with the goal of capturing and bringing to life for the student the personalities of history's greatest mathematicians.

Cooperative learning: One of the primary goals of this module is also to promote cooperative learning, so that students share knowledge and skills through a lot of group discussions and group activities given at each of new ideas introduced.

Module Objectives:

At the end of this module students will be able to:

- Define sequences, types of sequences, infinite series and power series.
- Identify basic properties of sequence whether it converges or divergence.
- Determine whether or not a given sequence is bounded and monotone.
- Identify the relation between sequence and series.
- Identify different types of tests for convergence of series and choose appropriate test of convergence
- Determine differentiation and integration of a Power Series.
- Determine the Taylor's series representation of a function.
- Apply the concept of sequence, real series and Taylor's formula in solving physical and real life problems.
- Determine domain and range of functions of two or three variables
- Determine limit and continuity of functions of two or three variables
- Determine differentiability of functions of two or three variables
- Determine directional derivative of functions of two or three variables
- Determine gradient of functions of two or three variables
- Determine maximum and minimum (extreme) values of functions of two or three variables on a given region
- Apply the concept of differentiability of functions of two or three variables in solving real life problems
- Define double and Triple integrals in different coordinates
- Determine double and multiple integrals of functions of several variables
- Apply multiple integrals in determining volume of a solid region, area of plane region, surface area and so on
- Find the mass of a planar lamina using a double integral
- Find the center of mass of a planar lamina using double integrals
- Find moments of inertia using double integrals

CHAPTER ONE

INFINITE SEQUENCES AND SERIES

Unit Introduction

In this chapter we will be concerned with infinite sequences and series. This unit is divided into four sections. The first section presents *definitions* and *notations* of sequence, *convergence* and d*ivergence* properties of Sequences and the basic properties of sequence, in particular *boundedness* and *monotoness* will also be treated in this section. The Second section presents *partial sum of a sequence, definition and notation of a series* and The third section deals with *Different types of tests for convergence, in particular Integral Test, Comparison Test, Root Test, and Ratio Tests*. *Alternating Series; Absolute and Conditional Convergences* will be treated in the fourth section.

Unit Objectives:

At the end of the unit students will be able to:

- Define different types of sequences.
- Identify basic properties of sequence.
- Determine whether a given sequence converges or not, bounded or not and monotone or not.
- Demonstrate how to differentiate increasing and decreasing sequences together with solving exercises.
- Apply the concept of sequence in solving real life problems.
- Identify the relation between sequence and series.
- Define the term series.
- Identify the two types of convergence of series.
- Choose appropriate test for convergence of infinite series.
- Demonstrate the application of different tests together with solving exercises.
- Apply the concept of series in solving real life problems

1.1 **Real Sequences**

Infinite sequences and series were introduced briefly in *A Preview of Calculus* in connection with Zeno's paradoxes and the decimal representation of numbers. Their importance in calculus stems from Newton's idea of representing functions as sums of infinite series. For instance, in finding areas he often integrated a function by first expressing it as a series and then integrating each term of the series. The main objective of this unit is to study about infinite series. To do so it is important to be familiar with the basic concepts of sequences and convergence of sequences primarily.

1.1.1 **Notations and Terminology**

We begin this section with two questions to remind readers their previous study about sequences and motivate (brainstorm) readers for their studies about sequence from the section:

1. Define: i) Arithmetic Sequence
   ii) Geometric Sequence
2. Give two examples of each sequence.

In everyday language, the term "sequence" means a succession of things in a definite order, chronological order, size order, or logical order. In mathematics, the term "sequence" is commonly used to denote a succession of numbers called terms in a definite order:

$$a_1,\ a_2, a_3\ , a_4 \ldots\ldots, a_n \ldots\ldots\ldots$$

The number a_1 is called the first term, the number a_2 is called the second term , a_3 is called the third term and in general the nth term is denoted by a_n.

Definition: A Real sequence is a real valued function whose domain is the set of positive integers greater or equal to a given integer m (usually 0 or 1).

Examples:

1. $\{2,4,6,8,\ldots\}$
2. $\left\{1,\frac{1}{2},\frac{1}{3},\frac{1}{4},\frac{1}{5},\ldots\right\}$
3. $\left\{\frac{1}{2},\frac{2}{3},\frac{3}{4},\frac{4}{5},\ldots\right\}$

4. $\left\{\frac{1}{2},\frac{1}{4},\frac{1}{8},\frac{1}{16},...\right\}$

For example, in the sequence {2,4,6,8,...} of example 1,We have the following:

Term number	1	2	3	4	…	n
Term	2	4	6	8	…	$2n$

Each term is twice the term number; thus the nth term is given by the formula $2n$. We denote this by writing the sequence as $2,4,6,8,...,2n,...$. We call the function $f(n)=2n$ the general term of the sequence**.** Similarly sequences of the above types can be defined by giving a formula for the nth-term.

Quick check Class Exercises 1.1.1

1: Find the general term of each sequences in example 2-4 above .by relating each term with their respective term numbers.*(Group Discussion in a Class)*

Notations:

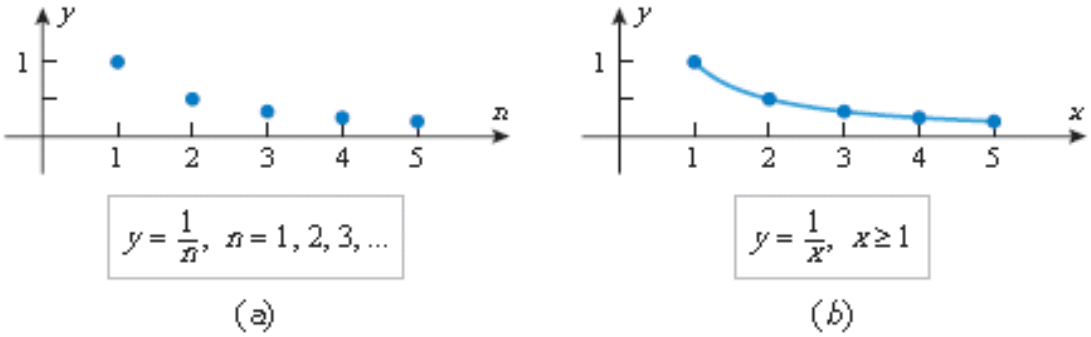
1. When the general term of the sequence with elements $a_m, a_{m+1}, a_{m+2}, \cdots, a_{m+n-1}, \cdots$ is known, it is usually denoted by $\{a_n\}_{n=m}^{\infty}$ or $\{a_m, a_{m+1}, a_{m+2}, \cdots, a_{m+n-1}, a_n \cdots\}$.
2. If $m=1, or\ m=0$ the sequence is written as $\{a_n\}_{n=1}^{\infty}$ *or* $\{a_n\}_{n=0}^{\infty}$.The letter n in this notation is called the index of the sequence and the element a_i is called the $i^{th}-term\ \ of\ the$ sequence.
3. Since sequence $\{a_n\}_{n=m}^{\infty}$ is a function, then we may also write $f(n)=a_n$.

Graphs of Sequences

Since sequences are functions, it makes sense to talk about the graph of a sequence. For example, the graph of the sequence $\left\{\frac{1}{n}\right\}_{n=1}^{\infty}$ is the graph of the equation

$y=\frac{1}{n}, n=1,2,3....$

Because the right side of this equation is defined only for positive integer values of *n*, the graph consists of a succession of isolated points (Figure *a*). This is different from the graph of $y = \frac{1}{x}, x \geq 1$ which is a continuous curve (Figure *b*)



Remarks:

1. There are sequences that do not have a simple defining equation.

For instance,

a) The sequence $\{p_n\}$, where p_n is the population of the world as of January 1 in the year n.

b) Let a_n be the digit in the nth decimal place of the number e, then $\{a_n\}$ is well defined sequence whose first few terms are

$\{7,1,8,2,8,1,8,2,8,4,5,\cdots\}$

2. Some sequences also arise from a formula or a set of formulas that specify how to generate each term in a sequence from terms that precede it; such sequences are said to be sequences defined recursively and the formulas are said to be recursion formulas. For instance,

a) The Fibonacci Sequence $\{f_n\}_1^{\infty}$ is defined by the recursion formulas:

$f_1 = 3, \quad f_2 = 5, \quad f_n = f_{n-1} + f_{n-2,} \quad n \geq 3$. The first few terms of the sequence are:

$\{3,5,8,13,21,34,55,89,144,\cdots\}$

Definition: A sequence $\{a_n\}_{n=m}^{\infty}$ where

$$a_n = (-1)^n \cdot b_n \ \textit{and either } b_n \geq 0 \quad \textit{or} \quad b_n \leq 0, \ \forall n \geq m$$

is called an oscillating sequence.

Example: Sequences $\{(-1)^n\}_{n=3}^{\infty}$ and $\left\{\frac{(-1)^n}{3n-4}\right\}_{n=1}^{\infty}$ are oscillating sequences because of the fact that the terms of the sequence alternate between positive and negative numbers.

Activity 1.1.1 (*Home work*)

1.List at least three elements of the sequence given below

(*Individual Exercises*)

a) $\{a_n\}_{n=1}^{\infty}$,where $a_n=\frac{n}{n+1}$, a_1 _____ , a_2 = _____, a_3 = _____

b) $\{a_n\}_{n=1}^{\infty}$, where $a_n=\frac{(-1)^n(n+1)}{3^n}$, a_1 _____ , a_2 = _____, a_3 = _____

c) $\{a_n\}_{n=3}^{\infty}$, where $a_n=\sqrt{n-3}$, a_1 = ____ , a_2 = _____, a_3 = _____

d) $\{a_n\}_{n=0}^{\infty}$, where $a_n=\cos\frac{n\pi}{6}$, a_1 _____ , a_2 = ____, a_3 = _____

e) $\{a_n\}_{n=1}^{\infty}$, where $a_n=\frac{2^n}{(n-1)!}$, a_1 = _____ , a_2 = _____, a_3 = _____

f) $\{a_n\}_{n=1}^{\infty}$, where $a_n=\frac{n}{2^{n-1}}$, a_1 = _____ , a_2 = _____, a_3 = _____

g) $\{s_n\}_{n=1}^{\infty}$, where $s_n=\sum_{k=1}^{n}\frac{1}{k}$, s_1 = _____ , s_2 = _____, s_3 = _____

h) $\{a_n\}_{n=1}^{\infty}$, where $a_n=\frac{x^{n-1}}{\sqrt{2n+1}}$, a_1 = _____ , a_2 = _____, a_3 = _____

3. Find the general formula a_n of the indicated sequence.**(**Group Discussion**)**

Sequence	a_n
a. { 2,9,16,23,30, …}	
b. {1,8,27,64,125, …}	
c. {1, $\frac{1}{3!}$, $\frac{1}{5!}$, $\frac{1}{7!}$, $\frac{1}{9!}$, …}	
d. $1, 1+\frac{1}{3}, 1+\frac{1}{3}+\frac{1}{5}, 1+\frac{1}{3}+\frac{1}{5}+\frac{1}{7},\ldots$	

e. $\left\{x, \frac{x^3}{3!}, \frac{x^5}{5!}, \frac{x^7}{7!}, \cdots\right\}$	
f. $\left\{\frac{e}{\sqrt{2}}, \frac{e^2}{2}, \frac{e^3}{\sqrt{6}}, \frac{e^4}{\sqrt{8}}, \frac{e^5}{\sqrt{10}}, \cdots\right\}$	

Assessment

- *Asking an answer for some of the questions.*
- *Check students' participation in the group activity.*
- *Give feedback to their answers*

1.1.2 Convergence and Divergence of Sequence

Since sequences are functions, we can inquire about their limits. However, because a sequence $\{a_n\}$ is only defined for integer values of n, the only limit that makes sense is the limit of $a_n$as $n \to \infty$.

Definition:(Limits of sequences)

1. A sequence $\{a_n\}_{n=m}^{\infty}$ is said to converge to some finite limit L, written as:

   $\lim_{n\longrightarrow\infty} a_n = L$,if and only if

   $$\forall \varepsilon > 0, \exists n_o \in N \ \ni \ n \geq n_o \ \Rightarrow \ |a_n - L| < \varepsilon \ .$$

   A sequence that does not converge to some finite limit *L* is said to diverge.

2. A sequence $\{a_n\}_{n=m}^{\infty}$ is said to diverge to ∞, written as $\lim_{n\longrightarrow\infty} a_n = \infty$, if and only if

   $\forall M > 0, \exists n_o \in N \ \ni \ n \geq n_o \ \Rightarrow \ a_n > M.$

3. Similarly , $\{a_n\}_{n=m}^{\infty}$ is said to diverge to $-\infty$ if for every negative integer M

   $\exists n_o \in N \ni n \geq n_o \Rightarrow a_n < M$ and written as $\lim_{n\longrightarrow\infty} a_n = -\infty$

Examples:

1. Let $a_n = c$, for $n \geq 1$,where c is a constant. show that $\lim_{n\longrightarrow\infty} a_n = c.$

Solution: Given any $\varepsilon > 0$, we need to find N such that $|a_n - L| < \varepsilon, \forall n \geq N$.

That is we need to find N which satisfies

$$|a_n - L| = |c - c| = 0 < \varepsilon, \forall n \geq N$$

Since this is always true we can choose N=1, therefore $|a_n - L| = 0 < \varepsilon, \forall n \geq 1$.

Thus $\lim_{n \longrightarrow \infty} a_n = c.$ Limit of constant $a_n = c$ sequence is constant.

2**.** Show that $\lim_{n \longrightarrow \infty} \frac{1}{n} = 0$.

Solution: Given any $\varepsilon > 0$, we need to find N such that $|a_n - L| < \varepsilon, \forall n \geq N$

That is we need to find N which satisfies $|a_n - L| = \left|\frac{1}{n} - 0\right| = \frac{1}{n} < \varepsilon, \forall n \geq N$

We can Choose $N > \frac{1}{\varepsilon}$,so that $\left|\frac{1}{N}\right| < \varepsilon$. Thus

$$|a_n - L| = \left|\frac{1}{n} - 0\right| < \left|\frac{1}{N}\right| < \varepsilon, \forall n \geq N$$

So by the definition of limit $\lim_{n \longrightarrow \infty} \frac{1}{n} = 0$.

3. Show that $\lim_{n \to \infty} -\sqrt{n} = -\infty$.

Solution: Here we want to show that for every negative integer $M < 0$

$$\exists n_o \in N \ni n \geq n_o \Rightarrow a_n = -\sqrt{n} < M$$

Thus for any number M we can find a number $n_0 = (M-1)^2 \in N$ such that

$$n \geq n_o \Rightarrow a_n = -\sqrt{n} < -\sqrt{n_0} = -\sqrt{(M-1)^2} = -|M-1| < M$$

Therefore by the above definition of limit $\lim_{n \to \infty} -\sqrt{n} = -\infty$.

Theorem 1.1.1: If a sequence $\{a_n\}_{n=m}^{\infty}$ converges then its limit is unique.

Proof: Suppose there exists two limits $L_1 \& L_2$.Therefore by the definition of limit for every positive number $\varepsilon > 0$ there exists $N_1, N_2 \in N$ such that

$|a_n - L_1| < \varepsilon, \forall n \geq N_1$ and $|a_n - L_2| < \varepsilon, \forall n \geq N_2$

So if we choose $n_0 = \max\{N_1, N_2\}$ we have

$$|L_1 - L_2| = |L_1 - a_n + a_n - L_2| < |a_n - L_1| + |a_n - L_2| < 2\varepsilon, \forall n \geq n_0$$

$$\Rightarrow |L_1 - L_2| = 0 \Rightarrow L_1 = L_2$$

Remark:

1. Convergence or divergence of a sequence $\{a_n\}_{n=m}^{\infty}$ is a property which does not depend on the initial terms of the sequence rather it is a result of the behavior of the general term eventually i.e. as $n \longrightarrow \infty$. For instance, see the sequences

- $100, -200, 400, -800, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \cdots$ is convergent.
- $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, 10, -10, 10, -10, \cdots$ is divergent.

Quick check Class Activity 1.1.2 :

1. Use the definition of limit of sequences to show that

a) $\lim_{n\to\infty} -n = -\infty$ b) $\{(-1)^n\}_{n=1}^{\infty}$ diverges c) $\lim_{n\to\infty} n^2 = \infty$

2. Given that $\lim_{n\to\infty} \frac{n}{n+1} = 1$. By using the definition of limit, find the smallest value of N for the given value of ε in each part.

a. $\varepsilon = 0.25$ b. $\varepsilon = 0.1$ c) $\varepsilon = 0.001$

Instructor's Role

- *Check and give feedback to their answers*

The above definitions of limit could not help us to evaluate the limit of a sequence, thus we seek for further properties of convergent sequences to evaluate their limiting value.

Theorem 1.1.2 (Properties of Convergent sequences)

Let $\{a_n\}_{n=m}^{\infty}$ and $\{b_n\}_{n=m}^{\infty}$ be convergent sequences. Then

a. $\{a_n + b_n\}_{n=m}^{\infty}$ converges and $\lim_{n\to\infty}(a_n + b_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n$

b. $\{r.a_n\}_{n=m}^{\infty}$ converges and $\lim_{n\to\infty}(r.a_n) = r.\lim_{n\to\infty} a_n$,where r is a constant.

c. $\{a_n.b_n\}_{n=m}^{\infty}$ converges and $\lim_{n\to\infty}(a_n.b_n) = \lim_{n\to\infty} a_n.\lim_{n\to\infty} b_n$

d. $\left\{\frac{a_n}{b_n}\right\}_{n=m}^{\infty}$ converges and $\lim_{n\longrightarrow\infty} \frac{a_n}{b_n} = \frac{\lim_{n\longrightarrow\infty} a_n}{\lim_{n\longrightarrow\infty} b_n}$, provided that $\lim_{n\longrightarrow\infty} b_n \neq 0$

Proof: Direct consequence of the above definitions of limits of sequences.

The above theorem which we stated without proof ensure that the algebraic techniques used to find limits of functions can also be applied to find limits of sequences.

Theorem 1.1.3: Let $\{a_n\}_{n=m}^{\infty}$ be a sequence and let f be a function defined on $[m,\infty)$ such that $f(n)=a_n,\ \forall n\geq m$. Then

a) If $\lim_{x\longrightarrow\infty} f(x)=L\in R$, then $\{a_n\}_{n=m}^{\infty}$ converges and $\lim_{n\longrightarrow\infty} a_n=L$.

b) If $\lim_{x\longrightarrow\infty} f(x)=\infty \ or\ -\infty$, then $\{a_n\}_{n=m}^{\infty}$ diverges and $\lim_{n\longrightarrow\infty} a_n=\infty \ or -\infty$

Proof: (Reading Assignment)

Examples:

1. Find the limit of the sequence $\{a_n\}_{n=1}^{\infty}$ where $a_n=\dfrac{\ln(n+1)}{n}$.

Solution: Let $f(x)=\dfrac{\ln(x+1)}{x}$ *for* x *in* $[1,\infty)$.

Since $\lim_{x\longrightarrow\infty} f(x)=\lim_{x\longrightarrow\infty}\dfrac{\ln(x+1)}{x}$ is $\left(\dfrac{\infty}{\infty}\right)$ form, then, by using L'Hopitals rule,

$$\lim_{x\longrightarrow\infty} f(x)=\lim_{x\longrightarrow\infty}\frac{\ln(x+1)}{x}=\frac{\lim_{x\to\infty}\frac{1}{x+1}}{\lim_{x\to\infty}1}=\lim_{x\to\infty}\frac{1}{x+1}=0,$$

which implies $\lim_{n\longrightarrow\infty}\dfrac{\ln(n+1)}{n}=0$.

2. Find the limit of the sequence $\{a_n\}_{n=1}^{\infty}$ where $a_n=\dfrac{n}{2n+1}$.

Solution: Dividing numerator and denominator by n and applying the above theorem:

$$\lim_{n\to\infty}\frac{n}{2n+1}=\lim_{n\to\infty}\frac{1}{2+\frac{1}{n}}=\frac{\lim_{n\to\infty}1}{\lim_{n\to\infty}2+\lim_{n\to\infty}\frac{1}{n}}=\frac{1}{2+0}=\frac{1}{2}$$

Thus the sequence converges $\dfrac{1}{2}$.

Theorem 1.1.4: Suppose that $\lim_{n\longrightarrow\infty} a_n = L$ and that for each n, a_n is in the domain of a function f. If f is continuous at L then

$$\lim_{n\longrightarrow\infty} f(a_n) = f(L).$$

Proof: Exercise.

Example 1: Find the limit of the sequence $\{a_n\}_{n=1}^{\infty}$ where

a. $a_n = \cos\left(\frac{\pi}{n}\right)$ b. $a_n = \ln\left(\frac{n}{n+1}\right)$

Solution: a. Since

$$\lim_{n\longrightarrow\infty} \frac{\pi}{n} = 0,$$

and the cosine function is continuous at 0,

$$\lim_{n\longrightarrow\infty} \cos\left(\frac{\pi}{n}\right) = \cos\left(\lim_{n\to\infty} \frac{\pi}{n}\right) = \cos(0) = 1.$$

b. Since

$$\lim_{n\to\infty} \frac{n}{n+1} = \lim_{n\to\infty} \frac{1}{1+\frac{1}{n}} = \frac{\lim_{n\to\infty} 1}{\lim_{n\to\infty} 1 + \lim_{n\to\infty} \frac{1}{n}} = \frac{1}{1+0} = 1$$

and logarithmic function is continuous at 1.

$$\lim_{n\to\infty} \ln\left(\frac{n}{n+1}\right) = \ln\left(\lim_{n\to\infty} \frac{n}{n+1}\right) = \ln 1 = 0.$$

Quick Check Class activity 1.1.3: Evaluate the limits of the following Sequences

a. $a_n = \frac{\ln n}{n}$ c. $a_n = \tan\sqrt{\frac{\pi^2 n^2 - 8}{16n^2}}$

b. $a_n = \sqrt{4 + \left(\frac{1}{n}\right)^2}$ d. $a_n = \frac{5n^2 + 1}{4 - 3n^2}$

Instructor's Role:

- *Check their answers and Give feedback for their answers*

Theorem 1.1.5: (The Version of Squeezing Theorem for Sequences)

Suppose $\{a_n\}_{n=m}^{\infty}, \{b_n\}_{n=m}^{\infty}$ and $\{c_n\}_{n=m}^{\infty}$ are sequences such that $a_n \le b_n \le c_n, \forall n \ge m$ and, $\lim_{n\longrightarrow\infty} a_n = \lim_{n\longrightarrow\infty} c_n = L$. Then $\lim_{n\longrightarrow\infty} a_n = \lim_{n\longrightarrow\infty} b_n = \lim_{n\longrightarrow\infty} c_n = L$.

Proof: Exercise

Example 1: Find the limit of the sequence $\{a_n\}_{n=1}^{\infty}$ where

a. $a_n = \dfrac{\sin n}{n}$ b. $a_n = \dfrac{n!}{n^n}$

Solutions:

a. Since $\dfrac{-1}{n} \le \dfrac{\sin n}{n} \le \dfrac{1}{n}$,and

$$\lim_{n\longrightarrow\infty}\left(\frac{-1}{n}\right) = 0 = \lim_{n\longrightarrow\infty}\frac{1}{n},$$

Then, the squeezing theorem implies, $\lim_{n\longrightarrow\infty}\dfrac{\sin n}{n} = 0$.

b. We have $0 \le \dfrac{n!}{n^n} = \dfrac{1\cdot 2\cdot 3\cdot \ldots \cdot n}{n\cdot n\cdot n\cdot \ldots \cdot n} = \left(\frac{1}{n}\right)\left(\frac{2}{n}\right)\left(\frac{3}{n}\right)\ldots\left(\frac{n}{n}\right) \le \frac{1}{n}$ (Why?)

Thus $0 \le a_n \le \dfrac{1}{n}$. However,

$$\lim_{n\to\infty} 0 = \lim_{n\longrightarrow\infty}\frac{1}{n} = 0.$$

Thus by squeezing theorem $\lim_{n\to 0}\dfrac{n!}{n^n} = 0..$

Theorem 1.1.6: If $\lim_{n\longrightarrow\infty}|a_n| = 0$ then $\lim_{n\longrightarrow\infty} a_n = 0.$

Proof: Depending on the size of a_n either $a_n = |a_n|$ or $a_n = -|a_n|$. Thus in both cases we have: $-|a_n| \le a_n \le |a_n|$. However the limit of the two outside terms is 0, hence the limit of a_n is 0 by squeezing theorem.

Example 1: Show that

a) $\left\{(-1)^n\dfrac{1}{n}\right\}_{n=1}^{\infty}$ converges to 0. b) $\left\{(-1)^n\dfrac{1}{2^n}\right\}_{n=1}^{\infty}$ converges to 0.

Solution: a) Since $\left|(-1)^n\dfrac{1}{n}\right| = \dfrac{1}{n}$ and $\dfrac{1}{n}$ converges to 0 the result follows by the above theorem.

b) Since $\left|(-1)^n\dfrac{1}{2^n}\right| = \dfrac{1}{2^n}$ and $\dfrac{1}{2^n}$ converges to 0 the result follows by the above theorem.

Quick Check Class activity 1.1.4: Evaluate the limits the following Sequences

a. $a_n = \dfrac{\sin^2 n}{n}$ b. $a_n = \dfrac{1-\cos n}{n}$

c. $a_n = \dfrac{\cos 2n}{n}$

d. $\lim\limits_{n\to\infty} \dfrac{1}{n!}$

Instructor's Role:

- *Check their answers and Give feedback for their answers*

Group Activity 1.1.2

1. *(Home Takes Group Assignment)*
   a. The current in an electric circuit is measured after each minute and found to be approximated by $i_n = 10.(1+e^{-n})$ .If the limit of this value is the steady state current, what is the steady state current?
   b. The height of an electronic "bouncing ball" is described by

$$h_n = \frac{7n+2}{5n+5}$$

   What is the limiting value of the height?
   c. Suppose the number of bacteria in a culture is growing exponentially, with a doubling time of 10 hours. Suppose also that there are 1000 bacteria in the culture. Find a formula for the number, a_n of bacteria in the culture after n hours..
2. Investigate the convergence or divergence of the following sequences by using appropriate method. (*Individual Exercises*)

i. $\left\{3-\dfrac{2}{n}\right\}_{n=1}^{\infty}$

ii. $\left\{\dfrac{(-1)^n}{n+1}\right\}_{n=1}^{\infty}$

iii. $\left\{\dfrac{n^4-1}{n^4+n-6}\right\}_{n=1}^{\infty}$

iv. $\left\{\tan\left(\dfrac{n\pi}{4n+1}\right)\right\}_{n=1}^{\infty}$

v. $\left\{\dfrac{4^n}{2^n+10^6}\right\}_{n=0}^{\infty}$

vi. $a_n = \left(n^2+n\right)^{\frac{1}{n}}$

vii. $\left\{\dfrac{2^n}{4^n+7}\right\}_{n=0}^{\infty}$

viii. $\left\{\dfrac{n^5}{n^3+6}\right\}_{n=0}^{\infty}$

3. (A*ssignment*). Show that

i. If $x>0$ *then* $\lim\limits_{n\longrightarrow\infty} x^{\frac{1}{n}} = 1$

ii. If $|x|<1$, then $\lim\limits_{n\longrightarrow\infty} x^n = 0$

iii. $\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = e^x$

v. $\lim_{n\longrightarrow\infty}\left(1+\frac{1}{n}\right)^n = e$

iv. For each x $\lim_{n\longrightarrow\infty}\frac{x}{n!}=0$

4. Evaluate the following limits

a. $\lim_{n\to\infty}\frac{n}{10^n}$

b. $\lim_{n\to\infty}\frac{n^{10}}{10^n}$

5. Consider the sequence: $a_1=\sqrt{6}, a_2=\sqrt{6+\sqrt{6}}, a_3=\sqrt{6+\sqrt{6+\sqrt{6}}},...$

.Find a recursion formula for a_{n+1}?

Assessment

- *Asking an answer for some of the questions.*
- *Check students' participation in the group activity.*
- *Give feedback to their answers*

1.1.3 Bounded and Monotonic Sequences

- **Bounded Sequences**

Definition: A sequence $\{a_n\}_{n=m}^{\infty}$ is called bounded sequence if there is a positive real number M such that $|a_n|\le M,$ for all $n\ge m$. Otherwise, it is unbounded.

Examples:

a. Consider the sequence $\{a_n\}_{n=1}^{\infty}$, where $a_n=\sin nx$.

Since, $-1\le\sin nx\le 1,\ \forall x\in R, n\ge 1$ then $\exists M=1>0$ ə $|a_n|\le M, n\ge 1$.

b. Since there is no M such that $|2^n|\le M, \forall n\in N$ the sequence is unbounded.

But since $0\le|2^n|, \forall n\in N$ the sequence is bounded below by 0 but not bounded above.

Remark:

Let $\{a_n\}_{n=m}^{\infty}$ be a sequence then,

a) M is called an upper bound if $a_n\le M, \forall n$ (for all n).

b) M is called a lower bound if $a_n\ge M, \forall n$

c) A sequence $\{a_n\}_{n=m}^{\infty}$ is said to be bounded if it is bounded above and below.

Examples:

1. Since $0 \le \frac{1}{n} \le 1$, *for all* $n \ge 1$. The sequence $\left\{\frac{1}{n}\right\}_{n=1}^{\infty}$ is bounded both below and above. Therefore the sequence is bounded.

2.Consider the sequence $\{3n+7\}_{n=0}^{\infty}$ .Then $0 < 3n+7, \forall n \in N$, & $\{3n+7\}_{n=0}^{\infty}$ increases without bound as n increases (not bounded above). Thus the sequence is unbounded.

Quick check Exercises 1.1.5:

1. Determine whether or not the following sequences are bounded.

a. $\left\{3-\frac{2}{n}\right\}_{n=1}^{\infty}$ b. $\left\{\frac{(-1)^n}{n+1}\right\}_{n=1}^{\infty}$

Instructor's Role:

- *Check their answers and Give feedback for their answers*

Theorem 1.1.6: Let $\{a_n\}_{n=m}^{\infty}$ be a sequence and $\lim_{n\longrightarrow\infty} a_n = L$, where L is a real number. Then $\{a_n\}_{n=m}^{\infty}$ is bounded.

Remark: The converse of the above theorem is false. For example, the sequence $\{(-1)^n\}_{n=m}^{\infty}$ is bounded, since $|a_n| \le 1$, *for all* n, but it is divergent.

- **Monotone Sequences**

Definition: A sequence $\{a_n\}_{n=m}^{\infty}$ is said to be

i. Increasing if $a_n \le a_{n+1}, \quad \forall n \ge m$

ii. Decreasing if $a_n \ge a_{n+1}, \quad \forall n \ge m$

iii. Strictly increasing if $a_n < a_{n+1}, \quad \forall n \ge m$

iv. Strictly decreasing if $a_n > a_{n+1}, \quad \forall n \ge m$

If a sequence $\{a_n\}_{n=m}^{\infty}$ is either increasing or decreasing, then it is said to be monotone sequences and if it is strictly increasing(decreasing) it is said to be strictly monotone sequence.

Frequently, one can *guess* whether a sequence is monotone by writing out some of its initial terms. However, to be certain that the guess is correct, one mustgive a precise mathematical argument.

Testing for Monotonocity:

Difference between successive terms	Ratio between successive terms	Classification
i) $a_{n+1}-a_n>0$	$\frac{a_{n+1}}{a_n}>1$	Strictly increasing
ii) $a_{n+1}-a_n<0$	$\frac{a_{n+1}}{a_n}<1$	Strictly decreasing
iii) $a_{n+1}-a_n\geq 0$	$\frac{a_{n+1}}{a_n}\geq 1$	Increasing
iv) $a_{n+1}-a_n\leq 0$	$\frac{a_{n+1}}{a_n}\leq 1$	Decreasing

Examples:1.Identify whether the following sequence increases or decreases.

a. $\left\{\frac{1}{n}\right\}_{n=1}^{\infty}$ b. $\left\{\frac{n}{n+1}\right\}_{n=1}^{\infty}$

c. $\left\{\frac{2^n}{n!}\right\}_{n=1}^{\infty}$ d. $\left\{\frac{10^n}{n!}\right\}_{n=1}^{\infty}$

Solutions: a. Since $a_n>0$, we can apply Ratio test. Thus

$$\frac{a_{n+1}}{a_n}=\frac{n}{n+1}<1,\forall n\geq 1$$

That is we have

$$a_{n+1}<a_n,\ \textit{for all positive integer } n\ .$$

Thus the sequence decreases.

b. Since $a_n>0$,using the difference of successive terms we have

$$a_{n+1}-a_n=\frac{n+1}{n+2}-\frac{n}{n+1}=\frac{1}{n^2+3n+1}>0,\forall n\geq 1$$

That is we have $a_{n+1}>a_n$, Thus the sequence increases.

c. Since $a_n>0$, we can apply Ratio test. Thus

$$\frac{a_{n+1}}{a_n}=\frac{2^{n+1}}{(n+1)!}\times\frac{n!}{2^n}=\frac{2}{n+1}<1,\forall n\geq 1$$

That is we have $a_{n+1}<a_n$, Thus the sequence decreases.

d. Since $a_n>0$, we can apply Ratio test. Thus

$$\frac{a_{n+1}}{a_n}=\frac{10^{n+1}}{(n+1)!}\times\frac{n!}{10^n}=\frac{10}{n+1}<1,\forall n\geq 9$$

That is we have $a_{n+1}<a_n$, *for all positive* int*eger* $n\geq 9$.

Thus the sequence decreases after the first nine terms, but notice that the first nine terms show that the sequence is increasing. We call such sequences Eventually decreasing.

Another third technique for testing monotonocity is using the derivative of the function obtained by replacing n by x in the general term of the sequence.

Derivative of f for $x\geq 1$	Conclusion for the sequence with $a_n=f(n)$
$f'(x)>0$	Strictly increasing
$f'(x)<0$	Strictly decreasing
$f'(x)\geq 0$	Increasing
$f'(x)\leq 0$	Decreasing

Example 1: Show that $a_n=\frac{n}{n^2+1}$ is decreasing sequence.

Solution: Consider the function $f(x)=\frac{x}{x^2+1}$

$$f'(x)=\frac{x^2+1-2x^2}{(x^2+1)^2}=\frac{1-x^2}{(x^2+1)^2}\leq 0,\forall x\geq 1$$

Thus f is decreasing on $(1,\infty)$and so $f(n)<f(n+1)$Therefore $\{a_n\}_{n=1}^{\infty}$ is decreasing.

Theorem 1.1.7:a. Every bounded and increasing sequence converges.
(to the least upper bound of its range)

b. Every bounded and decreasing sequence converges.
(to the greatest lower bound of its range)

Proof: Exercise.

Examples:1. Show that the sequence $\{a_n\}_{n=m}^{\infty}$ converges, where

a. $a_n=\frac{2^n}{n!}$ b. $a_n=1+\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\cdots+\frac{1}{n!}$

Solution: a. i) Since $\frac{a_{n+1}}{a_n}=\frac{2}{n+1}\leq 1$

we have $a_{n+1} \le a_n$, *for all positive* int*eger n* .Thus *the sequence is decreasing*.

ii) Again since

$$\left|\frac{2^n}{n!}\right| \le 2, \text{ for all } n \ge 1 \quad ,$$

the sequence is bounded.

Therefore, the above theorem implies the sequence converges

b. i) Since $a_{n+1} - a_n > 0$, the sequence is increasing.

ii) But since there is no $M, \ni |a_n| \le M$ the sequence is unbounded.

Therefore, the above theorem implies the sequence diverges.

Group Activity 1.1.3: (*They will discuss some of the questions in their respective groups and present the result for the whole group*)

1. Determine whether or not the indicated sequences are bounded, monotonic or strictly monotonic. (*Group Discussion*)

a. $\left\{\frac{(-1)^n}{n}\right\}_{n=1}^{\infty}$

b. $\{(0.09)^n\}_{n=1}^{\infty}$

c. $\left\{\frac{n+(-1)^n}{n}\right\}_{n=1}^{\infty}$

d. $\left\{\frac{n-1}{n}\right\}_{n=1}^{\infty}$

e. $\{\sqrt{n^2+1}\}_{n=0}^{\infty}$

f. $\left\{\frac{2^n}{4^n+10{,}000}\right\}_{n=5}^{\infty}$

g. $\left\{\frac{(n+1)^2}{n^2}\right\}_{n=1}^{\infty}$

h. $\{(-1)^n\sqrt{n}\}_{n=0}^{\infty}$

i. $\left\{\sin\left(\frac{\pi}{n+1}\right)\right\}_{n=0}^{\infty}$

n. $\left\{\frac{n}{e^n}\right\}_{n=1}^{\infty}$

2. State whether or not the sequence converges, if it converges, find its limit.

c. $\left\{\frac{(-1)^n}{n}\right\}_{n=1}^{\infty}$

d. $\left\{\tan\left(\frac{n\pi}{4n+3}\right)\right\}_{n=0}^{\infty}$

e. $\left\{\ln\left(\frac{2n}{5n+1}\right)\right\}_{n=1}^{\infty}$

f. $\left\{\frac{n^2}{7n^4+12}\right\}_{n=0}^{\infty}$

g. $\left\{e^{\frac{1}{\sqrt{n}}}\right\}_{n=1}^{\infty}$

h. $\left\{\sqrt{4-\frac{1}{n}}\right\}_{n=1}^{\infty}$

h.

i. $\left\{\left(4^n+5^n\right)^{\frac{1}{n}}\right\}_{n=0}^{\infty}$

j. $\{(\frac{n+3}{n+1})^n\}_{n=1}^{\infty}$

3 State whether the following sequence converges or not, if it does find the limit.

a. $\left\{\left(\frac{2}{n}\right)^n\right\}_{n=1}^{\infty}$

b. $\left\{\frac{\log_{10}^n}{n}\right\}_{n=1}^{\infty}$

c. $\left\{n^{\frac{1}{n+2}}\right\}_{n=1}^{\infty}$

d. $\left\{\int_{-n}^{n}\frac{dx}{1+x^2}\right\}_{n=0}^{\infty}$

i. $\left\{\left(1+\frac{x}{n}\right)^{5n}\right\}_{n=1}^{\infty}$

e. $\left\{\left(1-\frac{1}{n}\right)^n\right\}_{n=1}^{\infty}$

f. $\left\{\int_{0}^{n}e^{-x}dx\right\}_{n=0}^{\infty}$

g. $\left\{n^2\sin(n\pi)\right\}_{n=0}^{\infty}$

h. $\left\{\frac{5^{n+1}}{4^{2n-1}}\right\}_{n=1}^{\infty}$

4 a) For convergent sequences, if $\lim_{n\longrightarrow\infty} a_n = L$ then what is $\lim_{n\longrightarrow\infty} a_{n+1}$?

b) Assuming the sequence defined recursively by

$$a_1=\sqrt{6}, a_2=\sqrt{6+\sqrt{6}}, a_3=\sqrt{6+\sqrt{6+\sqrt{6}}},...$$

converges find its limit.

Assessment

- *Asking an answer for some of the questions.*
- *Check students participation in the group activity.*
- *Give feedback to their answers*

1.2 INFINITE SERIES

In this section we will be concerned with infinite series, which are sums that involve infinitely many terms. Since it is impossible to add up infinitely many numbers directly, one goal will be to define exactly what we mean by the sum of an infinite series and identify the basic properties about convergence and divergence of a series. However, unlike finite sums, it turns out that not all infinite series actually have a sum, so we will need to develop tools for determining which infinite series have sums and which do not. To do so it is important be familiar with the basic concepts of partial sums of infinite series and convergence and divergence properties of a series.

1.2.1 SUMS OF INFINITE SERIES

The most familiar examples of such sums occur in the decimal representations of real numbers.

For example, when we write $\frac{1}{3}$ in the decimal form $\frac{1}{3}=0.3333...$, we mean

$$\frac{1}{3}=.0.333\ldots..=0.3+0.03+0.003+0.0003+\bullet\bullet\bullet$$

which suggests that the decimal representation of $\frac{1}{3}$ can be viewed as a sum of infinitely many terms.

Definition: A sum $\sum_{n=0}^{\infty} a_n = a_0 + a_1 + a_3 + ...$ of infinitely many terms of a sequence is called an *infinite series*.

For instance,

$$\text{a.}\sum_{n=1}^{\infty}\frac{1}{3^n}=\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\frac{1}{81}\ \cdots,$$

$$\text{b.}\sum_{n=0}^{\infty}(-1)^n=1+(-1)+1+(-1)+\cdots$$

are examples of infinite series.

NB: The symbol $"\sum"$ is called sigma notation.

Sums of infinitely many terms of a sequence are defined and computed by indirect limiting process as follows.

For a sequence $\{a_k\}_{k=m}^{\infty}$,

$$\sum_{k=m}^{n} a_k = a_m + a_{m+1} + a_{m+2} + \cdots + a_n$$

is the sum of the first $(n-m)+1$ terms of the sequence.

In particular, for $m=0$,

$$\sum_{k=0}^{n} a_k = a_0 + a_1 + a_2 + \cdots + a_n,$$

which is called the nth *partial sum of the sequence,* and is usually denoted by s_n. Thus

$$s_0 = a_0 = \sum_{k=0}^{0} a_k ,$$

$$s_1 = a_0 + a_1 = \sum_{k=0}^{1} a_k ,$$

$$s_2 = a_0 + a_1 + a_2 = \sum_{k=0}^{2} a_k ,$$

$$s_3 = a_0 + a_1 + a_2 + a_3 = \sum_{k=0}^{3} a_k ,$$

$$\vdots$$

$$s_n = a_0 + a_1 + a_2 \cdots + a_n = \sum_{k=0}^{n} a_k ,$$

$$\vdots$$

For instance,

a. $s_3 = \sum_{K=0}^{3}(3k+1) = 1+4+7+10$

b. $s_5 = \sum_{K=0}^{5} 2^k = 1+2+2^2+2^3+2^4+2^5$

c. $\sum_{K=3}^{5} \frac{(-1)^k}{k!} = \frac{-1}{3!} + \frac{1}{4!} - \frac{1}{5!}$

d. $\sum_{K=3}^{5} \frac{1}{r^k} = \frac{1}{r^3} + \frac{1}{r^4} + \frac{1}{r^5}$ where r is a constant, are sequences of partial sums.

In the sequence of partial sums, if $n \to \infty, s_n$ includes more and more terms of the series and we can conclude that:

$$\lim_{n\to\infty} S_n = \sum_{n=0}^{\infty} a_n$$

Many of the functions that arise in mathematical physics and chemistry, such as Bessel functions, are defined as sums of series. For determining which infinite series have sums

and which do not, it is important be familiar with the basic concepts of convergence of infinite series.

1.2.2 Convergence and Divergence of Infinite Series

Definition: An infinite series $\sum_{n=1}^{\infty} a_n$, with the sequence of partial sum $\{s_n\}_{n=1}^{\infty}$, is said to be convergent if $\lim_{n\longrightarrow\infty} s_{.n}$ exists.

Otherwise the series diverges.

Remark: If the sequence of partial sums $\{s_n\}_{n=1}^{\infty}$ converges to L , then the series

$$\lim_{n\longrightarrow\infty} s_{.n} = \lim_{n\longrightarrow\infty} \sum_{k=0}^{n} a_k = \sum_{k=1}^{\infty} a_k = L .$$

The number L is called the sum of the series.

Example:

1. Show that the series

   a. $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$,*Known as Telescoping series* converges and find its sum.

   b. $\sum_{k=1}^{\infty} \frac{1}{k}$, *Known as Harmonic series,* diverges.

2. Determine whether the series $\sum_{k=1}^{\infty} (-1)^k$ converges or diverges (exercise!!!)

Solutions:

1. We know first write S_n in closed form that means we need an expression for S_n in which the number of terms in its expression do not vary.

   a. Since $\frac{1}{k(k+1)} = \frac{1}{K} - \frac{1}{k+1}$,by partial fractions

   we can see that:

$$s_n = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots + \frac{1}{(n-1)n} + \frac{1}{n(n+1)}$$

$$= \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n}\right) + \left(\frac{1}{n} - \frac{1}{n+1}\right)$$

$$=\frac{1}{1}-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\cdots+\frac{1}{n-1}-\frac{1}{n}+\frac{1}{n}-\frac{1}{n+1}$$

$$=1-\frac{1}{n+1}.$$

Now,

$$\lim_{n\longrightarrow\infty} s_n = \lim_{n\longrightarrow\infty}\left(1-\frac{1}{n+1}\right)=1.$$

This means that the series converges to 1 and

$$\sum_{k=1}^{\infty}\frac{1}{k(k+1)}=1.$$

b. $s_1=1$

$s_2=1+\frac{1}{2}$

$s_{2^2}=s_4=1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}$

$\geq 1+\frac{1}{2}+\underbrace{\frac{1}{4}+\frac{1}{4}}_{\frac{1}{2}}=1+2.\left(\frac{1}{2}\right)$

$s_{2^3}=s_8=1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}$

$\geq 1+\frac{1}{2}+\underbrace{\frac{1}{4}+\frac{1}{4}}_{\frac{1}{2}}+\underbrace{\frac{1}{8}+\frac{1}{8}+\frac{1}{8}+\frac{1}{8}}_{\frac{1}{2}}=1+3.\left(\frac{1}{2}\right)$

$\vdots$

$s_{2^n}\geq 1+n.\left(\frac{1}{2}\right)$

$\therefore \lim_{n\to\infty} s_{2^n}\geq \lim_{n\to\infty}\left[1+n.\left(\frac{1}{2}\right)\right]=1+\left(\frac{1}{2}\right)\lim_{n\to\infty} n=\infty$, that is, the series is not bounded above.

Thus the series diverges.

Quick check Class Exercises 1.2.1:

1. Determine whether the series converges and if so find its sum.

a) $\sum_{k=3}^{\infty}\frac{1}{(k+1)(k+2)}$ b) $\sum_{k=3}^{\infty}\frac{1}{k^2-k}$

Instructor's Role:

- *Check their answers and Give feedback for their answers*

One important example of an infinite series is the geometric series which is useful for expressing repeating decimals as fractions.

Definition: - A series of the form $\sum_{n=m}^{\infty} c \cdot r^{n}$,

where r and c are constants and $c \neq 0$, is called a *geometric series***.**

Theorem 1.2. 1: Let r be a real number and $c \neq 0$. Then the geometric series

$$\sum_{n=m}^{\infty} c\, r^{n} = \begin{cases} \dfrac{c\, r^{m}}{1-r} & \text{if } |r| < 1 \\ \text{diverges if } |r| \geq 1 \end{cases}$$

Proof: To be discussed in the class

Note that the number *r* in the above theorem is called the *ratio* of the geometric series.

Example:

1. Determine the convergence or divergence of the following series.

   a. $\sum_{n=2}^{\infty} \left(\frac{4}{7}\right)^{n}$

   b. $\sum_{n=2}^{\infty} 23(0.7)^{n}$

Solutions:

1.

a. Taking $c = 1,\ r = \frac{4}{7}$ *and* $m = 2$, we have

$$\sum_{n=2}^{\infty} \left(\frac{4}{7}\right)^{n} = \frac{\left(\frac{4}{7}\right)^{2}}{1-\frac{4}{7}} = \frac{16}{21}.$$

b. Taking $c = 23,\ r = 0.7$ *and* $m = 2$, we have

$$\sum_{n=2}^{\infty} 23(0.7)^{n} = \frac{23.(0.7)^{2}}{1-0.7} = 7.889.$$

Geometric series allows us to express any repeating decimal as an infinite series and hence as a rational number.

Examples: a) $0.333\ldots. = 0.3 + 0.03 + 0.003 + \cdots$

$$= 3(\tfrac{1}{10})^1 + 3(\tfrac{1}{10})^2 + 3(\tfrac{1}{10})^3 + 3(\tfrac{1}{10})^4 + \cdots$$

$$= \sum_{n=1}^{\infty} 3\left(\frac{1}{10}\right)^n = \frac{3\left(\frac{1}{10}\right)}{1-\frac{1}{10}} = \frac{1}{3}$$

b) $0.45454545\ldots. = 0.45 + 0.0045 + 0.000045 + \ldots.$

$$= \sum_{n=1}^{\infty} 45\left(\frac{1}{100}\right)^n = \frac{45\left(\frac{1}{100}\right)}{1-\frac{1}{100}} = \frac{45}{99}$$

Quick check Class Exercises 1.2.2:

1.Find the rational number represented by the following repeating decimals

a) 0.99999.....

b) 5.373737.....

c) 0.44444.....

d) 0.451141414.....

2. Suppose that a ball dropped from a height h hits the floor and rebounds to a height proportional to h, that is, to the height αh (assume $\alpha < 1$). It then falls from the height αh, hits the floor, and rebounds to the height $\alpha(\alpha(h)) = \alpha^2 h$, and so on. Find the total distance traveled by the ball.(Exercise)

Teachers' role:

- *Observe while they work and answer for the raised questions.*
- *Check and give feedback for their answers*

Theorem 1.2.2: If the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge, then

i. $\sum_{n=1}^{\infty} (a_n + b_n)$ converges and

$$\sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$$

ii. For a constant α, $\sum_{n=1}^{\infty} \alpha\, a_n$ converges and

$$\sum_{n=1}^{\infty} \alpha . a_n = \alpha . \sum_{n=1}^{\infty} a_n$$

Proof: Exercise.

Example 1: - Show that the series $\sum_{n=1}^{\infty}\left(\frac{5}{3^n} - \frac{7}{n(n+1)}\right)$ converges, and find its sum.

Solutions: Since

$$\sum_{n=1}^{\infty}\frac{5}{3^n} = \sum_{n=1}^{\infty}5\left(\frac{1}{3}\right)^n = \frac{5\left(\frac{1}{3}\right)}{1-\frac{1}{3}} = \frac{5}{2},$$

and

$$\sum_{n=1}^{\infty}\frac{7}{n(n+1)} = 7,$$

then

$$\sum_{n=1}^{\infty}\left(\frac{5}{3^n} - \frac{7}{n(n+1)}\right) = \sum_{n=1}^{\infty}\frac{5}{3^n} - \sum_{n=1}^{\infty}\frac{7}{n(n+1)} = \frac{5}{2} - 7 = -\frac{9}{2}$$

2. Find the sum of the following series

a) $\sum_{k=0}^{\infty}\frac{3^k + 4^k}{5^k}$ b) $\sum_{k=0}^{\infty}\frac{2^{k+3}}{3^k}$

Solution:

a. Since, $\frac{3^k + 4^k}{5^k} = \left(\frac{3}{5}\right)^k + \left(\frac{4}{5}\right)^k$

$$\sum_{k=0}^{\infty}\frac{3^k + 4^k}{5^k} = \sum_{k=0}^{\infty}\left(\frac{3}{5}\right)^k + \sum_{k=0}^{\infty}\left(\frac{4}{5}\right)^k$$

$$= \frac{1}{1-\frac{3}{5}} + \frac{1}{1-\frac{4}{5}}$$

$$= \frac{5}{2} + 5 = \frac{15}{2}$$

b. We have $\sum_{k=0}^{\infty}\frac{2^{k+3}}{3^k} = \sum_{k=0}^{\infty}2^3\left(\frac{2}{3}\right)^k = \frac{8}{1-\frac{2}{3}} = 24$

Remark: (Change of Base):

For a series $\sum_{n=m}^{\infty}a_n$, let $b_n = a_{m+n}, n \geq 0$, $\sum_{n=m}^{\infty}a_n = \sum_{n=0}^{\infty}b_n$

Example: - For the series $\sum_{n=5}^{\infty}\frac{1}{n!,}$, $\sum_{n=5}^{\infty}\frac{1}{n!} = \sum_{n=0}^{\infty}\frac{1}{(n+5)!}$

Theorem 1.2.3: Let m be a positive integer. The series $\sum_{n=0}^{\infty} a_n$ converges if and only if the series $\sum_{n=m}^{\infty} a_n$ converges. Moreover, if $\sum_{n=0}^{\infty} a_n = L$, then

$$\sum_{n=m}^{\infty} a_n = L - (a_0 + a_1 + a_2 \cdots + a_{m-1}); \text{ Or}$$

If $\sum_{n=m}^{\infty} a_n = M$, then

$$\sum_{n=0}^{\infty} a_n = a_0 + a_1 + a_2 \cdots + a_{m-1} + M$$

Remark:

i. Notice that the convergence or divergence of an infinite series is not affected by where you start the summation.

ii. From the above theorem ; if the series is convergent, then the sum *does depend* on where you begin the summation.

Example: Observe that $\sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n = 4$,

but

$$\sum_{n=3}^{\infty} \left(\frac{3}{4}\right)^n = \frac{27}{16}.$$

Theorem 1.2.4: If $\sum_{n=1}^{\infty} a_n$ converges, then

$$\lim_{n\to\infty} a_n = 0.$$

Proof: By using the sequence $\{a_n\}_{n=0}^{n}$, $s_n = a_1 + a_2 + \cdots + a_{n-1} + a_n$ and

$s_{n-1} = a_1 + a_2 + \cdots a_{n-1}$.

Since the series $\sum_{n=1}^{\infty} a_n$ converges and

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty} s_{n-1} = \sum_{n=1}^{\infty} a_n$$

Therefore

$$a_n = s_n - s_{n-1}$$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty}(s_n - s_{n-1})$$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} s_n - \lim_{n\to\infty} s_{n-1}$$

$$\lim_{n\to\infty} a_n = 0$$

Remark:

1. The contra positive of the above theorem is important, that is, if $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ diverges (*sometimes called divergence test*).

   For instance,

   a. Since

   $$\lim_{n\to\infty} \frac{n}{n+1} = 1 \neq 0,$$

   the series $\sum_{n=1}^{\infty} \frac{n}{n+1}$ diverges.

   b. Since

   $$\lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n = e \neq 0,$$

   the series $\sum_{n=1}^{\infty} \left(1+\frac{1}{n}\right)^n$ diverges.

2. The converse of the above theorem is false, that is, "If $\lim_{n\to\infty} a_n = 0$, then the series $\sum_{n=1}^{\infty} a_n$ converges" is false.

   For instance, $\lim_{n\to\infty} \frac{1}{n} = 0$, but the series

   $$\sum_{n=1}^{\infty} \frac{1}{n}$$ a divergent harmonic series.

Quick check Class Exercises 1.2.3: Test for divergence of the following series.

a) $\sum_{n=1}^{\infty}(1+\frac{1}{n})$ b) $\sum_{n=1}^{\infty} n\sin\frac{1}{n}$

Instructor's Role

- *Observe while they work*
- *Check and give feedback for their answers*

Group Activity 1.2.1. *(Group Discussion and assignment)*

1. Find the sum of the following series, if it converges.

   a. $1+\frac{3}{2}+\frac{9}{4}+\frac{27}{8}+\frac{81}{16}+\cdots$

   b. $\sum_{k=0}^{\infty}\frac{3}{10^k}$

   c. $\sum_{n=0}^{\infty}\frac{1-2^n}{3^n}$

   d. $\sum_{n=0}^{\infty}\left(\frac{25}{100^n}-\frac{6}{100^n}\right)$

   e. $\sum_{n=0}^{\infty}\frac{(-1)^n}{5^n}$

   f. $\sum_{n=1}^{\infty}\ln\left(\frac{n+1}{n}\right)$

2. Express the following decimals as an infinite series and find its sum if it converges.

   a. $0.5555555\cdots$

   b. $0.898989\cdots$

   c. $12.273273273\cdots$

   d. $0.112112112\cdots$

   e. $0.314231423142\cdots$

   f. $0.62454545\cdots$

3. Show that:

   a. $\sum_{k=0}^{\infty}(-1)^k x^k=\frac{1}{1+x},\ |x|<1.$

   b. $\sum_{k=0}^{\infty}(-1)^k x^{2k}=\frac{1}{1+x^2},\ |x|<1.$

4. Find a series expansion for the given expression.

   a. $\frac{x}{1+x^2}$ *for* $|x|<1.$

   b. $\frac{x}{1-x}$ *for* $|x|<1.$

   c. $\frac{x}{1+x}$ *for* $|x|<1.$

5. Let $\{d_k\}_1^{\infty}$ be a sequence of real numbers that converges to 0. Show that $\sum_{k=1}^{\infty}(d_k-d_{k+1})=d_1$

6. Prove that the series $\sum_{k=1}^{\infty}(a_{k+1}-a_k)$ converges if and only if the sequence $\{a_k\}_1^{\infty}$ converges.

Assessment

- *Asking an answer for some of the questions*
- *Give feedback to answers*
- *Check students participation in the group activity*

1.3 Tests for Convergence of Non-Negative Term Series

Unlike finite sums, it turns out that not all infinite series actually have a sum as seen in the previous section, so we will need to develop tools for determining which infinite series have sums and which do not. So in this section we will define Non-negative term series and discuss some techniques (tests) for determining their convergence and divergence.

Definition: A series $\sum_{n=m}^{\infty} a_n$ is said to be a non-negative terms series if and only if $a_n \geq 0, \forall n \geq m$.

Remark: For a positive term series $\sum_{n=m}^{\infty} a_n$, it holds that

$$s_m < s_{m+1} < s_{m+2} < \ldots < s_j < \ldots$$

That is, *the sequence of partial sum* $\{s_j\}_m^{\infty}$ *is an increasing.*

Theorem 1.3.1 : A series with non-negative terms converges if and only if its sequence of partial sums is bounded.

Proof: Exercise

I. The Integral Test

Theorem 1.3.2 : (***The Integral Test***)

If f is continuous, decreasing and positive on $[m, \infty)$, then the series $\sum_{k=m}^{\infty} f(k)$ converges iff $\int_m^{\infty} f(x)dx$ converges, where $f(k) = a_k$.

Proof: *Reading Assignment*

Examples:

1. Show that $\sum_{k=1}^{\infty}\frac{\ln k}{k}$ diverges.

Solution:

Let $f(x)=\frac{\ln x}{x}\Rightarrow f'(x)=\frac{1-\ln x}{x^2}\le 0,\forall x\ge 1$. So $f(x)$ is positive, decreasing on $[1,\infty)$ and since

$$\int_1^{\infty} f(x)dx=\lim_{t\to\infty}\int_1^t\frac{\ln x}{x}dx=\lim_{t\to\infty}\left((\ln x)^2\right)_1^t=\lim_{t\to\infty}\left[(\ln t)^2-(\ln 1)^2\right]$$

$$=\lim_{t\longrightarrow\infty}(\ln t)^2=\infty$$

Therefore by Integral Test $\sum_{k=1}^{\infty}\frac{\ln k}{k}$ diverges.

2. Show that The *Harmonic Series,* $\sum_{k=1}^{\infty}\frac{1}{k}$, diverges.

Solution:

Let $f(x)=\frac{1}{x}$. Clearly f is continuous, decreasing and positive on $[1,\infty)$, and since

$$\int_1^{\infty} f(x)dx=\lim_{t\longrightarrow\infty}\int_1^t\frac{1}{x}\,dx=\lim_{t\longrightarrow\infty}\ln x\Big|_{x=1}^{t}=\lim_{t\longrightarrow\infty}(\ln t-\ln 1)$$

$$=\lim_{t\longrightarrow\infty}\ln t=\infty$$

Therefore, the improper integral $\int_1^{\infty} f(x)dx$ diverges. Thus,the series $\sum_{k=1}^{\infty}\frac{1}{k}$ diverges.

Theorem 1.3.3: The *P-series,*

$$\sum_{k=1}^{\infty}\frac{1}{k^p}=1+\frac{1}{2^p}+\frac{1}{3^p}+\frac{1}{4^p}+\cdots,$$

converges if and only if $p>1$ and diverges otherwise.

Proof: Let $f(x)=\frac{1}{x^p}$ .Clearly f is continuous, decreasing and positive on $[1,\infty)$, and

$$\int_1^{\infty} f(x)dx=\lim_{t\longrightarrow\infty}\int_1^t\frac{1}{x^p}dx=\lim_{t\to\infty}\left(\frac{x^{1-p}}{1-p}\right)_{x=1}^{x=t}=\lim_{t\to\infty}\frac{1}{1-p}\left[t^{1-p}-1\right]$$

$$=\begin{cases}\dfrac{1}{p-1} & \text{if } p>1\\ \infty, & \text{if } p\le 1\end{cases}.$$

Then, by the above theorem, $\sum_{k=1}^{\infty}\frac{1}{k^p}$ converges for $p>1$ and diverges otherwise.

The integral test is most effective when the function to be used is easily integrated.

Quick check Class Exercises 1.3.1: Use the integral test to determine whether the following series converge or diverge.

a) $\sum_{k=1}^{\infty}\frac{1}{k\ln k}$

b) $\sum_{k=1}^{\infty}\frac{\tan^{-1}k}{1+k^2}$

c) $\sum_{k=1}^{\infty}\frac{1}{(4+2k)^{\frac{3}{2}}}$

d) $\sum_{k=1}^{\infty}\frac{k}{1+k^2}$

Instructor's Role:

- *Observe while they work and answer for the raised questions.*
- *Check and give feedback for their answers*

II. The Basic Comparison Test

Theorem 1. 3.4: **(The Basic Comparison Test*)***

Let $\sum a_n$ and $\sum b_n$ be a series with non-negative terms and Suppose $0 \le a_n \le b_n$ for some $n \ge N$. Then

i. If $\sum b_n$ converges then $\sum a_n$ converges.

ii. If $\sum a_n$ diverges then $\sum b_n$ diverges.

Proof: Exercise.

Remarks:

1. If $0 \le a_n \le b_n$ for sufficiently large n, then the series $\sum a_n$ is said to be dominated by $\sum b_n$.
2. Every infinite series dominated by a convergent series is also convergent.
3. There are two steps required for using the comparison test to determine whether a series u_k with positive terms converges:

Step 1. Guess at whether the series $\sum u_k$ converges or diverges.

Step 2. Find a series that proves the guess to be correct. That is, if we guess that $\sum u_k$ diverges, we must find a divergent series whose terms are "smaller" than the

corresponding terms of $\sum u_k$ and if we guess that $\sum u_k$converges, we must find a convergent series whose terms are "bigger" than the corresponding terms**.**

Examples. Determine whether the following series converges or not.

a. $\sum_{k=1}^{\infty}\frac{1}{2k^3+1}$

b. $\sum_{k=1}^{\infty}\frac{1}{3k+1}$

c. $\sum_{k=1}^{\infty}\frac{1}{3^{-k}-1}$

Solution:

a.Since

$\frac{1}{2k^3+1} \le \frac{1}{k^3}$, for all positive integer k, and

$\sum_{k=1}^{\infty}\frac{1}{k^3}$ converges *(p -series with* $p=3>1$*)* then, the series $\sum_{k=1}^{\infty}\frac{1}{2k^3+1}$ converges by the basic comparison test.

b.Since

$\frac{1}{3k+1} \ge \frac{1}{3(k+1)}$, for all positive integer $k \ge 1$, and

$\sum_{k=1}^{\infty}\frac{1}{3(k+1)}$ diverges by integral test, then the series $\sum_{k=1}^{\infty}\frac{1}{3k+1}$ diverges by the basic comparison test.

c. Since

$\frac{1}{3^{-k}-1} \ge \frac{1}{3^{-k}}, = 3^k$ for all positive integer $k \ge 1$, and

$\sum_{k=1}^{\infty}3^k$ is divergent geometric series, then the series $\sum_{k=1}^{\infty}\frac{1}{3^{-k}-1}$ diverges by comparison test.

Quick check Class activity 1.3.2 :

1. Use the comparison test to determine whether the following series converge or diverge.

a) $\sum_{k=1}^{\infty}\frac{1}{2^k+1}$

b) $\sum_{k=1}^{\infty}\frac{1}{\ln k}$

Instructor's Role:

- *Check their answers and Give feedback for their answers*

III. The Limit Comparison Test

Theorem 1.3.5: (The Limit Comparison Test)

Let $\sum_{n=m}^{\infty} a_n$ and $\sum_{n=m}^{\infty} b_n$ be series with positive terms. If

$$\lim_{k\to\infty} \frac{a_k}{b_k} = L,$$

where L is some positive number, then either both series converge or both series diverge.

Proof: Exercise

Example: Determine whether the following series converges or not.

a) $\sum_{k=1}^{\infty} \sin\frac{\pi}{k}$ b) $\sum_{k=1}^{\infty} \frac{3k^2+2k+1}{k^3+1}$ c) $\sum_{k=1}^{\infty} \frac{1}{\sqrt[3]{8k^2-5k}}$

Solution: a. Here $a_k = \sin\frac{\pi}{k}$.

Let $b_k = \frac{\pi}{k}$, since

$$\sum_{k=1}^{\infty} \frac{\pi}{k} = \pi\sum_{k=1}^{\infty} \frac{1}{k} \text{ diverges,}$$

and

$$\lim_{k\to\infty}\left(\frac{\sin\frac{\pi}{k}}{\frac{\pi}{k}}\right) = 1 > 0,$$

then the series: $\sum_{k=1}^{\infty} \sin\frac{\pi}{k}$ diverges.

b. Here, we have $a_k = \frac{3k^2+2k+1}{k^3+1}$. Taking only terms with the highest power of k both in the numerator and denominator choose

$$b_k = \frac{3k^2}{k^3} = \frac{3}{k} \text{ and since}$$

$\sum_{k=1}^{\infty} \frac{3}{k} = 3\sum_{k=1}^{\infty} \frac{1}{k}$ diverges (a constant times divergent p *-series with* $p=1$) and

$$\lim_{k\to\infty}\frac{a_k}{b_k}=\lim_{k\to\infty}\left(\frac{3k^2+2k+1}{k^3+1}\right)\left(\frac{k}{3}\right)=1>0$$

By Limit comparison test the series $\sum_{k=1}^{\infty}\frac{3k^2+2k+1}{k^3+1}$ also diverges.

c. Here $a_k=\frac{1}{\sqrt[3]{8k^2-5k}}$.

Taking only terms with the highest power of k both in the numerator and denominator choose

$$b_k=\frac{1}{\sqrt[3]{8k^2}}=\frac{1}{2k^{\frac{2}{3}}}$$

$\sum_{k=1}^{\infty}\frac{1}{2k^{\frac{2}{3}}}=\frac{1}{2}\sum_{k=1}^{\infty}\frac{1}{k^{\frac{2}{3}}}$ diverges(a constant times divergent *p -series with* $p=\frac{2}{3}<1$) and

$$\lim_{k\to\infty}\frac{a_k}{b_k}=\lim_{k\to\infty}\left(\frac{8k^2}{8k^2-5k}\right)^{\frac{1}{3}}=1>0$$

By Limit comparison test the series $\sum_{k=1}^{\infty}\frac{1}{\sqrt[3]{8k^2-5k}}$ also diverges.

Remark: It is often important to apply these two informal principles to help with guessing in the first step of Comparison tests:

i) Constant terms in denominator of a_n can usually be deleted without affecting the convergence or divergence

ii) If a polynomial in n appears as a factor in the numerator or denominator of , all but the highest power of n in the polynomial may usually be deleted without affecting convergence or divergence of the series

Quick check Class activity 1.3.3 :

1. Use the limit comparison test to determine whether the following series converge or diverge.

a) $\sum_{k=1}^{\infty}\frac{1}{2k^2-k}$ b) $\sum_{k=1}^{\infty}\frac{3k^3-2k^2+4}{k^5-k^3+k^2}$ c) $\sum_{k=1}^{\infty}\frac{1}{\sqrt{k^3+2k}}$

Instructor's Role

- *Observe while they work*
- *Check their answers and give feedback to their answers*

Group Activity 1.3.1

1. Determine whether or not the following series converges

(*Group Discussion*)

a. $\sum_{k=1}^{\infty} \frac{k}{k^3+1}$

b. $\sum_{k=0}^{\infty} \frac{1}{3k+2}$

c. $\sum_{k=1}^{\infty} \frac{1}{(2k+1)^2}$

d. $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k+1}}$

e. $\sum_{k=0}^{\infty} \frac{1}{k^2+1}$

f. $\sum_{k=1}^{\infty} \left(\frac{5}{2}\right)^{-k}$

g. $\sum_{k=1}^{\infty} \frac{\ln k}{k^3}$

h. $\sum_{k=1}^{\infty} \frac{2}{k(\ln k)^2}$

i. $\sum_{k=1}^{\infty} \frac{1}{1+2\ln k}$

j. $\sum_{k=1}^{\infty} \frac{\ln \sqrt{k}}{k}$

k. $\sum_{k=1}^{\infty} \frac{2k-1}{\sqrt{k^4+1}}$

l. $\sum_{k=1}^{\infty} \frac{k^4-1}{3k^2+5}$

m. $\sum_{k=0}^{\infty} k\, e^{-k^2}$

n. $\sum_{k=1}^{\infty} \frac{1}{k\ln(k+1)}$

o. $\sum_{k=1}^{\infty} \frac{k^3}{k^5+5k^4+7}$

p. $\sum_{k=1}^{\infty} \frac{5\sqrt{k}+100}{2k^2\sqrt{k}+9\sqrt{k}}$

q. $\sum_{k=1}^{\infty} \frac{1}{1+2+3+\cdots+k}$

Assessment:

- *Asking an answer for some of the questions*
- *Check students participation in the group activity.*
- *Answer for the raised questions*

The comparison test and the limit comparison test hinge on first making a guess about convergence and then finding an appropriate series for comparison, both of which can be difficult tasks in cases where the two informal principles cannot be applied. In such cases the next tests can often be used.

IV. The Root Test

Theorem 1.3.6: (***The Root Test)***

Let $\sum_{n=m}^{\infty} a_n$ be a series with non-negative terms such that

$\lim_{n\to\infty}(a_n)^{\frac{1}{n}} = L$ (Possibly ∞). Then

i. If $L<1$, then $\sum_{n=m}^{\infty} a_n$ converges

ii. If $L>1$, then $\sum_{n=m}^{\infty} a_n$ diverges

iii. If $L=1$, then the test is inconclusive; the series may either converge or diverge.

Example: Determine the convergence or divergence of the following Series

a. $\sum_{k=1}^{\infty}\left(1-\frac{1}{k}\right)^k$ b. $\sum_{k=2}^{\infty}\frac{1}{(\ln k)^k}$

Solution

a. Since

$$\lim_{k\to\infty}(a_k)^{\frac{1}{k}} = \lim_{k\to\infty}\left(1-\frac{1}{k}\right)=1,$$

then the root test is inconclusive.

However, since

$\lim_{k\to\infty}\left(1-\frac{1}{k}\right)^k = \frac{1}{e}\neq 0$, then the series diverges.

b. Since

$$\lim_{k\to\infty}(a_k)^{\frac{1}{k}} = \lim_{k\to\infty}\left(\frac{1}{\ln k}\right)=0<1,$$

then by root test the series $\sum_{k=2}^{\infty}\frac{1}{(\ln k)^k}$ converges.

It is often advisable to try root test first when the terms in the series are power of k.

Quick check Class activity 1.3.4:

1. Use the Root test to determine whether the following series converge or diverge.

a) $\sum_{k=2}^{\infty}\left(\frac{4k-5}{2k+1}\right)^k$ b) $\sum_{k=1}^{\infty}\left(\frac{1}{\ln(k+1)}\right)^k$ c) $\sum_{k=2}^{\infty}\left(\frac{1}{\ln k}\right)^k$

Instructor's Role

- *Observe while they work*
- *Check their answers and give feedback to their answers*

V. The Ratio Test

Theorem 1.3.7: (*The Ratio Test)*

Let $\sum_{k=m}^{\infty} a_k$ be a series with non-negative terms such that

$\lim_{k\to\infty}\frac{a_{k+1}}{a_k} = L$ (Possibly ∞).Then

i. If $L<1$, then $\sum_{k=m}^{\infty} a_n$ converges.

ii. If $L>1$, then $\sum_{k=m}^{\infty} a_n$ diverges.

iii. If $L=1$, then the test is inconclusive; the series may either converge or diverge.

Example: Determine the convergence or divergence of the following

a. $\sum_{k=0}^{\infty}\frac{1}{k!}$ b. $\sum_{k=0}^{\infty}\frac{k^k}{k!}$

Solution

a. Since

$$\lim_{k\to\infty}\frac{a_{k+1}}{a_k} = \lim_{k\to\infty}\frac{1}{k+1} = 0 < 1,$$

then the series converges.

b. Since

$$\lim_{k\to\infty}\frac{a_{k+1}}{a_k} = \lim_{k\to\infty}\frac{(k+1)^k}{k^k} = \left(1+\frac{1}{k}\right)^k = e > 1,$$

then the series $\sum_{k=0}^{\infty}\frac{k^k}{k!}$ diverges.

It is often advisable to use ratio test when the terms in the series involves factorials and powers of k.

Quick check Class activity 1.3.4:

1. Use the Ratio test to determine whether the following series converge or diverge.

a) $\sum_{k=0}^{\infty}\frac{2^k}{k!}$

b) $\sum_{k=1}^{\infty}\frac{2^k}{k^2}$

c) $\sum_{k=1}^{\infty}\frac{k!}{k^k}$

d) $\sum_{k=1}^{\infty}\frac{k!}{(k+2)!}$

Instructor's Role

- *Check their answers and give feedback to their answers*

Group Activity 1.3.2

1. Determine whether the series converges or diverges

(*Group Discussion*)

a. $\sum_{k=0}^{\infty}\frac{10^k}{k!}$

b. $\sum_{k=1}^{\infty}\frac{1}{k.2^k}$

c. $\sum_{k=1}^{\infty}\frac{1}{k^k}$

d. $\sum_{k=0}^{\infty}\left(\frac{4}{2k+1}\right)^k$

e. $\sum_{k=0}^{\infty}\frac{k!}{100^k}$

f. $\sum_{k=1}^{\infty}\frac{(\ln k)^2}{k}$

g. $\sum_{k=2}^{\infty}\frac{1}{(\ln k)^{10}}$

h. $\sum_{k=0}^{\infty}\frac{1}{1+\sqrt{k}}$

i. $\sum_{k=1}^{\infty}\frac{2k+\sqrt{k}}{k^3+\sqrt{k}}$

j. $\sum_{k=1}^{\infty}\frac{k!}{10^{4k}}$

k. $\sum_{k=1}^{\infty}\left(\frac{k}{k+100}\right)^k$

l. $\sum_{k=1}^{\infty}\frac{\ln k}{k^2}$

m. $\sum_{k=1}^{\infty}\frac{(2k+1)^{2k}}{(5k^2+1)^k}$

n. $\sum_{k=0}^{\infty}\frac{k!(2k)!}{(3k)!}$

o. $\sum_{k=1}^{\infty}\frac{2^k}{k^3}$

p. $\sum_{k=1}^{\infty}\frac{k!}{1\cdot 3\cdot\cdots\cdot(2k-1)}$

2. Let r be a positive number. Prove that the sequence $\left\{\frac{r^n}{n!}\right\}_{n=0}^{\infty}$ has limit 0.

3. Let $\{a_k\}$ be a sequence of positive numbers and take $r > 0$. By using the *root test* show that, if $\lim_{k\to\infty}(a_k)^{\frac{1}{k}} = L$ and $L < \frac{1}{r}$, then $\sum a_k r^k$ converges.

Assessment:

- *Asking an answer for some of the questions*
- *Check students' participation in the group activity.*

1.4 Alternating Series; Absolute and Conditional Convergence

Definition: A series of the form

$$\sum_{k=1}^{\infty}(-1)^{k+1}a_k = a_1 - a_2 + a_3 - a_4 + \cdots (-1)^{k+1}a_k + \cdots, \text{ or}$$

$$\sum_{k=1}^{\infty}(-1)^{k}a_k = -a_1 + a_2 - a_3 + a_4 + \cdots (-1)^{k}a_k + \cdots$$

where each $a_k > 0$, having alternatively positive and negative terms, is called an *alternating series.*

Examples: a. $\sum_{k=1}^{\infty}(-1)^{k+1} = 1-1+1-1+\ldots$ is an alternating series.

b. $\sum_{k=1}^{\infty}(-1)^{k}\frac{1}{2k+1} = -\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\frac{1}{9}+\ldots$ is also an alternating series.

Theorem 1.4.1: (**Alternating Series Test)**

Suppose the alternating series $\sum_{k=1}^{\infty}(-1)^{k+1}a_k$ satisfies the conditions

1. $a_{k+1} \le a_k, \forall k \in N$, that is, the sequence $\{a_k\}_1^{\infty}$ is decreasing, and
2. $\lim_{k\to\infty} a_k = 0$,

then the series $\sum_{k=1}^{\infty}(-1)^{k+1}a_k$ converges.

Proof: *Reading Assignment*

Examples: Determine the convergence or divergence of the following series

a. $\sum_{k=1}^{\infty}\frac{(-1)^k}{k}$ b. $\sum_{k=2}^{\infty}\frac{(-1)^k}{k\ln k}$

Solution

a. Since $a_k = \frac{1}{k}$, $a_{k+1} = \frac{1}{k+1}$,

such that $\frac{1}{k+1} < \frac{1}{k}$, for all positive integer k, and $\lim_{k\to\infty} \frac{1}{k} = 0$,

then the series $\sum_{k=1}^{\infty} \frac{(-1)^k}{k \ln k}$ converges.

b. Since

$$a_k = \frac{1}{k \ln k}, \qquad a_{k+1} = \frac{1}{(k+1)\ln(k+1)},$$

$$\Rightarrow a_k = \frac{1}{k \ln k} < a_{k+1} = \frac{1}{(k+1)\ln(k+1)}, \text{ for all positive integer } k, \text{ and}$$

$\lim_{k\to\infty} \frac{1}{k \ln k} = 0$, then the series $\sum_{k=1}^{\infty} \frac{(-1)^k}{k}$ converges.

Notice that:

i. If $S_1 > 0$ then $S_1 = a_1 > 0$, $S_2 = a_1 - a_2 < 0$, $S_3 = a_1 - a_2 + a_3 < 0$ and so on.

So if $S_1 > 0$, then $S_1 > S_2 < S_3 > S_4 < S_5 \ldots$.

ii. Again if $S_1 < 0$, then $S_1 < S_2 > S_3 < S_4 > S_5 \ldots$.

Theorem 1.4.2 (Approximating Sums of an Alternating Series):

If an alternating series satisfies the hypotheses of the alternating series test, and if S is the sum of the series, then:

i) S lies between any two successive partial sums that is

$$S_n \le S \le S_{n+1}$$

or

$$S_{n+1} \le S \le S_n$$

depending on which partial sum is larger.

ii) If S is approximated by S_n, then the absolute error $|S - S_n| < a_{n+1}$.

Moreover, the sign of the error $S - S_n$ is the same as that of a_{n+1}.

Quick Check Class Exercises 1.4.1:(*Group Work*)

1. Use the Alternating Series test to determine whether the series converge or diverge.

a) $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{k+3}{k(k+1)}$ b) $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{1}{e^k}$

2. Assuming that $\sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k}=\ln 2$,

a) Find a partial upper bound on the magnitude of the error that results if $ln2$ is approximated by the sum of the first eight terms of the series.

b) Find a particular sum that approximates $ln2$ to one decimal place accuracy (the nearest tenth).

Instructor's role

- *Observe while they work*
- *Answer for the raised questions.*
- *Check and give feedback to their answers*

Definition: A series $\sum_{k=m}^{\infty}a_k$ is said to be absolutely convergent if the series obtained by using the absolute value of the terms,

$$\sum_{k=m}^{\infty}|a_k|=|a_m|+|a_{m+1}|+|a_{m+2}|\cdots,$$

converges and diverges absolutely if the series of absolute values diverges.

Example:

1. Show that the series $\sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k^2}$ is absolutely convergent.

2. Show that $\sum_{k=1}^{\infty}(-1)^{k+1}\frac{1}{k}$ diverges absolutely.

Solution:

1.Since the series of absolute values is

$$\sum_{k=1}^{\infty}\left|\frac{(-1)^{k+1}}{k^2}\right|=\sum_{k=1}^{\infty}\frac{1}{k^2},$$

which is convergent (p-series with $p=2>1$), the series $\sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k^2}$ converges absolutely.

2. Since the series of absolute values becomes

$$\sum_{k=1}^{\infty}\left|\frac{(-1)^{k+1}}{k}\right|=\sum_{k=1}^{\infty}\frac{1}{k}$$

which is divergent harmonic series. So the given series diverges.

Theorem 1.4.3: Every absolutely convergent series $\sum_{k=m}^{\infty} a_k$ is convergent, that is,

If $\sum_{k=m}^{\infty} |a_k|$ converges, then so does $\sum_{k=m}^{\infty} a_k$.

Examples: Show that the following series converges

a) $\sum_{k=1}^{\infty} \frac{\sin k}{k^2}$ b) $\sum_{k=1}^{\infty} \frac{\cos k}{k^3}$ c) $\sum_{k=0}^{\infty} (-1)^k \frac{1}{2^k}$

Solution: For a-c we have no convergence test that can be applied directly but since all of them converge absolutely then we can conclude that they are convergent.

a) Since

$0 \le \left|\frac{\sin k}{k^2}\right| \le \frac{1}{k^2}$ and $\sum_{k=1}^{\infty} \frac{1}{k^2}$ is convergent P-series,then

$\sum_{k=1}^{\infty} \left|\frac{\sin k}{k^2}\right|$ converges by the basic comparison test.

Thus $\sum_{k=1}^{\infty} \frac{\sin k}{k^2}$ is absolutely convergent hence converges by the above theorem.

b) Similarly done as a.

c) Since the series of absolute values is

$$\sum_{k=1}^{\infty} |a_k| = \sum_{k=0}^{\infty} \left|\frac{(-1)^k}{2^k}\right| = \sum_{k=0}^{\infty} \frac{1}{2^k}$$

which is convergent geometric series thus $\sum_{k=0}^{\infty} (-1)^k \frac{1}{2^k}$ converges absolutely.

Hence, converges by the above theorem.

Remark: If a series diverges absolutely it may converge or diverge.

For example, $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{1}{k}$ diverges absolutely but converges by alternating series test.

As a consequence, we have the following definition:

Definition: If $\sum_{k=m}^{\infty} a_k$ converges, but $\sum_{k=m}^{\infty} |a_k|$ diverges then $\sum_{k=m}^{\infty} a_k$ is called conditionally convergent.

Examples:

a) $\sum_{k=1}^{\infty}(-1)^{k+1}\frac{1}{k}$, is conditionally convergent as stated in remark above.

b) $\sum_{k=1}^{\infty}\frac{\cos k\pi}{k}=-1+\frac{1}{2}-\frac{1}{3}+\frac{1}{4}....$ is conditionally convergent because,

$$\sum_{k=1}^{\infty}\frac{\cos k\pi}{k}=-1+\frac{1}{2}-\frac{1}{3}+\frac{1}{4}....=\sum_{k=1}^{\infty}(-1)^{k}\frac{1}{k}$$

is convergent alternating series but, the series of absolute values becomes

$$\sum_{k=1}^{\infty}\left|\frac{\cos k\pi}{k}\right|=\sum_{k=1}^{\infty}\frac{1}{k}$$

which is the divergent harmonic series.

Quick check Class activity 1.4.2: Classify as absolutely convergent or conditionally convergent.

a) $\sum_{k=1}^{\infty}(-1)^{k}\frac{k!}{(2k)!}$

b) $\sum_{k=1}^{\infty}\frac{\cos k}{k^{2}}$

c) $\sum_{n=1}^{\infty}(-1)^{n}\frac{(\ln n)^{2}}{n}$

d) $\sum_{n=1}^{\infty}\frac{\cos n\pi}{\sqrt{n}}$

Instructor's Role

- *Observe while they work*
- *Answer for the raised questions.*
- *Check and give feedback to their answers*

Group Activity 1.4.1

1. Test the series for
   i. absolutely convergence,
   ii. Conditionally convergence.

a. $\frac{1}{4}-\frac{1}{6}+\frac{1}{8}-\frac{1}{10}+\cdots+\frac{(-1)^{k}}{2k}+\cdots$

b. $\sum_{k=1}^{\infty}(-1)^{k}\frac{\ln k}{k}$

c. $\sum_{k=1}^{\infty}(-1)^{k}\frac{k}{\ln k}$

d. $\sum_{k=1}^{\infty}\left(\frac{1}{k}-\frac{1}{k!}\right)$

e. $\sum_{k=0}^{\infty}(-1)^{k}\frac{(k!)^{2}}{(2k)!}$

f. $\sum_{k=0}^{\infty}\sin\left(\frac{\pi}{4}k\right)$

g. $\sum_{k=1}^{\infty} \frac{(-1)^K}{\sqrt{k(k+1)}}$

h. $\sum_{k=0}^{\infty} (-1)^k \frac{k}{2^k}$

i. $\sum_{k=1}^{\infty} \left(\frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}} \right)$

j. $\sum_{k=0}^{\infty} (-1)^k \frac{(3k+2)(3k+3)}{(3k+4)(3k+5)}$

k. $\sum_{k=1}^{\infty} \frac{\sin(\pi k/4)}{k^2}$

l. $\sum_{k=0}^{\infty} (-1)^{k+1} \frac{k^k}{k!}$

m. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n+1}$

p. $\sum_{n=2}^{\infty} (-1)^n \frac{1}{\ln n}$

q. $\sum_{n=1}^{\infty} (-1)^n \frac{n^2}{2n+1}$,

r. $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k!}$

s. $\sum_{n=1}^{\infty} n \left(\frac{4}{5} \right)^n$

t. $\sum_{n=2}^{\infty} (-1)^n \frac{1}{n \ln n}$

u. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n(n-2)}$

y. $\sum_{n=2}^{\infty} (-1)^{n+1} \frac{1}{(\ln n)^n}$

2. Prove that if $\sum a_k$ is absolutely convergent and $|b_k| \le |a_k|$ for all k, then $\sum b_k$ is absolutely convergent.

Assessment

- *Asking Answer for the raised questions and for the given assignment.*
- *Check students participation in the group activity and make to present*
- *Giving feedback for their work*

1.4.1 Generalized Convergence Tests for Absolute Convergence

Theorem: Let $\sum_{n=1}^{\infty} a_n$ be a series.

a. *Generalized Comparison Test*:

If $|a_n| \le |b_n|$, for all $n \ge 1$ and if $\sum_{n=1}^{\infty} |b_n|$ converges, then $\sum_{n=1}^{\infty} a_n$

converges (absolutely).

b. *Generalized Limit Comparison Test*:

If $\lim_{n \to \infty} \left| \frac{a_n}{b_n} \right| = L$, where L is a positive real number, then either both

series $\sum_{n=1}^{\infty} b_n$ and $\sum_{n=1}^{\infty} a_n$ converge (absolutely) or both series diverge.

c. *Generalized Ratio Test* :

Suppose that $a_n \neq 0$ *for* $n \ge 1$ *and*

$\lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right| = r$ (Possibly ∞)

If $r < 1$, then $\sum_{n=1}^{\infty} a_n$ converges (absolutely).

If $r > 1$, then $\sum_{n=1}^{\infty} a_n$ diverges.

If $r = 1$, then the test fails.

d. *Generalized Root Test*:

Suppose that

$\lim_{n \to \infty} \sqrt[n]{|a_n|} = r$ (Possibly∞)

If r < 1, then $\sum_{n=1}^{\infty} a_n$ converges (Possibly ∞).

If r > 1, then $\sum_{n=1}^{\infty} a_n$ diverges.

If r = 1, then the test fails

Proof: (Reading Assignment)

Examples: For what value of x does the following series

a. Converges absolutely?

b. Converges conditionally?

c. Diverges?

i. $\sum_{n=1}^{\infty}\frac{x^n}{n}=x+\frac{x^2}{2}+\frac{x^3}{3}+\frac{x^4}{4}+---$

ii. $\sum_{n=0}^{\infty}\frac{(-1)^n}{2n+1}x^{2n+1}=x-\frac{x^3}{2}-\frac{x^5}{5}-\frac{x^7}{7}+\ldots$

Solution: i. Here $a_n=\frac{x^n}{n}, a_{n+1}=\frac{x^{n+1}}{n+1}$.Thus

$$r=\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim_{n\to\infty}\left|\frac{x^{n+1}\cdot n}{x^n\cdot n+1}\right|$$

$$r=|x|\lim_{n\to\infty}\left|\frac{n}{n+1}\right|=|x|$$

Hence by the generalized ratio test the series converges absolutely for $|x|<1$ i.e for $-1<x<1$ and diverges for $x>1$.

- At $x=-1$ the series reduces to $\sum_{n=0}^{\infty}\frac{(-1)^n}{n}$ which converges conditionally.
- At $x=1$ the series reduces to $\sum_{n=0}^{\infty}\frac{(1)^n}{n}$ a divergent harmonic series.

Therefore the series converges absolutely on (-1,1),converges conditionally at $x=-1$ and diverges for $x\geq 1$.

ii. Similarly , $a_n=\frac{(-1)^n}{2n+1}x^{2n+1}, a_{n+1}=\frac{(-1)^{n+1}}{2n+3}x^{2n+3}$

$$r=\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim_{n\to\infty}\left|-\frac{x^{2n+3}\cdot 2n+1}{x^{2n+1}\cdot 2n+3}\right|=\left|x^2\right|$$

Therefore by generalized ratio test the series converges absolutely for $|x^2|<1$ i.e for $-1<x<1$ and diverges for $x>1$.

- At $x=1$ the series reduces to $\sum_{n=0}^{\infty}\frac{(-1)^n}{2n+1}$ and thus converges by ALST but not absolutely.
- At $x=-1$ the series becomes $\sum_{n=0}^{\infty}\frac{(1)^n}{2n+1}$ which is divergent (check?).

Therefore the series converges absolutely on (-1,1),converges conditionally at $x = -1$ and diverges for $x \geq 1$.

Corollary: Let $\{a_n\}_{n=1}^{\infty}$ be a sequence. If

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = r < 1 \text{ or } \lim_{n\to\infty}\sqrt[n]{|a_n|} = r < 1,$$

Then $\lim_{n\to\infty} a_n = 0$

Note that by the above corollary since $\sum_{n=0}^{\infty}\frac{x^n}{n!}$ is convergent for all x by Ratio test

we see that, $\lim_{n\to\infty}\frac{x^n}{n!} = 0$

Group Activity 1.4.2: Determine whether the following series converges.

a. $\sum_{n=2}^{\infty}(-1)^n\frac{1}{\ln n}$

b. $\sum_{n=1}^{\infty}(-1)^n\frac{(\ln n)^2}{n}$

c. $\sum_{n=1}^{\infty}(-1)^n\frac{n^2}{2n+1}$,

d. $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n^{\frac{1}{n}}}$

e. $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{3n+4}$,

f. $\sum_{n=1}^{\infty}n\left(\frac{4}{5}\right)^n$

g. $\sum_{n=1}^{\infty}(-1)^n\frac{n^n}{n!}$

h. $\sum_{n=2}^{\infty}(-1)^n\frac{1}{n\ln n}$

i. $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n(n-2)}$

j. $\sum_{n=2}^{\infty}(-1)^{n+1}\frac{1}{(\ln n)^n}$

Assessment

- *Asking Answer for the raised questions and for the given assignment.*
- *Check students participation in the group activity and make to present*
- *Giving feedback for their work*

1.5 Unit Summary

- A sequence, denoted by $\{a_n\}_{n=m}^{\infty}$, is a function whose domain is the set of natural numbers.
- A sequence $\{a_n\}_{n=m}^{\infty}$ that has a limit, which is a finite real number, is called convergent. Otherwise it is divergent.
- Convergence or divergence of a sequence $\{a_n\}_{n=m}^{\infty}$ is the property which is the result of the behavior of the general term as eventually.
- If a sequence $\{a_n\}_{n=m}^{\infty}$ converges, its limit is unique.
- A sequence $\{a_n\}_{n=m}^{\infty}$ is said to be bounded if it is bounded above and below.
- A sequence is said to be monotone if it is either increasing or decreasing.
- Every bounded and monotonic sequence converges.
- A sum of infinitely many terms of a sequence denoted by

  $\sum_{n=m}^{\infty} a_n = a_1 + a_2 + a_2 + \cdots$ is known as an infinite series.
- An infinite series $\sum_{n=m}^{\infty} a_n$ converges, if and only if its sequence of partial sum converges.
- If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.
- Let r be a real number and $c \neq 0$. Then the geometric

  Series, $\sum_{n=m}^{\infty} c\,r^{\,n} = \begin{cases} \dfrac{c\,r^{m}}{1-r} & \text{if } |r| < 1 \\ \text{diverges if } |r| \geq 1 \end{cases}$
- The convergence or divergence of an infinite series is not affected by where you start the summation.
- A series of the form $\sum_{k=1}^{\infty}(-1)^{k+1}a_k$ or $\sum_{k=1}^{\infty}(-1)^{k}a_k$, where each $a_k > 0$, is called an alternating series.
- A series $\sum_{k=m}^{\infty} a_k$ is said to be absolutely convergent if the series $\sum_{k=m}^{\infty}|a_k|$ converges.
- Recall that there are different types of tests; such as *Integral, Basic Comparison, Limit Comparison, Root, Ratio, Alternating Series Tests, Absolutely and Conditionally Convergent series* tests or Generalized convergence tests.

1.6 Review Exercises

SEQUENCES

1. Write the first four terms of the following sequence

   a. $\left\{\frac{\sqrt{n}}{n+1}\right\}_{n=0}^{\infty}$

   b. $\left\{\frac{(-1)^{n+1}}{n!}\right\}_{n=0}^{\infty}$

   c. $\left\{\frac{(2x)^{n-1}}{(2n-1)^5}\right\}_{n=1}^{\infty}$

   d. $\left\{\frac{(-1)^n \cdot x^{2n-1}}{1.3.5.\cdots.(2n-1)}\right\}_{n=1}^{\infty}$

   e. $\left\{\frac{\cos nx}{x^2+n^2}\right\}_{n=1}^{\infty}$

2. Find a possible formula for the sequence whose first 5 terms are indicated and find the 6 th term;

   a. $\frac{-1}{5}, \frac{3}{8}, \frac{-5}{8}, \frac{7}{14}, \frac{-9}{17}, \cdots$

   b. $1, 0, 1, 0, 1, \cdots$

   c. $\frac{2}{5}, 0, \frac{3}{4}, 0, \frac{4}{5}, \cdots$

3. Consider a circle. Take two points on the circle and connect them with a line segment, now the circle is divided into $a_1 = 2$ regions. Add a third point, connect all points and show that there are $a_2 = 4$ regions. Add a fourth point, connect all points and show that there are $a_3 = 8$ regions. Show that $a_5 = 32$ and find a general formula for n distinct points on the circle.

4. Suppose that a ball is launched from the ground with initial velocity v. Ignoring all resistance it will rise to a height $\frac{v^2}{2g}$ and fall back to the ground at time $t = \frac{2v}{g}$. Depending on how "lively" the ball is, the next bounce will only rise to a fraction of the previous height. The coefficient of restitution, r, defined as the ratio of landing velocity to bound velocity, measure the liveliness of the ball. The Second bounce has launch velocity rv, the third bounce has launch velocity $r^2 v$ and so on.

Thus

$$a_2 = \frac{2v}{g}(1+r),\ \ a_3 = \frac{2v}{g}(1+r+r^2),\ \ \cdots \ etc.$$

Find the general expression for a_n where $r \in (-1,1)$. And determine the limit of the sequence.

5. If \$1000 is invested at 6% interest, compounded annually then after years the investment is worth $a_n = 1000(1.06)^n$ dollars.
   (a) Find the first five terms of the sequence $\{a_n\}$.
   (b) Is the sequence convergent or divergent? Explain.

SERIES

1. Find the sum of the following series, if it converges
   a. $\sum_{k=1}^{\infty}(1+\frac{1}{k})^k$
   b. $\sum_{k=0}^{\infty}\frac{3}{10^k}$
   c. $\sum_{k=0}^{\infty}\frac{12}{100^k}$
   d. $\sum_{n=0}^{\infty}\frac{1-2^n}{3^n}$
   e. $\sum_{n=0}^{\infty}\frac{2^{n+3}}{3^n}$
   f. $\sum_{n=0}^{\infty}\frac{(-1)^n}{5^n}$
   g. $\sum_{n=1}^{\infty}\ln\left(\frac{n+1}{n}\right)$
2. Express the following decimals as an infinite series and find its sum if it converges
   a. $0.36585858\cdots$
   b. $0.367536753675\cdots$
   c. $32.794548548\cdots$
3. Find a series expansion for the expression:

$$\frac{x}{1+x^2} \ \ for\ \ |x|<1.$$

4. Determine whether or not the following series converges
   a. $\sum_{k=1}^{\infty}\frac{k}{k^3+1}$
   b. $\sum_{k=1}^{\infty}\frac{\ln k}{k}$

c. $\sum_{k=1}^{\infty} \frac{2k-1}{\sqrt{k^4+1}}$

d. $\sum_{k=1}^{\infty} \frac{k^3}{k^5+5k^4+7}$

e. $\sum_{k=1}^{\infty} \frac{\ln k}{k^3}$

f. $\sum_{k=1}^{\infty} \frac{1}{k\ln(k+1)}$

g. $\sum_{k=1}^{\infty} \frac{\ln k}{k^3}$

h. $\sum_{k=1}^{\infty} \frac{1}{k^{\frac{2}{3}}}$

i. $\sum_{k=0}^{\infty} \frac{10^k}{k!}$

j. $\sum_{k=1}^{\infty} \frac{1}{k.2^k}$

k. $\sum_{k=1}^{\infty} \frac{k!}{10^{4k}}$

l. $\sum_{k=1}^{\infty} \frac{(2k+1)^{2k}}{(5k^2+1)^k}$

m. $\sum_{k=0}^{\infty} (-1)^k \frac{(k!)^2}{(2k)!}$

n. $\sum_{n=1}^{\infty} (-1)^n \frac{(\ln n)^2}{n}$

o. $\sum_{n=1}^{\infty} (-1)^n \frac{n^2}{2n+1}$

p. $\sum_{k=0}^{\infty} (-1)^{k+1} \frac{k^k}{k!}$

q. $\sum_{n=2}^{\infty} (-1)^{n+1} \frac{1}{(\ln n)^n}$

CHAPTER TWO
POWER SERIES

Introduction

Power series play a fundamental role in both mathematics and science they are used, for example, to approximate trigonometric functions and logarithms, to solve differential equations, to evaluate difficult integrals, to create new functions, and to construct mathematical models of physical laws. Physicists also use power series in another way: In studying fields as diverse as optics, special relativity, and electromagnetism, they analyze phenomena by replacing a function with the first few terms in the series that represents it.

This unit is divided into three sections. The first section presents *definition and notation of a power series; radius and interval of convergence of power series. Differentiation and integration of power series* are parts of the second section. *Taylor polynomials, Taylor series and the application* will be treated in the third section.

Objectives:

At the end of the unit, students will be able to:

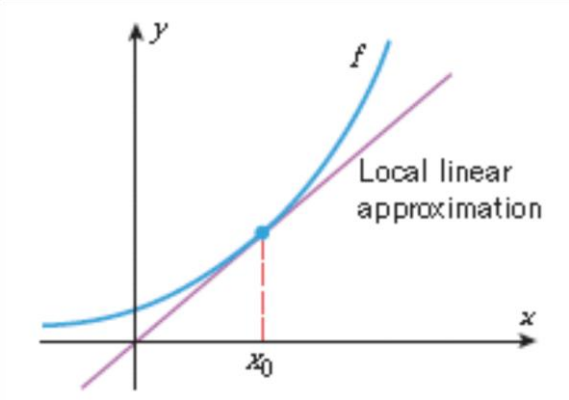
- Define power and Taylor series.
- Identify the relation between power and Taylor series.
- Determine differentiation and integration of a Power Series.
- Find the Maclaurin and Taylor polynomials for functions
- Express a function in the form of a power series.
- Determine the Taylor's series representation of a function.
- Demonstrate the application of power and Taylor series together with solving exercises.
- Use Taylor's theorem to approximate function to the desired level of accuracy.
- Apply the concept of power series and Taylor's formula in solving real life problems.

2.1 Maclaurian and Taylor Polynomials

Recall that the local linear approximation of a function f at x_0 is

$$f(x) \cong f(x_0) + f'(x_0)(x - x_0)$$

In this formula the approximating function $P(x) = f(x_0) + f'(x_0)(x - x_0)$ is a first degree polynomial satisfying $P(x_0) = f(x_0)$ and $P'(x_0) = f'(x_0)$ (Verify).

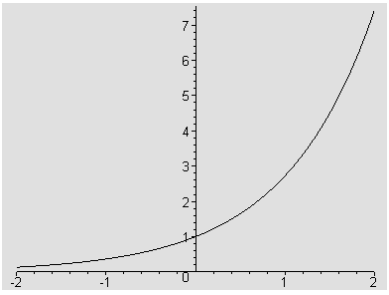


If the graph of f has a pronounced "bend" at x_0 ,then we can expect that the accuracy of the local approximation of f at x_0 will decrease rapidly as we progress away from x_0. This leads us to the following general problem

Problem: *Given a function that can be differentiated function f that can be differentiated n times at* $x = x_0$ *find a polynomial P of degree n to approximate* $f(x)$.

One way to deal with this problem is to approximate the function f by a polynomial $P(x)$of degree n with the property that the value of $P_n(x)$ and the values of its first n derivatives match those of f at $x = x_0$. This ensures that the graph of $f(x)$ and $P_n(x)$not only have the same tangent line at x_0 , but they also bend in the same direction at x_0 (either both concave upward or concave down).

The classic example in this regard is the transcendental function $f(x) = e^x$ near c = 0, where it looks like this:



We began to look at ways of finding polynomials which looked like this $f(x)$ near $x = 0$ and we were able to find functions of degrees 1, 2, 3, and 4 which agreed with $f(x) = e^x$ to the specified degree of derivatives, and we got:

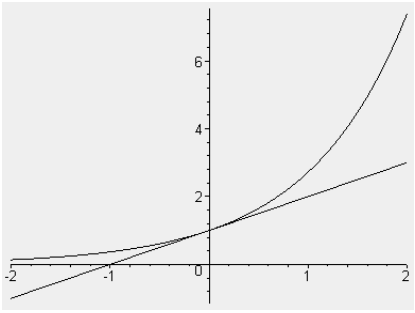
$$P_1(x) = 1 + x,$$

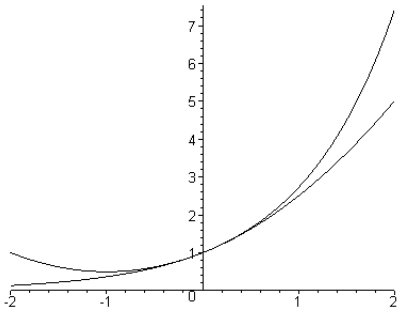
$$P_2(x) = 1 + x + \frac{x^2}{2},$$

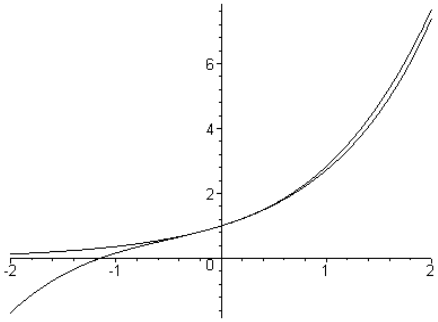
$$P_3(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6},$$

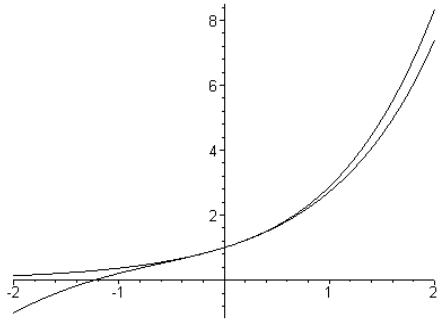
$$P_4(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}.$$

The graphs of these functions, with $f(x) = e^x$.









You'll notice that the polynomial 'hugs the curve' closer as the degrees (and thus the degrees of agreement of the derivatives) increase.

We discovered two facts:

(1) The higher the degree of the polynomial, the better the 'fit'.

(2) We could construct a polynomial $P_n(x)$ of given degree *n*, by specifying that the first *n* derivatives of $f(x)$ and $P_n(x)$ agree at $x = c$.

This was noticed by a guy named Taylor that bringing the successive derivatives of a function $f(x)$ and the associated polynomial $P_n(x)$ into agreement generated a pattern in the coefficients of $P_n(x)$. This gave rise to the *Taylor Polynomial*:

Definition: -(***Taylor and Maclaurin Polynomial***)

Let f be a function for which the nth derivative, $f^{(n)}(x)$, exists at some number c.

Then the polynomial:

$$P_n(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \frac{f^{(3)}(c)}{3!}(x-c)^3 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n$$

$$= \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k$$

of degree n is called the nth *Taylor polynomial for* f *in power of* $(x-c)$.

In particular if $c=0$, then the nth Taylor polynomial for f in power of x is given by:

$$P_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \cdots + \frac{f^{(n)}(0)}{n!}x^n = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!}x^k \text{ and}$$

is called the nth Maclaurin polynomial for f.

Examples:

1. Find the Taylor Polynomial of degree 5 for $f(x) = \ln x$ with center $c=1$.

Solution:

We need the first five derivatives, w/ 1 plugged into them.

$$f(x) = \ln x,\ f(1) = 0$$

$$f'(x) = \frac{1}{x} = x^{-1},\ f'(1) = 1$$

$$f''(x) = -\frac{1}{x^2} = (-1)x^{-2},\ f''(1) = -1$$

$$f'''(x) = \frac{2}{x^3} = (-2)(-1)x^{-3},\ f'''(1) = 2! = 2$$

$$f^{(4)}(x) = \frac{(-3)(-2)(-1)}{x^4} = (-3)(-2)(-1)x^{-2},\ f^{(4)}(1) = -(3!) = -6$$

$$f^{(5)}(x) = \frac{(-4)(-3)(-2)(-1)}{x^5} = (-4)(-3)(-2)(-1)x^{-2},\ f^{(5)}(1) = 4! = 24$$

Therefore the 5th Taylor polynomial of $f(x) = \ln x, c = 1$ is

$$P_5(x) = f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \frac{f^{(3)}(0)}{3!}(x-1)^3 + \frac{f^{(4)}(1)}{4!}(x-1)^4 + \frac{f^{(5)}(1)}{5!}(x-1)^5$$

$$P_5(x) = 0 + 1(x-1) - \frac{1(x-1)^2}{2!} + \frac{2(x-1)^3}{3!} - \frac{6(x-1)^4}{4!} + \frac{24(x-1)^5}{5!}$$

$$= (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \frac{(x-1)^5}{5}$$

2. Find the MacLaurin Polynomial of degree 5 for $f(x) = \cos x$.

Solution:

MacLaurin means $c = 0$, and we're going to need five derivatives.

$$f(x) = \cos x,\ f(0) = \cos(0) = 1$$

$$f'(x) = -\sin x, \Rightarrow f'(0) = 0$$

$$f''(x) = -\cos x, \Rightarrow f''(0) = -1$$

$$f'''(x) = \sin x, \Rightarrow f'''(0) = 0$$

$$f^{(4)}(x) = \cos x, \Rightarrow f^{(4)}(0) = 1$$

$$f^{(5)}(x) = -\sin x, \Rightarrow f^{(5)}(0) = 0$$

$P_5(x) = 1 + 0x - \frac{1x^2}{2!} + \frac{0x^3}{3!} + \frac{1x^4}{4!} - \frac{0x^5}{5!}$, note that the odd-degreed terms are all 0.

$P_5(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!}$.

3. Find the seventh degree Taylor polynomial of

i. $f(x) = \sin x \quad at\ c = 0$.

ii. $f(x) = e^x \quad at\ c = 0$.

Solution:

i) Since $f(x) = \sin x$ is infinitely differentiable in its domain

$$f(x) = \sin x \Rightarrow f(0) = 0$$

$$f'(x) = \cos x, \Rightarrow f'(0) = 1$$

$$f''(x) = -\sin x, \Rightarrow f''(0) = 0$$

$$f'''(x) = -\cos x, \Rightarrow f'''(0) = -1$$

$$f^{(4)}(x) = \sin x, \Rightarrow f^{(4)}(0) = 0$$

$$f^{(5)}(x) = \cos x, \Rightarrow f^{(5)}(0) = 1$$

$$f^{(6)}(x) = -\sin x, \Rightarrow f^{(6)}(0) = 0$$

$$f^{(7)}(x) = -\cos x, \Rightarrow f^{(7)}(0) = -1$$

Thus the seventh degree Taylor polynomial of $f(x) = \sin x$ at c=0 is

$$P_7(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \cdots + \frac{f^{(7)}(0)}{7!}x^7 = \sum_{k=0}^{7} \frac{f^{(k)}(0)}{k!}x^k$$

$$P_7(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$$

ii. Since the derivative of $f(x) = e^x$ is itself, we have

$$f'(x) = f''(x) = f'''(x) = f^{(4)}(x) = f^{(5)}(x) = f^{(6)}(x) = f^{(7)}(x) = e^x \text{ and}$$

$$f'(0) = f''(0) = f'''(0) = f^{(4)}(0) = f^{(5)}(0) = f^{(6)}(0) = f^{(7)}(0) = e^0 = 1.$$

Thus the seventh degree Taylor polynomial of $f(x) = e^x$ at $c = 0$ is

$$P_7(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \cdots + \frac{f^{(7)}(0)}{7!}x^7 = \sum_{k=0}^{7} \frac{f^{(k)}(0)}{k!}x^k$$

$$P_7(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \frac{x^7}{7!} \ .$$

Quick check Class Exercises 2.1.1:

1.Find the first four Taylor Polynomials for $\ln x$ about $x = 2$.

2. Find the seventh degree Taylor polynomial of

i. $f(x) = \sin x$ *at* $c = \frac{\pi}{2}$.

ii. $f(x) = e^{2x}$ *at* $c = 1$

Theorem 2.1.1: *(Taylor's Remainder Formula)*

Let f be a function such that $f^{(n+1)}(x)$ exists on an open interval I containing c. Then, there exists some number z between x and c such that

$$f(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n + \frac{f^{(n+1)}(z)}{(n+1)!}(x-c)^{n+1}$$

$$= \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}x^k + \frac{f^{(n+1)}(z)}{(n+1)!}(x-c)^{n+1} \quad = P_n(x) + R_n(x),$$

where $R_n(x) = \frac{f^{(n+1)}(z)}{(n+1)!}(x-c)^{n+1}$, for $x \in I$.

Proof: Beyond this level.

Note:

a. The term $R_n(x) = \dfrac{f^{(n+1)}(z)}{(n+1)!}(x-c)^{n+1}$

in the Taylor's formula is called the ***Lagrange remainder*** of f at c. Thus

$$f(x) = P_n(x) + R_n(x).$$

b. If $c = 0$, the Taylor's formula becomes

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k + \frac{f^{(n+1)}(z)}{(n+1)!} x^{n+1},$$

where z is between 0 and x (*i.e.* $z \in [0, x]$ *or* $z \in [x, 0]$). This Taylor formula is called ***Maclaurine formula.***

c. If $P_n(x)$ *and* $R_n(x)$ are the nth degree Taylor's polynomial and the nth degree Lagrange remainder of f at x, respectively, then

$$f(x) = P_n(x) + R_n(x)$$

$$\Leftrightarrow R_n(x) = f(x) - P_n(x)$$

If we can find a number $d > o$ such that

$$|R_n(x)| < d, \text{ then } f(x) - P_n(x) < d$$

$$\Leftrightarrow f(x) = P_n(x) \pm d, \quad i.e.; \quad f(x) \cong P_n(x).$$

Example 1. Apply taylor's approximation formula to $f(x) = e^x$

In this case $f^{(n)}(x) = e^x$ for all n and $f^{(n)}(0) = 1$ for all n.

Therefore the nth polynomial approximation becomes

$$e^x \approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots + \frac{x^n}{n!}$$

The error in the approximation is

$$R = f(x) - \left[f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f^{(3)}(0)x^3}{3!} + \ldots + \frac{f^{(n)}(0)x^n}{n!} \right]$$

Applying remainder theorem to $f(x) = e^x$. In this case

$$R = e^x - \left[1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots + \frac{x^n}{n!} \right] = \frac{e^{\xi} x^{n+1}}{(n+1)!}$$

It turns out that the error $R_n(x)$ is approximately a constant times x^{n+1}. Thus the error is small when *x* is small, but increases rapidly with *x*.

Example 2. Use third Maclaurin polynomial to approximate $e^{-0.5}$. Find an upper bound for the error in this approximation.

Solution: Putting *x* = - 0.5 and *n* = 3 in (2) and (3) we get

$$e^{-0.5} \approx \left[1 - 0.5 + \frac{(-0.5)^2}{2!} + \frac{(-0.5)^3}{3!}\right] = 0.60416...$$

$$R = \frac{e^{\xi}(-0.5)^4}{4!}$$

where ξ is between 0 and – 0.5. Therefore $0 \le e^{\xi} \le 1$. So $0 \le R \le$ **Error!** = 0.0026...

Group Activity 2.1.1

1) a. Find the 8th degree Taylor series approximation to $f(x) = \tan x$ centered at $x_o = 0$.

b. Use this to estimate the value of tan(0.1). Estimate the error in this approximation.

c. Estimate the truncation error in part a as *x* varies over the interval $|x| \le 0.2$.

d. For the approximation in part a, find δ so that $|R| \le 10^{-4}$ for $|x| \le \delta$.

e. If in part a we take the terms up to order *n*, find *n* so that $|R| \le 10^{-3}$ for $|x| \le 0.2$.

f. Approximate e with an error (remainder) less than 0.001 to five decimal places.

Assessment

- *Check students participation in the group activity.*
- *Check and give feedback to their answers*

2.2 Maclaurin and Taylor Series; Power Series

2.2.1 Definition and Notation of Power Series

Definition: - An infinite series of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots + c_n(x-a)^n + \cdots \quad (1)$$

is called a *power series* in $x - a$.

If $a = 0$, then the power series in x is given

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots + c_n x^n + \cdots$$ and is called power series in x.

Examples: i) $\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + x^4 + \ldots$

ii) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots$

iii) $\sum_{n=0}^{\infty} \frac{(x-1)^n}{n+1} = 1 + \frac{(x-1)}{2} + \frac{(x-1)^2}{3} + \frac{(x-1)^3}{4} + \ldots$

iv) $\sum_{n=0}^{\infty} (-1)^n (x+3)^n \frac{1}{n!} = 1 - (x+3) + \frac{(x+3)^2}{2!} - \frac{(x+3)^3}{3!} + \ldots$

(i) and (ii) are power series in x where as (iii) and (iv) are power series in $x - 1$ and $x + 3$ respectively.

Quick Check Class Activity: 2.2.1 Find "a" and the coefficients c_0, c_1, c_2 *and* c_3 for each of the following power series

a. $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$ $a =$ _____, $c_0 =$ _____, $c_1 =$ _____, $c_2 =$ _____, $c_3 =$ _____

b. $\sum_{n=0}^{\infty} (-1)^n \frac{(x-5)^n}{n.5^n}$ $a =$ _____, $c_0 =$ _____, $c_1 =$ _____, $c_2 =$ _____, $c_3 =$ _____

c. $\sum_{n=0}^{\infty} (2n)! \left(\frac{x}{2}\right)^n$ $a =$ _____, $c_0 =$ _____, $c_1 =$ _____, $c_2 =$ _____, $c_3 =$ _____

2.2.2 Radius and Interval of Convergence of Power Series

Definition: - A power series $\sum_{n=0}^{\infty} c_n x^n$ in x is said to converge

a. At d if and only if $\sum_{n=0}^{\infty} c_n d^n$ converges;

b. On the set S if and only if $\sum_{n=0}^{\infty} c_n x^n$ converges for each $x \in S$.

The most useful method of our disposal for determining the interval of convergence of a power series is the ***Generalized Ratio Test*** **and** we may sometimes also use **Generalized Root test.**

Example: - Determine the values of x, for which the following series converge.

a. $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{2^n.x^n}{n.3^n}$ *b.* $\sum_{n=0}^{\infty}\frac{x^n}{n}$ *c.* $\sum_{n=0}^{\infty}\frac{x^n}{n!}$

Solution: -

a. By using generalized ratio test with $a_n=(-1)^{n+1}\frac{2^n.x^n}{n.3^n}$, the series $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{2^n.x^n}{n.3^n}$ converges whenever

$$r=\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|<1$$

$$\Rightarrow\ r=\lim_{n\to\infty}\left|\frac{2^{n+1}.3^n.\ n.\ x^{n+1}}{2^n.3^{n+1}.\ (n+1).x^n}\right|=\frac{2}{3}\lim_{n\to\infty}\frac{n}{n+1}.|x|<1$$

$$\Rightarrow\quad |x|<\frac{3}{2}$$

Thus the series $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{2^n.x^n}{n.3^n}$ converges whenever $-\frac{3}{2}<x<\frac{3}{2}$

- If $x=-\frac{3}{2}$, then $\sum_{n=1}^{\infty}(-1)^{2n+1}\frac{2^n.\left(\frac{3}{2}\right)^n}{n.3^n}=-\sum_{n=1}^{\infty}\frac{1}{n}$ is a divergent series because $\sum_{n=1}^{\infty}\frac{1}{n}$ is a divergent Harmonic series.

- If $x=\frac{3}{2}$, then $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{2^n.\left(\frac{3}{2}\right)^n}{n.3^n}=\sum_{n=1}^{\infty}(-1)^{n+1}.\frac{1}{n}$ is a convergent alternating series.

Hence the series $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{2^n.x^n}{n.3^n}$ converges whenever $-\frac{3}{2}<x\le\frac{3}{2}$.

b. Using generalized ratio test with $a_n=\frac{x^n}{n}$, the series $\sum_{n=0}^{\infty}\frac{x^n}{n}$ converges whenever

$$r=\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|<1$$

$$\Rightarrow \lim_{n\to\infty}\left|\frac{x^{n+1}.n}{(n+1).x^n}\right|<1$$

$$\Rightarrow |x|.\lim_{n\to\infty}\frac{n}{n+1}<1 \Leftrightarrow |x|<1$$

Thus the series $\sum_{n=0}^{\infty}\frac{x^n}{n!}$ converges whenever $-1<x<1$

- If $x=-1$, then $\sum_{n=0}^{\infty}\frac{x^n}{n!}=\sum_{n=0}^{\infty}(-1)^n\frac{1}{n}$ is a convergent alternating series.
- If $x=1$, then $\sum_{n=0}^{\infty}\frac{x^n}{n}=\sum_{n=0}^{\infty}\frac{1}{n}$ which is a divergent harmonic series.

Hence the series $\sum_{n=0}^{\infty}\frac{x^n}{n}$ converges whenever $-1\le x<1$.

(c) $\sum_{n=0}^{\infty}\frac{x^n}{n!}$. Note $c=0$. Take the ratio limit with $a_n=\frac{x^n}{n!}$, the series $\sum_{n=0}^{\infty}\frac{x^n}{n!}$ converges whenever

$$r=\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|<1$$

$$\Rightarrow \lim_{n\to\infty}\left|\frac{x^{n+1}.}{(n+1)!.}*\frac{n!}{x^n}\right|<1$$

$$\Rightarrow |x|.\lim_{n\to\infty}\frac{1}{n+1}=0<1 \text{ for every x.}$$

Since $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=0<1$ always, so the ROC is R = ∞. This means that the IOC is the entire real line R.

Quick check Class Exercises 2.2.2:

Determine the radius and interval of convergence of the following power series.

a. $\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{(2n+1)!}$

b. $\sum_{n=0}^{\infty}n^3.x^n$

Lemma 2.2.1:

a. If $\sum_{n=0}^{\infty} c_n x^n$ converges, then $\sum_{n=0}^{\infty} c_n t^n$ converges absolutely for $|t| < |x|$.

b. If $\sum_{n=0}^{\infty} c_n x^n$ diverges, then $\sum_{n=0}^{\infty} c_n t^n$ diverges for $|t| > |x|$

Theorem 2.2.1: Let $\sum_{n=0}^{\infty} c_n (x-a)^n$ be a power series. Then exactly one of the following conditions holds;

i. $\sum_{n=0}^{\infty} c_n (x-a)^n$ converges only at $x = a$

ii. $\sum_{n=0}^{\infty} c_n (x-a)^n$ converges for all real numbers x

iii. There is a number $R > 0$ such that

a. $\sum_{n=0}^{\infty} c_n (x-a)^n$ converges for $|x-a| < R$

b. $\sum_{n=0}^{\infty} c_n (x-a)^n$ diverges for $|x-a| > R$

Proof: Exercise

Remark:

a. The number **R** in the above theorem is called the ***radius of convergence*** of the series $\sum_{n=0}^{\infty} c_n (x-a)^n$ .And

If the series satisfies case (i), $\mathbf{R} = 0$.

If the series satisfies case (ii), $\mathbf{R} = \infty$.

Thus, every power series has the radius of convergence **R**, which is either *zero,* a *positive number* or ∞.

b. The collection of all values of x for which $\sum_{n=0}^{\infty} c_n (x-a)^n$ converges is called the ***interval of convergence*** of the power series and is always centered at x=a.

c. If R is the radius of convergence of the series $\sum_{n=0}^{\infty} c_n (x-a)^n$, then the interval of convergence is one of the following intervals; $(-R+a, R+a)$, $[-R+a, R+a]$, $[a-R, a+R)$, $(a-R, a+R]$.

d. At $x=a$, $\sum_{n=0}^{\infty} c_n (x-a)^n = 0$. Therefore $\sum_{n=0}^{\infty} c_n (x-a)^n$ is always convergent at x=a.

e. A Power series $\sum_{n=0}^{\infty} a_n x^n$ defines a function whose domain is the interval of convergence of the series.

Examples: (1) Determine the interval of convergence for the power series:

a. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x-1)^n}{2^n}$ b. $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2}$

Solution:

(1) (a) Before we start, note that the center of the series is the number being subtracted from *x*, that is, $c=1$. Apply Generalized Ratio Test , $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x-1)^n}{2^n}$ converges if

$$r = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$$

$$\Rightarrow \lim_{n\to\infty} \left| -1.\frac{(x-1)^{n+1}.2^n}{2^{n+1}.(x-1)^n} \right| < 1$$

$$\Rightarrow |x-1|. \lim_{n\to\infty} \frac{1}{2} < 1 \Leftrightarrow |x-1| < 2$$

Since we have $c=1$, $R=2$, the IOC (not including endpoints is the interval from $1-2 = -1$ to $1 + 2 = 3$.

Thus the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x-1)^n}{2^n}$ converges whenever $-1 < x < 3$.

Now we test the endpoints by plugging them in and checking convergence:

- If $x=-1$, then $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x-1)^n}{2^n} = \sum_{n=1}^{\infty} -1$ is a divergent series.
- If $x=3$, then $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x-1)^n}{2^n} = \sum_{n=1}^{\infty} (-1)^{n+1}$ which is a divergent alternating series.

So the IOC is (-1,3), which contains neither end point.

Hence the series $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}(x-1)^n}{2^n}$ converges only whenever $-1<x<3$.

(b) Again, take the limit of ratios (note that the center is *c = 2*).

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim_{n\to\infty}\left|\frac{(x-2)^{n+1}}{(n+1)^2}\cdot *\frac{n^2}{(x-2)^n}\right|=|x-2|<1$$

when it converges, so here the ROC is 1. (1: Quotient of lead coeff's.)

The IOC would run from $2-1=1$ to $2+1=3$, so we need to check the endpoints:

- If $x=1$, then $\sum_{n=1}^{\infty}\frac{(x-2)^n}{n^2}=\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}$ converges by AST.
- If $x=3$, then $\sum_{n=1}^{\infty}\frac{(x-2)^n}{n^2}=\sum_{n=1}^{\infty}\frac{1}{n^2}$ is a convergent *p-series*.

The IOC includes both end points: [1,3].

Quick check Class Exercises 2.2.3:

Determine the radius and interval of convergence of the following power series.

a. $\sum_{n=1}^{\infty}\frac{(x-2)^n}{3^n n^2}$ b. $\sum_{n=1}^{\infty}\frac{(x+2)^n}{2^n(n+1)}$

Group Activity 2. 2.1 (*The students will discuss in their respective groups)*

1. Find the radius and interval of convergence of the following power series.

a. $\sum_{n=1}^{\infty}\frac{2^n x^n}{n^2}$

b. $\sum_{n=1}^{\infty}\frac{x^n}{2^n\sqrt{n}}$

c. $\sum_{n=1}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!}$

d. $\sum_{n=1}^{\infty}\frac{(n+1)}{n^{2n}}x^n$

e. $\sum_{n=0}^{\infty}\frac{x^n}{(n+1).5^n}$

f. $\sum_{n=1}^{\infty}\frac{(2n)!}{(3n)!}x^n$

g. $\sum_{n=1}^{\infty}(\sinh 2n).x^n$

h. $\sum_{n=1}^{\infty}\frac{\ln n}{n+1}.(x-5)^n$

i. $\sum_{n=1}^{\infty}n^n(x-3)^n$

j. $\sum_{n=1}^{\infty}\frac{n!}{n^n}x^n$

k. $\sum_{n=0}^{\infty}\frac{(x-3)^n}{2^n}$

l. $\sum_{n=0}^{\infty}\left(\frac{3}{4}\right)^n(x+5)^n$

m. $\sum_{n=0}^{\infty}n!x^n$

n. $\sum_{n=0}^{\infty}\left(\frac{3}{4}\right)^n(2x+5)^n$

2. a. If a and b are positive integers, find the radius of convergence of

$$\sum_{n=1}^{\infty} \frac{(n+a)!}{n!(n+b)!} x^n$$

b. If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$, then show that the radius of convergence of

$$\sum_{n=0}^{\infty} a^n x^n = \begin{cases} \frac{1}{\mathrm{L}}, & \text{if } L \in \Re \setminus \{0\} \\ \infty, & \text{if } L = 0 \\ 0, & \text{if } L = \infty \end{cases}$$

Assessment

- *Answer for the raised questions and for the given assignment.*
- *Observation their work in groups and present*
- *Giving feedback for their work*

2.2.3 Maclaurin's and Taylor Series

Functions defined by power series:

If a function is expressed as a power series on some interval, then we say that f is represented by power series on that interval.

Examples:(Geometric Power series)

1) Consider the function $f(x) = \frac{1}{1-x}$. The form of f is closely similar to the sum of geometric series $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}, |r| < 1$.In other words, if you let $a = 1$and $r = 1$,then a geometric series $\sum_{n=0}^{\infty} x^n$, converges to $\frac{1}{1-x}$ for $|x| < 1$.

Thus, $\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots + x^n + \cdots = \frac{1}{1-x}, \quad if \quad |x| < 1 \ldots$ (*)

In other words, the power series representation for $f(x) = \frac{1}{1-x}$ centered at $x = 0$ is

$$f(x) = \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \text{ on (-1,1)}$$

This infinite geometric series represents the function only on (-1,1).

- If x is replaced by $-x$ in (*) we obtain

$$\sum_{n=0}^{\infty}(-1)^n x^n = 1-x+x^2-x^3+\cdots+(-1)^n x^n+\cdots=\frac{1}{1+x},\quad if\quad |x|<1$$

- If x is replaced by x^2 in (*) we obtain

$$\sum_{n=0}^{\infty}x^{2n} = 1+x^2+x^4+x^6+\cdots+x^{2n}+\cdots=\frac{1}{1-x^2},\quad if\quad |x|<1$$

- If x is replaced by $-x^2$ in (*) we obtain

$$\sum_{n=0}^{\infty}(-1)^n x^{2n} = 1-x^2+x^4-x^6+\cdots+(-1)^n x^{2n}+\cdots=\frac{1}{1+x^2},\quad if\quad |x|<1$$

2. Find the power series representation of $f(x)=\frac{1}{1-x}$ centered at -1.

Solution: We can write

$$f(x)=\frac{1}{1-x}=\frac{1}{2-(x+1)}=\frac{1}{1-\frac{(x+1)}{2}}$$

which has the form $\frac{a}{1-r}$ where $a=1, r=\frac{x+1}{2}$.

Thus for $\left|\frac{x+1}{2}\right|<1\Rightarrow|x+1|<2\Rightarrow -3<x<2$,we have

$$\frac{1}{1-x}=\sum_{n=0}^{\infty}(\frac{1}{2})(\frac{x+1}{2})^n \text{ on (-3,1)}$$

Quick check Group Activity 2.2.4:

Represent the rational function as a Geometric power series with the specified center *c*:

a) $f(x)=\frac{4}{x+2}$ centered at $x=0$.

c) $f(x)=\frac{1}{x}$ centered at $x=1$.

(e) $\frac{3}{2x-1}, c=2$

(d) $\frac{1}{2-x}, c=4$

Recall that if f has derivatives of n-orders then its nth Taylor approximation is given by

$$P_n(x)=\sum_{k=0}^{n}a_k(x-a)^k \text{ and } f^{(n)}(a)=n!a_n$$

Then if $f(x)$ has derivatives of all orders on some interval around point a we have the Taylor Series of $f(x)$ about a.

Theorem 2.2.2 (Taylor Series)

If $f(x)$ is infinitely differentiable at $x=a$ and on some open interval around a then f can be represented by the power series

$$f(x)=\sum_{n=0}^{\infty}\frac{f^{(n)}(a)}{n!}(x-a)^n$$

on the interval of convergence of this power series.

This power series representation is said to be the *Taylor series of* $f(x)$ about a.

In particular if $a=0$, then power series representation becomes

$$f(x)=\sum_{n=0}^{\infty}\frac{f^{(n)}(0)}{n!}x^n \quad ,$$

and is called *Maclaurin Series of* f.

Examples: Find the Taylor series representation of the following functions

a) $f(x)=\ln x, c=1$ b) $f(x)=\cos x$ centered at $x=0$.

Solution: a) Since lnx is infinitely differentiable at $x=1$.

$$f(x)=\ln x,\ f(1)=0$$

$$f'(x)=\frac{1}{x}=x^{-1},\ f'(1)=1$$

$$f''(x)=-\frac{1}{x^2}=(-1)x^{-2},\ f''(1)=-1$$

$$f'''(x)=\frac{2}{x^3}=(-2)(-1)x^{-3},\ f'''(1)=2!=2$$

$$f^{(4)}(x)=\frac{(-3)(-2)(-1)}{x^4}=(-3)(-2)(-1)x^{-2},\ f^{(4)}(1)=-(3!)=-6$$

$$f^{(5)}(x)=\frac{(-4)(-3)(-2)(-1)}{x^5}=(-4)(-3)(-2)(-1)x^{-2},\ f^{(5)}(1)=4!=24$$

$$\vdots \qquad\qquad \vdots$$

$$f^{(n)}(x)=\frac{(-1)^{n+1}(n-1)!}{x^n}\Rightarrow f^{(n)}(1)=(-1)^{n+1}(n-1)!$$

Therefore the Power (Taylor's) series of $f(x)=\ln x, c=1$ is

$$\ln x=0+1(x-1)-\frac{1(x-1)^2}{2!}+\frac{2(x-1)^3}{3!}-\frac{6(x-1)^4}{4!}+\frac{24(x-1)^5}{5!}+....+\frac{(-1)^{n+1}(n-1)!(x-1)^n}{n!}+...$$

$$\ln x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \frac{(x-1)^5}{5} + \ldots .$$

In Sigma notation we have, $\ln x = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(x-1)^n}{n}$

Since this power series representation is valid only on its interval of convergence we need to find this interval. Therefore by the generalized ratio test this representation is true for:

$$r = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = |x-1| < 1$$

Therefore the Radius of Convergence is $-1 < x - 1 < 1 \Longrightarrow 0 < x < 2$ *or* $(0,2)$.

To check at the end points of this interval:

- Let $x = 0$, the series becomes:

  $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}(-1)^n}{n} = -1 - \frac{1}{2} - \frac{1}{3} - \frac{1}{4} - \ldots = -\sum_{n=1}^{\infty} \frac{1}{n}$ which is divergent harmonic series

- Let $x = 2$ then the series becomes $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n}$ which is convergent alternating series as seen in the first chapter.

Therefore $\ln x = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (x-1)^n$ on $(0,2]$.

b) Since $f(x) = \cos x$ is infinitely differentiable at $x = 0$.

$$f(x) = \cos x,\ f(0) = \cos(0) = 1$$
$$f'(x) = -\sin x, \Rightarrow f'(0) = 0$$
$$f''(x) = -\cos x, \Rightarrow f''(0) = -1$$
$$f'''(x) = \sin x, \Rightarrow f'''(0) = 0$$
$$f^{(4)}(x) = \cos x, \Rightarrow f^{(4)}(0) = 1$$
$$\vdots \qquad\qquad \vdots$$

In general we have $f^{(2n+1)}(0) = 0$ and $f^{(2n)}(0) = (-1)^n, n = 0,1,2,3,\ldots$

Therefore the power series of $f(x) = \cos x$ is

$$\cos x = 1 + 0x - \frac{1x^2}{2!} + \frac{0x^3}{3!} + \frac{1x^4}{4!} - \frac{0x^5}{5!} + \ldots,$$(note that the odd-degreed terms are all 0).

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots + \frac{(-1)^n x^{2n}}{(2n)!} + \ldots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

To find its interval of convergence

$$r=\lim_{n\to\infty}\left|\frac{(-1)^{n+1}x^{2n+2}(2n)!}{(-1)^{n}x^{2n}(2n+2)!}\right|=\lim_{n\to\infty}\frac{x^{2}}{(2n+2)(2n+1)}=0<1 \text{ for every x.}$$

Thus interval of convergence is $(-\infty,\infty)$ *and* the radius of convergence is $R=\infty$.

Quick check Class Group activity 2.2.5:

1. a. Find the Maclaurine Series of

   i. $f(x)=e^{x}$ iii. $f(x)=\sin x$

   ii. $f(x)=e^{2x}$

   b. Find the Taylor Series of:

   i. $f(x)=e^{x}$, at $c=1$

   ii. $f(x)=\sin x$, at $c=\frac{\pi}{2}$

   iii. $f(x)=\frac{1}{x}, c=1$

2) Approximate

   a. $e^{0..2}$ with in 0.0005

   b. $\sin(0.5)$ with in 0.001

   c. $\sin x$ by the fourth degree polynomial in x if $0<x<0.2$.

   d. $\ln x$ by the fifth degree polynomial in x, if $1<x<1.2$.

Instructor's Role

- *Observation while they work in groups*
- *Check and Give feedback for their answers*

2.3 Operations on Convergent Power Series

Theorem 2.3.1: Let $f(x)=\sum_{n=0}^{\infty}a_{n}x^{n}$ and $g(x)=\sum_{n=0}^{\infty}b_{n}x^{n}$ are convergent power series centered at 0. Then

1. $f(kx)=\sum_{n=0}^{\infty}a_n k^n x^n$, for $k \in R$

2. $f(x^N)\sum_{n=0}^{\infty}a_n x^{Nn}$

3. $f(x)\pm g(x)=\sum_{n=0}^{\infty}(a_n \pm b_n)x^n$

Note: The interval of convergence for sum is the intersection of the interval of convergence of the two original series.

Examples:

1. $\sum_{n=0}^{\infty}x^n+\sum_{n=0}^{\infty}\left(\frac{x}{2}\right)^n=\sum_{n=0}^{\infty}\left(1+\frac{1}{2^n}\right)x^n$ and the interval of convergence for the sum is (-1,1) which is the intersection interval of convergences (-1,1) and (-2,2) of the two series.

2. Use the power series representation $\sum_{n=0}^{\infty}\frac{x^n}{n!}$ of $f(x)=e^x$ to find the power series representation of $g(x)=e^{-x^2}$.

Solution: Since the power series representation of $f(x)=e^x=\sum_{n=0}^{\infty}\frac{x^n}{n!}$ then using 2 and 3 of the above theorem $\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{n!}$.

Quick check Class activity 2.3.1:

(1) Given that the Taylor Series for *sin*(*x*) = $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{x^{2n+1}}{(2n+1)!}$, find a Taylor series for $f(x)=\sin(4x)$.

(2) Given that the Taylor Series centered at $x=1$ for *ln*(*x*) = $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}(x-1)^n}{n}$, find a Taylor series for $\ln(x^2)$ and $\ln(3x)$.

2.3.1 Differentiation and Integration of Power Series

Theorem 2.3.2: If $\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots + c_n x^n + \cdots$

is a power series having radius of convergence $\mathbf{R} > 0$,

then the derivative of the series is

$$\frac{d}{dx}\sum_{n=0}^{\infty}(c_n x^n) = \sum_{n=0}^{\infty}\frac{d}{dx}(c_n x^n) = \sum_{n=1}^{\infty} n c_n x^{n-1}$$

$$= c_1 + 2c_2 x + 3c_3 x^2 + \cdots + nc_n x^{n-1} + \cdots$$

and also has the same radius of convergence $\mathbf{R} > 0$.

Remark: Repeated application of the above theorem shows that, $\sum_{n=0}^{\infty}\frac{d^2}{dx^2}(c_n x^n)$, $\sum_{n=0}^{\infty}\frac{d^3}{dx^3}(c_n x^n)$ and $\sum_{n=0}^{\infty}\frac{d^4}{dx^4}(c_n x^n)$, and so on, have the same radius of convergence $\mathbf{R} > 0$.

Example: Since the geometric series $\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots + x^n + \cdots$

has the radius of convergence $R = 1$ and converges on $(-1,1)$, the series

$$\sum_{n=0}^{\infty}\frac{d}{dx}x^n = \sum_{n=1}^{\infty} n x^{n-1} = 1 + 2x + 3x^2 + 4x^3 + 5x^4 + 6x^5 + \cdots,$$

$$\sum_{n=0}^{\infty}\frac{d^2}{dx^2}x^n = \sum_{n=2}^{\infty} n(n-1)x^{n-2} = 2 + 6x + 12x^2 + 20x^3 + 30x^4 + \cdots,$$

$$\sum_{n=0}^{\infty}\frac{d^3}{dx^3}x^n = \sum_{n=3}^{\infty} n(n-1)(n-2)x^{n-3} = 6 + 24x + 60x^2 + 120x^3 + \cdots$$

$$\vdots$$

all have the same radius of convergence and converges on $(-1,1)$,

Theorem 2.3.3: If $\sum_{n=0}^{\infty} c_n x^n$ is a power series with radius of convergence R and

$$f(x) = \sum_{n=0}^{\infty} c_n x^n \qquad \text{for all } x \text{ in } (-R, R),$$

then f is differentiable on $(-R, R)$ and

$$f'(x) = \sum_{n=0}^{\infty} \frac{d}{dx}(c_n x^n) \qquad \text{for all } x \text{ in } (-R, R),$$

Example 1: Find the radius and interval of convergence of

$$\sum_{n=0}^{\infty} \frac{x^{n+1}}{(n+1)^2}$$

and its derivative.

Solution:

Let $a_n = \frac{x^{n+1}}{(n+1)^2}$. Then by Generalized Ratio Test the series converges when

$$r = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$$

$$\Rightarrow \lim_{n\to\infty} \left|\frac{x^{n+2}}{(n+2)^2} \cdot \frac{(n+1)^2}{x^{n+1}}\right| = \lim_{n\to\infty} \frac{(n+1)^2}{(n+2)^2}|x| < 1$$

$$\Rightarrow \quad |x| < 1$$

Therefore the radius of convergence is $\mathbf{R} = 1$.

Obviously, the series converges at $x = 1$ *and* $x = -1$. Then the interval of convergence is $[-1, 1]$.

Let $f(x) = \sum_{n=0}^{\infty} \frac{x^{n+1}}{(n+1)^2}$. Then

$$f'(x) = \frac{d}{dx}\sum_{n=0}^{\infty} \frac{x^{n+1}}{(n+1)^2} = \sum_{n=0}^{\infty} \frac{d}{dx}\frac{x^{n+1}}{(n+1)^2}$$

$$= \sum_{n=1}^{\infty} \frac{x^n}{(n+1)}.$$

Thus $f'(x) = \sum_{n=1}^{\infty} \frac{x^n}{(n+1)}$

Example 2:

a) Obtain a power series representation of the function

$$f(x)=\frac{1}{(1-x)^2},\quad if\quad |x|<1$$

b) Show that

$$\sum_{n=0}^{\infty}\frac{x^n}{n!}$$

is the power series representation of e^x.

Solution:

a. Since the function

$g(x)=\frac{1}{1-x},\ |x|<1$ has power series representation $g(x)=\sum_{n=1}^{\infty}x^n$, and $f(x)=-g'(x)$ then the power series representation of the function

$$f(x)=-\frac{dg(x)}{dx}=\frac{1}{(1-x)^2}=\frac{d}{dx}\left(-\sum_{n=1}^{\infty}x^n\right)=-\sum_{n=1}^{\infty}n\,.x^{n-1}$$

b. Let $f(x)=\sum_{n=0}^{\infty}\frac{x^n}{n!}$. Then

$$f'(x)=\sum_{n=1}^{\infty}\frac{x^{n-1}}{(n-1)!},\ \text{let}\quad m=n-1$$

$$\Rightarrow f'(x)=\sum_{m=0}^{\infty}\frac{x^m}{m!}=f(x)$$

$$f''(x)=\sum_{n=2}^{\infty}\frac{x^{n-2}}{(n-2)!},\ \text{let}\quad k=n-2$$

$$\Rightarrow f''(x)=\sum_{k=0}^{\infty}\frac{x^k}{k!}=f(x)$$

$$\vdots$$

Since the only function which is the derivative of the function is itself is e^x, then $f(x)=e^x$, that is $e^x=\sum_{n=0}^{\infty}\frac{x^n}{n!}$.

2.3.2 Integration of Power Series

Theorem 2.3.4: Integration of Power Series

If

$f(x)=\sum_{n=0}^{\infty}c_n x^n$ converges on $(-R,R)$, then

$g(x)=\sum_{n=0}^{\infty}\frac{c_n}{n+1}x^{n+1}$ converges on $(-R,R)$ and

$\int f(x)dx=g(x)$, that is,

$$\int\left(\sum_{n=0}^{\infty}c_n\,x^n\right)dx=\sum_{n=0}^{\infty}\left(\int c_n x^n dx\right)=\sum_{n=0}^{\infty}\frac{c_n}{n+1}x^{n+1} \quad \text{on}\,(-R,R).$$

Proof: Exercise

Remark: If a power series converges at a and converges at b, then it converges at all numbers in between a and b..

$$\int_a^b\left(\sum_{n=0}^{\infty}c_n\,x^n\right)dx=\sum_{n=0}^{\infty}\left(\int_a^b c_n x^n dx\right)=\sum_{n=0}^{\infty}\frac{c_n}{n+1}\left(b^{n+1}-a^{n+1}\right)$$

Example: Show that

a) $\ln(1+x)=\sum_{n=0}^{\infty}\frac{(-1)^n}{n+1}x^{n+1}$, *for* $|x|<1$

b) $\arctan x=\sum_{n=0}^{\infty}\frac{(-1)^n}{2n+1}x^{2n+1}$, *for* $|x|<1$

c) $\ln\left(\frac{1}{1-x}\right)=-\sum_{n=1}^{\infty}\frac{x^n}{n}$, *for* $|x|<1$

Solution:

a. Since $\ln(1+x)=\int\frac{1}{1+x}\,dx$ and $\frac{1}{1+x}=\sum_{n=0}^{\infty}(-1)^n x^n$, *for* $|x|<1$.

$$\Rightarrow \ln(1+x)=\int\sum_{n=0}^{\infty}(-1)^n x^n\,dx=\sum_{n=0}^{\infty}\int\sum_{n=0}^{\infty}(-1)^n x^n\,dx$$

$$\Rightarrow \ln(1+x)=\sum_{n=0}^{\infty}\frac{(-1)^n}{n+1}\,x^{n+1}$$

Thus $\ln(1+x)=\sum_{n=0}^{\infty}\frac{(-1)^n}{n+1}\,x^{n+1}$ *for* $|x|<1$

b. Since $\arctan x = \int \frac{1}{1+x^2}\ dx$ and $\frac{1}{1+x^2} = \sum_{n=0}^{\infty}(-1)^n x^{2n},\ \ for\ |x|<1$

$$\Rightarrow \arctan x = \int \sum_{n=0}^{\infty}(-1)^n x^{2n}\,dx\ = \sum_{n=0}^{\infty}\int \sum_{n=0}^{\infty}(-1)^n x^{2n}\ dx$$

$$\Rightarrow \arctan x = \sum_{n=0}^{\infty}\frac{(-1)^n}{2n+1}\ x^{2n+1}$$

Thus $\arctan x = \sum_{n=0}^{\infty}\frac{(-1)^n}{2n+1}\ x^{2n+1}$, $for\ |x|<1$//

c. Since $\ln\left(\frac{1}{1-x}\right) = -\ln(1-x) = -\int \frac{1}{1-x}\ dx$ and $\frac{1}{1-x} = \sum_{n=0}^{\infty}x^n$ $for\ |x|<1$, then

$$\ln\left(\frac{1}{1-x}\right) = -\ \int \sum_{n=0}^{\infty}x^n\ dx = -\sum_{n=0}^{\infty}\int x^n\ dx\ = -\sum_{n=0}^{\infty}\frac{x^{n+1}}{n+1}$$

$$= -\sum_{n=1}^{\infty}\frac{x^n}{n}$$

$$\text{Thus}\ln\left(\frac{1}{1-x}\right) = -\sum_{n=1}^{\infty}\frac{x^n}{n}$$

Quick Check Class Activity 2.3.2:

1. Find the derivative $f'(x)$ and integral $\int f(x)\,dx$ of $f(x)$:

(a) $f(x) = \sum \frac{(-1)^{n+1}(x-4)^{n+1}}{n+1}$ (b) $f(x) = \sum_{n=1}^{\infty}\frac{(-1)^{n+1}(x-3)^n}{n5^n}$

2. Calculate, accurate to four decimal places.

a. $\int_0^1 \cos(x^2)dx$ *b.* $\int_0^{0.5} \sin\sqrt{x}\,dx$

Instructor's Role

- *Observation while they work in groups*
- *Check and Give feedback for their answers*

Activity: (*Individual Assignment exercises*)

1. Find the Taylor's (or Maclaurine series) of

   a. $f(x) = \ln(1+x)$

   b. $f(x) = \dfrac{x-1}{x+1}$, at $c = 2$

   c. $f(x) = x^2 \cos x^3$

   d. $h(x) = \sin 2x$, at $c = \pi$

   e. $f(x) = \sin x.\cos x$

   f. $h(x) = \sqrt{x}$, at $c = 1$

Group Activity 2.3.1 (*They will discuss in their respective groups and present the result for the whole group*)

1. Let $f(x)$ be the sum of the series. Find $f'(x)$ *and* $\int_0^x f(t)\,dt$

   a. $\sum_{n=0}^{\infty} \dfrac{(-1)^n}{n+1}\ x^n$

   b. $\sum_{n=0}^{\infty} \dfrac{1}{n^2+1}\ x^{n+1}$

   c. $\sum_{n=1}^{\infty} (n+1)\ x^n$

   d. $\sum_{n=1}^{\infty} \dfrac{5}{n}\ x^{n^2}$

2. Show that

   a. $\cosh x = \sum_{n=0}^{\infty} \dfrac{1}{(2n)!} x^{2n}$, *for all* x

   b. $\sinh x = \sum_{n=0}^{\infty} \dfrac{1}{(2n+1)!} x^{2n+1}$, *for all* x

   c. $\sqrt{e} = \sum_{n=0}^{\infty} \dfrac{1}{2^n n!}$

   d. $\dfrac{\pi}{4} = 1 - \dfrac{1}{3} + \dfrac{1}{5} - \dfrac{1}{7} + \dfrac{1}{9} - \cdots$

3. Find a power series representation of

   a. $f(x) = e^{3x^3}$

   b. $f(x) = \dfrac{e^x - 1}{x}$

   c. $r(x) = \dfrac{e^x - x - 1}{x^2}$

   d. $g(x) = \ln\left(\dfrac{1+x}{1-x}\right)$, $|x| < 1$

   e. $h(x) = \ln(2-3x)$, $|x| < \dfrac{2}{3}$

   f. $f(x) = \sinh x^2$

   g. $h(x) = \dfrac{\arctan x}{x}$

   h. $s(x) = \ln(1+x^2)$, $|x| < 1$

   i. $g(x) = \dfrac{x^2}{1+x^2}$, $|x| < 1$

j. $\int_0^x e^{-t^2}\,dt$

k. $\int_0^1 e^{-x^2}\,dx$

l. $f(x) = x^2 \tan^{-1} x$

4. Evaluate (*Assignment*)

a. $\int_0^x \frac{e^t - 1}{t}\,dt$

b. $\lim_{x\to 0} \frac{e^x - x - 1}{x^2}$

c. $\int_0^x \frac{\ln(1+t)}{t}\,dt$

Assessment

- *Answer for the raised questions and for the given assignment.*
- *Observation while they work in groups and present.*
- *Give feedback for their answers*

2.4 Binomial Series

Let p be any real number and let $f(x) = (1+x)^p$. Then we define the binomial coefficient $\binom{p}{n}$ by the formulas:

$$\binom{p}{0} = 1 \text{ and } \binom{p}{n} = \frac{p(p-1)(p-2)\ldots(p-(n-1))}{n!} \text{ for } n \geq 1.$$

In particular, $\binom{p}{1} = p$ and $\binom{p}{2} = \frac{p(p-1)}{2}$

Moreover if p is a positive integer, then

$$\binom{p}{n} = \frac{p!}{(p-k)!\,k!}$$

Theorem 2.4.1: (Binomial series)

For any real number $p \geq 0$ and for all x $|x| < 1$,

$$(1+x)^p = 1 + \binom{p}{1}x + \binom{p}{2}x^2 + \binom{p}{3}x^3 + \cdots + \binom{p}{n}x^n + \cdots = \sum_{n=0}^{\infty}\binom{p}{n}x^n$$

Proof: Exercise.

Example:

1. Find the Maclaurin series of
   i. $\sqrt{1+x}$
   ii. $(1-x^2)^{\frac{1}{2}}$,
2. Approximate $\sqrt{1.1}$ to five decimal places.

Solutions:

i. $(1+x)^{\frac{1}{2}} = \sum_{n=0}^{\infty}\binom{\frac{1}{2}}{n}x^n$

$$(1+x)^{\frac{1}{2}} = \binom{\frac{1}{2}}{0}+\binom{\frac{1}{2}}{1}x+\binom{\frac{1}{2}}{2}x^2+\binom{\frac{1}{2}}{3}x^3+\cdots+\binom{\frac{1}{2}}{n}x^n+\cdots$$

$$=1+\frac{1}{2}x+\frac{1}{2!}\left(\frac{1}{2}\right)\left(\frac{-1}{2}\right)x^2+\frac{1}{3!}\left(\frac{1}{2}\right)\left(\frac{-1}{2}\right)\left(\frac{-3}{2}\right)x^3+.....$$

ii. $\left(1-x^2\right)^{\frac{1}{2}} = \sum_{n=0}^{\infty}\binom{\frac{1}{2}}{n}(-1)^n x^n$

$$=1-\frac{1}{2}x^2+\frac{1}{2!}\left(\frac{1}{2}\right)\left(\frac{-1}{2}\right)x^4-\frac{1}{3!}\left(\frac{1}{2}\right)\left(\frac{-1}{2}\right)\left(\frac{-3}{2}\right)x^6+.....$$

Quick Check Activity 2.4.1 (*Group Discussion*)

1. Find the Maclaurine series of
   i. $(1-x)^{-2}$ *for* $|x|<1.$
   ii. $\sqrt[4]{1+x},\ \ |x|<1$
2. Approximate $\sqrt{1.4}$ to five decimal places.

Assessment

- *Answer for raised questions and for the given assignments.*
- *Observation while they work in groups and present.*
- *Giving feedback for their work.*

2.5 Unit Summary

- A series of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots + c_n(x-a)^n + \cdots$$

is said to be a power series in $x-a$.

- Every power series has the radius of convergence R, which is either 0, ∞, or a non-negative number.
- The interval in which the power series converges is called the interval of convergence.
- If R is the radius of convergence of the power series given by (*), then the interval of convergence is one of the following intervals: $(a-R,\ a+R)$, $[a-R, a+R]$, $[a-R, a+R)$, $(a-R, a+R]$
- Suppose f has derivatives of all order at **a,** then the power Series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n \qquad (**)$$

is called the Taylor series of f at a. In particular if $a=0$, then (**) becomes

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n \ ,$$

and is called Maclaurin Series of f.

- The partial sum of Taylor series,

$$f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a) + \frac{f^{(3)}(a)}{3!}(x-a)^3 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \frac{f^{(n+1)}(z)}{(n+1)!}(x-a)^{n+1},$$

where z is a number between **a** and x, is called Taylor formula.

In the Taylor formula,

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$$

is called the n[th] degree Taylor polynomial of f at **a** and

$$R_n(x) = \frac{f^{(n+1)}(z)}{(n+1)!}(x-a)^{n+1}$$

is called the Lagrange formula of the remainder of f at a.

For any real number $p \geq 0$ and for all x $|x| < 1$,

$$(1+x)^p = 1+\binom{p}{1}x+\binom{p}{2}x^2+\binom{p}{3}x^3+\cdots+\binom{p}{n}x^n+\cdots=\sum_{n=0}^{\infty}\binom{p}{n}x^n$$

Common Taylor series include:

$$\frac{1}{x}=1-(x-1)+(x-1)^2-(x-1)^3+(x-1)^4-\ldots+(-1)^n(x-1)^n$$

$$\frac{1}{1+x}=1-x+x^2-x^3+x^4-x^5+\ldots+(-1)^n(x-1)^n$$

$$\ln x=(x-1)-\frac{(x-1)^2}{2}+\frac{(x-1)^3}{3}-\frac{(x-1)^4}{4}+\ldots+\frac{(-1)^{n-1}}{n}\frac{(x-1)^n}{}$$

$$e^x=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\frac{x^5}{5!}+\ldots+\frac{x^n}{n!}$$

$$\sin x=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\ldots+\frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\cos x=1-\frac{x^2}{3!}+\frac{x^4}{5!}-\frac{x6}{7!}+\ldots+\frac{(-1)^n x^{2n}}{(2n)!}$$

$$\arctan x=x-\frac{x^3}{3}+\frac{x^5}{5}-\frac{x^7}{7}+\frac{x^9}{9}-\ldots+\frac{(-1)^n}{2n+1}\frac{x^{2n+1}}{}$$

2.6 Review Exercise

1. Determine the radius and interval of convergence of the following power series.

   a. $\sum_{n=1}^{\infty}\frac{x^n}{2^n(n+1)}$

   b. $\sum_{n=1}^{\infty}\frac{x^n}{n^2+1}$

   c. $\sum_{n=1}^{\infty}\frac{x^n}{2^n\sqrt{n}}$

   d. $\sum_{n=1}^{\infty}\frac{n!}{n^n}x^n$

   e. $\sum_{n=1}^{\infty}\frac{(2n)!}{(3n)!}x^n$

   f. $\sum_{n=0}^{\infty}\left(\frac{3}{4}\right)^n(2x+5)^n$

   g. $\sum_{n=1}^{\infty}\frac{(x+2)^n}{(n+1).2^n}$

   h. $\sum_{n=1}^{\infty}(\sinh 2n).x^n$

2. Let $f(x)$ be the sum of the series. Find $f'(x)$ *and* $\int_0^x f(t)\,dt$

a. $\sum_{n=0}^{\infty} \frac{1}{n^2+1}\ x^{n+1}$

b. $\sum_{n=1}^{\infty} \frac{5}{n}\ x^{n^2}$

3. Find a power series representation of

a. $f(x) = e^{3x^3}$

b. $r(x) = \frac{e^x - x - 1}{x^2}$

c. $g(x) = \ln\left(\frac{1+x}{1-x}\right),\ |x| < 1$

d. $h(x) = \frac{\arctan x}{x}$

e. $s(x) = \ln(1+x^2),\ |x| < 1$

f. $g(x) = \frac{x^2}{1+x^2},\quad |x| < 1$

g. $\int_0^1 e^{-x^2} dx$

h. $f(x) = x^2 \tan^{-1} x$

4. Find the Taylor's expansion of

a. $h(x) = \arctan x$ at $c = 0$

b. $f(x) = \frac{x-1}{x+1}$, at $c = 2$

c. $f(x) = \sin x.\cos x$ at $c = 0$

CHAPTER THREE

DIFFERENTIAL CALCULUS OF FUNCTIONS OF SEVERAL VARIABLES

Unit Introduction

In this chapter we will extend many of the basic concepts of calculus to functions of two or more variables, commonly called functions of several variables. We will begin by discussing limits and continuity for functions of two and three variables, then we will define derivatives of such functions, and then we will use these derivatives to study tangent planes, rates of change, slopes of surfaces, and maximization and minimization problems. This unit is divided into four sections. The first section presents basic definitions of functions of several variables. Limit and continuity of functions of several variables are parts of the second section. Partial derivatives will be treated in third section and extreme values in the fourth.

Unit Objectives

At the end of this unit students would be able to

- Determine domain and range of functions of two or three variables.
- Determine combination and composition of functions of two or three variables.
- Find out limit and continuity of functions of two or three variables.
- Compute partial derivatives of functions of two or three variables.
- Determine gradient of functions of two or three variables.
- Determine directional derivative of functions of two or three variables in the direction of a given non-zero vector.
- Identify the relation between directional derivative and gradient of functions of two or three variables.
- Apply the concept of partial derivatives of functions of two or three variables in solving real life problems.

3.1 Notations and Terminologies

Many quantities depend on more than one variable. For example, the area A of a triangle depends on the base length b and height h by the formula $A = \frac{1}{2}bh$; the volume V of a rectangular box depends on the length l, width w, and height h by the formula $V = lwh$; and the arithmetic average x of n real numbers, $(x_1, x_2, ..., x_n)$, depends on those numbers by the formula

$$\bar{x} = \frac{1}{n}(x_1 + x_2 + ... + x_n)$$

Thus, we say that

A is a function of the two variables b and h;

V is a function of the three variables l, w, and h;

$\bar{x}$ is a function of the n variables .

The terminology and notation for functions of two or more variables is similar to that for functions of one variable. For example, the expression

$$z = f(x, y)$$

means that z is a function of x and y in the sense that a unique value of the dependent variable z is determined by specifying values for the independent variables x and y. Similarly,

$$w = f(x, y, z)$$

expresses w as a function of x, y, and z, and

$$u = f(x_1, x_2, ..., x_n)$$

expresses u as a function of $x_1, x_2, \ldots, x_n$.

The following definitions summarize this discussion.

Definition: -(Functions of two variables)

Let D be a non-empty subset of the xy-plane. A rule f that assigns a real number $f(x, y)$ to each point (x, y) in D is called a ***function of two variables.*** The set D is called ***the domain of*** f and the set of all values $f(x, y)$ is called ***the range of*** f .

Similarly,

Definition (Functions of three variables)

Let D be a non-empty subset of a three dimensional space. A rule f that assigns a real number $f(x,y,z)$ to each point (x,y,z) in D is called a *function of three variables.* The set D is called *the domain of* f and the set of all values $f(x,y,z)$ is called *the range of* f .

Unless stated explicitly, the **domain** of functions of several variables is taken to be the set of all points for which the function is defined or yields a real value for the dependent variable. We call this the **natural domain** of the function

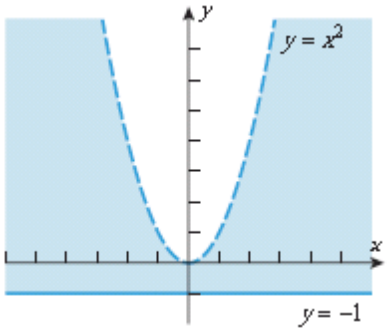
Example 1: Let $f(x,y)=\sqrt{y+1}+\ln\left(x^2-y\right)$**.** Find $f(e,0)$ and sketch the natural domain of f

Solution. By substitution,

$$f(e,0)=\sqrt{0+1}+\ln\left(e^2-0\right)=1+2=3$$

To find the natural domain of f, we note that $\sqrt{y+1}$ is defined only when $y\geq -1$, while $\ln\left(x^2-y\right)$is defined only when $0<x^2-y$ or $y<x^2$. Thus, the natural domain of f consists of all points in the *xy*-plane for which $-1\leq y<x^2$ i.e. $D=\left\{(x,y)\in R^2 / -1\leq y<x^2\right\}$.

To sketch the natural domain, we first sketch the parabola $y=x^2$ as a "dashed" curve and the line $y=-1$ as a solid curve. The natural domain of f is then the region lying above or on the line $y=-1$ and below the parabola $y=x^2$.



Example 2: Find and sketch the natural domains of the following functions

a) $f(x,y)=\sqrt{x+y}$

b $f(x,y)=\dfrac{x+y}{x-y}$

c) $f(x,y)=\ln(2x-y)$

d) $f(x,y,z)=\sqrt{x^2+y^2+z^2}$

Solutions:

a) Because of the radical for the function $f(x,y)=\sqrt{x+y}$ the inequality $x+y\geq 0$ must hold true. Therefore the domain of $f(x,y)=\sqrt{x+y}$ is all x such that $x\geq -y$. That is, $D=\{(x,y)\in R^2 / x\geq -y\}$.

b) Since the denominator of a rational function cannot be zero, the domain of $f(x,y)=\dfrac{x+y}{x-y}$ is all x such that $x-y\neq 0$ or $x\neq y$ .Thus $D=\{(x,y)\in R^2 / x\neq y\}$

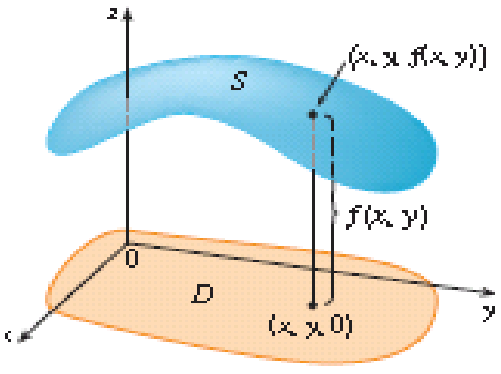
c) Since the argument of a logarithmic function must always be greater than zero, that is $2x-y>0$, the domain of $f(x,y)=\ln(2x-y)$ is all x such that $2x>y$. Thus $D=\{(x,y)\in R^2 / 2x>y\}$

d) Since the function $f(x,y,z)=\sqrt{x^2+y^2+z^2}$ is defined for all points, the domain is $D=\{(x,y,z)\in R^3\}$

Exercise: Sketch the graphs of the natural domains of each of the functions in example (a)-(c).

3.2 Graphs of Functions of Two Variables

Just as the graph of a function of one variable is a curve with equation $y=f(x)$, so the graph of a function of two variables is the graph of the equation $z=f(x,y)$. In General, such graph will be a Surface in 3-space. We can visualize the graph S of f as lying directly above or below its domain D in the *xy*-plane.

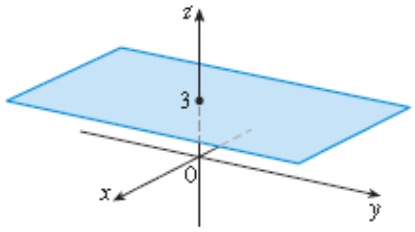


Therefore, the **graph** of a function of two variables is the surface consisting of all points (x,y,z) in space such that $z=f(x,y)$.

Examples:

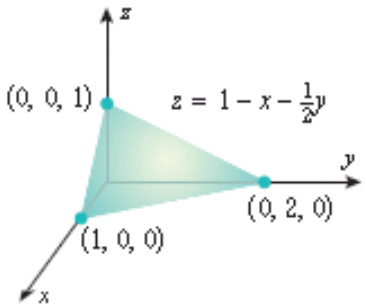
1. Sketch the graph of the constant function $f(x, y) = 4$

Solution: The equation of the graph of $f(x, y) = 4$ is $z = 4$. So the graph is the surface consisting of all points (x, y, z) whose z-coordinate is 4. This is the horizontal plane that is parallel to the xy-plane and four units above it, as shown in Figure below.



2. Sketch the graph of $f(x, y, z) = 1 - x - \frac{1}{2}y$

Solution: By definition, the graph of the given function is the graph of the equation $z = 1 - x - \frac{1}{2}y$ which is a plane. A triangular portion of the plane can be sketched by plotting the intersections with the coordinate axes and joining them with line segments as below.



3: Describe the graph of the function $f(x, y) = \sqrt{1 - x^2 - y^2}$ in an xyz-coordinate system.

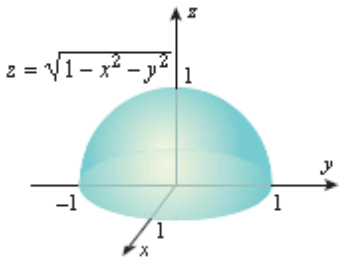
***Solution* :** By definition, the graph of the given function is the graph of the equation

$$z = \sqrt{1 - x^2 - y^2}$$

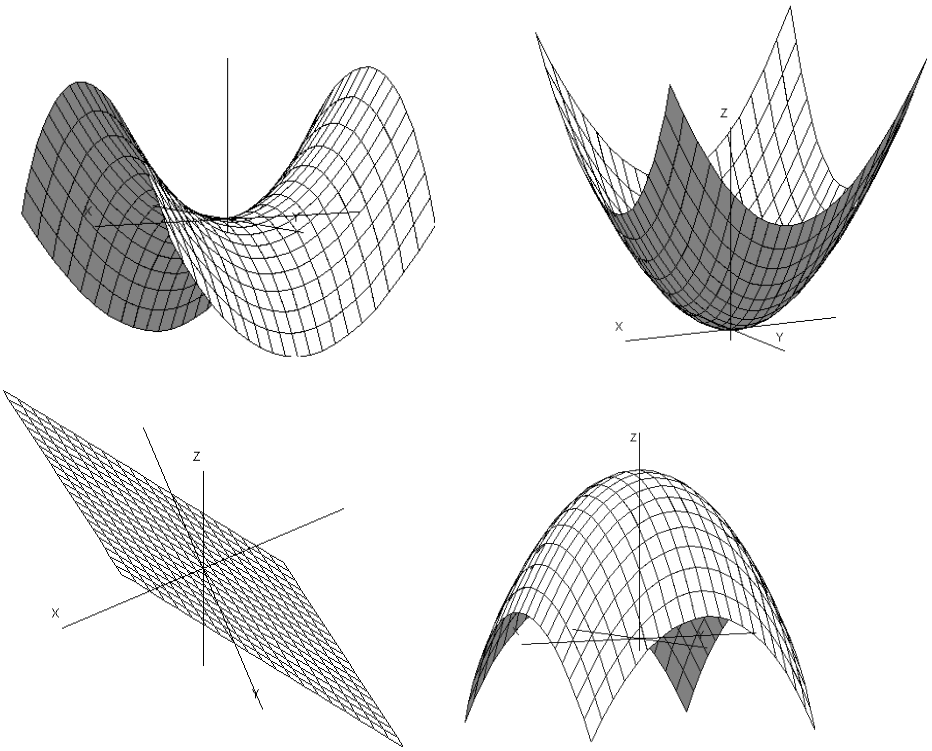
After squaring both sides, this can be rewritten as

$$x^2 + y^2 + z^2 = 1$$

which represents a sphere of radius 1, centered at the origin. Since the equation imposes the added condition that $z \geq 0$, the graph is just the upper hemisphere (Figure below).



Here are Some common graphs of functions of two variables



3.2.1 Level Curves

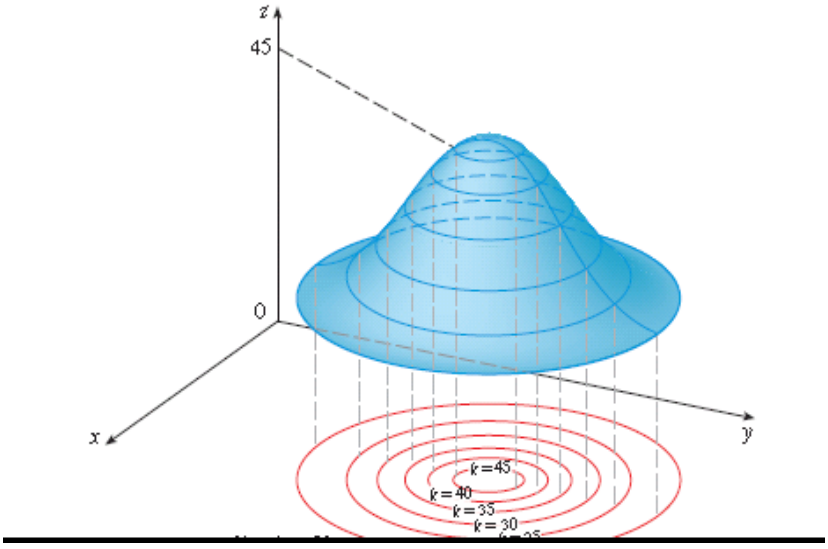
One method for visualizing a function of two variables is to look at its graph. Another method, borrowed from mapmakers, is a **contour map** on which points of constant elevation are joined to form *contour curves*, or *level curves*.

- The level curves of a function of two variables are the curves with equatins $f(x, y) = k$ where k is a constant (in the range of f).

A level curve $f(x, y) = k$ is the set of all points in the domain of f at which f takes on a given value k. In other words, it shows where the graph of f has height k .You can see from Figure below the relation between level curves and horizontal traces, which are obtained by slicing the graph of with horizontal planes.

The level curves $f(x, y) = k$ are just the traces of the graph of f in the horizontal plane $z = k$ projected down to the *xy*-plane. So if you draw the level curves of a function and visualize

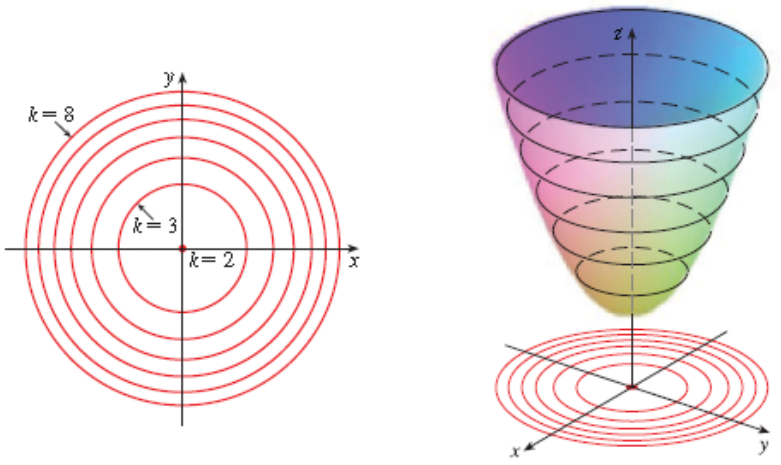
them being lifted up to the surface at the indicated height, then you can mentally piece together a picture of the graph. The surface is steep where the level curves are close together. It is somewhat flatter where they are farther apart



Example 1: Sketch a contour map of the function $h(x, y) = x^2 + y^2 + 2$.

Solution:

Recall that a contour map consists of a collection of level curves. Each level curve has the equation $x^2 + y^2 + k = 2$ or $x^2 + y^2 = k - 2$ If $k > 2$, we recognize this as the equation of a circle with center the origin and radius $\sqrt{k-2}$. We sketch these circles for several values of k in Figure (a). Then Figure (b) shows these level curves lifted up to form the graph of h, which has the equation $z = x^2 + y^2 + 2$.This surface is called a paraboloid because vertical traces have the equations $z = x^2 + k^2 + 2$ and $z = y^2 + k^2 + 2$, which are parabolas.



(a) Contour map of $h(x, y) = x^2 + y^2 + 2$ (b) Graph of h

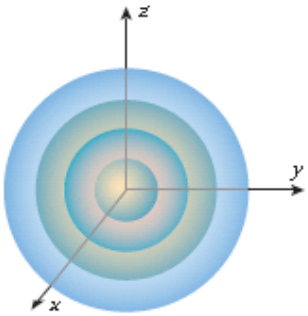
3.2.2 Level Surfaces

We close this section by looking at the graphs of functions of three variables $f(x, y, z)$. We won't actually graph any such functions, since a true graph require four dimensions. However we can gain important information from looking at level surfaces of a function *f*.

Level Surfaces of a function $f(x, y, z)$ are the graphs of the equation $f(x, y, z) = c$, for different choices of the constant c. Much as level curves do for functions of two variables, level surfaces can help you identify symmetries and regions of rapid or slow change in functions of three variables.

Example: Describe the level surfaces of $f(x, y, z) = x^2 + y^2 + z^2$

Solution: The level surfaces are described by the equation $x^2 + y^2 + z^2 = c$. Of course these are spheres of radius $\sqrt{c}$ centered at the origin; for $c > 0$. For c = 0 the graph is the single point *(*0, 0, 0*)*; and for c < 0 there is no level surface.(see fig below).



Quick Check Class Activity (Group work)

1. Sketch contour plots for (a) $f(x, y) = -x^2 + y$ and (b) $f(x, y) = x^2 + y^2$

2.Sketch the indicated traces and graph $z = f(x, y)$

a) $f(x, y) = x^2 + y^2, z = 1, z = 4, z = 9, x = 0$

b) $f(x, y) = x^2 - y^2; z = 0, z = 1, y = 0, y = 2$

c) $f(x, y) = x - 2y; z = 0, z = 1, x = 0, y = 0$

3) Sketch level surfaces of a) $f(x, y, z) = x^2 - y^2 + z^2$

b) $f(x, y, z) = x^2 + y^2 - z$

Instructor's Role

- *Observation while they work in groups*
- *Give feedback for their answers*

Group Activity 3.2.1

1. Find and sketch the domain D of the following functions of two variables:

   a. $f(x,y) = xy$

   b. $f(x,y) = \dfrac{x^2 - y}{x^2 + 7}$

   c. $f(x,y) = \tan^{-1}(\dfrac{x}{y})$

   d. $f(x,y) = \dfrac{2x^2 - y^2}{x^3}$

   e. $f(x,y) = \dfrac{1}{1-\sqrt{x^2+y^2}}$

2. Find and sketch the domain of the following functions

   a. $f(x,y,x) = x.y.z$

   b. $h(x,y,z) = \cos\left(\dfrac{1}{x+y-z}\right)$

   c. $g(x,y,z) = \dfrac{1}{\sqrt{1-(x^2+y^2+z^2}}$

Applied Problems (Functions of Sveral Variables)

3. Suppose that the concentration *C* in mg/L of medication in a patient's bloodstream is modeled by the function $C(x,t) = 0.2x(e^{-0.2t} - e^{-t})$ *C(x, t)* = 0.2*x*(*e*–0.2*t* − *e*–*t*), where *x* is the dosage of the medication in mg and *t* is the number of hours since the beginning of administration of the medication.

(a) Estimate the value of *C(*25, 3*)* to two decimal places. Include appropriate units and interpret your answer in a physical context.

(b) If the dosage is 100 mg, give a formula for the concentration as a function of time *t* .

(c) Give a formula that describes the concentration after 1 hour in terms of the dosage

4. If *T (x, y)* is the temperature at a point *(x, y)* on a thin metal plate in the *xy*-plane, then the level curves of *T* are called ***isothermal curves***. All points on such a curve are at the same temperature. Suppose that a plate occupies the first quadrant and *T (x, y) = xy*.

(a) Sketch the isothermal curves on which *T* = 1, *T* = 2, and *T* = 3.

(b) An ant, initially at *(*1, 4*)*, wants to walk on the plate so that the temperature along its path remains constant. What path should the ant take and what is the temperature along that path?

4. If $V(x,y)$ is the voltage or potential at a point *(x, y)* in the *xy*-plane, then the level curves of *V* are called ***equipotential curves***. Along such a curve, the voltage remains constant. Given that

$$V(x,y) = \frac{8}{\sqrt{16 + x^2 + y^2}}$$

sketch the equipotential curves at which $V = 2.0, V = 1.0$, and $V = 0.5$.

3.3 Limit and Continuity of Functions of Two Variables

Definition: - Let f be a function defined throughout a set containing a disc centered at (x_0, y_0) except possibly at (x_0, y_0)itself, and let L a number. Then L is said to be the limit of f at (x_0, y_0)if for every $\varepsilon > 0$ there exists $\delta > 0$such that

$$0 < \sqrt{(x-x_0)^2} + \sqrt{(y-y_0)^2} < \delta \Rightarrow |f(x,y) - L| < \varepsilon$$

$$\Leftrightarrow \lim_{(x,y)\to(x_0,y_0)} f(x,y) = L$$

Similarly, for functions of three variables

Definition: Let f be a function defined throughout a set containing a sphere centered at (x_0, y_0, z_0) except possibly at (x_0, y_0, z_0)itself, and let L a number. Then L is said to be the limit of f at (x_0, y_0, z_0) if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < \sqrt{(x-x_0)^2} + \sqrt{(y-y_0)^2} + \sqrt{(z-z_0)^2} < \delta \Rightarrow |f(x,y,z) - L| < \varepsilon$$

$$\Leftrightarrow \lim_{(x,y,z)\to(x_0,y_0,z_0)} f(x,y,z) = L$$

Note that in both definitions δ is not unique and L is unique.

Example: - Show that $\lim_{(x,y)\to(x_0,y_0)} x = x_0 \quad \wedge \quad \lim_{(x,y)\to(x_0,y_0)} y = y_0$

Solution:- Given any number $\varepsilon > 0$, we must find another number $\delta > 0$ so that $|f(x,y) - L| = |x - x_0| < \varepsilon$ whenever $0 < \sqrt{(x-x_0)^2} + \sqrt{(y-y_0)^2} < \delta$.

Notice that

$$\sqrt{(x-x_0)^2 + (y-y_0)^2} > \sqrt{(x-x_0)^2} = |x-x_0|$$

and so taking $\delta = \varepsilon$ we have that

$$|x-x_0| = \sqrt{(x-x_0)^2} < \sqrt{(x-x_0)^2 + (y-y_0)^2} < \varepsilon$$

whenever $0 < \sqrt{(x-x_0)^2} + \sqrt{(y-y_0)^2} < \delta$. Likewise, we can show that $\lim_{(x,y)\to(x_0,y_0)} y = y_0$

With these definitions of limit we can prove the following results for sums of combinations of functions.

Theorem 3.3.1- If $\lim_{(x,y)\to(x_0,y_0)} f(x,y)$ *and* $\lim_{(x,y)\to(x_0,y_0)} g(x,y)$ exists, then

a. $\lim_{(x,y)\to(x_0,y_0)} \big(f(x,y) \pm g(x,y)\big) = \lim_{(x,y)\to(x_0,y_0)} f(x,y) \pm \lim_{(x,y)\to(x_0,y_0)} g(x,y)$

b. $\lim_{(x,y)\to(x_0,y_0)} f(x,y).g(x,y) = \lim_{(x,y)\to(x_0,y_0)} f(x,y) \cdot \lim_{(x,y)\to(x_0,y_0)} g(x,y)$

c. $\lim_{(x,y)\to(x_0,y_0)} \dfrac{f(x,y)}{g(x,y)} = \dfrac{\lim_{(x,y)\to(x_0,y_0)} f(x,y)}{\lim_{(x,y)\to(x_0,y_0)} g(x,y)}$, Provided $\lim_{(x,y)\to(x_0,y_0)} g(x,y) \neq (0,0)$

A polynomial in two variables is any sum of terms of the form $cx^n y^m$ where c is a constant and m and n are non-negative integers. Therefore by the above theorem we can conclude that the limit of any polynomial always exists and is found by simple substitution.

Examples: 1. Show that $\lim_{(x,y)\to(-1,2)} 5x^3y^2 = -20$.

Solution: Since $f(x,y)$ is a polynomial function of two variables, using simple substitution

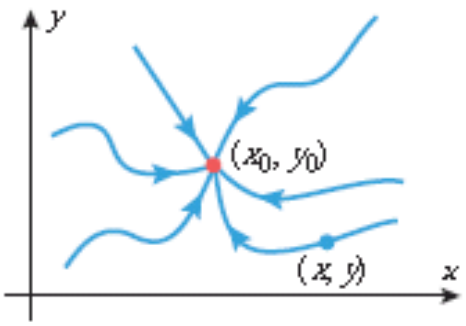
$$\lim_{(x,y)\to(-1,2)} 5x^3y^2 = 5(-1)^3(2)^2 = -20$$

2. Evaluate $\lim_{(X,y)\to(2,1)} \dfrac{2x^2y+3xy}{5xy^2+3y}$

Solution: Since $\lim_{(x,y)\to(2,1)} (5xy^2+3y) = 10+3 = 13 \neq 0$. Applying property (c) in the theorem we have $\lim_{(X,y)\to(2,1)} \dfrac{2x^2y+3xy}{5xy^2+3y} = \dfrac{8+6}{10+3} = \dfrac{14}{13}$.

Remarks

1. Since we are in two dimensions, there are infinite number of paths along which we approach any given point (x_0,y_0) and By the definition $\lim_{(x,y)\to(x_0,y_0)} f(x,y) = L$ implies that $f(x,y)$ must approach L along each line or curve through (x_0,y_0).



2. If $f(x,y)$ approaches L_1 as (x,y) approaches (x_0,y_0) along path P_1 and $f(x,y)$ approaches $L_2 \neq L_1$ along path P_2, then $\lim_{(x,y)\to(x_0,y_0)} f(x,y)$ does not exist.

3.Obviously, you can't check each path individually. In practice, if you suspect that a limit does not exist, you should check the limit along the simplest paths through that point.

4. The simplest paths to try are:

(i). $x = a, y \to b$ (vertical line)

(ii). $y = b, x \to a$ (horizontal line)

(iii). A curve $y = g(x), x \to a$ where $g(a) = b$

(iv). A curve $x = g(y), y \to b$ where $g(b) = a$

Example:1. - Show that $\lim_{(x,y)\to(0,0)} \dfrac{x^2 - y^2}{x^2 + y^2}$ does not exist.

Solution: -

- If $(x,y) \to (0,0)$ along the $x-axis$, then

$$\lim_{(x,y)\to(x,0)} \frac{x^2 - y^2}{x^2 + y^2} = \lim_{(x,y)\to(x,0)} \frac{x^2}{x^2} = 1$$

- If $(x,y) \to (0,0)$ along the $y-axis$, then

$$\lim_{(x,y)\to(0,y)} \frac{x^2 - y^2}{x^2 + y^2} = \lim_{(x,y)\to(0,y)} \frac{-x^2}{x^2} = -1$$

- If $(x,y) \to (0,0)$ along the *the line* $y = x$, then

$$\lim_{(x,y)\to(0,0)} \frac{x^2 - y^2}{x^2 + y^2} = \lim_{x\to 0} \frac{0}{x^2} = 0$$

Since $f(x,\ y) = \dfrac{x^2 - y^2}{x^2 + y^2}$ has more than one limit at $(0,\ 0)$, then $\lim_{(x,y)\to(0,0)} \dfrac{x^2 - y^2}{x^2 + y^2}$ does not exist.

Example 2. Show that $\lim_{(x,y)\to(0,0)} \dfrac{xy^2}{x^2 + y^4}$ doesn't exist**.**

Solution: -

- If $(x,y) \to (0,0)$ along the line $x = 0$, then

$$\lim_{(x,y)\to(x,0)} \frac{x\,y^2}{x^2 + y4} = \lim_{(x,y)\to(x,0)} \frac{0.y^2}{0^2 + y^4} = \lim_{(x,y)\to(0,0)} 0 = 0$$

- If $(x,y)\to(0,0)$ along the line $y=0$, then

$$\lim_{(x,y)\to(x,0)}\frac{xy^2}{x^2+y4}=\lim_{(x,y)\to(x,0)}\frac{x.0^2}{x^2+0^4}=\lim_{(x,y)\to(0,0)}0=0$$

- If $(x,y)\to(0,0)$ along the *the line* $y=kx(x=0\Rightarrow y=0)$, then

$$\lim_{(x,y)\to(0,0)}\frac{xy^2}{x^2+y4}=\lim_{(x,y)\to(x,0)}\frac{x.(kx)^2}{x^2+(kx)^4}=\lim_{(x,y)\to(0,0)}0=0$$

Still the limit is zero we must find another path through (0,0) along which the limit is nonzero. Finally,

- If $(x,y)\to(0,0)$ along the $x=y^2(y=0\Rightarrow x=0)$, then

$$\lim_{(y^2,y)\to(0,0)}\frac{y^2(y^2)}{(y^2)^2+y4}=\lim_{(y^2,y)\to(0,0)}\frac{y^4}{2y^4}=\lim_{y\to(0,0)}\frac{1}{2}=\frac{1}{2}$$

Since $f(x,\ y)$ has more than one limit at $(0,\ 0)$, then $\lim_{(x,y)\to(0,0)}\frac{xy^2}{x^2+y^4}$ does not exist.

Example 3 Find $\lim_{(x,y)\to(0,0)}(x^2+y^2)\ln(x^2+y^2)$

Solution. Let (r,θ) be polar coordinates of the point (x,y) with $r\geq 0$. Then we have $x=r\cos\theta, y=r\sin\theta, r^2=x^2+y^2$.

Moreover, since $r\geq 0$ we have $r=\sqrt{x^2+y^2}$, so that $r\to 0^+$ if and only if $(x,y)\to(0,0)$. Thus, we can rewrite the given limit as

$$\lim_{(x,y)\to(0,0)}(x^2+y^2)\ln(x^2+y^2)=\lim_{r\to 0^+}r^2\ln r^2=\lim_{r\to 0^+}\frac{\ln r^2}{\frac{1}{r^2}}$$

This converts the limit to an indeterminate form of type ∞/∞.

Applying L'Hôpital's rule

$$\lim_{r\to 0^+}\frac{\ln r^2}{\frac{1}{r^2}}=\lim_{r\to 0^+}-r^2=0$$

Theorem 3.3.2: Suppose that $|f(x,y)-L|\leq g(x,y)$ for all (x,y) in the interior of some circle centered at (x_0,y_0) except possibly at (x_0,y_0). If $\lim_{(x,y)\to(x_0,y_0)}g(x,y)=0$ then

$\lim_{(x,y)\to(x_0,y_0)}f(x,y)=L$.

Example: 1.Show that $\lim_{(x,y)\to(0,\,0)} \dfrac{x^3+y^3}{x^2+y^2} = 0$

Solution: Observe that $0 \le \left|\dfrac{x^3}{x^2+y^2}\right| \le \left|\dfrac{x^3}{x^2}\right| = |x|$ .Since $\lim_{x\to 0} x = 0$,by squeezing theorem $\lim_{(x,y)\to(0,\,0)} \dfrac{x^3}{x^2+y^2} = 0$.

Again observe that

$0 \le \left|\dfrac{y^3}{x^2+y^2}\right| \le \left|\dfrac{y^3}{y^2}\right| = |y|$ and since $\lim_{x\to 0} y = 0$,by squeezing theorem $\lim_{(x,y)\to(0,\,0)} \dfrac{y^3}{x^2+y^2} = 0$.

Thus we can use the sum formula to get

$$\lim_{(x,y)\to(0,\,0)} \frac{x^3+y^3}{x^2+y^2} = \lim_{(x,y)\to(0,\,0)} \frac{x^3}{x^2+y^2} + \lim_{(x,y)\to(0,\,0)} \frac{x^3}{x^2+y^2} = 0$$

Another alternative to show this uses polar coordinates: If $x = r\cos\theta, y = r\sin\theta, r^2 = x^2+y^2$ then $(x,y)\to(0,0)$ if and only if $r \to 0^+$. Thus after substitution of $x = r\cos\theta, y = r\sin\theta,$ the limit becomes

$$\lim_{(x,y)\to(0,0)} \frac{x^3+y^3}{x^2+y^2} = \lim_{r\to 0} r(\cos^3\theta + \sin^3\theta) = 0$$

Quick Check Class Group Activity 3.3.1

1.Show that

a. $\lim_{(x,y)\to(-1,\,2)} \dfrac{x^3+y^3}{x^2+y^2} = \dfrac{7}{5}$

b. $\lim_{(x,y)\to(1,0)} \dfrac{(x-1)^2 \ln x}{(x-1)^2+y^2} = 0$

c) $\lim_{(x,y)\to(0,\,0)} \dfrac{x^2 y}{x^2+y^2} = 0$

2. Show that the following limit doesn't exist

a. $\lim_{(x,y)\to(0,0)} \dfrac{y}{x}$

b. $\lim_{(x,y)\to(0,0)} \dfrac{xy}{x^2+y}$

c. $\lim_{(x,y)\to(0,0)} \dfrac{2x^2\sin y}{2x^2+y^2}$

d. $\lim_{(x,y)\to(0,0)} \dfrac{x^3+4x^2+2y^2}{2x^2+y^2}$

Instructor's Role

- *Observation while they work in groups*
- *Check and Give feedback for their answers*

Definition: -

1. A function f of two variables is ***continuous*** at (x_0, y_0) if

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = f(x_0,y_0)$$

2. A function f of three variables is ***continuous*** at (x_0, y_0) if

$$\lim_{(x,y,z)\to(x_0,y_0,z_0)} f(x,y,z) = f(x_0,y_0,z_0)$$

3. A function of several variables is continuous if it ***continuous*** at each point in its domain.

In addition, if f is continuous at every point in an open set *D*, then we say that *f* is ***continuous on D***, and if *f* is continuous at every point in the *xy*-plane, then we say that *f* is ***continuous everywhere***.

The following theorem, which we state without proof, illustrates some of the ways in which continuous functions can be combined to produce new continuous functions.

Theorem 3.3.3:

(a). If $g(x)$ is continuous at x_0 and $h(y)$ is continuous at y_0, then $f(x,y) = g(x)h(y)$ is continuous at (x_0, y_0)

(b). If $h(x,y)$ is continuous at (x_0, y_0) and $g(u)$ is continuous at $u = h(x_0, y_0)$, then the composition $f(x,y) = g(h(x,y))$ is continuous at (x_0, y_0).

(c). If $f(x,y)$ is continuous at (x_0, y_0) and if $x(t)$ and $y(t)$ are continuous at t_0 with $x(t_0) = x_0$ and $y(t_0) = y_0$ then the composition $f(x(t), y(t))$ is continuous at t_0.

A similar result holds when f is a function is of three variables.

Example 1: Show that $f(x,y) = \dfrac{x^3+y^3}{x^2+y^2}$ is not continuous at $(0,0)$.

Solution: i) From Example 1 above we have $\lim_{(x,y)\to(0,0)} \dfrac{x^3+y^3}{x^2+y^2} = 0$

ii) $f(0,0)$ is not defined iii) $\lim_{(x,y)\to(0,0)} f(x,y) \neq f(0,0)$

Thus by the definition of continuity the function $f(x,y) = \dfrac{x^3+y^3}{x^2+y^2}$ is not continuous at $(0,0)$

Example 2: Show that the functions $f(x,y)=3x^2y^2$ and $f(x,y)=\sin(3x^2y^2)$ are continuous every where.

Solution: The polynomials $g(x)=3x^2$ and $h(y)=y^2$ are continuous at every real number, and therefore by part (a) of Theorem 13.2.4, the function $f(x,y)=3x^2y^2$ is continuous at every point in the *xy*-plane. Since $3x^2y^2$ is continuous at every point in the *xy*-plane and *sinu* is continuous at every real number *u*, it follows from part (b) that the composition $f(x,y)=\sin(3x^2y^2)$ is continuous everywhere.

Quick check Class Activity 3.3.1

1.Show that the following functions are not continuous at the given point.

a) $f(x,y)=\dfrac{x^2y}{x^2+y^2},(0,0)$ b) $f(x,y)=\dfrac{(x-1)^2\ln x}{(x-1)^2+y^2}$ at $(1,0)$

2. Show that $\lim\limits_{(x,y)\to(e,1)}\ln\left(\dfrac{x}{y}\right)=1$

Group Activity 3.3.1

1. Exhibit the limit and continuity of the following functions at the given points.
   a. $f(x,y)=x^3-4xy+5y^2$ at (1,-2)
   b. $g(x,y)=\dfrac{x^2-xy+1}{x^2+y^2}$ at (1, 0)
   c. $h(x,y)=\dfrac{3x^2+2xy^2-y^4}{1+y^2}$ at (-1,1)
   d. $r(x,y)=\dfrac{\sin(x^2+y^2)}{x^2+y^2}$ at (0, 0)
   e. $k(x,y)=\begin{cases}0 & if\ (x,y)=(0,0)\\ \dfrac{2xy}{x^2+y^2} & if\ (x,y)\neq(0,0)\end{cases}$ at (0, 0)

2. Show that the function

$$f(x,y)=\begin{cases}\dfrac{xy^2}{x^3+y^3} & , \quad if\ (x,y)\neq(0,0)\\ 0 & , \quad if\ (x,y)=(0,0)\end{cases}$$

is continuous in each variable except at $(0\ ,\ 0)$, that is, $f(x,0)$ is continuous function of x at 0 and $f(0,y)$ is a continuous function of y at 0. And f is not continuous at (0, 0).

3. Let $f(x,y)=\begin{cases}\dfrac{\sin xy}{x^2+y^2} & ,\ if\ (x,y)\neq(0,0)\\ 0 & ,\ if\ (x,y)=(0,0)\end{cases}$ Show that f is not continuous at (0, 0).

4. Let $f(x,y)=\begin{cases}\dfrac{x^3.y^3}{x^{12}+y^4} & ,\ if\ (x,y)\neq(0,0)\\ 0 & ,\ if\ (x,y)=(0,0)\end{cases}$ Show that f is not continuous at $(0\ ,\ 0)$.

Instructor's Role

- *Answer for the raised questions.*
- *Observation while they work in groups*
- *Give feedback for their answers*

3.4. Partial Derivatives

If $z=f(x,y)$,then one can inquire how the value of z changes if y is held fixed and x is allowed to vary, or if x is held fixed and y is allowed to vary. For example, the ideal gas law in physics states that under appropriate conditions the pressure exerted by a gas is a function of the volume of the gas and its temperature. Thus, a physicist studying gases might be interested in the rate of change of the pressure if the volume is held fixed and the temperature is allowed to vary, or if the temperature is held fixed and the volume is allowed to vary.

Suppose that (x_0,y_0) is a point in the domain of a function $f(x,y)$. If we fix $y=y_0$ then $f(x,y_0)$ is a function of the variable x alone. The value of the derivative

$$\frac{d[f(x,y_0)]}{dx}$$

at x_0 then gives us a measure of the instantaneous rate of change of f with respect to x at the point (x_0,y_0). Similarly, the value of the derivative

$$\frac{d[f(x_0,y)]}{dy}$$

at y_0 gives us a measure of the instantaneous rate of change of f with respect to y at the point (x_0, y_0). These derivatives are so basic to the study of differential calculus of multivariable functions that they have their own name and notation. We now define a derivative that describes such rates of change.

Definition: - Let f be a function of two variables and (x_0, y_0) in the domain of f.

The partial derivative of f with respect to x at (x_0, y_0) is defined by

$$f_x(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}$$

Similarly, the partial derivative of f with respect to y at (x_0, y_0) is defined by

$$f_y(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0, y_0 + h) - f(x_0, y_0)}{h}$$

provided these limit exist.

Moreover,

Definition: The partial derivative of f with respect to x at any point (x, y) in its domain is also defined by $f_x(x, y) = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}$ and the partial derivative of f with respect to y is defined by $f_y(x, y) = \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}$ provided these limits exist.

Both definitions can be extended analogously for functions of three variables.

Notice that: -

- The partial derivative of f with respect to x, usually denoted by $\frac{\partial f}{\partial x}$ *or* f_x is obtained by differentiating f with respect to x, treating y as a constant.
- The partial derivative of f with respect to y, usually denoted by $\frac{\partial f}{\partial y}$ *or* f_y is obtained by differentiating f with respect to y treating x as a constant.

Partial Derivatives of Combinations and Composition of Functions

Theorem: Let f *and* g be functions of two variables in x and y having partial derivatives, then

i. $(f \pm g)_x (x,y) = f_x(x,y) \pm g_x(x,y)$

ii. $(f.g)_x (x,y) = f_x(x,y) \cdot g(x,y) + f(x,y) \cdot g_x(x,y)$

iii. $\left(\frac{f}{g}\right)_x (x,y) = \frac{f_x(x,y) \cdot g(x,y) - f(x,y) \cdot g_x(x,y)}{(g(x,y))^2}$

By similar fashion it is possible to determine, $(f \pm g)_y (x,y)$, $(f.g)_y (x,y)$ and $\left(\frac{f}{g}\right)_y (x,y)$

Theorem 3.4.1: Let f be a function of two variables and g be a function of one variable. For the composition function $h(x,y) = g(f(x,y))$,

$$h_x(x,y) = g'(f(x,y)).f_x(x,y)$$

and

$$h_y(x,y) = g'(f(x,y).f_y(x,y)$$

In Leibniz's notation, *let* $u = f(x,y)$. Then

$$\frac{\partial h}{\partial x} = \frac{dg}{du} \cdot \frac{\partial f}{\partial x} = \frac{dg}{du} \cdot \frac{\partial u}{\partial x} \quad \textit{and} \quad \frac{\partial h}{\partial y} = \frac{dg}{du} \cdot \frac{\partial u}{\partial y}$$

Similarly, one can extend for combinations and composition function of three variables.

Example: 1. Given $f(x,7) = 3x^2y - 2y^3x + 100$ find $f_x(4,7)$ and $f_y(4,7)$

Solution: When finding $f_x(4,7)$ we need to substitute y = 7 into $f(x,y)$, take a partial derivative with respect to x and then substitute x = *4* into the resulting function.

$$f(x,7) = 3x^2(7) - 2(7)^3 x + 100 = 21x^2 - 686 + 100$$

$$f_x(4,7) = 42x - 686$$

$$f_x(4,7) = 42(4) - 686 = -518$$

Now to find $f_y(4,7)$ we need to substitute x = *4* into $f(x,y)$, take a partial derivative with respect to y and then substitute y = 7 into the resulting function.

$$f(4,y) = 3(4)^2 y - 2y^3(4) + 100 = 48y - 8y^3 + 100$$

$$f_y(4,y) = 48 - 24y^2$$

$$f_y(4,7) = 48 - 24(7)^2 = -1128$$

Example:2. Find $f_x(x,y)$ **and** $f_y(x,y)$ **for** $f(x,y)=e^{2y}y^3+\ln(x^2y^2)$

Solution: To find $f_x(x,y)$ we need to treat y as a constant and take the derivative of each term with respect to x. To do this we will need to apply the derivative rule for logarithmic functions.

$$f_x(x,y)=\frac{d}{dx}\left(e^{2y}y^3\right)+\frac{d}{dx}\left(\ln\left(x^2y^2\right)\right)$$

$$f_x(x,y)=\left(e^{2y}y^3\right)\frac{d}{dx}(1)+\frac{1}{x^2y^2}\times\frac{d}{dx}\left(x^2y^2\right)$$

$$f_x(x,y)=\left(e^{2y}y^3\right)(0)+\frac{1}{x^2y^2}\times y^2\frac{d}{dx}\left(x^2\right)$$

$$f_x(x,y)=\frac{1}{x^2y^2}\times 2xy^2=\frac{2}{x}$$

Now let us find $f_y(x,y)$. To find $f_y(x,y)$ we need to apply the product rule to find $\frac{d}{dy}\left(e^{2y}y^3\right)$as well as use derivative rules for exponential and logarithmic functions.

$$f_y(x,y)=\frac{d}{dy}\left(e^{2y}y^3\right)+\frac{d}{dy}\left(\ln\left(x^2y^2\right)\right)$$

$$=y^3\frac{d}{dy}\left(e^{2y}\right)+e^{2y}\frac{d}{dy}\left(y^3\right)+\frac{d}{dy}\left(\ln\left(x^2y^2\right)\right)$$

$$=y^3\left(2e^{2y}\right)+e^{2y}\left(3y^2\right)+\frac{2x^2y}{x^2y^2}$$

$$=2y^3\left(e^{2y}\right)+e^{2y}\left(3y^2\right)+\frac{2}{y}$$

We are now ready to find partial derivatives of $f(x,y)$ without reducing it to a function of one variable.

Example 3: - Determine the partial derivatives of

a. $f(x,y)=x\tan^{-1}xy$

b. $f(x,y)=x^2y\sec xy$

Solution: -

a. For $f(x,y)=x\tan^{-1}xy$, applying product rule and composition rule we obtain

$$f_x(x,y) = \tan^{-1}xy + \frac{xy}{1+x^2y^2} \text{ and}$$

$$f_y(x,y) = \frac{x^2}{1+x^2y^2}$$

b. For $f(x,y) = x^2 y \sec xy$, applying product rule and composition rule we obtain

$$f_x(x,y) = 2xy\sec xy + x^2y^2\tan xy\sec xy$$

and

$$f_y(x,y) = x^2\sec xy + x^3y\tan xy\sec xy$$

Example 4: Let $f(x,y) = y^2e^{xz}$. Find $f_x(1,1,1)$, $fy(1,1,1)$, $f_z(1,1,1)$

Solution : To find $f_x(x,y,z)$ we hold y and z to be constant and derivate with respect to x. Thus

$$f_x(x,y,z) = zy^2e^{xz} \Rightarrow f_x(1,1,1) = 1.(1)^2.e^{1.1} = e$$

To find $f_y(x,y,z)$ we hold x and z to be constant and derivate with respect to y. Thus

$$f_y(x,y,z) = 2ye^{xz} \Rightarrow f_y(1,1,1) = 2.1..e^{1.1} = 2e$$

To find $f_z(x,y,z)$ we hold x and y to be constant and derivate with respect to z. Thus

$$f_z(x,y,z) = xy^2e^{xz} \Rightarrow f_z(1,1,1) = 1.(1)^2.e^{1.1} = e$$

In an applied problems, the interpretations of $f_x(x_0,y_0)$ and $f_y(x_0,y_0)$ must be accompanied by the proper units. See following Example .

Example 5 : Recall that the wind chill temperature index is given by the formula

$$W = 35.74 + 0.6215T + (0.4275T - 35.75)v^{0.16}$$

Compute the partial derivative of *W* with respect to *v* at the point *(T , v) = (*25, 10*)* and interpret this partial derivative as a rate of change.

Solution. Holding *T* fixed and differentiating with respect to *v* yields

$$\frac{\partial W}{\partial v}(T,v) = 0 + 0 + (0.4275T - 35.75)(0.16)v^{0.16-1}$$

$$= (0.4275T - 35.75)(0.16)v^{-0.84}$$

Since *W* is in degrees Fahrenheit and *v* is in miles per hour, a rate of change of *W* with respect to *v* will have units ◦F/(mi/h) (which may also be written as $F \cdot h/mi$). Substituting

$T = 25$ and $v = 10$ gives

$$\frac{\partial W}{\partial v}(25, 10) = (4.275 - 35.75)(0.16)10^{-0.84} \approx -0.58\, F \cdot h/mi$$

as the instantaneous rate of change of W with respect to v at $(T, v) = (25, 10)$. We conclude that if the air temperature is a constant 25◦F and the wind speed changes by a small amount from an initial speed of 10 mi/h, then the ratio of the change in the wind chill index to the change in wind speed should be about −0.58◦F/(mi/h).

Quick Check Class activity 3.4.1: Find first order partial derivatives for each of the following

a) $f(x,y) = 3x^2y + 6x^2 + y$

b) $f(x,y) = \left(3x^2y + e^{x^2y}\right)^2$

c) $f(x,y) = \dfrac{xy}{x^2 + y^2}$

d) $f(x,y,z) = \sqrt{x^2 + y^2 + z^2}$

Instructor's Role

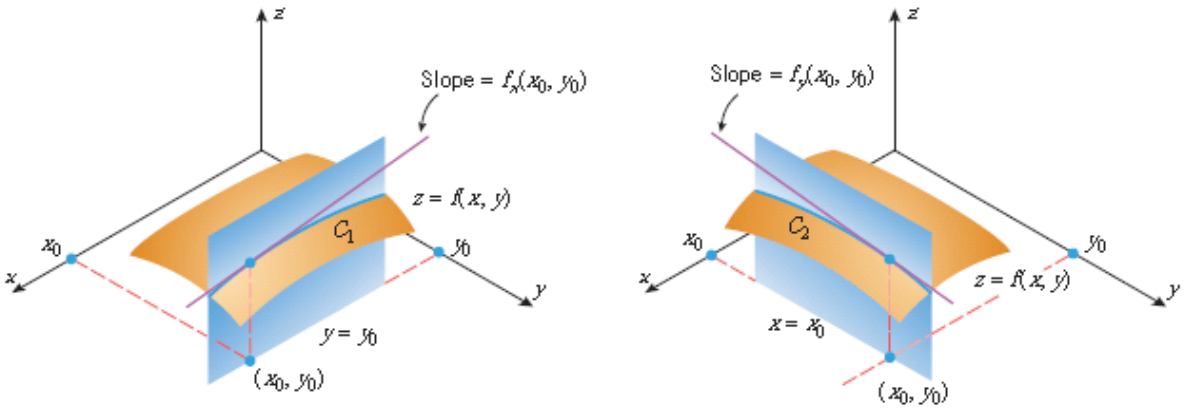
- *Answer for the raised questions and give feed back for their answers*

3.4.1 Geometrical Interpretation of Partial Derivatives

Geometrically, partial derivatives $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ describes the rate of change of $f(x,y)$ in the directions parallel to the $x-axis$ and $y-axis$ respectively. Equivalently, $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ describe how the graph of f is slanted near $(x_0, y_0, f(x_0, y_0))$.

More specifically, $f_x(x_0, y_0)$ is the slope of the line tangent to the curve C(determined by the intersection of the graph of f and the plane $y = y_0$) at $(x_0, y_0, f(x_0, y_0))$. This implies that the vector $\hat{i} + f_x(x_0, y_0)\hat{k}$ is tangent to C at $(x_0, y_0, f(x_0, y_0))$. We call $f_x(x_0, y_0)$ the slope of the surface in the x-direction at (x_0, y_0).

Similarly, $f_y(x_0, y_0)$ is the slope of the line tangent to the curve C(determined by the intersection of the graph of f and the plane $x = x_0$) at $(x_0, y_0, f(x_0, y_0))$. This implies that the vector $\hat{j} + f_y(x_0, y_0)\hat{k}$ is tangent to C at $(x_0, y_0, f(x_0, y_0))$. We call $f_y(x_0, y_0)$ the slope of the surface in the y-direction at (x_0, y_0).



Example 1: Let $f(x,y) = x^2y + 5y^3$

(a) Find the slope of the surface $z = f(x,y)$ in the x-direction at the point(1,-2).

(b) Find the slope of the surface $z = f(x,y)$ in the y-direction at the point (1,-2).

Solution:

(a).Differentiating f with respect to x with y held fixed yields

$$f_x(x,y) = 2xy$$

$$f_x(1,-2) = -4$$

Thus, the slope in the x-direction is -4; that is,z is decreasing at the rate of 4x units per unit increase in x.

(b).Differentiating f with respect to y with x held fixed yields

$$f_y(x,y) = x^2 + 15y^2$$

Thus, the slope in the y-direction is $f_y(1,-2) = 61$; that is z is increasing at the rate of 61y units per unit increase in y.

Group Activity 3.4.1

1. Determine the partial derivatives of

   a. $f(x,y) = e^{(x-y)} - e^{(y-x)}$

   b. $f(x,y) = \dfrac{Ax + By}{Cx + Dy}$

   c. $f(x,y,z) = xy + xz + yz$

   d. $f(x,y) = (x+y)\sin(x-y)$

   e. $f(x,y,z) = \ln\dfrac{x}{y} - y\,e^{xz}$

   f. $f(x,y) = \dfrac{x\sin y}{y\cos x}$

   g. $f(x,y) = \ln(x^2y)$

2. a. Find $g_x(1,2)$ *and* $g_y(1,2)$ given that $g(x,y) = \dfrac{y}{x+y^2}$

   b. Find $g_x(0,e)$ *and* $g_y(0,e)$ given that $g(x,y) = e^x \ln y$

3. Consider the formula for determining area of a triangle

$$A = \frac{1}{2}\, b\, c \sin\theta \ .$$

At time t_0 we have $b_0 = 10$ *inches and* $c_0 = 20$ *inches and* $\theta_0 = \frac{\pi}{2}$ *radian* . Find

a. The area of the triangle at t_0 .

b. The rate of change of the area with respect to b *at* t_0 provided that c *and* θ remain constant.

c. The rate of change of the area with respect to θ *at* t_0 provided that c *and* b remain constant.

d. The rate of change of c with respect to b *at* t_0 provided that the area and θ remain constant.

4. Let g be a differentiable function of one variable.

a. Show that $b\,\frac{\partial w}{\partial x} = a\,\frac{\partial w}{\partial y}$, where $w = g(ax + by)$

b. Show that $nx\,\frac{\partial w}{\partial x} = my\,\frac{\partial w}{\partial y}$, where $w = g(x^m . y^n),\ \ m, n \in \mathbf{Z} \setminus \{0\}$

5. Find f_x *and* f_y of

a. $f(x, y) = 24xy - 6x^2 y$

b. $f(x, y) = \frac{x^3 y - x y^3}{x^2 + y^2}$

6. Show that $f_x(0,0) = f_y(0,0) = 0$, where

$$f(x, y) = \begin{cases} \frac{x^3 y - x y^3}{x^2 + y^2}, & \text{if } (x, y) \neq (0,0) \\ 0 & \text{if } (x, y) = (0,0) \end{cases}$$

7. Let $f(x, y, z) = e^{2x} . \cos z + e^{3y} . \sin z$. *Find* f_x, f_y *and* f_z

Instructor's Role

- *Answer for the raised questions.*
- *Observation while they work in groups*
- *Give feedback for their answers*

3.4.2. Higher Order Partial Derivatives

Let f be a function of two variables x *and* y. Then the partial derivatives:

$(f_x)_x$, usually denoted by f_{xx} *or* $\dfrac{\partial^2 x}{\partial x^2}$

$(f_x)_y$, usually denoted by f_{xy} *or* $\dfrac{\partial^2 x}{\partial y\,\partial x}$

$(f_y)_x$, usually denoted by f_{xy} *or* $\dfrac{\partial^2 x}{\partial x\,\partial y}$

$(f_y)_y$, usually denoted by f_{yy} *or* $\dfrac{\partial^2 x}{\partial y^2}$

are called **Second Order Partial Derivatives of** f**,** in particular f_{xy} and f_{yx} are usually called ***Mixed Partial Derivatives of*** f .We can continue taking derivatives for computing third, fourth, or even higher order partial derivatives and the extension to functions of three variables are completely the same.

Example: -1.Find the all second partial derivatives of $f(x,y) = \sin x^2 y$

Solution: - To find the second order partial derivatives we first need to find $f_x(x,y)$ and $f_y(x,y)$.

$$f_x(x,y) = 2xy\cos x^2 y \qquad \text{and} \qquad f_y(x,y) = x^2 \cos x^2 y$$

To find $f_{xx}(x,y)$ we will take the partial derivative of $f_x(x,y) = 2xy\cos x^2 y$ with respect to *x*.

$$f_{xx}(x,y) = 2y\cos x^2 y + 2xy(-2xy\ \sin x^2 y) = 2y\cos x^2 y - (2xy)^2 \sin x^2 y$$

To find $f_{xy}(x,y)$ we will take the partial derivative of $f_x(x,y) = 2xy\cos x^2 y$ with respect to *y*.

$$f_{xy}(x,y) = 2x\cos x^2 y + 2xy\ (-x^2\ \sin x^2 y) = 2x\cos x^2 y - 2x^3 y\ .\ \sin x^2 y$$

To find $f_{yx}(x,y)$ we take the partial derivative of $f_y(x,y) = x^2 \cos x^2 y$ with respect to *x*.

$$f_{yx}(x,y) = 2x\ \cos x^2 y - 2x^3 \sin x^2 y$$

To find $f_{yy}(x,y)$ we will take the partial derivative of $f_y(x,y) = x^2 \cos x^2 y$ with respect to *y*

$$f_{yy}(x,y) = 2x\ \cos x^2 y - 2x^3 \cos x^2 y$$

2. Given $f(x,y)=3x^2y-2y^3x+40$ find all four second order partial derivatives.

Solution: We first need to find $f_x(x,y)$ and $f_y(x,y)$.Thus

$$f_x(x,y)=6xy-2y^3 \qquad \text{and} \qquad f_y(x,y)=3x^2-6y^2$$

To find $f_{xx}(x,y)$ we take the partial derivative of $f_x(x,y)=6xy-2y^3$ with respect to *x*.

$$f_{xx}(x,y)=6y$$

To find $f_{xy}(x,y)$ we take the partial derivative of $f_x(x,y)=6xy-2y^3$ with respect to *y*.

$$f_{xy}(x,y)=6x-6y^2$$

To find $f_{yy}(x,y)$ we take the partial derivative of $f_y(x,y)=3x^2-6y^2$ with respect to *y*.

$$f_{yy}(x,y)=-12yx$$

To find $f_{yx}(x,y)$ we take the partial derivative of $f_y(x,y)=3x^2-6y^2$ with respect to *x*.

$$f_{yx}(x,y)=6x-6y^2$$

In the examples above we notice that $f_{xy}(x,y)=f_{yx}(x,y)$. Although this does not hold true for all second order partial derivatives, but we have the following theorem of equality of mixed partial derivatives.

Theorem 3.4.2: - Let f be a function of two variables. Suppose that f_{xy} and f_{yx} are continuous at (x_0,y_0).Then $f_{xy}(x_0,y_0)=f_{yx}(x_0,y_0)$

Similarly, let f be a function of three variables. Suppose that f_{xy} and f_{yx} are continuous at (x_0,y_0,z_0).Then $f_{xyz}(x_0,y_0,z_0)=f_{yxz}(x_0,y_0,z_0)$.

Quick Check Class Activity 3.4.2:

1.Find all second order partial derivatives of

a) $f(x,y)=x^2y-y^3x+\ln x$

b) $f(x,y)=\cos(xy^2)$

c) $f(x,y,z)=x^3y^2+\sin(yz)$

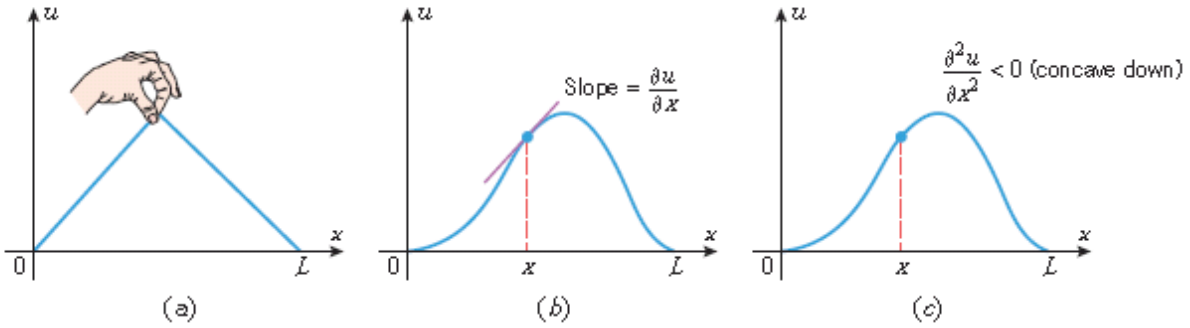
d) $f(x,y,z)=\ln(xyz^2)$

Instructor's Role

- *Answer for the raised questions.*
- *Check their Answers and*
- *Give feedback for their answers*

PHYSICAL APPLICATION (THE WAVE EQUATION)

Consider a string of length L that is stretched taut between $x = 0$ and $x = L$ on an x-axis, and suppose that the string is set into vibratory motion by "plucking" it at time $t = 0$ (Figure*a*). The displacement of a point on the string depends both on its coordinate x and the elapsed time t , and hence is described by a function $u(x,t)$of two variables. For a fixed value t , the function $u(x,t)$ depends on x alone, and the graph of u versus x describes the shape of the string—think of it as a "snapshot" of the string at time t (*b*). It follows that at a fixed time t , the partial derivative $\frac{\partial u}{\partial x}$ represents the slope of the string at x, and the sign of the second partial derivative $\frac{\partial^2 u}{\partial x^2}$ tells us whether the string is concave up or concave down at x (Figure below *c*).



For a fixed value of x, the function $u(x,t)$ depends on t alone, and the graph of u versus t is the position versus time curve of the point on the string with coordinate x. Thus, for a fixed value of x, the partial derivative $\frac{\partial u}{\partial t}$ is the velocity of the point with coordinate x, and $\frac{\partial^2 u}{\partial t^2}$ is the acceleration of that point.

It can be proved that under appropriate conditions the function $u(x,t)$satisfies an equation of the form

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

where c is a positive constant that depends on the physical characteristics of the string. This equation, which is called the ***one-dimensional wave equation***, involves partial

derivatives of the unknown function $u(x,t)$ and hence is classified as a ***partial differential equation***.

Techniques for solving partial differential equations are studied in advanced courses and will not be discussed in this module.

Example : Show that the funct ion $u(x,t)=\sin(x-ct)$ is a solution of the wave equation.

Solution. We have

$$\frac{\partial u}{\partial x}=\cos(x-ct)$$

$$\frac{\partial^2 u}{\partial x^2}=-\sin(x-ct)$$

$$\frac{\partial u}{\partial t}=-c\cos(x-ct)$$

$$\frac{\partial^2 u}{\partial t^2}=-c^2\sin(x-ct)$$

Thus, $u(x,t)$ satisfies wave equation..

APPLIED PROBLEMS (*Group Discussion***)**

1. According to the ideal gas law, the pressure, temperature, and volume of a gas are related by $P = kT/V$, where k is a constant of proportionality. Suppose that V is measured in cubic inches (in3), T is measured in kelvins (K), and that for a certain gas the constant of proportionality is k = 10 in·lb/K.

   (a) Find the instantaneous rate of change of pressure with respect to temperature if the temperature is 80 K and the volume remains fixed at 50 in3.

   (b) Find the instantaneous rate of change of volume with respect to pressure if the volume is $50in3$ and the temperature remains fixed at 80 K.

2. The temperature at a point *(x, y)* on a metal plate in the *xy*-plane is

$$T(x,y)=x^3+2y^2+x$$

in degrees Celsius. Assume that distance is measured in centimeters and find the rate at which temperature changes with respect to distance if we start at the point *(*1, 2*)* and move

(a) to the right and parallel to the *x*-axis

(b) upward and parallel to the *y*-axis.

3. Show that the function satisfies Laplace's equation $\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial y^2}=0$.

   a) $z=x^2-y^2+2xy$ b) $z=e^x\sin y+e^y\cos x$

Group Activity 3.4.2

1. Determine Second Order Partial Derivatives of

   a. $f(x,y)=\sin(xy^2)$.

   b. $f(x,y)=x^2\cos y+y^2\sin x$

   c. $f(x,y)=\sqrt{x+y^2}$

   d. $f(x,y)=\ln\left(\dfrac{x}{x+y}\right)$

   e. $f(x,y)=\sin\left(x^3y^2\right)$

   f. $f(x,y)=\cos^2(xy)$

   g. $f(x,y)=x\,e^y+y\,e^x$

   h. $f(x,y,z)=(x+y)(x+z)(y+z)$

   i. $f(x,y,z)=xy\sin z-xz\sin y$

   j. $f(x,y,z)=x\ e^y+y\ e^z+z\,e^x$

2. Show that $x^2\dfrac{\partial^2u}{\partial x}+2xy\dfrac{\partial^2u}{\partial x\partial y}+y^2\dfrac{\partial^2u}{\partial y^2}=0$ *where* $u(x,y)=\dfrac{xy}{x+y}$.

3. Let

$$f(x)=\begin{cases}\dfrac{x^3y-xy^3}{x^2+y^2} & \text{if } (x,y)\neq 0\\ 0 & \text{if } (x,y)=0\end{cases}$$

   Show that $f_{xy}(0,0)\neq f_{yx}(0,0)$

4. Let $f(x,y)=y^2\ e^x+y$. Find f_{xyy}

Assessment

- *Answer for the raised questions.*
- *Check their Answers and Give feedback for their answers*

3.4.3 The Chain Rule and Implicit differentiation

1. Let $z=f(x,y),\ \ x=g_1(t)$ *and* $y=g_2(t)$. Then $z=f(g_1(t),g_2(t))$ and

$$\frac{dz}{dt}=\frac{\partial z}{\partial x}\cdot\frac{dx}{dt}+\frac{\partial z}{\partial y}\cdot\frac{dy}{dt}$$

2. Let $z=f(x,y),\ \ x=g_1(u,v)$ *and* $y=g_2(u,v)$ Then $z=f(g_1(u),g_2(v))$, and

$$\frac{\partial z}{\partial u}=\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial u}$$

and

$$\frac{\partial z}{\partial v}=\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial v}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial v}$$

In the same way you can extend to functions of three or more variables.

Examples:1. For $f(x,y)=x^2\ e^y,\quad x(t)=t^2-1,\ \ y(t)=\sin t$, find $f'(t)$.

Solution: We first compute

$$\frac{\partial f}{\partial x}=2xe^y,\frac{\partial f}{\partial y}=x^2\ e^y,\quad x'(t)=2t\ ,y'(t)=\cos t$$

The chain rule then gives us

$$f'(t)=\frac{\partial f}{\partial x}\cdot\frac{dx}{dt}+\frac{\partial f}{\partial y}\cdot\frac{dy}{dt}=2xe^y(2t)+x^2e^y\cos t$$

$$=2\left(t^2-1\right)e^{\sin t}(2t)+\left(t^2-1\right)^2e^{\sin t}\cos t$$

Example 2: Find $\frac{\partial w}{\partial u}$ *and* $\frac{\partial w}{\partial v}$ where $w=x\ln y,\ x=u^2+v^2a$ *and* $y=u^2-v^2$

Solution: We compute first

$$\frac{\partial w}{\partial x}=\ln y\ ,\ \frac{\partial w}{\partial y}=\frac{x}{y}\ ,\ \frac{\partial x}{\partial u}=2u\ \ ,\ \frac{\partial y}{\partial u}=2u,\ \frac{\partial x}{\partial v}=2u\ \ and\ \ \frac{\partial y}{\partial v}=-2v$$

The chain rule then gives:

$$\frac{\partial w}{\partial u}=\frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial u}+\frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial u}=\ln y(2u)+\frac{x}{y}(2u)$$

$$=2u\ln\left(u^2-v^2\right)+2u\left(\frac{u^2+v^2}{u^2-v^2}\right)$$

and

$$\frac{\partial w}{\partial v}=\frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial v}+\frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial v}=\ln y(2v)+\frac{x}{y}(-2v)$$

$$=2v\ln\left(u^2-v^2\right)-2v\left(\frac{u^2+v^2}{u^2-v^2}\right)$$

Implicit Differentiation:

Suppose that a function $f(x,y)=0$ implicitly defines a differentiable function $y=g(x)$ of x so that $f(x,g(x))=0$. If we let $w=f(x,y)$ then

$$\frac{dw}{dx}=\frac{d}{dx}\left[f\left(x,g(x)\right)\right]=\frac{d}{dx}(0)=0$$

Further we have from the chain rule $0=\frac{dw}{dx}=\frac{\partial w}{\partial x}+\frac{\partial w}{\partial y}\cdot\frac{dy}{dx}$

Finally if $\frac{\partial w}{\partial y}\neq 0$, the by solving for $\frac{dy}{dx}$ we obtain $\frac{dy}{dx}=\frac{-\frac{\partial w}{\partial x}}{\frac{\partial w}{\partial y}}$

Example: Let $x^3 + y^3 - 2xy = 0$. Find $\frac{dy}{dx}$

Solution: Let $w = x^3 + y^3 - 2xy$. Then

$$\frac{\partial w}{\partial x} = 3x^2 - 2y, \quad \frac{\partial w}{\partial y} = 3y^2 - 2x$$

$$\Rightarrow \frac{dy}{dx} = \frac{-\frac{\partial w}{\partial x}}{\frac{\partial w}{\partial y}} = \frac{2y - 3x^2}{3y^2 - 2x}$$

Suppose that a function $F(x, y, z) = 0$ implicitly defines a differentiable function $z = f(x, y)$ of x and y. If we let $w = F(x, y, z)$ then

$$\frac{\partial w}{\partial x} = \frac{\partial}{\partial x}(0) = 0$$

Further we have from the chain rule

$$0 = \frac{\partial w}{\partial x} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial x} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial x} + \frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial x}$$

Finally if $\frac{\partial w}{\partial z} \neq 0$, the by solving for $\frac{\partial w}{\partial x}$ we obtain

$$\frac{\partial z}{\partial x} = \frac{-\frac{\partial w}{\partial x}}{\frac{\partial w}{\partial z}} = \frac{-F_x}{F_z}$$

Likewise differentiating with respect to y leads us to

$$\frac{\partial z}{\partial y} = \frac{-\frac{\partial w}{\partial y}}{\frac{\partial w}{\partial z}} = \frac{-F_y}{F_z}$$

Quick Class Exercises 3.4.3:

1. Find $\frac{\partial F}{\partial x}$ *and* $\frac{\partial F}{\partial y}$ given that $F(x, y, z) = xy^3 + z^3 + \sin(xyz) = 0$

2. Find $\frac{\partial z}{\partial u}$ *and* $\frac{\partial z}{\partial v}$ of $z = x^2 - xy$, $x = u \cos v$ *and* $y = v \sin u$.

3. Find the slope of the sphere $x^2 + y^2 + z^2 = 1$ in the y-direction at the points $\left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)$ and $\left(\frac{2}{3}, \frac{1}{3}, \frac{-1}{3}\right)$.

Group Activity 3.4.3

1. Find $\frac{du}{dt}$ of

   a. $u(x,y) = x^2 + 3xy + 2y^2, \quad x = \cos t, \quad y = \sin t$

   b. $u(x,y) = e^x \sin y + e^y \sin x, \quad x = \frac{t}{2}, \quad y = 2t$

   c. $u(x,y) = x^2\ e^y, \quad x = \sin t, \quad y = t^2$

   d. $u(x,y,z) = z \ln\left(\frac{y}{x}\right), \quad x = t^2 + 1, \quad y = \sqrt{t}, \quad z = t\ e^t$

   e. $u(x,y,z) = xy + yz + zx, \quad x = t^2, \quad y = t(1-t)$

   f. $u(x,y,z) = x\cos yz^2, \quad x = \sin t \ \ and \ \ y = t^2, \ z = e^t$

2. Find $\frac{\partial z}{\partial u}$ *and* $\frac{\partial z}{\partial v}$ of

   a. $z = \sin(x-y) + \cos(x+y), \ x = u\,v \ \ and \ y = u^2 - v^2$

   b. $z = x^2 \tan y, \ x = u^2 v \ and \ y = u + v^2$

   c. $z = w^2 \sec xy, \ x = 2uv, \ y = u - v^2 \ and \ w = u^2\ v$

   d. $z = xe^{\,yw^2}, \ x = \ln(uv), \quad y = v^3 \ and \quad w = u^2 + v^2$

3. Show that $\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} = 0$, where $z = f(u-v, v-u)$.

4. Suppose that a particle moving at along a metal plate in the xy-plane has velocity $V = i - 4j(\frac{cm}{s})$ at the point (3,2). Given that the temperature of the plate at poimts in the xy-plane is $T(x,y) = y^2 \ln x, x \geq 1$ in degree celsius,find $\frac{dT}{dt}$ at the point (3,2).

Assessment

- *Answer for the raised questions.*
- *Check their Answers*
- *Give feedback for their answers*

3.4.4 Differentiability; Differentials and Linear Approximations

Definition: A function f of two variables is said to be differentiable at (x_0, y_0) if $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ exist and Δf can be written in the form

$$\Delta f = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + \varepsilon_1 \Delta x + \varepsilon_2 \Delta y$$

where ε_1 and ε_2 are functions of Δx and Δy such that $\varepsilon_1 \to 0$ and $\varepsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0,0)$ and $\varepsilon_1 = \varepsilon_2 = 0$ if $(\Delta x, \Delta y) = (0,0)$.

Moreover, f is said to be differentiable function if it is differentiable at each point of its domain.

Sufficient Conditions for Differentiability

Theorem 3.4.3: If f has partial derivatives at each point in some circular region centered at (x_0, y_0) and if these partial derivatives are continuous at (x_0, y_0) then f is differentiable at (x_0, y_0).

Example: Show that $f(x, y) = x^3 y^4$ is a differentiable function.

Solution: $f_x(x, y) = 3x^2 y^4$ and $f_y(x, y) = 4x^3 y^3$ are defined and continuous everywhere in the $xy - plane$. Thus by the above theorem $f(x, y) = x^3 y^4$ is everywhere differentiable.

Differentials

From the definition of differentiability it follows if Δx and Δy are close to 0 , we have the approximations

$$\Delta f \approx f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y$$

for a function of two variables and the approximation

$$\Delta f \approx f_x(x_0, y_0, z_0)\Delta x + f_y(x_0, y_0, z_0)\Delta y + f_z(x_0, y_0, z_0)\Delta z$$

for a function of three variables. and have a convenient formulation in the language of differentials.

If $z = f(x, y)$ is differentiable at a point (x_0, y_0) , we let

$$df = f_x(x_0, y_0)dx + f_y(x_0, y_0)dy$$

denote a new function with dependent variable dz and independent variables dx and dy.

We refer to this function (also denoted df) as the *total differential of* z at (x_0, y_0) or as the *total differential of* f at (x_0, y_0)

Similarly, for a function $w = f(x, y, z)$ of three variables we have the *total differential of f* at (x_0, y_0, z_0),

$$df = f_x(x_0, y_0, z_0)dx + f_y(x_0, y_0, z_0)dy + f_z(x_0, y_0, z_0)dz .$$

It is common practice to omit the subscripts and write total differentials as

$$df = f_x(x, y)dx + f_y(x, y)dy$$

and

$$df = f_x(x, y, z)dx + f_y(x, y, z)dy + f_z(x, y, z)dz$$

Linear approximations

In the two-variable case, the approximation $\Delta f \approx f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y$ can be written in the form

$$\Delta f \approx df$$

for $dx = \Delta x$ and $dy = \Delta y$. Equivalently, we can write the approximation as

$$\Delta z \approx dz$$

In other words, we can estimate the change Δz in *z* by the value of the differential dz where dx is the change in *x* and dy is the change in *y*. Then approximation Δf can also be written in the form

$$f(x_0 + \Delta x, y_0 + \Delta y) \approx f(x_0 + y_0) + f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y$$

If we let $\Delta x = x - x_0$ and $\Delta y = y - y_0$, this approximation becomes

$$f(x, y) \approx f(x_0 + y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Definition: The approximation

$$f(x, y) \approx f(x_0 + y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

is called the *local linear approximation* of f.

Example : Approximate the change in $z = xy^2$ from its value at $(0.5, 1.0)$ to its value at $(0.503, 1.004)$. Compare the magnitude of the error in this approximation with the distance between the points *(*0.5, 1.0*)* and *(*0.503, 1.004*)*.

Solution: For $z = xy^2$ we have $dz = y^2dx + 2xy\,dy$. Evaluating this differential at $(x, y) = (0.5, 1.0)$, $dx = \Delta x = 0.503 - 0.5 = 0.003$, and

$dy = \Delta y = 1.004 - 1.0 = 0.004$ yields

$$dz = 1.02(0.003) + 2(0.5)(1.0)(0.004) = 0.007$$

Since *z* = 0.5 at *(*x, y) = (0.5, 1.0) and *z* = 0.507032048 at (x, y) = (0.503, 1.004), we have

$$\Delta z = 0.507032048 - 0.5 = 0.007032048$$

and the error in approximating Δz by dz has magnitude

$$|dz - z| = |0.007 - 0.007032048| = 0.000032048$$

Thus, the magnitude of the error in our approximation is less than 1.

3.5 Directional Derivatives and Gradient

3.5.1 The Gradients; Normal Vectors; Tangent Lines and Tangent Plane

Definition: -

Let f be a function of two variables that has partial derivatives at (x_0, y_0).

Then the ***gradient*** of f at (x_0, y_0), usually denoted by **grad** $f\ (x_0, y_0)$ or $\nabla f(x_0, y_0)$, is defined as

$$\nabla f(x_0, y_0, z_0) = f_x(x_0, y_0)i + f_y(x_0, y_0)j.$$

Let f be a function of three variables that has partial derivatives at (x_0, y_0, z_0). Then the gradient of f at (x_0, y_0, z_0) is defined by

$$\nabla f(x_0, y_0, z_0) = f_x(x_0, y_0, z_0)\ i + f_y(x_0, y_0, z_0)j + f_z(x_0, y_0, z_0)k.$$

Notice that Gradient is a vector valued function.

Example 1: - Let $f(x,y) = \sin xy$. *Find* $\nabla f, \nabla f\left(\frac{\pi}{3}, 1\right)$

Solution: - $f_x(x,y) = y\ \cos xy, \quad f_y(x,y) = x\ \cos xy$.

Thus $\nabla f(x,y) = (y\ \cos xy)\hat{i} + (x\ \cos xy)\hat{j}$.

$$f_x\left(\frac{\pi}{3},1\right) = \frac{1}{2}, \quad f_y\left(\frac{\pi}{3},1\right) = \frac{\pi}{6}$$

$$\nabla f\left(\frac{\pi}{3}, 1\right) = \frac{1}{2}\, i + \frac{\pi}{6}\, j$$

Example 2: Let $f(x,y,z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$. Find a formula for the gardient and evaluate $\nabla f\left(2\sqrt{2}, 2\sqrt{2}, -3\right)$.

Solution: The partial derivatives are

$$f_x(x,y,z)=\frac{-x}{\left(x^2+y^2+z^2\right)^{\frac{3}{2}}},\ f_y(x,y,z)=\frac{-y}{\left(x^2+y^2+z^2\right)^{\frac{3}{2}}},\ f_z(x,y,z)=\frac{-z}{\left(x^2+y^2+z^2\right)^{\frac{3}{2}}}$$

Thus $f_x(x,y,z)=\dfrac{-(x\hat{i}+y\hat{j}+z\hat{k})}{\left(x^2+y^2+z^2\right)^{\frac{3}{2}}}$ and $\nabla f\left(2\sqrt{2},2\sqrt{2},-3\right)=\dfrac{1}{125}\left(-2\sqrt{2}i-2\sqrt{2}j+3k\right).$

Remark: - If f is a function of two variables that is differentiable at (x_0,y_0), and $u=a_1i+a_2j$ is a vector in the $xy-$ plane, then

$$\nabla f(x_0,y_0)\cdot u = grad.\, f(x_0,y_0).(a_1i+a_2i)$$

$$\nabla f(x_0,y_0)\cdot u = f_x(x_0,y_0).a_1 + f_y(x_0,y_0).a_2$$

Quick check group Activity 3.5.1:

Find a formula for the gradient and evaluate $\nabla f(P)$ for the following functions.

a) $f(x,y)=x\,y\,e^{-(x^2+y^2)}$ *at* $(1,-1)$ b) $f(x,y)=xy^2e^{-x}$ at (-1,1)

c) $f(x,y,z)=e^{-x}\sin(z+2y)$ *at* $\left(0,\frac{\pi}{4},\frac{\pi}{4}\right)$ d) $f(x,y,z)=cos\,(xyz^2)$ *at* $\left(\pi,\frac{1}{4},-1\right)$

Reconstructing a Function from Its Gradient

Example 1: - Find f such that $\nabla f(x,y)=y^2i+(2xy-1)j$.

Solution: - $\nabla f(x,y)=f_x(x,y)i+f_y(x,y)j$

$$\Rightarrow \frac{\partial f}{\partial x}(x,y)=y^2 \quad and \quad \frac{\partial f}{\partial y}(x,y)=2x-1$$

$$\Rightarrow f(x,y)=xy^2+\phi(y)\cdots\cdots\cdots(*),$$

where $\phi(y)$ *is* a function on y or not but independent of x.

Differentiate (*) with respect to y, yields as

$$f_y(x,y)=2xy+\phi'(y)$$

$$\Rightarrow 2xy+\phi'(y)=2xy-1$$

$$\Rightarrow \phi'(y)=-1$$

$$\Rightarrow \phi(y)=-y+C,$$ where C is a constant.

Thus $f(x,y)=xy^2-y+C$

Example 2: - Find f such that

$$\nabla f(x,y)=\left(\sqrt{y}-\frac{y}{2\sqrt{x}}+2x\right)i+\left(\frac{x}{2\sqrt{y}}-\sqrt{x}+1\right)j$$

Solution: - $\nabla f(x,y)=f_x(x,y)i+f_y(x,y)j$

$$\Rightarrow \frac{\partial f}{\partial x}(x,y)=\sqrt{y}-\frac{y}{2\sqrt{x}}+2x \quad and \quad \frac{\partial f}{\partial y}(x,y)=\frac{x}{2\sqrt{y}}-\sqrt{x}+1$$

$$\Rightarrow f(x,y)=x\sqrt{y}+y\sqrt{x}+x^2+\phi(y)\cdots\cdots\cdots(*) \text{ where } \phi(y) \text{ is a function on } y \text{ or not but independent of } y$$

Differentiate (*) with respect to y , yields as

$$\Rightarrow f_y(x,y)=\frac{x}{2\sqrt{y}}+\sqrt{x}+\phi'(y)$$

$$\Rightarrow \frac{x}{2\sqrt{y}}+\sqrt{x}+\phi'(y)=\frac{x}{2\sqrt{y}}+\sqrt{x}+1$$

$$\Rightarrow \phi'(y)=1 \Leftrightarrow \phi(y)=y+C$$

Thus $f(x,y)=x\sqrt{y}+y\sqrt{x}+x^2+y+C$.

Group Activity 3.5.1

1. Find the gradient of

a. $f(x,y)=x\,e^{x\,y}$

b. $f(x,y)=3x^2-xy+y$

c. $f(x,y)=x^2\,y^{-2}$

d. $f(x,y)=(x+y)\sin(x-y)$

e. $f(x,y)=e^x.\ln y$

f. $f(x,y)=3x^2-xy+y$

g. $f(x,y,z)=xye^x+ye^z-e^y\sin zx$

2. Find the gradient vector of

a. $f(x,y)=2x(x-y)^{-1}$ *at* $(3,\ 1)$

b. $f(x,y)=x\tan^{-1}(\frac{y}{x})$ *at* $(1,\ 1)$

c) $f(x,y)=\ln(x^2+y^2)$ *at* $(2,\ 1)$

3. Find a function f such that

a. $\nabla f(x,y)=2xy\ i+\left(1+x^2\right)j$

b. $\nabla f(x,y)=\left(2xy+x\right)i+\left(y+x^2\right)j$

c. $\nabla f(x,y)=xy^2\ i+x^2y\ j$

d. $\nabla f(x,y)=(y^2e^x-y)\ i+(2ye^x-x)\ j$

e. $\nabla f(x,y)=\left(y^3+x\right)i+\left(x^2+y\right)j$

f. $\nabla f(x,y)=(\cos x-y\sin x)\ i+\cos x\ \ j$

g. $\nabla f(x,y) = (x + \sin y)i + (x\cos y - 2y)\, j$ h. $\nabla f(x,y,z) = y\,z\, i + (xz + 2yz)j + (xy + y^2)k$

i. $\nabla f(x,y) =$

$$\left(\frac{\tan^{-1} y}{\sqrt{1-x^2}} + \frac{x}{y}\right) i + \left(\frac{\sin^{-1} x}{-+y^2} - \frac{x^2}{2y^2} + 1\right) j$$

2. Find the points (x,y) such that $\nabla f(x,y) = 0$ where

a. $f(x,y) = 1 + x^2 + y^2$ b. $f(x,y) = \dfrac{-2y}{1 + x^2 + y^2}$

Instructor's Role

- *Answer for the raised questions.*
- *Observation while they work in groups*
- *Give feedback for their answers*

The Gradient as Normal Vectors; Tangent Lines and Tangent Planes.

Theorem 3.5.1: Assume that $f(x,y)$ has continuous first order partial derivatives in an open disk centered at (x_0, y_0) and $\nabla f(x_0, y_0) \neq 0$. Then

$$\nabla f(x_0, y_0) = \frac{\partial f}{\partial x}(x_0, y_0)i + \frac{\partial f}{\partial y}(x_0, y_0)j$$

is perpendicular to the curve C of f through (x_0, y_0). We call it a ***normal vector***.

Proof: Reading Assignment

The vector

$$t(x_0, y_0) = \frac{\partial f}{\partial x}(x_0, y_0)i - \frac{\partial f}{\partial y}(x_0, y_0)j$$

is perpendicular to the gradient because

$$t(x_0, y_0).\nabla f(x_0, y_0) = \frac{\partial f}{\partial x}(x_0, y_0)\frac{\partial f}{\partial y}(x_0, y_0) - \frac{\partial f}{\partial x}(x_0, y_0)\frac{\partial f}{\partial y}(x_0, y_0) = 0.$$

It is called a ***tangent vector***.

The line through (x_0, y_0) which is perpendicular to the gradient is called the ***tangent line***. It is determined by the formula:

$$[(x - x_0)i + (y - y_0)j].\nabla f(x_0, y_0) = 0$$

$$\Leftrightarrow \frac{\partial f}{\partial x}(x_0, y_0).(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0).(y - y_0) = 0$$

And the equation of the normal line to the curve C at (x_0, y_0) is

$$\frac{\partial f}{\partial y}(x_0, y_0).(x - x_0) - \frac{\partial f}{\partial x}(x_0, y_0).(y - y_0) = 0.$$

Example 1: - Find the equations of tangent and normal lines to the graph of

$$x^2 + xy + y^2 = 3 \quad at \ (-1, -1).$$

Solution: - Let $f(x, y) = x^2 + xy + y^2$. Then

$$\frac{\partial f}{\partial x}(-1,-1) = (2x + y)\Big|_{(x,y)=(-1,-1)} = -3 \ \ and \ \ \frac{\partial f}{\partial y}(-1,-1) = (x + 2y)\Big|_{(x,y)=(-1,-1)} = -3$$

Thus $\frac{\partial f}{\partial x}(-1,-1).(x+1) + \frac{\partial f}{\partial y}(-1,-1).(y+1) = 0$

$\Leftrightarrow -3.(x+1) - 3.(y+1) = 0$

$\Leftrightarrow x + y + 2 = 0, \quad$ *which is the equation of the* tan *gent line*

Similarly, the equation of the normal line is

$\Leftrightarrow -3.(x+1) + 3.(y+1) = 0$

$\Leftrightarrow x - y = 0$

Example 2: Find a unit vector perpendicular to the level curve

$x^2 - xy + 3y^2 = 5 \quad at \ (-1, \ 1)$

Solution

Let $f(x, y) = x^2 - xy + 3y^2$ so that the given level curve is $f(x, y) = 5$, Since $f(-1,1) = 5$ the point $(-1,1)$ lies on the level curve. Therefore, $gradf(-1,1)$ is perpendicular to the given curve at $(-1,1)$. We find that

$$gradf(x, y) = (2x - y)j + (-x + 6y)j$$

$$\Rightarrow gradf(-1,1) = 3i - 7j$$

Therefore the unit vector $\frac{1}{\|3i - 7j\|}(3i - 7j) = \frac{1}{\sqrt{58}}(3i - 7j)$ is the desired unit vector perpendicular to the curve at $(-1,1)$.

For functions of three variables:

Definition: Let $f(x, y, z)$ has continuous first order partial derivatives and that $P_0(x_0, \ y_0, \ z_0)$ is a point on the level surface S: $f(x, y, z) = c$. If $\nabla f(x_0, \ y_0, \ z_0) \neq 0$, then

$\hat{n} = \nabla f(x_0, y_0, z_0)$ is a ***normal vector*** S at $P_0(x_0, y_0, z_0)$, in a sense that $\nabla f(x_0, y_0, z_0)$ is perpendicular to the tangent vectors of all smooth curves on the surface that pass through (x_0, y_0, z_0). Therefore the plane through (x_0, y_0, z_0) whose normal is $gradf(x_0, y_0, z_0)$ is said to be the **tangent plane** to S at (x_0, y_0, z_0).

The equation of tangent plane to the surface $f(x,y,z) = C$ at a point $\mathbf{X}_0 = (x_0, y_0, z_0)$ is thus given by:

$$\nabla f(\mathbf{X}_0) \bullet (\mathbf{X} - \mathbf{X}_0) = 0$$

$$\Leftrightarrow \frac{\partial f}{\partial x}(x_0, y_0, z_0).(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0, z_0).(y - y_0) + \frac{\partial f}{\partial x}(x_0, y_0, z_0).(z - z_0) = 0$$

The normal line to the surface $f(x,y,z) = C$ at a point $\mathbf{X}_0 = (x_0, y_0, z_0)$ is the line that passes through $\mathbf{X}_0 = (x_0, y_0, z_0)$ and parallel to $\nabla f(x_0, y_0, z_0)$. Thus, $\nabla f(x_0, y_0, z_0)$ is a direction vector for the normal line and

$$r(t) = r_0 + \nabla f(\mathbf{X}_0)\, t, \qquad where \;\; r_0 = x_0 i + y_0 j + z_0 k$$

is a vector equation for the line. In scalar parametric form of equation, the normal line are

$$\begin{cases} x = x_0 + \dfrac{\partial f}{\partial x}(x_0, y_0, z_0)\, t \\ y = y_0 + \dfrac{\partial f}{\partial y}(x_0, y_0, z_0)\, t \\ z = z_0 + \dfrac{\partial f}{\partial z}(x_0, y_0, z_0)\, t \end{cases}$$

Example: - Find an equation for the tangent plane and scalar parametric equations for the normal line to the elliptic cone:

$$x^2 + 4y^2 = z^2 \;\; at \;\; (3, 2, 5).$$

Solution: - The surface is of the form $f(x,y,z) = C$, *where* $f(x,y,z) = x^2 + 4y^2 - z^2$ *and* $C = 0$.

Now,

$$\frac{\partial f}{\partial x}(x, y, z) = 2x, \quad \frac{\partial f}{\partial y}(x, y, z) = 8y \qquad \frac{\partial f}{\partial z}(x, y, z) = -2z$$

$$\frac{\partial f}{\partial x}(3, 2, 5) = 6, \quad \frac{\partial f}{\partial y}(3, 2, 5) = 16 \qquad \frac{\partial f}{\partial z}(3, 2, 5) = -10$$

Thus the equation of the tangent plane is

$$6(x-3)+16(y-2)-10(z-5)=0$$

$$\Leftrightarrow \quad 6x+16y-10z=0$$

$$\Leftrightarrow \quad 3x+8y-5z=0$$

Since both $(3,8,-5)=\frac{1}{2}(6,16,-10)$ and $(6,16,-10)$ is normal to the curve at $(3,2,5)$, then $(3,8,-5)$ is also normal to the curve at $(3,2,5)$. Thus, the parametric equations for the normal line are

$$x=3+3t, \quad y=2+8t, \quad \textit{and} \quad z=5-5t$$

A surface of the form

$$z=g(x,y)$$

can be written in the form

$$f(x,y,z)=0 \quad \textit{where} \quad f(x,y,z)=g(x,y)-z.$$

If g is differentiable, so is f. Moreover,

$$\frac{\partial f}{\partial x}(x,y,z)=\frac{\partial g}{\partial x}(x,y,z),\ \frac{\partial f}{\partial y}(x,y,z)=\frac{\partial g}{\partial y}(x,y,z), \text{ and } \frac{\partial f}{\partial z}(x,y,z)=\frac{\partial g}{\partial z}(x,y,z)=-1.$$

Thus equation of the tangent plane to a surface $z=g(x,y)$ at $P_0(x_0,y_0,f(x_0,y_0))$ is

$$\frac{\partial g}{\partial x}(x_0,y_0)(x-x_0)+\frac{\partial g}{\partial y}(x_0,y_0)(y-y_0)-(z-z_0)=0$$

$$\Leftrightarrow \frac{\partial g}{\partial x}(x_0,y_0)(x-x_0)+\frac{\partial g}{\partial y}(x_0,y_0)(y-y_0)=z-z_0$$

$$\Leftrightarrow \frac{\partial g}{\partial x}(x_0,y_0)(x-x_0)+\frac{\partial g}{\partial y}(x_0,y_0)(y-y_0)+z_0=z$$

If $\nabla g(x_0,y_0)=0$, then $\frac{\partial g}{\partial x}(x_0,y_0)=\frac{\partial g}{\partial y}(x_0,y_0)=0$ *and* $z=z_0$. In this case the tangent plane is horizontal. And, scalar parametric equations for the normal line to the surface $z=g(x,y)$ at the point $\mathbf{X}_0=(x_0,y_0,z_0)$ are

$$x=x_0+\frac{\partial g}{\partial x}(x_0,y_0)\,t, \quad y=y_0+\frac{\partial g}{\partial y}(x_0,y_0)\,t \ \textit{and} \ z=z_0+(-1)\,t.$$

Example: - Find an equation for the tangent plane and symmetric equations for the normal line to the surface:

$$z = \ln(x^2 + y^2) \quad at \quad (-2,\ 1,\ \ln 5) \text{ on the surface.}$$

Solution: -Let $g(x, y) = \ln(x^2 + y^2)$. Then

$$\frac{\partial g}{\partial x}(x, y) = \frac{2x}{x^2 + y^2} \quad and \quad \frac{\partial g}{\partial y}(x, y) = \frac{2y}{x^2 + y^2}$$

$$\Rightarrow \frac{\partial g}{\partial x}(-2, 1) = -\frac{4}{5} \quad and \quad \frac{\partial g}{\partial y}(x, y) = \frac{2}{5}$$

Therefore, the tangent plane equation is

$$z = \ln 5 = -\frac{4}{5}(x+2) + \frac{2}{5}(y-1)$$

The symmetric equation for the normal line is

$$\frac{x+2}{-\frac{4}{5}} = \frac{y-1}{\frac{2}{5}} = \frac{z-\ln 5}{-1}$$

Note that

$$\frac{x+2}{-\frac{4}{5}} = \frac{y-1}{\frac{2}{5}} = \frac{z-\ln 5}{-1} = t$$

The gradient arises in many physical situations:

For instance, If $T(x,y,z)$is the temperature at any point (x,y,z) then level surfaces of T are isothermal surfaces. On isothermal surface the temperature is constant, and no heat flows along such a surface. Instead heat flows in a direction perpendicular to an isothermal surface; more precisely, it flows in the direction of the gradient.

If $V(x,y,z)$ represents the voltage at the point (x,y,z) then $-gradV(x,y,z)$ turns out to be the electric force that would be exerted on a positive unit charge at (x,y,z). This force is perpendicular to the equipotential surface at (x,y,z)

Quick check group Activity 3.5.2:

1. Find equations of the normal line and the plane tangent to the following level surfaces at point P.

a) $z = 6 - x^2 - y^2$, at $(1, 2, 1)$. b) $z = x^3 + y^3 + \frac{x^2}{y}$, at $(2, 1, 13)$

2. Find a vector normal to the graph of f and find an equation of the tangent plane at the indicated points.

a) $f(x,y)=3x^2+4y^2;(-2,1,16)$

b) $f(x,y)=4x^2+y^2-6;(2,1,16)$

Instructor's Activity

- *Asking an answer for some of the questions.*
- *Observation of whether they work in groups.*
- *Answering to the raised questions*

Group Activity 3.5.2

1. Find normal and tangent vectors at the indicated point and write equations for both tangent and normal lines:
   a. $(y-x)^2=2x$; $(2,4)$
   b. $y^3+x^3=9$; $(1,2)$
   c. $xy^2-2x^2+y+5x=6$; $(4,2)$
   d. $2x^3-x^2y^2=3x-y-7$; $(1,-2)$
   e. $x^3+y^2+2x=6$; $(-1,3)$
2. Find equations for both tangent plane and scalar parametric equations for normal lines at the indicated point:
   a. $z=(x^2+y^2)^2$, $(1,1,4)$
   b. $x^3+y^3+3xyz=18$, $\left(1,2,\frac{3}{2}\right)$
   c. $\sqrt{x}+\sqrt{y}+\sqrt{z}=4$, $(1,4,1)$
   d. $z=\sin x+\sin y+\sin(x+y)$, $(0,0,0)$

Instor's Activity

- *Asking an answer for some of the questions.*
- *Observation of whether they work in groups.*
- *Answering to the raised questions*

3.5.2 Directional Derivatives

The slope of a surface $z = f(x, y)$ at a point (x_0, y_0) depends on the point and varies with the direction of the unit vector that has its initial point at (x_0, y_0). In this section we determine how to find the slope of a surface $z = f(x, y)$ at a point (x_0, y_0) in an arbitrary specified direction.

Definition: - Let f be a function defined on a set containing a disk **D** centered at (x_0, y_0), and $u = a_1 i + a_2 j$ be a unit –vector. Then the ***Directional Derivative of f at*** (x_0, y_0) in the direction of u, denoted by $D_u f(x_0, y_0)$ is defined as

$$D_u f(x_0, y_0) = \lim_{h \to 0} \frac{f(x + ha_1, y_0 + ha_2) - f(x_0, y_0)}{h}$$

Note: -

1. If $u = i$, then

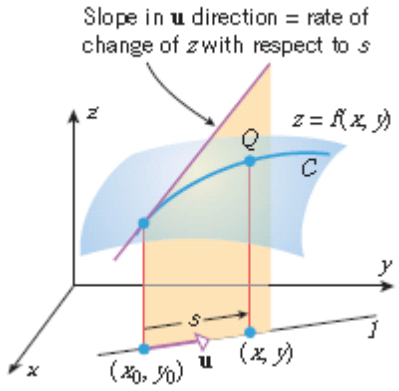
$$D_u f(x_0, y_0) = \lim_{h \to 0} \frac{f(x + ha_1, y_0) - f(x_0, y_0)}{h} = f_x(x_0, y_0)$$

2. If $u = j$, then

$$D_u f(x_0, y_0) = \lim_{h \to 0} \frac{f(x, y_0 + ha_2) - f(x_0, y_0)}{h} = f_y(x_0, y_0)$$

Geometrical Interpretation of Directional Derivatives

Geometrically, $D_u f(x_0, y_0)$ can be interpreted as the ***slope of the surface*** $z = f(x, y)$ ***in the direction of*** **u** at the point $(x_0, y_0, f(x_0, y_0))$ (Figure below). Usually the value of $D_u f(x_0, y_0)$ will depend on both the point (x_0, y_0) and the direction of **u**. Thus, at a fixed point the slope of the surface may vary with the direction (Figure). Analytically, the directional derivative represents the ***instantaneous rate of change of*** $f(x, y)$ ***with respect to distance in the direction of*** **u** at the point (x_0, y_0).



Theorem 3.5.2: - Let f be differentiable at (x_0, y_0). Then f has a directional derivative at (x_0, y_0) in every direction. Moreover, if $u = a_1 i + a_2 j$ is a unit vector, then

$$D_u f(x_0, y_0) = f_x(x_0, y_0)a_1 + f_y(x_0, y_0)a_2$$

Proof: - Reading Assignment

Example 1: - Find $D_u f(1,2)$ where $f(x,y) = 6 - 3x^2 - y^2$ *and* $u = \frac{1}{\sqrt{2}}(i - j)$.

Solution: - $f_x(x,y) = -6x$ *and* $f_y(x,y) = -2y$

$\Rightarrow f_x(1,2) = -6$ *and* $f_y(1,2) = -4$

Thus $D_u f(1,2) = f_x(1,2)\frac{1}{\sqrt{2}} + f_y(1,2)\left(-\frac{1}{\sqrt{2}}\right) = -3\sqrt{2} - 2\sqrt{2} = -5\sqrt{2}$ ///

Remark: - If "a" is a non-zero vector, the directional derivative in the direction of "a" is defined to be

$$D_u f(x_0, y_0), \quad \text{where } u = \frac{a}{\|a\|}.$$

Example 2: - Find the directional derivative of $f(x,y) = 3y^2 + x^2$ at $(1, 1)$ in the direction of $a = i + 2j$.

Solution: - $f_x(x,y) = 2x$ *and* $f_y(x,y) = 6y$

$\Rightarrow f_x(1, 1) = 2$ *and* $f_y(1, 1) = 6$ and $u = \frac{1}{\sqrt{5}}(i + 2j) = \frac{1}{\sqrt{5}}i + \frac{2}{\sqrt{5}}j$

Thus $D_a f(1,1) = f_x(1,1)\frac{1}{\sqrt{5}} + f_y(1,1)\frac{2}{\sqrt{5}} = \frac{2}{\sqrt{5}} + \frac{12}{\sqrt{5}} = \frac{14}{\sqrt{5}} = \frac{14}{5}\sqrt{5}$.

Remark: - Let f be a function of three variables x, y and z and $u = a_1 i + a_2 i + a_3 k$ a unit vector. Then $D_u f(x_0, y_0, z_0)$ is defined by

$$D_u f(x_0, y_0, z_0) = \lim_{h \to 0} \frac{f(x_0 + ha_1, y_0 + ha_2, z_0 + ha_3) - f(x_0, y_0, z_0)}{h}$$

Provided this limit exists. And, if f is differentiable at (x_0, y_0, z_0), then

$$D_u f(x_0, y_0, z_0) = f_x(x_0, y_0, z_0)a_1 + f_y(x_0, y_0, z_0).a_2 + f_z(x_0, y_0, z_0).a_3.$$

Example 3: - Find the directional derivative of $f(x, y, z) = xy + yz + zx$ at $(1, -1, 1)$ in the direction of $a = i + 2j + k$.

Solution: - $f_x(x, y, z) = y + z \quad f_y(x, y, z) = x + z$ *and* $f_z(x, y, z) = x + y$

$$\Rightarrow f_x(1,-1,1) = 0 \quad f_y(1,-1,1) = 2 \text{ and } \quad f_z(1,-1,1) = 0$$

$$u = \frac{1}{\sqrt{7}}(i + 2j + k)$$

Thus $D_a f(1,-1,1) = f_x(1,-1,1)\frac{1}{\sqrt{7}} + f_y(1,-1,1)\frac{2}{\sqrt{7}} + f_z(1,-1,1)\frac{1}{\sqrt{7}}$

$$= \frac{4}{\sqrt{7}}$$

If f is a function of two variables that is differentiable at (x_0, y_0), and $u = a_1 i + a_2 j$ is a unit vector in the $xy-$ plane, then

$$D_u f(x_0, y_0) = f_x(x_0, y_0).a_1 + f_y(x_0, y_0).a_2$$

$$D_u f(x_0, y_0) = grad.\, f(x_0, y_0).(a_1 i + a_2 i)$$

$$D_u f(x_0, y_0) = \nabla f(x_0, y_0)\ .\ u$$

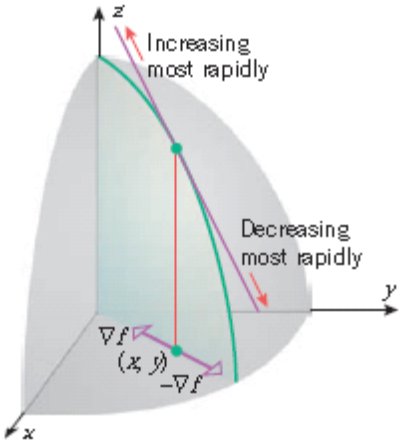
We can rewrite this as

$$D_u f(x, y) = \nabla f(x, y) \cdot u = \|\nabla f(x, y)\| \|u\| \cos\theta = \|\nabla f(x, y)\| \cos\theta$$

where θ is the angle between $\boldsymbol{\nabla} f(x, y)$ and $\boldsymbol{u}$. This equation tells us that the maximum value of $D_u f(x, y)$ at the point (x, y) is $\|\nabla f(x, y)\|$, and this maximum occurs when $\theta = 0$, that is, when **u** is in the direction of $\boldsymbol{\nabla} f(x, y)$. Geometrically, this means:

Property 1*: At (x, y), the surface $z = f(x, y)$ has its maximum slope in the direction of the gradient,and the maximum slope is $\|\nabla f(x, y)\|$.*

That is, the function $f(x, y)$ increases most rapidly in the direction of its gradient.



Similarly, it tells us that the minimum value of $D_u f$ at the point (x, y) is $-\|\nabla f(x, y)\|$ and this minimum occurs when $\theta = \pi$, that is, when **u** is oppositely directed to $\nabla f(x, y)$.

Geometrically, this means:

***Property 2**: At (x, y), the surface $z = f(x, y)$ has its minimum slope in the direction that is opposite to the gradient, and the minimum slope is* $-\|\nabla f(x, y)\|$

That is, the function $f(x, y)$ decreases most rapidly in the direction opposite to its gradient

Finally, in the case where $\nabla f(x, y) = 0$, it follows that $D_u f(x, y) = 0$ in all directions at the point (x, y) This typically occurs where the surface $z = f(x, y)$ has a "relative maximum," a "relative minimum," or a saddle point.

A similar analysis applies to functions of three variables and we have that

$$D_u f(x, y, z) = \nabla f(x, y, z) \cdot u = \|\nabla f(x, y, z)\| \|u\| \cos\theta = \|\nabla f(x, y, z)\| \cos\theta$$

For precisely the same reason as in two dimensions the direction of maximum increase of $f(x, y, z)$ at any point is given by the gradient at that point. As a consequence, we have the following result:

Theorem 3.5.3: Let f be a function of either two or three variables, and let P denote the point $P(x_0, y_0)$ or $P(x_0, y_0, z_0)$, respectively. Assume that f is differentiable at P:

(a) If $\nabla f = 0$ at P, then all directional derivatives of f at P are zero.

(b) If $\nabla f \neq 0$ at P, then among all possible directional derivatives of f at P, the derivative in the direction of ∇f at P has the largest value. The value of this largest directional derivative is $\|\nabla f\|$ at P.

(c) If $\nabla f \neq 0$ at P, then among all possible directional derivatives of f at P, the derivative

in the direction opposite to that of ∇f at P has the smallest value. The value of this smallest directional derivative is $\|\nabla f\|$ at P.

Examples: 1. Find the maximum and minimum rates of change of the following functions and the directions in which those values occur

a) $f(x,y)=x^2e^y; P(-2,0)$

b) $f(x,y,z)=x^2y-yz^3+z; P(1,-2,0)$

Solution. (a) Since

$$\nabla f(x,y)=f_x(x,y)i+f_y(x,y)j=2xe^yi+x^2e^yj$$

The gradient of *f* at *(*−2, 0*)* is

$$\nabla f(-2,0)=-4i+4j$$

By Theorem above, the maximum value of the directional derivative is

$$\|\nabla f(-2,0)\|=\sqrt{(-4)^2+4^2}=4\sqrt{2}$$

This maximum occurs in the direction of $\nabla f(-2,0)=-4i+4j$.

Again, the minimum value of the directional derivative is

$$-\|\nabla f(-2,0)\|=-\sqrt{(-4)^2+4^2}=-4\sqrt{2}$$

This minimum occurs in the direction of $-\nabla f(-2,0)=4i-4j$

b) Since

$$\nabla f(x,y,z)=f_x(x,y,z)i+f_y(x,y,z)j+f_z(x,y,z)k$$

$$\nabla f(x,y,z)=2xyi+\left(x^2-z^3\right)j+\left(-3z^2y+1\right)k$$

$$\nabla f(1,-2,0)=-4i+j+k$$

$-4i+j+k$ is the direction of maximum increase and $4i-j-k$ is the direction of slow change of $f(x,y,z)$ with both maximum rate of change and minimum rate of change having magnitude of but opposite in sign

$$\|\nabla f(1,-2,0)\|=\sqrt{(4)^2+(-1)^2+(-1)^2}=\sqrt{18}=3\sqrt{2}$$

2. (**Application**)**:** A heat-seeking particle is located at the point *(*2, 3*)* on a flat metal plate whose temperature at a point (x,y) is

$$T(x,y)=10-8x^2-2y^2$$

Find an equation for the trajectory of the particle if it moves continuously in the direction of maximum temperature increase.(*Group discussion in a class*)

Quick Check Class Exercises 3.5.3:

1. Find the directional derivative of

   a. $f(x,y) = x + \sin(x+y)$ *at* $(0, 0)$ in the direction of $a = 2i + j$

   b. $f(x,y,z) = yz.2^x$ *at* $(1,-1, 1)$ in the direction of $a = 2j - k$

2. Find a unit vector in the direction in which f increases and decreases most rapidly and the maximum and minimum rates of change at the given points.

   a. $f(x,y) = x^2 y - 4y^3; P(2,1)$

   b. $f(x,y,z) = \dfrac{x-z}{y+z}; P(-1,1,3)$

Group Activity 3.5.3

1. Find the directional derivative of

   a. $f(x,y) = x\,e^y - y\,e^x$ *at* $(1, 0)$ in the direction of $a = 3i + 4j$

   b. $f(x,y) = x.y^2$ *at* $(1, 0)$ in the direction of $a = i - 2j$

   c. $f(x,y) = \dfrac{ax + by}{x+y}$ *at* $(1, 1)$ in the direction of $a = i - j$

   d. $f(x,y) = \ln(x^2 + y^2)$ *at* $(0,1)$ in the direction of $a = 8i + j$

   e. $f(x,y,z) = x^2y + y^2z + z^2x$ *at* $(1,0, 1)$ in the direction of $a = 3j - k + i$

   f. $f(x,y,z) = x\tan(y+z)$ *at* $(1,0, 1)$ in the direction of $a = i + j - k$

2. Find a unit vector in the direction in which

   a. f increases most rapidly at **P** and give the rate of change of f in that direction.

   b. f decreases most rapidly at **P** and give the rate of change of f in that direction.

      i. $f(x,y) = y^2 . e^{2x}, \quad P = (0,1)$

      ii. $f(x,y) = x + \sin(x+2y)$; $P = (0, 0)$

      iii. $f(x,y,z) = \sqrt{x^2 + y^2 + z^2}$; $P = (1, -2, 1)$

      iv. $f(x,y,z) = x^2\ z\ e^y + xz^2$; $P = (1, \ln 2, 2)$

3.**(Application):** The temperature (in degrees Celsius) at a point *(x, y, z)* in a metal solid is

$$T(x,y,z) = \frac{xyz}{1 + x^2 + y^2 + z^2}$$

(a) Find the rate of change of temperature with respect to distance at $(1,1,1)$ in the direction of the origin.

(b) Find the direction in which the temperature rises most rapidly at the point (1, 1, 1). (Express your answer as a unit vector.)

(c) Find the rate at which the temperature rises moving from (1, 1, 1) in the direction obtained in part (b).

Assessment

- *Raising questions*
- *Observation while they work in groups.*
- *Answering to the raised questions*

3.6 Extreme Values of Functions of Several Variables

Definition: - Let f be a function of two variables and R a set contained in the domain of f. Then

a. f has maximum(absolute maximum) value on R at (x_0, y_0) if $f(x,y) \le f(x_0, y_0), \forall (x,y) \in R$

b. f has minimum(absolute minimum) value on R at (x_0, y_0) if $f(x,y) \ge f(x_0, y_0), \forall (x,y) \in R.$

Definition: If **R** is the domain of f, we say that f has a relative maximum value (respectively, a relative minimum value) at (x_0, y_0) if there is a disk **D** centered at (x_0, y_0) and contained in the domain of f such that

$f(x,y) \le f(x_0, y_0)$ (respectively, $f(x_0, y_0) \le f(x,y)$, for all (x, y) $\in$ D)

Note: -

1) Absolute Extreme values comprise both absolute maximum and absolute minimum values, where as relative extreme values compromise a relative minimum and a relative maximum values.

2) Further Absolute Extreme values can also be regarded as relative extreme values, but since a function can have more than one relative extreme values, not all relative extreme values are absolute extreme values.

3.6.1 First and Second Partial Derivative Method of Determining Extreme Values

Theorem 3.6.1: - Let f have a relative extreme value at (x_0, y_0). If f has partial derivatives at (x_0, y_0), then

$$f_x(x_0, y_0) = f_y(x_0, y_0) = 0$$

Definition: Let $f(x, y)$ be a function of two variables. Then points (x_0, y_0) in the domain of f where either $f_x = f_y = 0$ or f_x *or* f_y does not exist are called **Critical Points**.

Example1. Let $f(x, y) = 3 - x^2 + 2x - y^2 - 4y$. Find all critical points of f.

Solution: - $f_x(x, y) = -2x + 2$ and $f_y(x, y) = -2y - 4$

$$\Rightarrow f_x(x, y) = fx(x, y) = 0 \Rightarrow -2x + 2 = 0 \quad \wedge \quad f_x(x, y) = 0 \quad \Rightarrow \quad -2y - 4 = 0$$

$$\Rightarrow x = 1 \qquad \textit{and} \qquad y = -2$$

Thus $(1, -2)$ is the only critical point of f.

Alternatively,

$$f(x, y) = 3 - x^2 + 2x - y^2 - 4y$$

$$= 8 - \underbrace{(x-1)^2}_{\geq 0} - \underbrace{(y+2)^2}_{\geq 0}$$

$$\leq 8$$

Hence $f(1, -2) = 8$ is an absolute maximum value of f.

Example 2 Let $f(x, y) = \sqrt{x^2 + y^2}$. Determine

i. All the critical points of f

ii. Extreme values of f

Solution: - $f(x, y) = \sqrt{x^2 + y^2}$;

$$f_x(x, y) = \frac{2x}{2\sqrt{x^2 + y^2}}, \qquad \wedge \qquad f_y(x, y) = \frac{2y}{2\sqrt{x^2 + y^2}}$$

$$= \frac{x}{\sqrt{x^2 + y^2}} \qquad\qquad = \frac{y}{\sqrt{x^2 + y^2}}$$

$$f_x(x,y) = 0 \Rightarrow \frac{x}{\sqrt{x^2+y^2}} = 0 \Rightarrow x=0$$

$$f_y(x,y) = 0 \Rightarrow \frac{y}{\sqrt{x^2+y^2}} = 0 \Rightarrow y=0$$

But since the partial derivatives at (0, 0) do not exist, then (0, 0) is the only critical point of f. Since $f(x,y) \geq 0 = f(0,0) \ \forall (x,y)$, then 0 is an absolute minimum value of f.

Example 3: Determine the critical point for $f(x,y) = x^2 + y^2 - 4xy + 8x - 10y + 5$

Solution: First we need to find $f_x(x,y)$ and $f_y(x,y)$ then set them equal to zero.

$$f_x(x,y) = 2x - 4y + 8 \qquad f_y(x,y) = 2y - 4x - 10$$

$$0 = 2x - 4y + 8 \qquad 0 = 2y - 4x - 10$$

After we set these two first partial derivatives equal to zero we will need to solve for *x* and *y* using the substitution method. Let's choose the partial derivative with respect to *x* $f_y(x,y) = 0$ and solve it for *x*.

$$0 = 2x - 4y + 8$$
$$4y - 8 = 2x$$
$$2y - 4 = x$$

Now we will take this expression for *x* and substitute it into $f_y(x,y) = -2y - 4$.

$$0 = 2y - 4(2y - 4) - 10$$
$$0 = 2y - 8y + 16 - 10$$
$$0 = -6y + 6$$
$$-6 = -6y$$
$$y = 1$$

Now substitute $y = 1$ into $2y - 4 = x$ and we get

$$x = 2(1) - 4 \quad = -2$$

Thus, the critical point for $f(x,y) = x^2 + y^2 - 4xy + 8x - 10y + 5$ is $(-2,1)$.

Example 4: Determine the critical point for $f(x,y) = \frac{1}{3}x^3 + x^2 + \frac{1}{2}y^2 - 4y + 50$.

Solution: We need to find both first order partial derivatives, set them equal to zero, and then solve for *x* and *y*.

$$f_x(x,y) = x^2 + 2x, \qquad f_y(x,y) = y - 4$$

$$0 = x^2 + 2x \Rightarrow x = 0, -2 \quad \text{and} \quad 0 = y - 4 \Rightarrow y = 4$$

Thus the critical points for $f(x,y)=\frac{1}{3}x^3+x^2+\frac{1}{2}y^2-4y+50$ are $(0,4)$ and $(-2,4)$.

It is sometimes difficult to determine where extreme values may occur by using first derivative technique. For instance for $f(x,y)=y^2-x^2$, $(0,0)$ is the only critical point. However neither relative maximum nor a relative minimum occurs at this point **(Check**!).We call such a point a ***saddle point***.

Definition: - For a function f, we say that f has a **saddle** point at (x_0,y_0) if

$$f_x(x_0,y_0)=0=f_y(x_0,y_0),$$

and if there is a disk **D** centered at (x_0,y_0).such that the following conditions hold:

i. f assumes its maximum value on a diameter of the disk only at (x_0,y_0) and

ii. f assumes its minimum value on another diameter of the disk only at (x_0,y_0).

For instance, the function $f(x,y)=y^2-x^2$ has a saddle point at (0,0).

In fact we have here also the second partial derivative test, as of functions of single variable.

Theorem 3.6.2 (*The Second Partial Derivative Test*):

Assume that f has a critical point (x_0,y_0) and f has continuous second partial derivatives in a disk catered at (x_0,y_0). Let

$$D(x_0,y_0)=f_{xx}(x_0,y_0,)\,f_{yy}(x_0,y_0)-\left[f_{xy}(x_0,y_0)\right]^2.$$

i. If $D(x_0,y_0) > 0$ *and* $f_{xx}(x_0,y_0) < 0$ *or* $f_{yy}(x_0,y_0) < 0$, then f has a relative maximum value at (x_0,y_0)

ii. If $D(x_0,y_0)>0$ *and* $f_{xx}(x_0,y_0)>0$ *or* $f_{yy}(x_0,y_0)>0$, then f has a relative minimum value at (x_0,y_0)

iii. If $D(x_0,y_0)<0$, then f has a saddle point at (x_0,y_0).

But, *if* $D(x_0,y_0)=0$, then f may or may not have a relative extreme value at (x_0,y_0).

Remark: -

The expression $D(x_0,y_0)$ in the above Theorem is called the discriminant of f at (x_0,y_0).It can also be given in determinant form:

$$D(x_0,y_0)=\begin{vmatrix} f_{xx}(x_0,y_0) & f_{xy}(x_0,y_0) \\ f_{yx}(x_0,y_0) & f_{yy}(x_0,y_0) \end{vmatrix}$$

$$= f_{xx}(x_0,y_0).f_{yy}(x_0,y_0) - f_{xy}(x_0,y_0).f_{yx}(x_0,y_0)$$

A critical point (x_0,y_0) is said to be degenerate if $D(x_0,y_0)=0$; Otherwise non-degenerate.

Example 1. Let $f(x,y)=x^4+y^4$. Find the extreme values of f.

Solution: - $f_x(x,y)=4x^3$ *and* $f_y(x,y)=4y^3$

$$f_x(x,y)=0 \quad \textit{and} \quad f_y(x,y)=0$$

$$\Rightarrow 4x^3=0 \quad \textit{and} \quad 4y^3=0$$

$$\Rightarrow x=0 \quad \textit{and} \quad y=0$$

Therefore $(0,0)$ is the only critical point for $f(x,y)=x^4+y^4$.

Now,

$$f_{xx}(x,y)=12x^2 \quad \textit{and} \quad f_{yy}(x,y)=12y^2$$

$$f_{xy}(x,y)=0 \quad \textit{and} \quad f_{yx}(x,y)=0$$

$$f_{xx}(0,0)=0 \quad \textit{and} \quad f_{yy}(0,0)=0$$

$$\Rightarrow f_{xy}(0,0)=0 \quad \textit{and} \quad f_{yx}(0,0)=0$$

$$D=\begin{vmatrix} f_{xx}(0,0) & f_{xy}(0,0) \\ f_{yx}(0,0) & f_{yy}(0,0) \end{vmatrix}=\begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix}=0$$

Therefore, by using second derivative test it is impossible to determine the extreme values of f. But $f(x,y)=x^4+y^4 \geq 0 = f(0,0), \quad \forall (x,y) \in \Re^2$

Thus, $f(0,0)=0$ is the minimum value of $f(x,y)=x^4+y^4$

Example 2. Let $f(x,y)=x^2-2xy+\frac{1}{3}y^3-3y$. Determine the point at which f has relative extreme values and saddle point, if any.

Solution: - $f(x,y)=x^2-2xy+\frac{1}{3}y^3-3y$

$$f_x(x,y)=2x-2y \quad \textit{and} \quad f_y(x,y)=-2x+y^2-3$$

$$f_x(x,y)=0 \quad \text{and} \quad f_y(x,y)=0$$

$$\Rightarrow 2x-2y=0 \quad \text{and} \quad -2x+y^2-3=0$$

$$\Rightarrow x=y \quad \text{and} \quad y^2-2y-3=0$$

$$\Rightarrow x=y \quad \text{and} \quad (y-3)(y+1)=0$$

$$\Rightarrow x=y \quad \text{and} \quad y=3 \ \text{or} \ y=-1$$

Therefore the only critical points of $f(x,y)=x^2-2xy+\frac{1}{3}y^3-3y$ are $(-1,-1)$ *and* $(3,3)$.

Now

$$f_{xx}(x,y)=2 \quad \text{and} \quad f_{xy}(x,y)=-2$$

$$f_{yx}(x,y)=-2 \quad \text{and} \quad f_{yy}(x,y)=2y$$

$$D=\begin{vmatrix} f_{xx}(-1,-1) & f_{xy}(-1,-1) \\ f_{yx}(-1,-1) & f_{yy}(-1,-1) \end{vmatrix}=\begin{vmatrix} 2 & -2 \\ -2 & -2 \end{vmatrix}=-8<0$$

Thus f has a saddle point at $(-1,-1)$. And

$$D=\begin{vmatrix} f_{xx}(3,3) & f_{xy}(3,3) \\ f_{yx}(3,3) & f_{yy}(3,3) \end{vmatrix}=\begin{vmatrix} 2 & -2 \\ -2 & 6 \end{vmatrix}=12-4=8>0$$

Since

$$D=\begin{vmatrix} f_{xx}(3,3) & f_{xy}(3,3) \\ f_{yx}(3,3) & f_{yy}(3,3) \end{vmatrix}=\begin{vmatrix} 2 & -2 \\ -2 & 6 \end{vmatrix}=12-4=8>0,$$

$f_{xx}(3,3)=2>0$ *and* $f_{yy}(3,3)=12>0$, then by second derivative test (II) $f(x,y)=x^2-2xy+\frac{1}{3}y^3-3y$ has local minimum value at $(3,3)$.

Quick Check Group Activity 3.6.1:

1. Find all relative extrema and/or saddle points for

   a) $f(x,y)=-2x^2-3y^2-4x+3y+5$

   b) $f(x,y)=x^3+6x^2+\frac{1}{2}y^2+4y+4$

2. **Blood flow-** The shape of a blood vessel (a vein or artery) can be modeled by a cylindrical tube with radius and length. The velocity of the blood is modeled by the **law of laminar flow** (discovered by Poiseuille):

$$f(P,\eta,L,R,r)=\frac{P}{4\eta L}(R^2-r^2)$$

where η is the viscosity of the blood, P is the pressure difference between the ends of the tube (in $dynes/cm^2$), r is the distance from the central axis of the tube, and r,R and L are measured in centimeters.

(a) Evaluate $f(4000,0.027,2,0.008,0.002)$ and interpret it. (These values are typical for some of the smaller human arteries.)

(b) Where in the artery is the flow the greatest? Where is it least?

SAMPLE QUIZ

1. Find the domain of $f(x,y)=\ln(2x-y)$
2. Given $f(x,y)=(e^{2y+x}+\ln x)^3$ find $f_x(x,y)$ and $f_y(x,y)$.
3. Given $f(x,y)=xe^{xy}$ find $f_{xy}(x,y)$.
4. Find the critical point(s) for $f(x,y)=x^3+y^3-6xy$.
5. Find all relative extrema and/or saddle point(s) for $f(x,y)=x^3+y^3-3xy-10$.

Group Activity 3.6.1

1. Let $f(x,y)=y^2-x^2$. Show that (0, 0) is the only critical point but that $f(0,0)$ is not a relative extreme value of f.
2. Complete the square to identify all local extreme values of
   a. $f(x,y)=x^2+2x+y^2-4y+1$
   b. $f(x,y)=x^4-6x^2+y^4+2y^2+1$
3. Find all critical points and determine local extreme values:
   a. $f(x,y)=2x+2y-x^2+y^2+5$
   b. $f(x,y)=x^2+xy+y^2+3x+1$
   c. $f(x,y)=x^3-3x+y$
   d. $f(x,y)=x^2+2xy+3y^2+2x+10y+1$
   e. $f(x,y)=x^3-6xy+y^3$
   f. $f(x,y)=3x^2+xy-y^2+5x-5y+4$
   g. $f(x,y)=x\sin y$
   h. $f(x,y)=y+x\sin y$
   i. $f(x,y)=(x+y)(xy+1)$
   j. $f(x,y)=xy+x^{-1}+8y^{-1}$
   k. $f(x,y)=(x-y)(xy-1)$
   l. $f(x,y)=\dfrac{-2x}{x^2+y^2+1}$
   m. $f(x,y)=(x^2+y^2)e^{x^2-y^2}$

3.6.2 The Maximum and Minimum Value Theorem for Functions Two Variables

Theorem 3.6.3 (**Maximum and Minimum Value Theorem for functions two variables)**: -

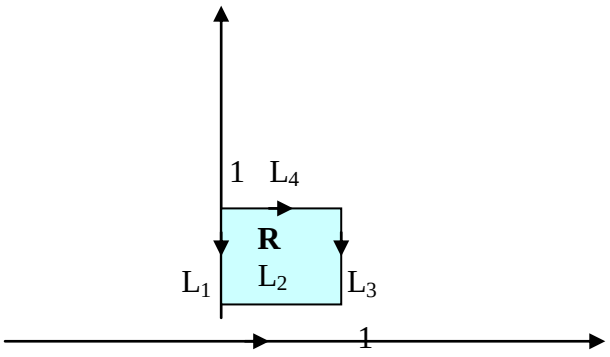
Suppose R be a bounded set in a plane that contains its boundary and f a function that is continuous on R. Then f have both a maximum and a minimum value on R.

Proof: Exercise

To determine extreme values on **R**:

- Find the critical points of f on **R** and compute the value of f at each of these points.
- Find the extreme values of f on the boundary of **R**
- The maximum value of f on **R** will be the largest of the values computed in steps (1) & (2) and the minimum value of f on R will be the smallest of these values.

Example 1: Let $f(x, y) = x.y - x^2$ and **R** a square region shown below



Solution: - Obviously, f is continuous on **R**. Then by Maximum-Minimum Theorem f have extreme values.

Now, $f_x(x\ ,\ y) = y - 2x \ \Rightarrow f_x(x\ ,\ y) = 0 \ \Rightarrow y = 2x$

And $f_y(x\ ,\ y) = x \ \Rightarrow f_y(x\ ,\ y) = 0 \ \Rightarrow x = 0$

Thus, the only critical point for f on **R** is $(0,\ 0)$, which is on the boundary of **R.**

i. On l_1, $x=0$ *and* $0\le y\le 1$; since $f(0,y)=0$, then the maximum and minimum value of f on l_1 are both 0.

ii. On l_2, $y=0$ *and* $0\le x\le 1$; since $f(x,0)=-x^2$, then the maximum value of f on l_2 is 0 and minimum value of f on l_2 is -1.

iii. On l_3, $x=1$ *and* $0\le y\le 1$; since $f(1,y)=y-1$, then the maximum value of f on l_3 is 0 and minimum value of f on l_3 is -1.

iv. On l_4, $y=1$ *and* $0\le x\le 1$; since $f(x,0)=x-x^2$, then the maximum value of f on l_4 is $\frac{1}{4}$ and minimum value of f on l_4 is 0.

v. $f(0,0)=0$

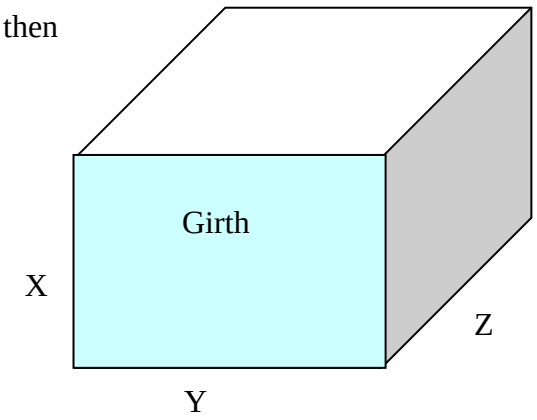
Thus the maximum and minimum values of f on **R** are $\frac{1}{4}$ and –1 respectively.

Example 2: - Under present post office regulations a package in the shape of a rectangular parallelepiped can be mailed parcel post only if the sum of the length and girth of the package is not more than 108 inches. Find the largest volume V of such a package.

Solution: - If x, y *and* z represent the dimensions of the package with z the length of the largest side, then 2x+2y is the girth and

$Volume = V = x.y.z \ \wedge\ 2x+2y+z\le 108$, where $x,y,z\ge 0$.

Since we wish to determine the largest possible volume, then



$$2x + 2y + z = 108 \quad (1)$$

$$z = 108 - 2x - 2y$$

Now,

$$V(x,y) = xy(108-2x-2y)$$
$$= 108xy - 2x^2y - 2y^2x$$

Since $x,y,z\ge 0$, then $2x+2y\le 108$ (2)

$$\Leftrightarrow\ x+y\le 54$$

$$V_x(x,y)=108y-4xy-2y^2 \ \wedge\ V_y(x,y)=108-2x^2-4xy$$

$$\Rightarrow V_x(x,y)=0 \wedge\ V_y(x,y)=0 \Rightarrow 108-4xy-2y^2=0\ \wedge\ 108x-2x^2-4xy=0$$

$$\Rightarrow y\,(108-4x-2y)=0\wedge x(108-2x-4y)=0$$

$$\Rightarrow y=0\,v(108-4x-2y)=0\wedge x=0\,v108-2x-4y=0$$

$$\Rightarrow y=0\,v\;\; y=54-2x\wedge x=0\,vy=27-\frac{1}{2}x$$

$$\Rightarrow y=0\wedge x=0\wedge 54-2x=27-\frac{1}{2}x$$

$$\Rightarrow y=0\;\wedge\;\; x=0\;\wedge\;\; 27=\frac{3}{2}x$$

$$\Rightarrow y=0\wedge x=0\wedge x=18\wedge y=18$$

Thus (0,0) and (18,18) are the only critical points of f

Then, $V(54,0)=0=V(0,0)=0$

$$V(18,18)=108(18)^2-2(18)^3-2(18)^3$$

$$=11{,}664\;\; cubic\;\; inch$$

Hence, $V(18,18)=$ 11,664 is the maximum volume of a mail able rectangular parallelepiped.

Extreme Values with Side Condition

Problem 1

Maximize $f(x,y)=x.y$

Subject to $x+y-1=0$.

That is, determine the maximum area of a rectangle with perimeter 2 units.

Solution: - From the side condition $x+y-1=0\Leftrightarrow y=1-x$. Now,

$$f(x,y)=x(1-x)\;\;\Leftrightarrow h(x)=x-x^2$$
$$\Rightarrow h'(x)=1-2x;h'(x)=0\Rightarrow 1-2x=0$$

$$\Leftrightarrow x=\frac{1}{2}$$

$$h''(x)=-2\,;\,h''\,(\frac{1}{2})=\;\;-2<0$$

Thus $f(\frac{1}{2},\;\frac{1}{2})=h(\frac{1}{2})=\frac{1}{4}$ is the maximum value of f.

Example 2. Find the maximum volume of a rectangular solid given that the sum of length of its edge is 12a.

Solution: - Let $x,\ y\ and\ z$ be dimensions of the solid. Then

$V = Volume\ of\ the\ solid = x\,.\ y.\ z \longrightarrow Max.$

Subject to

$$4(x+y+z) = 12.a$$

$$\Leftrightarrow x+y+z = 3.a$$

From the side condition

$$z = 3.a-(x+y)$$

Now, $V = x\,.\ y.(3.\ a-(x+y))$

$\Rightarrow V_x = y.(3.\ a-x-y)-x.y = 3ay-2xy-y^2$ and

$V_y = y.(3.\ a-x-y)-x.y = 3ax-2xy-x^2$

Then $V_x = 0$ and $V_y = 0$

$\Rightarrow 3ay-2xy-y^2 = 0 \ \wedge 3ax-2xy-x^2 = 0$

$\Rightarrow y\,(3a-2x-y) = 0 \ \wedge x\,(3\,a-2y-x) = 0$

$\Rightarrow [y = 0 \ \vee \ (3a-2x-y) = 0] \ \wedge [x = 0 \vee (3\,a-2y-x) = 0]$

But since $(0\,,\,0)$ is a boundary point, then

$\Rightarrow 3a-2x-y = 0 \ \wedge \ 3\,a-2y-x = 0$

$\Rightarrow 3a = 2x+y \ \wedge \ 3\,a = 2y+x$

$\Rightarrow 2x+y = 2y+x$

$\Rightarrow y = x$

$\Rightarrow 3\,.a = 3.x$

$\Rightarrow y = x = a$

Thus (a, a, a) is the only critical point of V, for $a \neq 0$.

Hence $V(a, a, a) = a^{\,3}$ *cubic unit* is the maximum volume of the solid.

Quick check group activity 3.6.4:

1.Find the absolute maximum and minimum values of

a) $f(x,y) = 5+4x-2x^2+3y-y^2$ on the region bounded by the lines $y = 2, y = x; x = 3$

b) $f(x,y) = x^2+y^2-4xy$ on the region bounded by the lines $y = -3, y = x; x = 3$

c) $f(x,y) = x^2+y^2-2x-4y$ on the region bounded by the lines $y = 3, y = x; x = 0$

d) $f(x,y) = x^2+y^2$ on the region bounded by $(x-1)^2+y^2 = 4$

2. **Writing:** Suppose that the second partials test gives no information about a certain critical point (x_0, y_0) because of $D(x_0, y_0) = 0$. Discuss what other steps you might take to determine whether there is a relative extremum at that critical point.

3.6.3 Lagrange Multiplier

Theorem: - Suppose that g is a continuously differentiable function of two or three variables defined on the subset of domain of f. If (x_0, y_0).maximizes (or minimizes) $f(x, y)$ subject to the side condition $g(x, y)$, then $\nabla f(x_0, y_0)$ is parallel to $\nabla g(x_0, y_0)$, that is, there is a scalar λ such that

$$\nabla f(x_0, y_0) = \lambda \ \nabla g(x_0, y_0)$$

Such a scalar λ is called a ***Lagrange Multiplier***.

Proof:- Exercise

Example 3: - Maximize and Minimize

$$f(x, y) = x.y$$

On the unit circle $x^2 + y^2 = 1$

Solution: - Since f is continuous and the unit circle is closed and bounded, then f has both maximum and minimum value.

Let $g(x, y) = x^2 + y^2 - 1$. Then we want to show that

$$f(x, y) = x.y \longrightarrow Max\ (Min.)$$

Subject to

$$g(x, y) = 0.$$

Now,

$$\nabla f(x, y) = y\, i + x\, j \ \wedge \nabla g(x, y) = 2x.i + 2y.j$$

By *Lagrange Multiplier method*

$$\nabla f(x, y) = \lambda . \nabla g(x, y)$$

$$\Rightarrow y = 2\lambda x \ \wedge \ x = 2\lambda y$$

$$\Rightarrow y^2 = (2\lambda x)y \ \wedge \ x^2 = (2\lambda y)x$$

$$\Rightarrow y^2 = (2\lambda y)x \ \wedge \ x^2 = (2\lambda x)y$$

$$\Rightarrow y^2 = x^2 \ \ldots\ldots\ldots(1)$$

From the side condition and (1) we have

$$2x^2 - 1 = 0 \Rightarrow x^2 = \frac{1}{2}$$

$$\Rightarrow x = \pm\sqrt{1/2} = \pm\frac{\sqrt{2}}{2}$$

Therefore, the only critical points of f are

$$\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right), \left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right), \left(\frac{\sqrt{2}}{2}, \frac{-\sqrt{2}}{2}\right), \left(\frac{-\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$$

Now, f $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right) = f\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right) = \frac{1}{2}$ *a* $f\left(\frac{-\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right) = f\left(\frac{\sqrt{2}}{2}, \frac{-\sqrt{2}}{2}\right) = -\frac{1}{2}$

Thus, $Max.(f) = \frac{1}{2}$ *and* $Min(f) = -\frac{1}{2}$.

Example 4 Find the minimum Value taken on by the function

$$f(x, y) = x^2 + (y-2)^2$$

On the hyperbola $x^2 - y^2 = 1$

Solution: - This minimum is simply the square of the distance from the point (0,2) to the hyperbola , clearly it exists.

Let $g(x, y) = x^2 - y^2 - 1$. Then

$$f(x, y) \longrightarrow Min.$$

Subject to $g(x, y) = 0$

Now,

$$\Rightarrow \nabla f(x, y) = 2xi + 2(y-2).j \quad \wedge \quad \nabla g(x, y) = 2xi - 2yj$$

By *Lagrange Multiplier method*

$$\nabla f(x, y) = \lambda . \nabla g(x, y)$$

$$\Rightarrow 2xi + 2(y-2)j = 2\lambda xi - 2\lambda yj$$

$$\Rightarrow 2x = 2\lambda x \ \wedge \ 2(y-2) = -2\lambda y$$

$$\Rightarrow \lambda = 1 \qquad 2y - 4 = -2(1)y$$

$$\Rightarrow \ 4y = 4 \Rightarrow y = 1$$

From side condition, $x^2 - y^2 - 1 = 0 \Rightarrow x^2 = 2 \ \Rightarrow \ x = \pm\sqrt{2}$

Thus, the only critical points of f are $(\sqrt{2},1)\&(-\sqrt{2},1)$ Now, $f(\sqrt{2},1)=3=f(-\sqrt{2},1)$.

Thus, f takes its minimum at $(\sqrt{2},1)$ *and* $(-\sqrt{2},1)$.

Remark: - Problem (4) could have been solved more simply by rewriting the side condition as $x^2=1+y^2$ and eliminating x from f(x, y) by substitution. Now, determine the minimum value of $f(y)=2y^2-4\ y+5$.

Example 5 : Find the Maximum value of $f(x,y)=x\,y\,z$ on $x^3+y^3+z^3=1,\ x,y,z\geq 0$.

Solution: - Let $g(x,y)=x^3+y^3+z^3-1$.

Obviously, the side condition is bounded and closed, and f is continuous on the side condition.

$$f_x(x,y)=yz,\ f_y(x,y)=x.z\ \wedge\ f_z(x,y)=xy$$

$$g_x(x,y)=3x^2,\ g_y(x,y)=3y^2\ \wedge\ g_y(x,y)=3z^2$$

Now, $\nabla f(x,y)=\lambda\nabla g(x,y)\Rightarrow yzi+xzj+xyk=3x^2\lambda i+3z^2\lambda k$

$$\Rightarrow yz=3\lambda x^2, xz=3\lambda y^2\ \wedge\ xy=3\lambda z^2$$

$$\Rightarrow xyz=3\lambda x^3, xyz=3\lambda y^3\ \wedge\ xyz=3\lambda z^3$$

$$\Rightarrow 3\lambda x^3=3\lambda y^3=3\lambda z^3$$

If λ = 0, obviously, $x=y=z=0$. But $f(0,0,0)$ is not the maximum value.

If λ ≠ 0, then

$$x^3=y^3=z^3$$

$$\Rightarrow x=y=z$$

Now, from side condition,

$$3x^3=1\Leftrightarrow x^3=\frac{1}{3}\Rightarrow x=\frac{1}{\sqrt[3]{3}}$$

Thus, the desired maximum value is

$$f\left(\frac{1}{\sqrt[3]{3}},\frac{1}{\sqrt[3]{3}},\frac{1}{\sqrt[3]{3}}\right)=\frac{1}{3}$$

A SHORT HISTORY OF LAGRANGE



Joseph Louis Lagrange (1736–1813) French–Italian mathematician and astronomer. Lagrange, the son of a public official, was born in Turin, Italy. (Baptismal records list his name as Giuseppe Lodovico Lagrangia.) Although his father wanted him to be a lawyer, Lagrange was attracted to mathematics and astronomy after reading a memoir by the astronomer Halley. At age 16 he began to study mathematics on his own and by age 19 was appointed to a professorship at the Royal Artillery School in Turin. The following year Lagrange sent Euler solutions to some famous problems using new methods that eventually blossomed into a branch of mathematics called calculus of variations. These methods and Lagrange's applications of them to problems in celestial mechanics were so monumental that by age 25 he was regarded by many of his contemporaries as the greatest living mathematician.

In 1776, on the recommendations of Euler, he was chosen to succeed Euler as the director of the Berlin Academy. During his stay in Berlin, Lagrange distinguished himself not only in celestial mechanics, but also in algebraic equations and the theory of numbers. After twenty years in Berlin, he moved to Paris at the invitation of Louis XVI. He was given apartments in the Louvre and treated with great honor, even during the revolution. Napoleon was a great admirer of Lagrange and showered him with honors—count, senator, and Legion of Honor. The years Lagrange spent in Paris were devoted primarily to didactic treatises summarizing his mathematical conceptions. One of Lagrange's most famous works is a memoir, *Mécanique Analytique*, in which he reduced the theory of mechanics to a few general formulas from which all other necessary equations could be derived. It is an interesting historical fact that Lagrange's father speculated unsuccessfully in several financial ventures, so his family was forced to live quite modestly. Lagrange himself stated that if his family had money, he would not have made mathematics his vocation. In spite of his fame, Lagrange was always a shy and modest man. On his death, he was buried with honor in the Pantheon.

Group Activity 3.6.4

1. Find the extreme value(s) of f subject to the given side conditions

   a. $f(x,y) = xy, \quad (x+1)^2 + y^2 = 1$

   b. $f(x,y,z) = xyz, \quad x^2 + y^2 + 4z^2 = 6$

   c. $f(x,y,z) = xy + yz, \quad x^2 + y^2 + z^2 = 8$

   d. $f(x,y,z) = 3z - x - 2y, \quad x^2 + 4y^2 = z$

   e. $f(x,y) = e^{2x+y}, \quad x^2 + y^2 = 5$

   f. $f(x,y) = x^3 + x^2 + \frac{y^2}{3}, \quad x^2 + y^2 \le 36$

   g. $f(x,y) = xy, \quad 2x^2 + y^2 \le 4$

   h. $f(x,y) = 16 - x^2 - 4y^2, \quad x^4 + 2y^4 \le 1$

   i. $f(x,y) = 3x^2 - xy + y^2, \quad 2x^2 + y^2 \le 4$

   j. $f(x,y,z) = \frac{x^2 + y^2}{z^2 + 5}, \quad x^2 + y^2 - 2 \le z \le 6$

   k. $f(x,y) = 4x^2 + y^3 + 3y + 7, \quad 2x^2 + \frac{3}{2}y^2 = \frac{3}{2}$

2. a) Suppose that the temperature of a metal plate is given by $T(x,y) = x^2 + 2x + y^2$ for points (x,y) on the elliptical plate defined by $x^2 + 4y^2 \le 24$. Find the maximum and minimum temperature on the plate.

3. Determine the maximum profit $P = 4x + 5y$ of a bossiness with given production possibility curve $2x^2 + 5y^2 \le 32,500$

4. Minimize $f(x,y,z) = x^2 + y^2 + z^2$ subject to the constraints $x + 2y + 3z = 6$ and $x + y = 0$

5. Maximize $f(x,y,z) = 3x + y + 2z$ subject to the constraints $y^2 + z^2 = 1$ and $x + y - z = 1$

3.7 Unit Summery

- Let f be a function defined throughout a set containing a disc centered at (x_0, y_0) except possibly at (x_0, y_0) itself, and let L a number. Then L is said to be the limit of f at (x_0, y_0) if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < \sqrt{(x-x_0)^2} + \sqrt{(y-y_0)^2} < \delta \Rightarrow |f(x,y) - L| < \varepsilon$$

$$\Leftrightarrow \lim_{(x,y)\to(x_0,y_0)} f(x,y) = L$$

- A function f of two variables is ***continuous*** at (x_0, y_0) if

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = f(x_0, y_0)$$

- A function f of three variables is ***continuous*** at (x_0, y_0) if

$$\lim_{(x,y,z)\to(x_0,y_0,z_0)} f(x,y,z) = f(x_0, y_0, z_0)$$

- Let f be a function of two variables and (x_0, y_0) in the domain of f. The partial derivative of f with respect to x at (x_0, y_0) is defined by

$$f_x(x_0, y_0) = \lim_{h\to 0} \frac{f(x_0+h, y_0) - f(x_0, y_0)}{h}$$

Provided this limit exists.

Similarly, the partial derivative of f w. r. t. y at (x_0, y_0) is defined by

$$f_y(x_0, y_0) = \lim_{h\to 0} \frac{f(x_0, y_0+h) - f(x_0, y_0)}{h}$$

Provided this limit exists.

- Let f *and* g be functions of two variables in x and y. Then

$$(f \pm g)_x (x,y) = f_x(x,y) \pm g_x(x,y)$$

$$(f.g)_x (x,y) = f_x(x,y) \cdot g(x,y) + f(x,y) \cdot g_x(x,y)$$

$$\left(\frac{f}{g}\right)_x (x,y) = \frac{f_x(x,y) \cdot g(x,y) - f(x,y) \cdot g_x(x,y)}{g(x,y)}$$

By similar fashion it is possible to determine, $(f \pm g)_y\ (x,y)$, $(f.g)_y\ (x,y)$ and $\left(\frac{f}{g}\right)_y (x,y)$.

- Let f be a function of two variables and g a function of one variable. For the composition function $h(x,y)=g(f(x,y))$,

$$h_x(x,y)=g'(f(x,y)).f_x(x,y)$$

&

$$h_y(x,y)=g'(f(x,y).f_y(x,y)$$

- Let f be a function of two variables x *and* y. Then the partial derivatives

$(f_x)_x$, usually denoted by f_{xx} *or* $\frac{\partial^2 x}{\partial x^2}$

$(f_x)_y$, usually denoted by f_{xy} *or* $\frac{\partial^2 x}{\partial y\ \partial x}$

$(f_y)_x$, usually denoted by f_{xy} *or* $\frac{\partial^2 x}{\partial x\ \partial y}$

$(f_y)_y$, usually denoted by f_{yy} *or* $\frac{\partial^2 x}{\partial y^2}$,

are called **Second Order Partial Derivatives of** f**,** in particular f_{xy} and f_{yx} are usually called ***Mixed Partial Derivatives of*** f.

- Let f be a function of two variables. Suppose that f_{xy} and f_{yx} are continuous at (x_0,y_0). Then

$$f_{xy}(x_0,y_0) = f_{yx}(x_0,y_0)$$

- Let f be a function defined on a set containing a disk **D** centered at (x_0,y_0), and $u=a_1i+a_2j$ be a unit –vector. Then the ***Directional Derivative of f at*** (x_0,y_0) in the direction of u, denoted by $D_u f(x_0,y_0)$ is defined as

$$D_u f(x_0,y_0)=\lim_{h\to 0}\frac{f(x+ha_1,y_0+ha_2)-f(x_0,y_0)}{h}$$

Provided this limit exists. $f_{xy}(x_0,y_0) = f_{yx}(x_0,y_0)$

❖

a. Let f be a function of two variables that has partial derivatives at (x_0, y_0). Then the ***gradient*** of f at (x_0, y_0), usually denoted by **grad** $f(x_0, y_0)$ or $\nabla f(x_0, y_0)$, is defined as

$$\nabla f(x_0, y_0, z_0) = f_x(x_0, y_0)i + f_y(x_0, y_0)j.$$

b. Let f be a function of three variables that has partial derivatives at (x_0, y_0, z_0). Then the gradient of f at (x_0, y_0, z_0) is defined by

$$\nabla f(x_0, y_0, z_0) = f_x(x_0, y_0, z_0)\ i + f_y(x_0, y_0, z_0)j + f_z(x_0, y_0, z_0)k.$$

❖ Gradients are normal vectors to level curves and level surfaces.

❖ A differential function f increases most rapidly in the direction of the gradient (the rate of change is $\|\nabla f(x, y)\|$) and it decreases most rapidly in the opposite direction of the gradient (the rate of change is -$\|\nabla f(x, y)\|$)

❖ Let f be a function of two variables and R a set contained in the domain of f. Then

a. f has maximum value on R at (x_0, y_0) if $f(x, y) \le f(x_0, y_0), \forall (x, y) \in R$

b. f has minimum value on R at (x_0, y_0) if $f(x, y) \ge f(x_0, y_0), \forall (x, y) \in R.$

❖ Let f have a relative extreme value at (x_0, y_0). If f has partial derivatives at (x_0, y_0), then

$$f_x(x_0, y_0) = f_y(x_0, y_0) = 0$$

Relative extreme values of f occur only at the points either

i. $f_x = f_y = 0$ or

ii. f_x *or* f_y does not exist.

Furthermore, points at which either condition (i) or (ii) holds are called **Critical Points**.

❖ **(*The Second Partial Derivative Test*):**

Assume that f has a critical point (x_0, y_0) and f has continuous second partial derivatives in a disk catered at (x_0, y_0). Let

$$D(x_0, y_0) = f_{xx}(x_0, y_0,)\, f_{yy}(x_0, y_0) - \left[f_{xy}(x_0, y_0)\right]^2.$$

i. If $D(x_0, y_0) > 0$ *and* $f_{xx}(x_0, y_0) < 0$ *or* $f_{yy}(x_0, y_0) < 0$, then f has a relative maximum value at (x_0, y_0)

ii. If $D(x_0, y_0) > 0$ *and* $f_{xx}(x_0, y_0) > 0$ *or* $f_{yy}(x_0, y_0) > 0$, then f has a relative minimum value at (x_0, y_0)

iii. If $D(x_0, y_0) < 0$, then f has a saddle point at (x_0, y_0).

But, *if* $D(x_0, y_0) = 0$, then f may or may not have a relative extreme value at (x_0, y_0).

❖ Suppose R be a bounded set in a plane that contains its boundary and f a function that is continuous on R. Then f have both a maximum and a minimum value on R.

❖ Suppose that g is a continuously differentiable function of two or three variables defined on the subset of domain of f. If (x_0, y_0).maximizes (or minimizes) $f(x, y)$ subject to the side condition $g(x, y)$, then $\nabla f(x_0, y_0)$ is parallel to $\nabla g(x_0, y_0)$, that is, there is a scalar λ such that

$$\nabla f(x_0, y_0) = \lambda \ \nabla g(x_0, y_0)$$

Such a scalar λ is called a ***Lagrange Multiplier***.

3.8 Review Exercise

1. Compute the indicated limit

   a. $\lim_{(x,y)\to(0,2)} \dfrac{3x}{y^2+1}$

   b. $\lim_{(x,y)\to(1,\pi)} \dfrac{xy}{\cos xy}$

2. Show that the indicated limits does not exist

   a. $\lim_{(x,y)\to(0,0)} \dfrac{3x^2y}{x^4+y^2}$

   b. $\lim_{(x,y)\to(0,0)} \dfrac{2xy^{3/2}}{x^2+y^3}$

   c. $\lim_{(x,y)\to(0,0)} \dfrac{x^2+y^2}{x^2+xy+y^2}$

   d. $\lim_{(x,y)\to(0,0)} \dfrac{x^2}{x^2+xy+y^2}$

3. Show that the indicated limits does not exist

   a. $\lim_{(x,y)\to(0,0)} \dfrac{x^3+xy^2}{x^2+y^3}$

   b. $\lim_{(x,y)\to(0,0)} \dfrac{3y^2\ln(x+1)}{x^2+3y^2}$

4. Find both first order partial derivatives of

   a. $f(x,y)=\dfrac{4x}{y}+xe^{xy}$

   b. $f(x,y)=xe^{xy}+3y^2$

   c. $f(x,y)=3xy^2\cos x-\sqrt{y}$

   d. $f(x,y)=\sqrt{x^3y}+3x-5$

5. Show that the **Lap lace's equation** $\dfrac{\partial^2 f}{\partial x^2}+\dfrac{\partial^2 f}{\partial y^2}=0$,where

   a. $f(x,y)=e^x\sin y$

   b. $f(x,y)=e^x\cos y$

6. Find the indicated derivatives

   a. $f(x,y) = 2x^4y + 3x^2y^2;\ f_{xx},\ f_{yy},\ f_{xy}$

   b. $f(x,y) = x^2\ e^{3y} - \sin y;\ f_{xx},\ f_{yy},\ f_{yyx}$

7. Find the equation of the tangent plane

   a. $z = x^2y + 2x - y^2$ at $(1, -1,\ 0)$

   b. $z = \sqrt{x^2 + y^2}$ at $(3,\ -4,\ 5)$

   c. $x^2 + 2xy + y^2 + z^2 = 5$ at $(0,\ 2,\ 1)$

   d. $x^2z + 3y - y^2x - z = -4$ at $(1, -1,\ 2)$

8. By using the Chain rule, find the indicated derivatives

   a. $g'(t) =$ *where* $g(t) = f(x(t),\ y(t))$, $f(x,y) = x^2y + y^2\ x(t) = e^{4t},\ y(t) = \sin t$

   b. $\dfrac{\partial g}{\partial u}$ *and* $\dfrac{\partial g}{\partial u}$ *where* $g(u,\ v) = f(x(u,\ v),\ y(u,\ v))$, $f(x,\ y) = 4x^2 - y$

   $x(u,\ v) = u^3v + \sin u,\quad y(u,\ v) = 4v^2$

9. State the general Chain rule for the general composition function

   a. $g(t) = f(x(t),\ y(t),\ z(t),\ w(t))$

   b. $g(u,\ v) = f(x(u,\ v),\ y(u,\ v))$

10. By using implicit differentiation, Find $\dfrac{\partial z}{\partial x}$ *and* $\dfrac{\partial z}{\partial y}$.

    a. $x^2 + 2xy + y^2 + z^2 = 1$

    b. $x^2z + 3y - y^2x - z = -4$

11. Find the gradient of the given function at the indicated point

    a. $f(x,\ y) = 3x\sin 4y - \sqrt{xy},\ \ (-\pi,\ \pi)$

    b. $f(x,\ y,\ z) = 4xz^2 - 3\cos x + 4y^2,\ \ at\ \ (0,\ 1,\ -1)$

12. Determine the directional derivative of f at the given point in the direction of the indicated vector:

    a. $f(x,\ y) = x^3y - 4y^2$ at $(-2,\ 3)$ *in the direction of* $\mathbf{u} = \left(\dfrac{3}{5},\ \dfrac{4}{5}\right)$

    b. $f(x,\ y) = x^2 + xy^2$ at $(2,\ 1)$ $\mathbf{u}$ *in the direction of* $(3,\ -2)$

    c. $f(x,\ y) = e^{3xy} - y^2$ at $(0,\ -1)$ $\mathbf{u}$ *in the direction from* $(2,\ 3)$ *to* $(3,1)$

    d. $f(x,\ y) = \sqrt{x^2 + xy^2}$ at $(2,\ 1)$ $\mathbf{u}$ *in the direction of* $(1,\ -2)$

13. Find the directions of maximum and minimum change of rate of f at the given point, and the values of the maximum and minimum rates of change.
    a. $f(x,y) = x^3y - 4y^2, \quad (-2,\ 3)$
    b. $f(x,y) = x^3 + xy^2, \quad (2,\ 1)$
    c. $f(x,y) = \sqrt{x^4 + y^4}, \quad (2,\ 0)$
    d. $f(x,y) = x^2 + xy^2, \quad (1,\ 2)$
14. Suppose that the elevation on a hill given by $f(x,y) = 100 - 4x^2 - 2y$. From the site at $(2,\ 1)$, in which direction will the rain run off?
15. If the temperature at the point $(x,\ y,\ z)$ is given by $T(x,\ y,\ z) = 70 + 5e^{-t^2}(4x + 3y^{-1})$, then the direction from the point $(1,\ 2,\ 1)$ in which the temperature decreases most rapidly.
16. Find all critical points of f
    a. $f(x,y) = 2x^4 - xy^2 + 2y^2$
    b. $f(x,y) = 2x^4 - x^2y + y^3$
    c. $f(x,y) = 4xy - x^3 - 2y^2$
    d. $f(x,y) = 3xy - x^3y + y^2 - y$
17. Find the absolute extreme of the function on the given region.
    a. $f(x,y) = 2x^4 - xy^2 + 2y^2, \ 0 \le x \le 4, \ 0 \le y \le 2$
    b. $f(x,y) = 2x^4 - x^2y + y^3$, *region bounded by* $y = 0,\ y = x$ *and* $x = 2$
18. By using the Lagrange Multiplier, find the maximum and minimum of the function $f(x,y)$ subject to the constraint $g(x,y) = C$, where C is a constant:
    a. $f(x,y) = x + 2y$, *subject to* $x^2 + y^2 = 5$
    b. $f(x,y) = 2x^2y$, *subject to* $x^2 + y^2 = 4$
    c. $f(x,y) = xy$, *subject to* $x^2 + y^2 = 1$
    d. $f(x,y) = x^2 + 2y^2 - 2x$, *subject to* $x^2 + y^2 = 1$

CHAPTER FOUR

MULTIPLE INTEGRALS

Unit Introduction

This chapter provides a very brief introduction to the major topic of multiple integration. This unit is divided into two sections. The first section presents basic definitions of double integrals, evaluation of double integrals in Cartesian coordinates and polar coordinates and its applications. And definitions of triple integrals, evaluation of triple integrals and its applications will be treated in second section. Uses of multiple integration include the evaluation of areas, volumes, masses, total charge on a surface and the location of a centre-of-mass.

Unit Objectives

At the end of this unit students should be able to

- Define double integrals of a given function on a rectangle and on an arbitrary plane region.
- Identify basic properties of double integrals.
- Develop different methods of evaluating double integrals.
- Apply the concept of double integrals in solving real life problems.
- Appreciate the role of polar coordinates in determining double integrals.
- Define triple integrals of a given function of three variables on a box and on an arbitrary solid region.
- Identify basic properties of triple integrals.
- Apply the concept of triple integrals in solving real life problems.
- Appreciate the role of spherical coordinates in determining triple integrals.
- Find the mass and center of a planar lamina using a double integral
- Find moments of inertia using double integrals.

4.1: Motivation and Definition of Double Integrals

Recall that, for any function $f(x)$ defined on the interval $[a,b]$ the definite integral of $f(x)$ on $[a,b]$ is defined by the reimann sum

$$\int_a^b f(x)dx = \lim_{\|P\|\to 0} \sum_{i=1}^{n} f(c_i)\Delta x_i$$

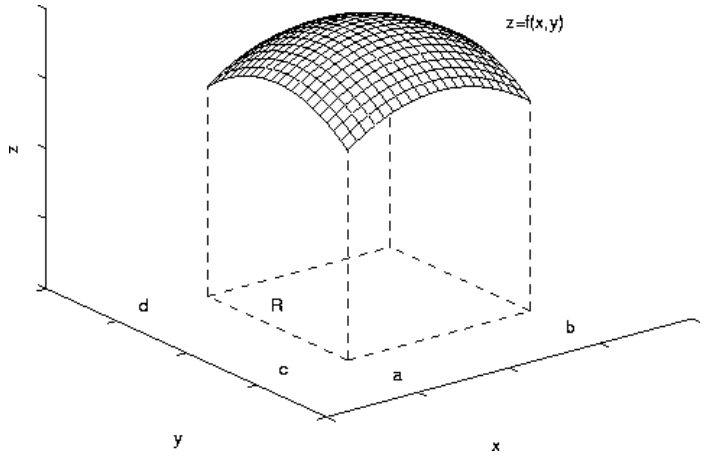
provided the limit exists and is the same for all values of the evaluation points $c_i \in [x_{i-1}, x_i]$ for $i = 1, 2, \ldots, n$. In this case, we say f is **integrable** on $[a,b]$. Further,

i) For $f(x) \geq 0$, $\int_a^b f(x)\ dx$ is the area of region bounded by the graph of $f(x)$, the vertical lines $x = a$ *and* $x = b$, and the $x-axis$.

ii) For general $f(x)$ the definite integral is equal to the area above the x-axis minus the area below the x-axis.

We developed the definite integral of $f(x)$ as a natural outgrowth of our method for finding area under a curve in the $xy-$plane. Likewise, we are guided in our development of the double integral with a following corresponding problem.

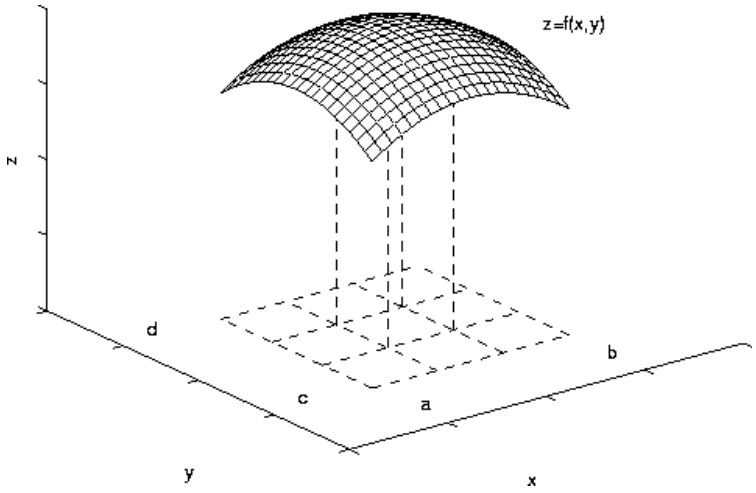
Motivation (The volume problem): *Given a function f of two variables that is continuous and nonnegative on a region R in the xy-plane, find the volume of the solid enclosed between the surface z = f(x, y) and the region R* (See Figure below).



Notation: The definite integral of f on the region of integration $R \in R^2$ is denoted by

$$\iint_R f(x,y)dA$$

Consider a function of two variables $z = f(x, y)$ which is continuous on the region $R \in R^2$. Suppose we subdivide the region **R** into sub rectangles as in the figure below (say there are M rectangles in the x direction and N rectangles in the y direction). Label the rectangles $\mathbf{R_{ij}}$ where $1 \le i \le M$ *and* $1 \le j \le N$.



Think of the definite integral as representing volume. The volume under the surface above rectangle R_{ij} is approximately $f(x_i, y_j).A_{ij}$, where A_{ij} is area of the rectangle and $f(x_i, y_j)$ is the approximate height of the surface in the rectangle. Here (x_i, y_j) is some point in the rectangle R_{ij}. If we sum over all n rectangles the volume is approximately:

$$V_i \approx \text{Height} \times \text{Base Area} \approx f(u_i, v_i)\Delta A_i$$

$$V \approx \sum_{i=1}^{n} V_i \approx \sum_{i=1}^{n} f(u_i, v_i)\Delta A_i$$

As the size of the rectangles goes close to 0, the sum on the right gives the exact total volume of the solid.

$$V = \lim_{\|P\| \to 0} \sum_{i=1}^{n} f(u_i, v_i)\Delta A_i$$

We call this limit the definite double integral of $f(x, y)$ on $R \in R^2$

Generally, the double Integral of $f(x, y)$ over any Bounded Region is defined as follow:

Definition 4.1 (Double integral)

For any function $f(x, y)$ defined and continuous on a bounded region $R \in R^2$, we define the double integral of *f* over *R* as:

$$\iint_R f(x, y)dA = \lim_{\|P\|\to 0} \sum_{i=1}^{n} f(u_i, v_i)\Delta A_i$$

provided the limit exists and is the same for all choices of the evaluation points $(u_i, v_i) \in R_i$ for $i = 1, 2, \ldots, n$. In this case, we say f is **integrable** over *R*.

The quantity $f(x, y)\, dA$ in the definite integral represents the volume in some infinitesimal region around the point (x, y). The region is so small that the function $f(x, y)$ only varies infinitesimally in the region.

Notice that for positive $f(x, y)$, the integral (Riemann Sum) is equal to the volume under the surface $z = f(x, y)$ and above xy-plane for x and y in the region R.

Definition 4.2:(Volume under a Surface)

For a function of the two variables $z = f(x, y) \geq 0$ defined and continuous over a region *R* in the $xy-$plane, the volume above *R* and under the surface S of $z = f(x, y)$ is defined by the double integral

$$\text{Volume under S} = \iint_R f(x, y)\, dA$$

For general $f(x, y)$, the definite integral is equal to the volume above the xy-plane minus the volume below the xy-plane.

The basic properties of the double integral are essentially the same as those for the definite integral of $f(x)$:

Basic Properties of the Double Integral:

a. $\iint_R [f(x, y) + g(x, y)]dxdy = \iint_R f(x, y)dxdy + \iint_R g(x, y)dxdy;$

b. $\iint_R C.f(x, y)\, dxdy = C.\iint_R f(x, y)dxdy,$ where C is a constant.

c. $\iint_R f(x, y)dxdy = \iint_{R_1} f(x, y)dxdy + \iint_{R_2} f(x, y)dxdy;$ where R is composed of two pieces $\mathbf{R_1}$ and $\mathbf{R_2}$ that are over lapping only at boundary points;

Iterated Integrals

As in the case of an integral of a function of one variable, a double integral is defined as a limit of a Riemann sum. Except in the simplest cases, it is impractical to obtain the value of a double integral from the limit in the definition of a double integral. However, we will now show how to evaluate double integrals by calculating two successive single integrals.

The symbols

$$\int_a^b f(x,y)dx$$

and

$$\int_c^d f(x,y)dx$$

denote *partial definite integrals*; the first integral, called the *partial definite integral with respect to x*, is evaluated by holding *y* fixed and integrating with respect to *x*, and the second integral, called the *partial definite integral with respect to y*, is evaluated by holding *x* fixed and integrating with respect to *y*.

Example 1 Evaluate

a) $\int_0^1 xy^2dx = y^2\left[\int_0^1 xdx\right] = y^2\left[\left(\frac{x^2}{2}\right)_0^1\right] = \frac{y^2}{2}$

b) $\int_0^1 xy^2dy = x\int_0^1 y^2dy = x\left[\frac{y^3}{3}\Big|_0^1\right] = \frac{x}{3}$

A partial definite integral with respect to x is a function of y and hence can be integrated with respect to y; similarly, a partial definite integral with respect to y can be integrated with respect to x. This two-stage integration process is called **iterated** (or repeated) integration.

In General, *Iterated* integrals, are double integrals of the form

$$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx$$

or

$$\int_c^d \int_{g_1(y)}^{g_2(y)} f(x,y)\,dx\,dy$$

Notes

1. The inside variable of integration can be a function of the outside.
2. The outside integral must have constant limits of integration and the inside limits of integration could be constants in both cases.

Example 1: Evaluate the iterated integral $\int_1^2 \int_0^1 (x^2 + y)\, dy\, dx$.

Solution:

$$\int_1^2 \int_0^1 (x^2 + y)\, dy\, dx = \int_1^2 (x^2 + \frac{1}{2}) dx = \frac{11}{2}$$

Example 2: Evaluate the iterated integral $\int_0^1 \int_0^{2x} (x + 2y)\, dy\, dx$.

Solution:

$$\int_0^1 \int_0^{2x} (x + 2y)\, dy\, dx = \int_0^1 \left(x\left(y\big|_0^{2x}\right) + y^2\big|_0^{2x} \right) dx$$

$$= \int_0^1 \left(2x^2 + 4x^2\right) dx = \left[2x^3\right]_0^1 = 2$$

Example 3: Evaluate the iterated integral $\int_0^2 \int_{3y^2-6y}^{2y-y^2} 4xy\, dx\, dy$

Solution:

$$\int_0^2 \int_{3y^2-6y}^{2y-y^2} 4xy\, dx\, dy = \int_0^2 \left[\frac{4yx^2}{2} \right]_{3y^2-6}^{2y-y^2} dy = \int_0^2 \left(\left[\frac{4y(2y-y^2)^2}{2} \right] - \left[\frac{4y(3y^2-6)^2}{2} \right] \right) dy$$

Exercise - Evaluate this integral to get the final answer

Quick check activity 4.1.1

1: Evaluate $\int_2^x \left(3x^2y^2 + 2x\right) dy$.

2a: Evaluate $\int_{y^2}^{4y+1} \left(3x^2y^2 + 2x\right) dx$.

2b: Evaluate $\int_{-2}^{1} \left[\int_{y^2}^{4y+1} \left(3x^2y^2 + 2x\right) dx \right] dy$.

3. Evaluate $\int_x^{x^2} \frac{y}{x} dy$.

A double integral $\iint_D f(x, y)\, dA$ may be separated into a pair of single integrals if

- The region *D* is a rectangle, with sides parallel to the coordinate axes; and
- The integrand is separable: $f(x, y) = g(x)\, h(y)$.
- The region D is either vertically or horizontally simple region.

I) Double Integrals over a Rectangular Region

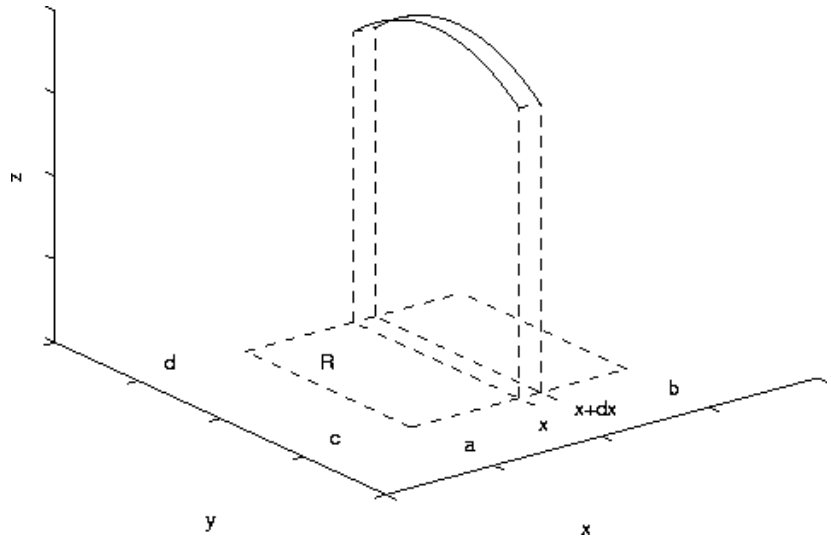
Theorem 4.1.1: Fubini's Theorem (Order of Integration is Interchangeable)

Suppose that $f(x,y)$ is continuous on a rectangular region $R = \{(x,y) \in R^2 : a \le x \le b\ , c \le y \le d\}$ in the xy-plane. The double integral, is given by

$$\iint_R f(x,y)dA = \int_c^d \left[\int_a^b f(x,y)dx\right]dy = \int_a^b \left[\int_c^d f(x,y)dy\right]dx$$

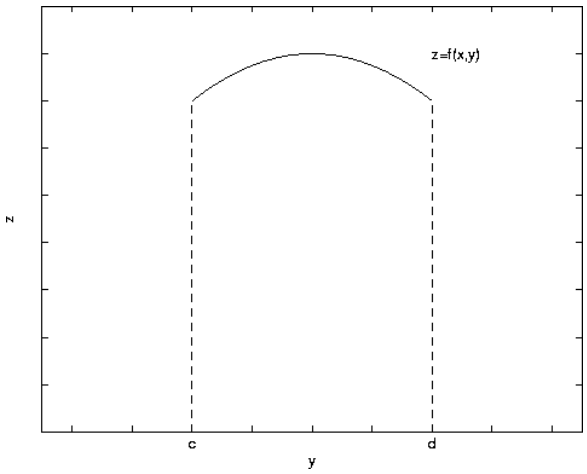
and represents the volume under the surface $z = f(x,y)$ above the plane region R.

Proof: We can compute the volume by slicing the three-dimensional region like a loaf of bread. Suppose the slices are parallel to the y-axis. An example of slice between x *and* $x + dx$ is shown in the figure.



In the limit of *infinitesimal* thickness dx, the volume of the slice is the product of the cross-sectional area and the thickness dx. The cross sectional area is the area under the curve $f(x,y)$ for fixed x *and* y varying between c and d. (Note that: if the thickness dx is infinitesimal, x varies only infinitesimally on the slice. We can assume that x is constant.)

The following picture shows the cross-sectional area.



The area is given by the integral

$$A(x) = \int_c^d f(x, y)dy$$

The variable of integation is y and x is a **constant**. The cross-sectional area depends on x and this is why we write $A = A(x)$. The volume of the slice between x and $x + dx$ is $A(x).dx$. The total volume is the sum of the volumes of all the slices between $x = a \quad and \quad x = b$:

$$V = \int_a^b A(x)\,dx$$

If we substitute for $A(x)$, we obtain:

$$V = \int_a^b \left[\int_c^d f(x, y)dy \right] dx = \int_a^b \int_c^d f(x, y)dy\ dx$$

This is an example of an *iterated* integral. One integrates with respect to y first, then x. The integrals with respect to y and x are called the inner and outer integrals, respectively. Alternatively, one can make slices that are parallel to the x-axis and obtain:

$$A(y) = \int_a^b f(x, y)dx$$

In this case the volume is given by

$$V = \int_c^d \left[\int_a^b f(x,y)dx \right] dy = \int_c^d \int_a^b f(x,y)dx\, dy$$

The inner integral corresponds to the cross-sectional area of a slice between y and $y+dy$

The quantities $f(x,y)\, dy\, dx$ and $f(x,y)\, dx\, dy$ represent the value of the double integral in the infinitesimal rectangle between x and $x+dx$ and, y and $y+dy$. The length and width of the rectangle are dx and dy, respectively. Hence $dy\ dx$ (or $dx\ dy$) is the area of the rectangle. We can make the connection $dA = dy\ dx$ (or $dA = dx\ dy$).

Example 1: Evaluate the iterated integral $\int_1^2 \int_0^1 (x^2 + y)\, dy\, dx$.

Solution:(Method 1)

$$\int_1^2 \int_0^1 (x^2+y)\, dy\, dx = \int_1^2 \left[x^2 y + \frac{y^2}{2} \right]_{y=0}^{y=1} dx$$

$$= \int_1^2 \left(x^2 + \frac{1}{2} \right) dx = \left[\frac{x^3}{3} + \frac{x}{2} \right]_{x=1}^{x=2}$$

$$= \frac{(2)^3}{3} + \frac{2}{2} - \left(\frac{(1)^3}{3} + \frac{1}{2} \right)$$

$$= \frac{17}{6}$$

Method 2: Solution: If you reverse the order and the limits of integration, $\int_1^2 \int_0^1 (x^2 + y)\, dy\, dx$, we obtain the integral $\int_0^1 \int_1^2 (x^2 + y)\, dx\, dy$. Then we have the following.

$$\int_0^1 \int_1^2 (x^2+y)\, dx\, dy = \int_0^1 \left[\frac{x^3}{3} + xy \right]_{x=1}^{x=2} dy$$

$$= \int_0^1 \left(\frac{(2)^3}{3} + 2y - \left(\frac{1}{3} + y \right) \right) dy$$

$$= \frac{17}{6}$$

Example 2: Evaluate the double integral

$$V = \iint_R (x^2 + xy^3)\ dA,$$

where R is the rectangle $0 \le x \le 1,\ \ 1 \le y \le 2$.

Solution: - Suppose we integrate with respect to y first. Then

$$V = \int_0^1 \int_1^2 (x^2 + xy^3)\ dy\ dx$$

The inner integral is

$$V = \int_0^1 (x^2.y + \frac{xy^4}{4})\Big|_{y=1}^{y=2} dx = \int_0^1 \left[(2x^2 + 4x) - (x^2 + \frac{x}{4}) \right].dx$$

$$= \int_0^1 \left[(x^2 + \frac{15x}{4}) \right].dx$$

$$= \left[\frac{x^3}{3} + \frac{15x^2}{8} \right]_0^1 = \frac{1}{3} + \frac{15}{8} = \frac{8+45}{24} = \frac{53}{24}$$

Note that we treat y as a constant when you integrate with respect to x vice versa resulting in the same answer.

Quick Check Exercises 4.1.2

1. Evaluate the double integral $\iint_R x\cos(x^2 + 2y)\, dA$, where $R = [0, \sqrt{\pi}] \times [0, \frac{\pi}{2}]$.
2. Let R = [-1, 1] x [0, 3]. Evaluate $\iint_R \left(4 - x^2 - y^2\right) dA$.
3. Sketch the solid whose volume is given by the iterated integral $\int_0^2 \int_{-1}^1 \left(2 + x + 2y\right) dy dx$.

Q. What if the region in the *x*-*y* plane is not a rectangle?

II) Double Integrals over General Regions

***a) Vertically simple region*:**

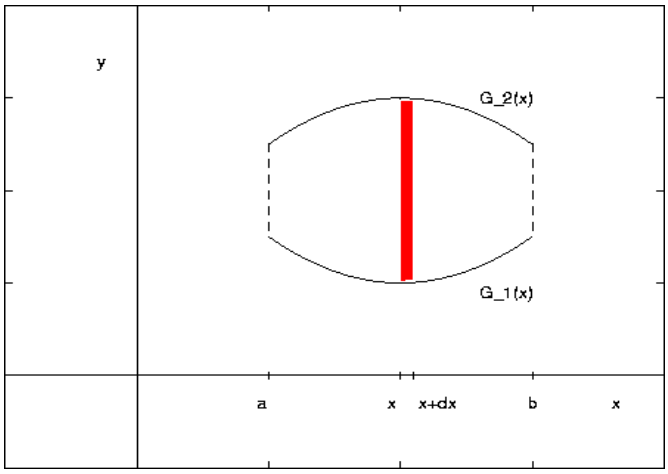
Theorem 4.1.2: If R has the form $R = \{(x, y) \mid a \le x \le b \text{ and } g_1(x) \le y \le g_2(x)\}$ the double integral on R is given by

$$\iint_R f(x, y)\, dA = \int_{x=a}^{x=b} \int_{y=g_1(x)}^{y=g_2(x)} f(x, y)\, dy\, dx$$

Here **R** is called a ***vertically simple region***.

Proof: Suppose that the region **R** is defined by $G_1(x) \le y \le G_2(x)$ with $a \le x \le b$.

To derive this formula we slice the three-dimensional region into slices parallel to the y-axis. The figure below shows a top view of slice between x and $x+dx$.



The inner integral

$$A(x) = \int_{G_1(x)}^{G_2(x)} f(x,y)\, dy$$

is the cross-sectional area of the slice between x and $x+dx$. The volume of the slice between x and $x+dx$ is $A(x).dx$. The total volume is the sum of the volumes of all the slices:

$$V = \int_a^b A(x)\, dx$$

Example 1: Compute $\iint_R \left(x^2 + y^2\right) dA$ over the region R bounded by the curves $y = 1 - x^2$ and $y = x^2 - 1$ in the *x*-*y* plane.

Solution: The intersection points of the two curves is found at the points satisfying

$1 - x^2 = x^2 - 1 \Rightarrow 2x^2 = 2 \Rightarrow x = 1; -1$

So the region is bounded by the concave up parabola $y = x^2 - 1$ and by the concave down parabola $y = 1 - x^2$ and can be described as the vertically simple region given by $R = \{(x,y): -1 \le x \le 1; 1 - x^2 \le y \le x^2 - 1\}$. Thus

$$\iint_R f(x,y)\, dA = \int_{x=a}^{x=b} \int_{y=g_1(x)}^{y=g_2(x)} f(x,y)\, dy\, dx$$

$$\iint_R f(x,y)\,dA = \int_{-1}^{1}\left(\int_{1-x^2}^{x^2-1}(x^2+y^2)\,dy\right)dx$$

$$= \int_{-1}^{1}\left(x^2y + \frac{y^3}{3}\Big|_{1-x^2}^{x^2-1}\right)dx$$

$$= \int_{-1}^{1}\left[x^2\left(x^2-1-\left(1-x^2\right)\right) + \frac{\left(x^2-1\right)^3}{3} - \frac{\left(1-x^2\right)^3}{3}\right]dx$$

$$= \frac{32}{21}$$

Example 2: Find the volume of the tetrahedron bounded by the coordinate axes and the plane $3x + 6y + 4z = 12$:

Solution: We have to find the volume of the tetrahedron S bounded by the plane

$$3x + 6y + 4z = 12 \Rightarrow z = \frac{12 - 3x - 6y}{4}$$

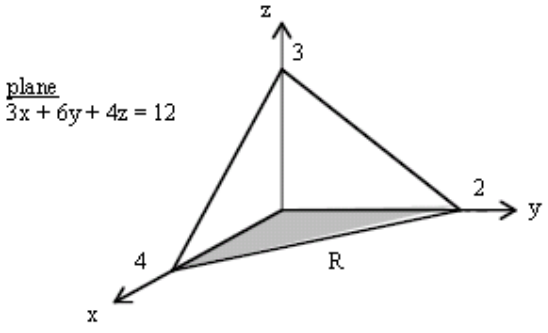
and the coordinate axes. This is the portion of the plane in the first octant, as one can see from graph (1) below. Then, we have

$$Volume(S) = \iint_R \left(\frac{12-3x-6y}{4}\right)dA$$

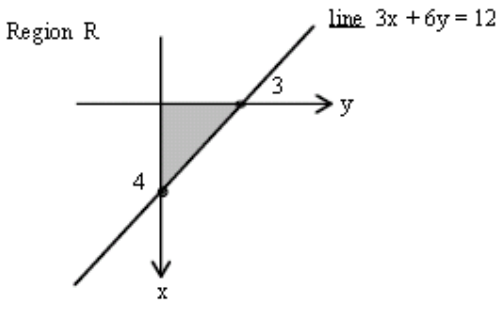
where R is the projection of the tetrahedron in the xy-plane.

Hence R can be described as the vertically simple region (graph (2) below) by

$$R = \left\{(x,y) \in R^2 : 0 \le x \le 4; \frac{12-3x}{6} \le y \le 3\right\}$$



(1)



(2)

Finally, this gives

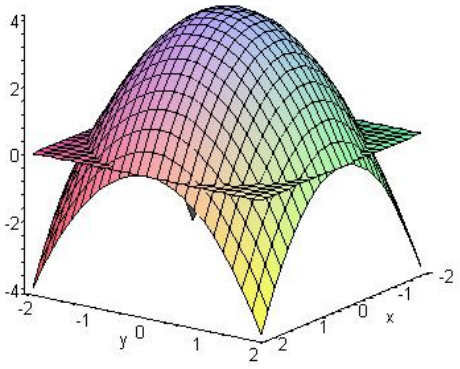
$$Volume(S) = \int_0^3 \left[\int_{\frac{12-3x}{6}}^{3} \left(\frac{12-3x-6y}{4} \right) dy \right] dx$$

$$= \frac{1}{4} \int_0^4 \left((12-3x)y - 3y^2 \right)_0^{2-\frac{x}{2}} dx$$

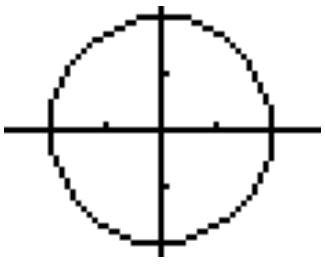
$$= \frac{1}{4} \int_0^4 (12-3x)\left(2-\frac{x}{2}\right) - 3\left(2-\frac{x}{2}\right)^2 dx$$

$$= \frac{1}{4} \left(\frac{x^3}{4} - 3x^2 + 12x \right)_0^4 = 4$$

Example 3: Determine the volume of the solid region bounded by the paraboloid $z = 4 - x^2 - y^2$ and the *xy*-plane.



By letting *z = 0*, we see that the base of the region in the *xy*-plane is the circle $x^2 + y^2 = 4$



Integrating over vertical strips, *y* goes from $y = -\sqrt{4-x^2}$ to $y = \sqrt{4-x^2}$

Thus, the inside integral is $\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \left(4 - x^2 - y^2\right) dy$.

There is a vertical strip for each x from $x = -2$ to $x = 2$.

Therefore, Volume = $\int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \left(4 - x^2 - y^2\right) dy\, dx$.

Inside Integral = $\left(4y - x^2 y - \frac{y^3}{3}\right)\Big|_{y=-\sqrt{4-x^2}=-M}^{y=\sqrt{4-x^2}=M}$

$= 4(M-(-M)) - x^2(M-(-M)) - \frac{1}{3}\left(M^3 - (-M)^3\right)$

$= 8M - 2x^2 M - \frac{2}{3}M^3$

$= 8M - 2x^2 M - \frac{2}{3}\left(4 - x^2\right)M$ since $M = \sqrt{4 - x^2}$

$= M\left(8 - 2x^2 - \frac{8}{3} + \frac{2}{3}x^2\right) = \sqrt{4-x^2}\left(\frac{16}{3} - \frac{4}{3}x^2\right)$

$= \sqrt{4-x^2} \cdot \frac{4}{3}\left(4 - x^2\right) = \frac{4}{3}\left(4 - x^2\right)^{3/2}$

Therefore, Volume = $\int_{-2}^{2} \frac{4}{3}\left(4 - x^2\right)^{\frac{3}{2}} dx = 8\pi$. **(** Use substitution $x = 2\sin\theta$)

Quick Check Class Activity 4.1.3

1. Evaluate the double integral $\iint_R \frac{y}{1+x^2} dA$ where $R = \left\{(x, y) \mid 0 \le x \le 4 \text{ and } 0 \le y \le \sqrt{x}\right\}$

2. Evaluate $\iint_R e^{x-y}\, dA$ where R is the interior of the triangle whose vertices are (0, 0), (1, 3) and (2, 2).

b) Horizontally simple region:

Theorem 4.1.3: If $R = \left\{(x, y) \mid c \le y \le d \text{ and } h_1(y) \le x \le h_2(y)\right\}$ the double integral on R is given by

$$\iint_R f(x, y)\, dA = \int_{y=c}^{y=d} \int_{x=h_1(y)}^{x=h_2(y)} f(x, y)\, dx\, dy$$

Here R is called **horizontally simple region**.

Proof: Let **R** be a **horizontally simple region** - that is, if the region is defined by $c \le y \le d$ and $H_1(y) \le x \le H_2(y)$. In this case the slices are parallel to the x-axis.

The inner integral

$$A(y) = \int_{H_1(y)}^{H_2(y)} f(x,y)\, dx$$

is the cross-sectional area of the slice between y and $y + dy$.

The volume of the slice between y and $y + dy$ is $A(y)\, dy$ .The total volume is

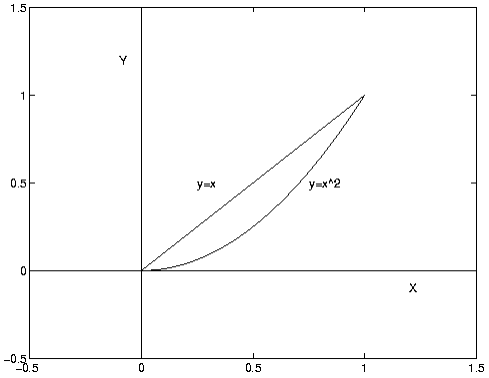
$$V = \int_c^d A(y)\, dy$$

Example: - Evaluate the double integral

$$\iint_{\mathbf{R}} 30\,.\,x.y\, dA$$

where R is bounded by $y = x$ *and* $y = x^2$.

Solution: - (**Method 1**) We can treat the region **R** as a vertically simple region as shown in the figure below. In this case the integral is given by



$$\iint_{\mathbf{R}} 30\,.\,x.y\, dA = \int_0^1 \int_{x^2}^{x} 30\,.\,x.y\, dy.dx$$

The inner integral is (remember x is a constant)

$$\int_{x^2}^{x} 30\,.\,x.y\, dy = \left[15xy^2\right]_{x^2}^{x} = 15x(x^2 - x^4) = 15\left(x^3 - x^5\right)$$

The outer integral is

$$\int_0^1 15\left(x^3 - x^5\right) dx = \frac{5}{4}$$

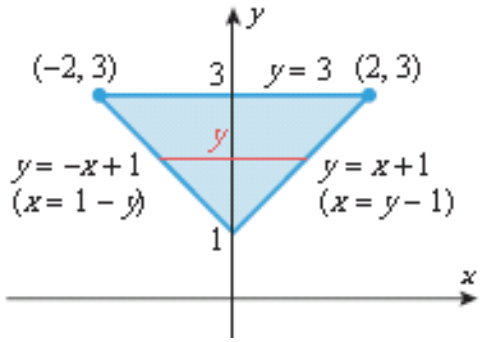
(Method 2) One can also treat the region as a horizontally simple region.

The left hand function $y = x$ can be written as $x = H_1(y) = y$. The right hand function $y = x^2$ can be written as $x = \sqrt{y}$. The iterated integral is

$$\iint_R 30\,.\,x.y\,dA = \int_0^1 \int_y^{\sqrt{y}} 30\,.\,x.y\,dx\,dy = \frac{5}{4} \quad \text{(check it)}$$

Example 2. Evaluate the double integral $\iint_R (2x - y^2)dA$ over the triangular region bounded by $y = -x + 3, y = x + 3 \,\&\, y = 3$

Solution: We view R as vertically simple region. A horizontal line meets the region R at its left hand boundary $x = 1 - y$ and its right boundary $x = y - 1$. These are the $x-$limits of integration. Moving this line first down and then up yields the $y-$limits, $y = 1, y = 3$.

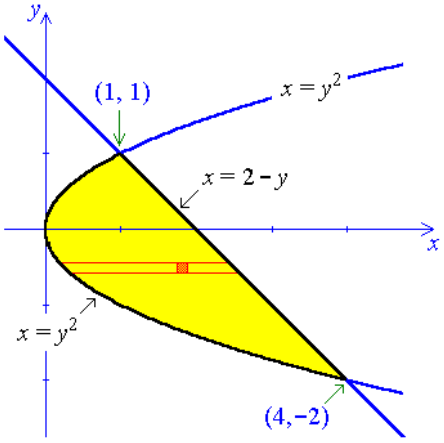


$$\iint_R (2x - y^2)dA = \int_1^3 \int_{1-y}^{y-1} (2x - y^2)dx dy = \int_1^3 \left[x^2 - y^2 x\right]_{x=1-y}^{x=y-1} dy$$

$$= 1\int^3 \left[(1 - 2y + 2y^2 - y^3) - (1 - 2y + y^3)\right]dy$$

$$= \int_1^3 (2y^2 - 2y^3)dy = \left[\frac{2y^3}{3} - \frac{y^4}{2}\right]_{y=1}^3 = -\frac{68}{3}$$

Example 3: Evaluate $I = \iint_R (6x + 2y^2)\,dA$ where R is the region enclosed by the parabola $x = y^2$ and the line $x + y = 2$.



Solution:

The upper boundary changes form at $x = 1$. The left boundary is the same throughout R. The right boundary is the same throughout R. Therefore choose horizontal strips.

$$I = \int_{-2}^{1}\int_{y^2}^{2-y}\left(6x + 2y^2\right)dx\,dy$$

$$I = \int_{-2}^{1}\left[3x^2 + 2xy^2\right]_{x=y^2}^{x=2-y}\,dy$$

$$= \int_{-2}^{1}\left(\left(3(2-y)^2 + 2(2-y)y^2\right) - \left(3y^4 + 2y^4\right)\right)dy$$

$$= \int_{-2}^{1}\left(\left(12 - 12y + 3y^2\right) + \left(4y^2 - 2y^3\right) - 5y^4\right)dy$$

$$= \int_{-2}^{1}\left(12 - 12y + 7y^2 - 2y^3 - 5y^4\right)dy$$

$$= \left[12y - 6y^2 + \frac{7}{3}y^3 - \frac{1}{2}y^4 - y^5\right]_{-2}^{+1}$$

Therefore

$$I = \underline{\underline{\frac{99}{2}}}$$

Quick Check Exercises 4.1.4

1. Let R be the region bounded by the y-axis and the parabola $x = 4y - y^2$. Find the integral over R of $f(x, y) = xy$.

2. Find the volume under the surface $z = 2x + y^2$ and above the region bounded by $x = y^2$ and $x = y^3$.

A region R is said to be ***Simple*** if it is both vertically simple and horizontally simple region.

Remarks:

1. In some cases a region may be neither vertically nor horizontally simple. However, in general, a region can be split up into vertically and horizontally simple regions.

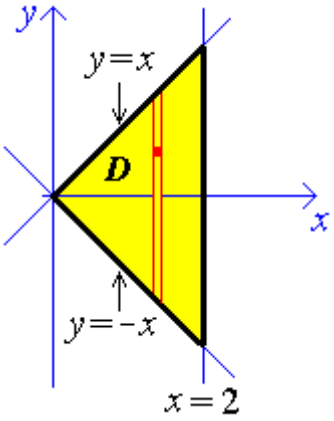
2. If a region is simple region we can use both iterated integrals to evaluate the double integral. For instance, rectangles, triangles and circles are simple regions and can be treated as either horizontally or vertically simple region.

4.1.1 Reversing the Order of Integration:

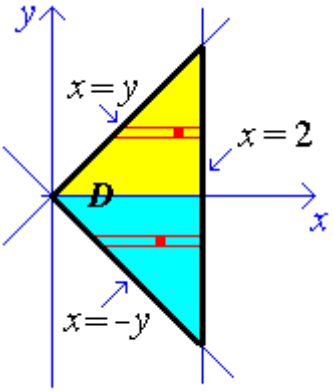
For a general region of integration, switching the order of integration requires substantial changes to the limits of integration. It sometimes happens that one iterated integral is either difficult or impossible to evaluate, where as the other iterated integral can be evaluated easily.

The change from one iterated integral to the other is called ***reversing the order of integration***, since it involves changing from $dy\,dx$ *to* $dx\,dy$, or vice versa. Choose the orientation of elementary strips that generates the simpler double integration.

For example,



is preferable to



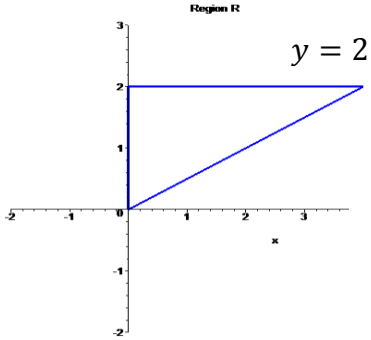
.

$$\int_{0}^{2}\int_{-x}^{x} f(x,y)\,dy\,dx = \int_{-2}^{0}\int_{-y}^{2} f(x,y)\,dx\,dy + \int_{0}^{2}\int_{+y}^{2} f(x,y)\,dx\,dy$$

Example 1: Evaluate the double integral $\iint_R e^{-x^2}\,dA$ where

$$R=\left\{(x,y)\,|\,0\le x\le 4\ and\ \frac{x}{2}\le y\le 2\right\}$$

Solution: The following graph shows the region *R* outlined.



$$y=\frac{x}{2}\Leftrightarrow x=2y$$

If we integrate with respect to *y* first and then with respect to *x*, the double integral would be evaluated as

$$\iint_R e^{-x^2}\,dA = \int_{x=0}^{x=4}\int_{y=\frac{x}{2}}^{y=2} e^{-y^2}\,dy\,dx$$

With respect to x, the region R changes from $x = 0$ to $x = 2y$. With respect to y, the region changes from $y = 0$ to $y = 2$. Thus, the double integral can be evaluated by computing the following iterated integral:

$$\iint_R e^{-x^2}\,dA = \int_{y=0}^{y=2}\int_{x=0}^{y=2y} e^{-y^2}\,dx\,dy$$

We compute this double integral as follows.

$$\iint_R e^{-x^2}\,dA = \int_{y=0}^{y=2}\int_{x=0}^{x=2y} e^{-y^2}\,dx\,dy$$

$$= \int_{y=0}^{y=2}\left[e^{-y^2}\,x\Big|_{x=0}^{x=2y}\right]dy \qquad \text{(With respect to } x,\ e^{-y^2} \text{ is treated as a constant)}$$

$$= \int_{y=0}^{y=2}\left[e^{-y^2}(2y) - e^{-y^2}(0)\right]dy \qquad \text{(Substitute in inner integration limits)}$$

$$= \int_{y=0}^{y=2} 2ye^{-y^2}\,dy \qquad \text{(Simplify)}$$

$$= -e^{-y^2}\Big|_{y=0}^{y=2}$$

Note we use $u-du$ substitution to integrate $\int 2ye^{-y^2}\,dy$

Let $u = -y^2, du = -2ydy$ or $-du = 2ydy$

Then $\int 2ye^{-y^2}\,dy = \int e^u(-du) = -e^u + C = -e^{-y^2} + C$

$$= -e^{-(2)^2} - -e^{-(0)^2} \qquad \text{(Substitute in outer integration limits)}$$

$$= -e^{-4} + 1 \qquad \text{(Simplify)}$$

$$= 1 - e^{-4}$$

Quick check Exercise 4.1.5 Evaluate by changing the order of integration.

a) $\int_0^1 \int_y^1 e^{x^2}\,dx\;dy$ b) $\int_0^3 \int_1^{\sqrt{4-y}} (x+y)\,dx dy$ c) $\int_0^1 \int_{y^2}^1 y e^{x^2}\,dx dy$

4.1.2 Application of Double Integrals

In applications, double integrals arise in computations of

Area: if $f(x,y)=1$, then the double integrals give the area of region R.

Volume: the double integral is equal to volume under the surface $z=f(x,y)$ above the region R.

Mass: if R is a plate and $f(x,y)$ is density per unit area of the plate, then the double integral is equal to the mass of the plate.

Force: if $f(x,y)$ is the force per unit area on the plate in the downward direction, then double integral is the total force on the plate.

Average: the double integral divided by the area of the region R is the the average of the function $f(x,y)$ on R.

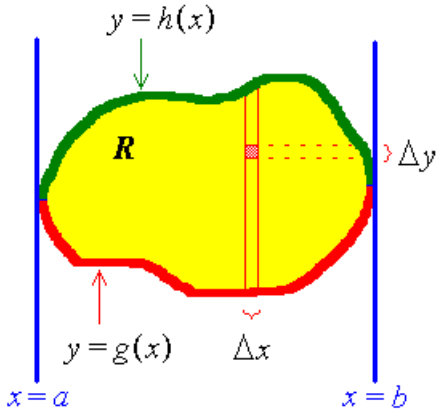
I. AREA OF PLANE REGION

In Cartesian coordinates on the *xy*-plane, the rectangular element of area is

$$\Delta A = \Delta x\,\Delta y.$$

Summing all such elements of area along a vertical strip, the area of the elementary strip is

$$\left(\sum_{y=g(x)}^{h(x)} \Delta y\right)\Delta x$$



Summing all the strips across the region *R*, the total area of the region is:

$$A \approx \sum_{x=a}^{b}\left(\left(\sum_{y=g(x)}^{h(x)} \Delta y\right)\Delta x\right)$$

In the limit as the elements Δx and Δy shrink to zero, this sum becomes

$$A = \int_{x=a}^{b}\int_{y=g(x)}^{h(x)} 1\,dy\,dx$$

Thus we define the area A of a plane region **R** by $area(R)=\iint_R 1dA$.

When **R** is the region between the graphs of two continuous functions g_1 and g_2 on [a, b] such that $g_1 \le g_2$, then

$$A=\int_a^b \int_{g_1(x)}^{g_2(x)} 1.dydx=\int_a^b [g_2(x)-g_1(x)]dx$$

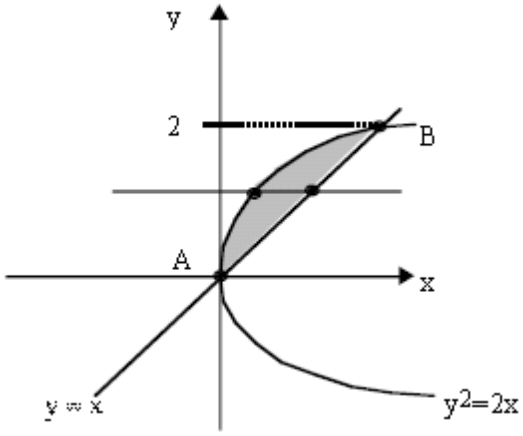
Example 1. Sketch the region R in the xy-plane bounded by the curves $y^2 = 2x$ and $y = x$, and find its area.

Solution

The region R is bounded by the parabola $x = \frac{1}{2}y^2$ and the straight line $y = x$. The points of intersection of the two curves are given by $y = \frac{y^2}{2}$

$$y^2 - 2y = 0 \Rightarrow y(y-2) = 0 \Longrightarrow y = 0, 2$$

This gives the two points $A = (0, 0)$ and $B = (2, 2)$.



This region is a horizontally simple region and can be described by

$$R = \left\{(x, y) \in R^2 : 0 \le y \le 2 \;\&\; \frac{y^2}{2} \le x \le y\right\}$$

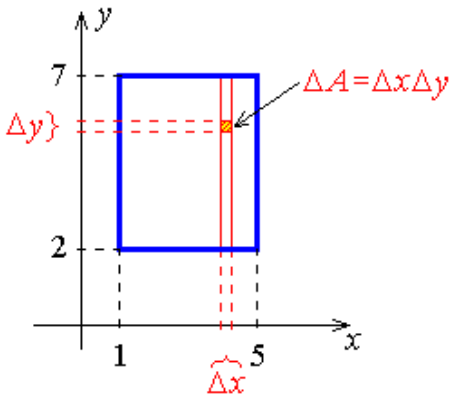
Then

$$area(R) = \iint_R 1dA = \int_1^2 \int_{\frac{y^2}{2}}^{y} 1dxdy$$

$$= \int_1^2 \left(y - \frac{y^2}{2} \right) dy = \left(\frac{y^2}{2} - \frac{y^3}{6} \right)\Big|_1^2$$

$$= 2 - \frac{8}{6} = \frac{2}{3}$$

Example 2 Find the area shown (assuming SI units).



$$\text{Area of strip} \approx \left(\sum_{y=2}^{7} \Delta y\right)\Delta x$$

$$\text{Total Area} \approx \sum_{x=1}^{5}\left(\left(\sum_{y=2}^{7} \Delta y\right)\Delta x\right)$$

As $\Delta x \to 0$ and $\Delta y \to 0$, the summations become integrals:

$$(\text{Total Area}) \;\to\; A = \int_{x=1}^{x=5}\left(\int_{y=2}^{y=7} 1\,dy\right)dx$$

The inner integral has no dependency at all on *x*, in its limits or in its integrand.

It can therefore be extracted as a "constant" factor from inside the outer integral.

$$\Rightarrow \quad A = \left(\int_{y=2}^{y=7} 1\,dy\right)\int_{x=1}^{x=5} 1\,dx \quad = [y]_2^7[x]_1^5 = (7-2)\times(5-1) = 5\times 4 = \underline{\underline{20\ \text{m}^2}}$$

II. MASS, CENTER OF MASS

Consider a LAMINA: A flat sheet so thin we consider it a 2D object.

The density (mass/unit area) varies throughout the plate. We want to find the Mass of the Lamina.

We know Mass = Density X Length so we chopped up the line into subintervals and created our Riemann Sum:

$$\text{Mass} \approx \sum \sigma(x_i, y_i)\Delta A_i$$

We can get the mass exactly by taking the limit as the norm of the partition goes to zero.

$$MASS = \iint_R \sigma(x,y)dA$$

$\boldsymbol{My}$ represents the moment around the y-axis. We will use it to calculate the x-coordinate of the center of mass.

$$\bar{x} = \frac{M_y}{m}$$

$\boldsymbol{Mx}$ represents the moment around the x-axis. We will use it to calculate the y-coordinate of the center of mass.

$$\bar{y} = \frac{M_x}{m}$$

$$M_y = \iint_R x\sigma(x,y)dA$$

$$M_x = \iint_R y\sigma(x,y)dA$$

If the surface density σ within the region is a function of location, $\sigma = f(x, y)$, then the mass of the region is

$$m = \int_{x=a}^{b} \left(\int_{y=g(x)}^{h(x)} f(x,y)\, dy \right) dx$$

The inner integral must be evaluated first.

Example 2

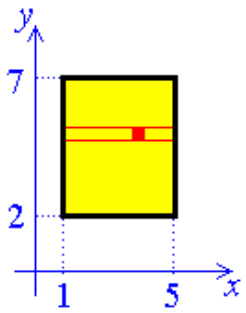
Suppose that the surface density on the rectangle is $\sigma = x^2 y$. Find the mass of the rectangle.

Solution: The element of mass is $\Delta m = \sigma \Delta A = \sigma \Delta x \, \Delta y$

$$\rightarrow \quad m = \int_1^5 \int_2^7 \sigma \, dy \, dx = \int_1^5 \int_2^7 x^2 y \, dy \, dx$$

$$= \int_1^5 x^2 \left(\int_2^7 y \, dy \right) dx = \left(\int_2^7 y \, dy \right) \int_1^5 x^2 \, dx$$

$$= \left[\frac{y^2}{2} \right]_2^7 \cdot \left[\frac{x^3}{3} \right]_1^5 = \frac{49-4}{2} \times \frac{125-1}{3} = 15 \times 62$$



Therefore the mass of the rectangle is m = 930 kg.

OR

We can choose to sum horizontally first:

$$m = \int_2^7 \int_1^5 x^2 y \, dx \, dy$$

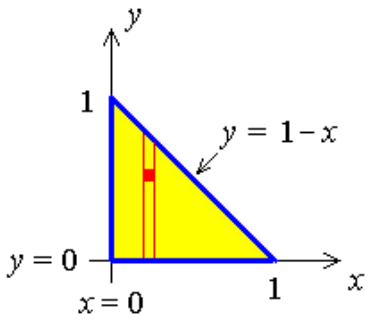
$$m = \int_2^7 y \left(\int_1^5 x^2 \, dx \right) dy$$

The inner integral has no dependency at all on y, in its limits or in its integrand. It can therefore be extracted as a "constant" factor from inside to the outer integral.

$$m = \left(\int_1^5 x^2 \, dx \right) \left(\int_2^7 y \, dy \right)$$

which is exactly the same form as before, leading to the same value of 930 kg.

Example 3 The triangular region (shown below) has surface density $\sigma = x + y$. Find the mass of the triangular plate.



Element of mass: $\Delta m = \sigma \Delta A = \sigma \Delta x \, \Delta y$

$$\text{Mass of strip} \approx \left(\sum_{y=0}^{1-x} \sigma \, \Delta y \right) \Delta x$$

$$\text{Total Mass} \approx \sum_{x=0}^{1} \left(\left(\sum_{y=0}^{1-x} \sigma \, \Delta y \right) \Delta x \right)$$

$$\rightarrow \quad m = \int_0^1 \int_0^{1-x} (x + y) \, dy \, dx$$

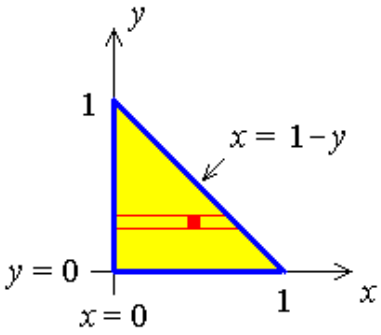
$$= \int_0^1 \left[xy + \frac{y^2}{2} \right]_0^{1-x} dx = \int_0^1 \left(x(1-x) + \frac{(1-x)^2}{2} - 0 - 0 \right) dx$$

$$= \int_0^1 \frac{1 - x^2}{2} dx = \left[\frac{x}{2} - \frac{x^3}{6} \right]_0^1 = \frac{1}{2} - \frac{1}{6} - 0 + 0 = \underline{\underline{\frac{1}{3} \text{ kg}}}$$

OR

We can choose to sum horizontally first (re-iterate):

Example 3 (continued)



$$m = \int_0^1 \int_0^{1-y} (x+y)\,dx\,dy$$

$$= \int_0^1 \left[\frac{x^2}{2} + xy \right]_0^{1-y} dy = \cdots = \underline{\underline{\frac{1}{3}\,\text{kg}}}$$

I. Moments of Inertia

M_x and M_y are the **first moments of inertia** about the x and y axis respectively. The units are the products of a mass times a distance.The **second moment**, the **second moment of area**, also known as the **area moment of inertia** or **second moment of inertia** is a property of a cross section that can be used to predict the resistance of beams to *bending* and *deflection*, and its units are the products of mass times the square of the distance. $I = md^2$

$$I_x = \iint_R y^2 \rho(x,y)\,dA \qquad I_y = \iint_R x^2 \rho(x,y)\,dA$$

The **polar moment of inertia** is the sum of these two moments

$$I_0 = I_x + I_y = \iint_R \left(x^2 + y^2\right) \rho(x,y)\,dA = \iint_R r^2 \rho(x,y)\,dA$$

.

Quick Check Exercises 4.1.6

1. Find the mass of the triangular lamina with vertices (0, 0), (0, 3), and (2, 3), given that the density at (x, y) is $\rho(x,y) = 2x + y$.
2. Find the mass of the lamina corresponding to the first-quadrant portion of the circle $x^2 + y^2 = 4$ where the density at the point (x, y) is proportional to the distance between the point and the origin.
3. Find the center of mass of the lamina corresponding to the parabolic region $0 \le y \le 4 - x^2$ where the density at the point (x, y) is proportional to the distance between (x, y) and the x-axis.
4. Find the moment of inertia about the x-axis of the lamina from exercise 3, above.

Group Activity 4.1.1

1. Evaluate the double integrals

   a. $\iint_R (x^2 - 2y)dA \quad where \quad R = \{(x,y): 0 \le x \le 2 \ , -1 \le y \le 1\}$

   b. $\iint_R (2xy + y^3)dA \quad where \quad R = \{(x,y): -1 \le x \le 2 \ , 0 \le y \le 2\}$

   c. $\iint_R 4xe^{2y}dA \quad where \quad R = \{(x,y): 2 \le x \le 4 \ , 0 \le y \le 1\}$

   d. $\iint_R (1 - ye^{xy})\ dA \quad where \quad R = \{(x,y): 0 \le x \le 2, \ 0 \le y \le 3\}$

   e. $\iint_R e^{x+y}\ dA \quad where \quad R = \{(x,y): 0 \le x \le 1, \ 0 \le y \le 1\}$

2. Evaluate the iterated integral

   a. $\int_0^1 \int_{-2}^{2} (6 - 2x - 3y)\ dxdy$

   b. $\int_0^2 \int_{-1}^{1} (2 + x + 2y)\ dydx$

   c. $\int_0^1 \int_0^3 x \,.\sqrt{x^2 + y}\ \ dy\ dx$

   d. $\int_0^1 \int_0^{2x} (x + 2y)\ dydx$

   e. $\int_0^2 \int_{-2}^{x^2} (x + 3)\ dydx$

   f. $\int_0^1 \int_0^{2y} (4x\sqrt{y} + y)\ dxdy$

   g. $\int_0^2 \int_0^{2y} e^{y^2})\ dxdy$

   h. $\int_1^2 \int_0^{\frac{2}{y}} e^{xy})\ dydx$

   i. $\int_0^1 \int_y^1 e^{(x^2)} dxdy$

   j. $\int_1^e \int_0^{1nx} ydydx$

   k. $\int_0^2 \int_0^{\sqrt{4-y^2}} \ y \,.\, dx\ dy$

   l. $\int_0^1 \int_0^y x \,.\sqrt{y^2 - x^2}\ \ dx\ dy$

   m. $\int_1^e \int_1^{\ln x} \ e^x\ dx\ dy$

   n. $\int_0^{\frac{\pi}{4}} \int_0^{\cos y} \ e^x\ \sin y\ dx\ dy$

   o. $\int_0^2 \int_{1+y^2}^{5} y\ e^{(x-1)^2} dx\ \ dy$

   p. $\int_0^1 \int_{\arcsin y}^{\frac{\pi}{2}} \sec^2(\cos x)\ dx\,.dy$

   q. $\int_1^e \int_{1/e}^{\frac{1}{y}} \cos\ (x - \ln x)\ dx\,.dy$

3.

a. Let R be the triangular region bounded by the lines y = 2x, x = 0, and y = 4. Find the area of R.

b. Let **R** be a region bounded by the lines x = 3, x = 5, y = 1 and y = x. Find $\iint_R x dA$.

c. Let **R** be a region bounded by $x = 0,\ y = 0,\ and\ x + y = 2$. Find $\iint_{\mathbf{R}} x(x-1)\, e^{xy}\ dA$.

d. Let **R** be a region bounded by the graphs of $y = x^2 + 1\ \ and\ \ y = 9 - x^2$. Find $\iint (4 + x^2)\ dA$.

4. Find the volume of the solid in the first octant ($x \geq 0,\ \ y \geq 0,\ \ z \geq 0$) bounded by

a. The circular parabola $z = x^2 + y^2$, the cylinder $4 = x^2 + y^2$, and the coordinate planes ($x = 0,\ \ y = 0,\ and\ \ z = 0$)

b. The parabola $y = x^2$ and the planes $x = 0,\ \ z = 0$ and $y + z = 1$.

5. Evaluate

a. $\iint_R xydA$, R is the region bold by y = x^2 and y = 1

b. $\iint_R \frac{2}{1+x^2} dA$; R is the triangular region with vertices (0, 0), (2, 2), (0, 2)

6. Since iterated integral represents the volume of solid region **D.** Sketch the region **D**

a. $\int_{-3}^{3}\int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} 5\ dy\ dx$

b. $\int_{-5}^{5}\int_{-\sqrt{25-x^2}}^{\sqrt{25-x^2}} \sqrt{25-x^2-y^2}\ dy\ d$

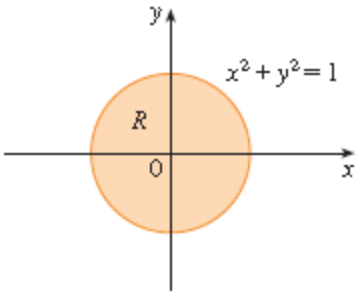
c. $\int_{0}^{5}\int_{0}^{\sqrt{25-x^2}} \sqrt{25-x^2-y^2}\ dy\ dx$

d. $\int_{-8}^{8}\int_{-\sqrt{64-x^2}}^{\sqrt{64-x^2}} (16-\sqrt{x^2+y^2}\ dy\ dx$

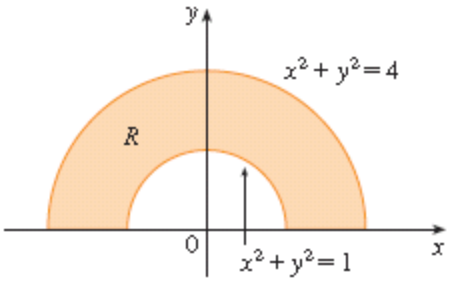
4.1.3 Double Integrals in Polar Coordinates

For some region **R** in $xy-plane$, sometimes it is convenient to convert to polar coordinates in order to evaluate double integral $\iint_R f(x,y)\ dA$

This is usually true if the region is bounded by a circle, a cardioid, a rose curve, a spiral, or, more generally, by any curve whose equation is simpler in polar coordinates than in rectangular coordinates. The two figures given below are examples of Polar Regions.

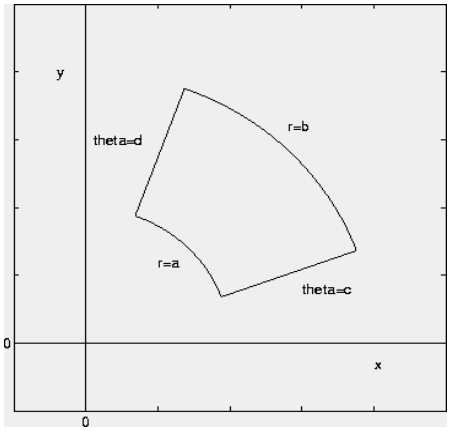


(a) $R=\{(r,\theta)\mid 0\leq r\leq 1, 0\leq \theta\leq 2\pi\}$



(b) $R=\{(r,\theta)\mid 1\leq r\leq 2, 0\leq \theta\leq \pi\}$

Consider the sector $a\leq r\leq b \text{ and } \quad c\leq\theta\leq d$ shown in the figure below.



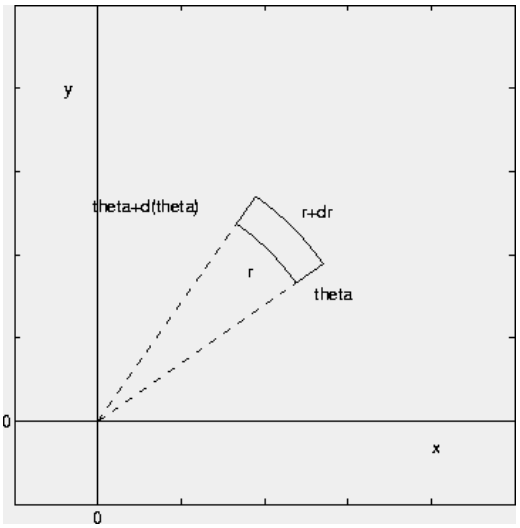
Recall that $x=r\cos\theta \quad and \quad y=r\sin\theta$. The double integral is given by:

$$\iint_R f(x,y)\ dA=\int_a^b\int_c^d f(r\cos\theta+r\sin\theta)\ r\ d\theta\ dr$$

In the above formula one integrates with respect to theta first, then with respect to r. Alternatively, one could integrate with respect to r first, then theta.

4.1.3.1 Discussion of the Iterated Integral in Polar Coordinates

In the case of double integral in polar coordinates we made the connection $dA = dy\ dx$. $dy\ dx$ is the area of an infinitesimal rectangle between x and $x + dx$ and, y and $y + dy$. In polar coordinates, $dA = r\,d\theta\ dr$ is the area of an infinitesimal sector between r and $r + dr$ and θ and $\theta + d\theta$. See the figure below.



The area of the region is the product of the length of the region in θ direction and the width in the r direction. The width is dr and the length is $r\ d\theta$, the arc length of a part of a circle of angle is $d\theta$. (The radius is essentially constant in the region since dr is infinitesimal.)

Example 1: Consider the integral with $f(x, y) = 2x + 3y^2$ where **R** is the region between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$. Now, $1 \le r \le 2 \text{ and } 0 \le \theta \le 2\pi$. We can convert the function $f(x, y)$ into polar coordinates with the substitutions $x = r\cos\theta$ *and* $y = r\sin\theta$. The iterated integral is

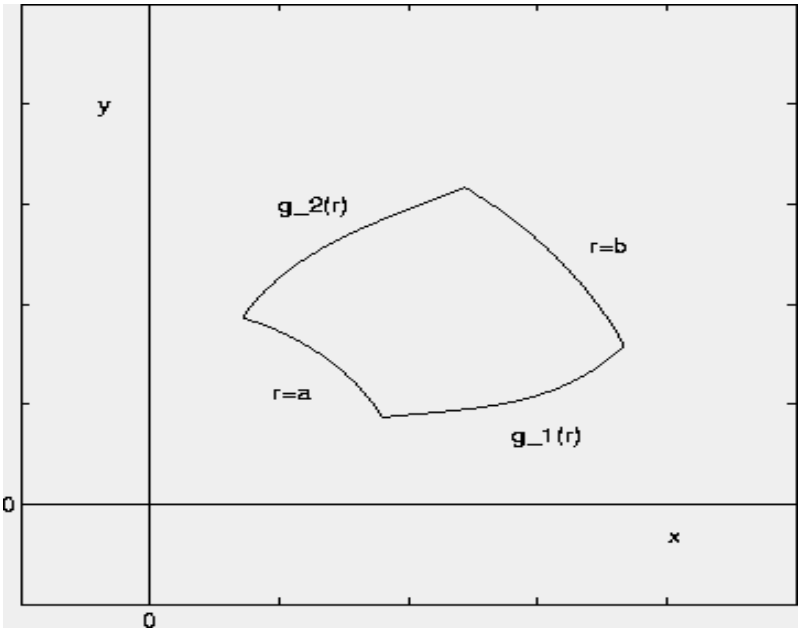
$$\iint_R f(x, y)\ dA = \int_1^2 \int_0^{2\pi} (2r\cos\theta + 3r^2\sin^2\theta)\ r\ d\theta\ dr$$

We integrate with respect to θ first, then with respect to r. Alternatively, we have

$$\iint_R f(x, y)\ dA = \int_0^{2\pi} \int_1^2 (2r\cos\theta + 3r^2\sin^2\theta)\ r\,dr\ d\theta$$

4.1.3.2 Iterated Integral in General Regions

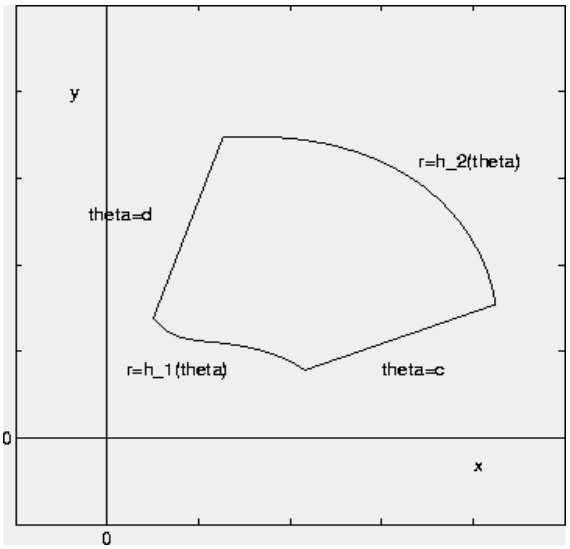
If the region **R** is of the form $g_1(r) \le \theta \le g_2(r)$ with $a \le r \le b$, as shown in the figure below,



then the double integral is given by the iterated integral

$$\iint_{\mathbf{R}} f(x,y)\ dA = \int_{a}^{b} \int_{g_1(r)}^{g_2(r)} f(r\cos\theta + r\sin\theta)\ r\ d\theta\ dr$$

If the region R is of the form $h_1(\theta) \le r \le g_2(\theta)$ with $c \le \theta \le d$, as shown in the figure below,



then the double integral is given by the iterated integral

$$\iint_{\mathbf{R}} f(x,y)\,dA = \int_c^d \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta, r\sin\theta)\; r\;\; dr\;\, d\theta$$

Remarks:

1. If f is non-negative function on **R**, the volume V of the region between the graph of f and R is given by

$$V = \int_{\alpha}^{\beta}\int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta, r\sin\theta) r dr d\theta.$$

2. If $f(x,y)=1$, the area A of the region R is given by

$$A = \int_{\alpha}^{\beta}\int_{h_1(\theta)}^{h_2(\theta)} r dr d\theta.$$

3. Every point in a plane has both Cartesian and polar coordinates. Suppose a point **p** in the plane has polar coordinates (r, θ) and Cartesian coordinates (x, y). Then from the definition of sine and the cosine we deduce that

$$x = r\cos\theta \qquad and \quad y = r\sin\theta$$

$$x^2 + y^2 = r^2 \quad and \quad \tan\theta = \frac{y}{x},\; x \neq 0$$

Example: - 1. Evaluate

a. $\iint_R ydA$; where R is the region in the first quadrant that is outside the circle r = 2 and inside the cardioids $r = 2(1+\cos\theta)$.

Solution: - $\iint_R ydA = \int_0^{\frac{\pi}{2}}\int_2^{2(1+\cos\theta)} (r\sin\theta) r dr d\theta = \int_0^{\frac{\pi}{2}}\int_2^{2(1+\cos\theta)} r^2\sin\theta dr d\theta$

$$\int_0^{\frac{\pi}{2}}\left[\left(\frac{r^3\sin\theta}{3}\right)\Bigg|_{r=2}^{r=2(1+\cos\theta)}\right]d\theta = \int_0^{\frac{\pi}{2}}\left[\left(\frac{8(1+\cos\theta)^3}{3} - \frac{8}{3}\right)\sin\theta\right]d\theta$$

$$= -\frac{8}{3}\left[\frac{(1+\cos\theta)^4}{4} - (1+\cos\theta)\right]_0^{\frac{\pi}{2}}$$

$$= -\frac{8}{3}\left(\frac{1}{4} - 1 - \left(\frac{1}{2} - 2\right)\right)$$

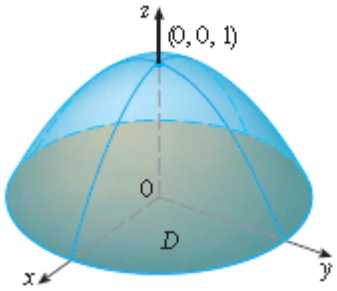
$$= 2$$

b. $\iint_R e^{x+y}dA$ R is the region enclosed by $x^2+y^2=4$.

Solution: - $\iint_R e^{x^2+y^2}\,dA=\int_0^2\int_0^{2\pi} r\,.\,e^{r^2}\,d\theta\,.dr = 2\pi.\int_0^2 r\,e^{r^2}dr=\pi.e^4$

Example 2: Find the volume of the solid bounded by the plane $z=0$ and the paraboloid $z=1-x^2-y^2$.

Solution: If we put $z=0$ in the equation of the paraboloid $z=1-x^2-y^2$ we get $x^2+y^2=1$. This means the paraboloid intersects the plane in the circle $x^2+y^2=1$, so the solid lies under the parabolid and above the circlular disk D given $x^2+y^2\le 1$(see fig below) .



In polar coordinates D is given by $0\le r\le 1, 0\le\theta\le\pi$. Since $1-x^2-y^2=1-r^2$, the volume is

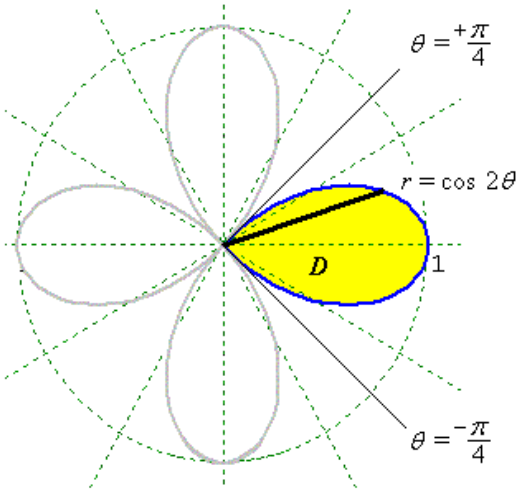
$$V=\iint_D\left(1-x^2-y^2\right)dA=\int_0^{2\pi}\int_0^1\left(1-r^2\right)r\,dr\,d\theta$$

$$V=\int_0^{2\pi}\left[\frac{r^2}{2}-\frac{r^4}{4}\right]_0^1 d\theta=\int_0^{2\pi}\frac{1}{4}d\theta=\frac{\pi}{2}$$

Example 3: Find the area enclosed by one loop of the curve $r = \cos 2\theta$.

Boundaries:

$0 \le r \le \cos 2\theta$; $-\frac{\pi}{4} \le \theta \le +\frac{\pi}{4}$



Area:

$$A = \iint_D 1\,dA = \int_{-\pi/4}^{+\pi/4}\int_0^{\cos 2\theta} 1\, r\, dr\, d\theta$$

$$= \int_{-\pi/4}^{+\pi/4}\left[\frac{r^2}{2}\right]_0^{\cos 2\theta} d\theta$$

$$= \int_{-\pi/4}^{+\pi/4}\left(\frac{\cos^2 2\theta}{2} - 0\right)d\theta$$

$$= \int_{-\pi/4}^{+\pi/4}\frac{\cos 4\theta + 1}{4}\,d\theta$$

$$= \left[\frac{\sin 4\theta}{16} + \frac{\theta}{4}\right]_{-\pi/4}^{+\pi/4} = \left(0+\frac{\pi}{16}\right) - \left(0-\frac{\pi}{16}\right)$$

Therefore

$$A = \underline{\underline{\frac{\pi}{8}}}$$

Quick Check Exercises 4.1.7

1. Find the area of the region that lies outside the circle $r=1$ and inside the circle $r = 2\sin\theta$
2. Use polar coordinates to evaluate the double integral $\iint_R x^2 + y^2 dA$ over the region R, which lies inside $x^2 + y^2 = y$ and in the first quadrant
3. Find the area A of the region between the spirals $r = e^{\theta}$ *and* $r = e^{2\theta}$ on $[0, 3\pi]$ by using iterated integrals in polar coordinates.

Group Activity 4.1.2

1. Evaluate

   a. $\int_0^{2\pi}\int_0^1 r.\sin\theta\, dr\, d\theta$ b. $\int_0^{2\pi}\int_0^1 r.\sqrt{1+r^2}\, dr\, d\theta$

2. Change the integral to an iterated integral in polar coordinates, and then evaluate

   a. $\int_1^3\int_0^x \frac{2}{x^2+y^2}\, dy\, dx$

   b. $\int_{\sqrt{3}}^3\int_0^{\sqrt{9-x^2}} \frac{1}{\sqrt{x^2+y^2}}\, dy\, dx$

c. $\int_{-2}^{2}\int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} e^{-(x^2+y^2)}\,dxdy$

d. $\int_{0}^{1}\int_{0}^{\sqrt{1-x^2}} e^{\sqrt{x^2+y^2}}\,dy\,dx$

e. $\int_{0}^{1}\int_{-\sqrt{x-x^2}}^{\sqrt{x-x^2}} (x^2+y^2)\;\;dy\;\;dx$

3. Evaluate

a. $\iint_R \frac{1}{x^2+y^2+1}dA,$ where R is the sector in the first quadrant of $x^2+y^2=4$

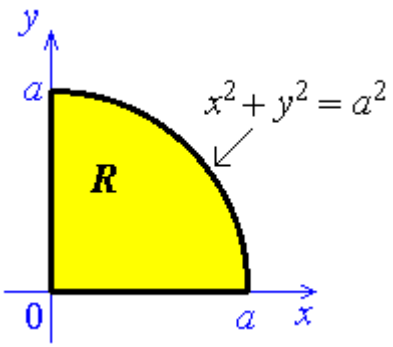
b. $\iint_R xy\;dA,$ where R is the region bounded by the circle $r=5$

c. $\iint_R x^2\;dA,$ where R is the region bounded by the circle $r=4\sin\theta$

FINDING MASS AND CENTER OF MASS USING DOUBLE INTEGRALS IN POLAR COORDINATE

Example 1. Find the centre of mass for a plate of surface density $\sigma=\frac{k}{\sqrt{x^2+y^2}}$, whose boundary is the portion of the circle $x^2+y^2=a^2$ that is inside the first quadrant. k and a are positive constants.

Solution: Use plane polar coordinates.



Boundaries:

The positive x-axis is the line $\theta=0$.

The positive y-axis is the line $\theta=\pi/2$.

The circle is $r^2=a^2$, which is $r=a$.

Mass:

Surface density $\sigma=\frac{k}{\sqrt{x^2+y^2}}=\frac{k}{r}$

$$m=\iint_R \sigma\,dA=\int_0^{\pi/2}\int_0^a \frac{k}{r}r\,dr\,d\theta$$

$$= k\int_0^{\pi/2}\int_0^{a} 1\,dr\,d\theta = k\left(\int_0^{a} 1\,dr\right)\left(\int_0^{\pi/2} 1\,d\theta\right) = k\,[r]_0^{a}\,[\theta]_0^{\pi/2}$$

$$m = \frac{k\pi a}{2}$$

Example 2. (Continued from example (1))

First Moments about the *x*-axis:

$$\Delta M_x = y\,\Delta m \quad \Rightarrow \quad M_x = \iint_R y\,\sigma\,dA$$

$$= \int_0^{\pi/2}\left(\int_0^{a}(r\sin\theta)\frac{k}{r}r\,dr\right)d\theta$$

$$= k\int_0^{a} r\,dr\int_0^{\pi/2}\sin\theta\,d\theta = k\left[\frac{r^2}{2}\right]_0^{a}\left[-\cos\theta\right]_0^{\pi/2}$$

$$= k\left(\frac{a^2}{2}-0\right)(-0+1)$$

$$\therefore \quad M_x = \frac{k\,a^2}{2}$$

But $M_x = m\bar{y} \quad \Rightarrow \quad \bar{y} = \frac{M_x}{m} = \frac{k\,a^2}{2}\cdot\frac{2}{k\pi\,a} = \frac{a}{\pi}$

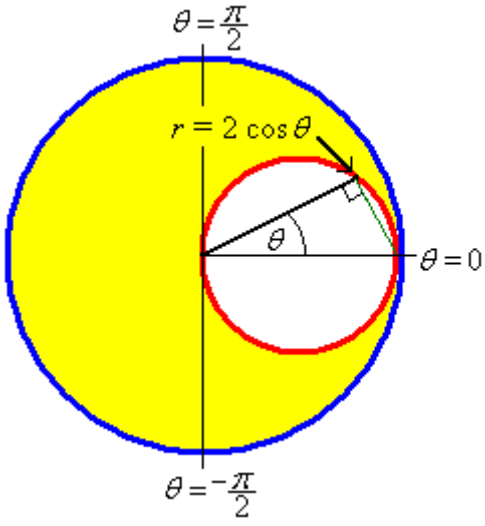
By sym., $\bar{x} = \bar{y}$

Therefore the centre of mass is at

$$(\bar{x},\bar{y}) = \underline{\underline{\left(\frac{a}{\pi},\frac{a}{\pi}\right)}}$$

Example 3:

Find the proportion of the mass removed, when a hole of radius 1, tangent to a diameter, is bored through a uniform sphere of radius 2.



Cross-section at right angles to the axis of the hole:

Use cylindrical polar coordinates, with the *z*-axis aligned parallel to the axis of the cylindrical hole. The plane polar equation of the boundary of the

hole is then

$$r = 2\cos\theta$$

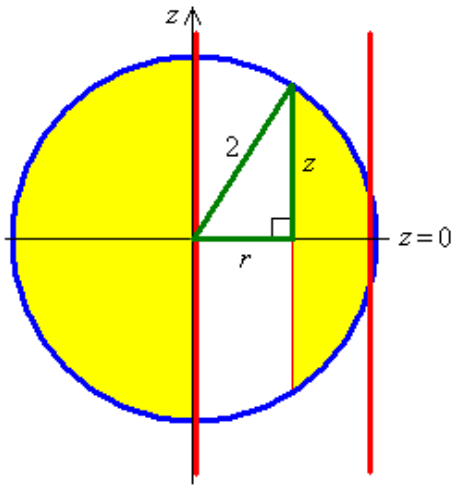
The entire circular boundary is traversed once for

$$-\frac{\pi}{2} \le \theta \le +\frac{\pi}{2}$$

Cross-section parallel to the axis of the hole:

At each value of r, the distance from the equatorial plane to the point where the hole emerges from the sphere is

$$z = \sqrt{2^2 - r^2}$$



The element of volume for the hole is therefore

$$dV = 2z\,dA = 2\sqrt{4-r^2}\,(r\,dr\,d\theta)$$

$$V = \int_{-\pi/2}^{+\pi/2}\int_{0}^{2\cos\theta} 2\sqrt{4-r^2}\; r\,dr\,d\theta$$

We can**not** separate the two integrals, because the upper limit of the inner integral, ($r = 2\cos\theta$), is a function of the variable of integration in the outer integral.

The geometry is entirely symmetric about $\theta = 0$

$$\Rightarrow\quad V = 4\int_{0}^{+\pi/2}\int_{0}^{2\cos\theta} \sqrt{4-r^2}\; r\,dr\,d\theta$$

$$= 4\int_{0}^{\pi/2}\left[\frac{(4-r^2)^{3/2}}{\frac{3}{2}\times(-2)}\right]_0^{2\cos\theta} d\theta$$

$$= -\frac{4}{3}\int_{0}^{\pi/2}\left(\left(4-4\cos^2\theta\right)^{3/2} - (4-0)^{3/2}\right)d\theta = +\frac{4}{3}\int_{0}^{\pi/2}\left(\left(-4\sin^2\theta\right)^{3/2} + 4^{3/2}\right)d\theta$$

$$= \frac{4}{3}\int_{0}^{\pi/2}\left(8 - 8\sin^3\theta\right)d\theta = \frac{32}{3}\int_{0}^{\pi/2}\left(1 - \sin^3\theta\right)d\theta$$

$$= \frac{32}{3}\left(\int_{0}^{\pi/2} 1\,d\theta - \int_{0}^{\pi/2} \sin^2\theta\sin\theta\,d\theta\right)$$

$$= \frac{32}{3}\left(\int_0^{\pi/2} 1\, d\theta \; - \; \int_0^{\pi/2} \left(1 \; - \; \cos^2\theta\right)\sin\theta \, d\theta \right)$$

Let $u = \cos\theta$, then $du = -\sin\theta\, d\theta$.

$\theta = 0 \;\Rightarrow\; u = 1$ and $\theta = \frac{\pi}{2} \;\Rightarrow\; u = 0$

$$V \; = \; \frac{32}{3}\left(\int_0^{\pi/2} 1\, d\theta \; - \; \int_{u=1}^{u=0} \left(1-u^2\right)\left(-du\right) \right) \; = \; \frac{32}{3}\left(\int_0^{\pi/2} 1\, d\theta \; - \; \int_0^1 \left(1-u^2\right) du \right)$$

$$= \frac{32}{3}\left(\left[\theta\right]_0^{\pi/2} \; - \left[u \; - \; \frac{u^3}{3} \right]_0^1 \right) \; = \; \frac{32}{3}\left(\left(\frac{\pi}{2} - 0\right) \; - \left(\frac{2}{3} - 0\right) \right)$$

$$= \frac{16\pi}{3} \; - \; \frac{64}{9}$$

The density is constant throughout the sphere. Therefore

$$\frac{m_{\text{hole}}}{m_{\text{sphere}}} \; = \; \frac{V_{\text{hole}}}{V_{\text{sphere}}} \; = \left(\frac{16\pi}{3} \; - \; \frac{64}{9} \right) \cdot \frac{3}{4\pi 2^3} \; = \; \frac{1}{2} \; - \; \frac{2}{3\pi}$$

therefore, the proportion of the sphere that is removed is

$$\frac{1}{2} \; - \; \frac{2}{3\pi} \; \approx \; 29\%$$

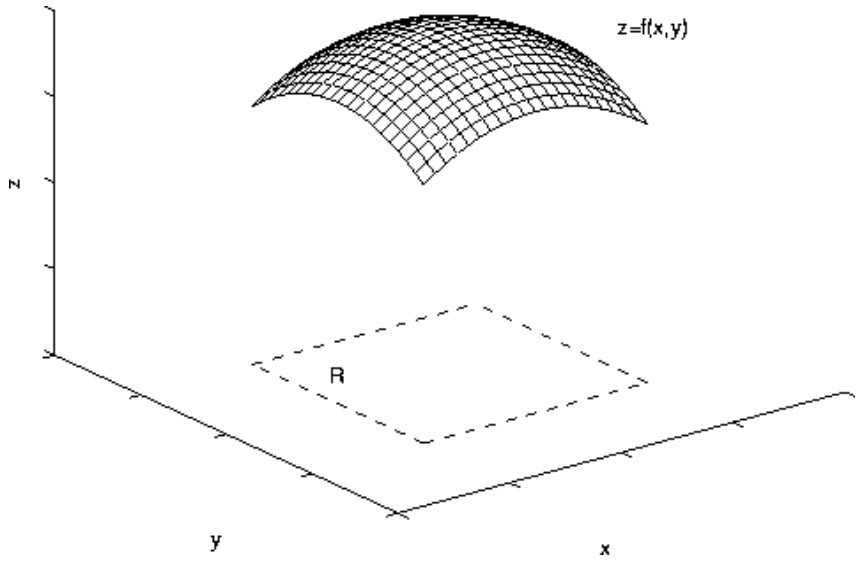
4.1.4 SURFACE AREA

Definition: -

Let R be a vertically or horizontally simple region, and let f have continuous partial derivatives on R. The **surface area** S of R is defined by the graph of f on R

$$S = \iint_R \sqrt{[f_x(x,y)]^2 + [f_y(x,y)]^2 + 1} \; dA$$

f on D.



Example 1 What is the surface area of the plane $z = 2x + 3y$ above the rectangle with $-1 \le x \le 2$ and $0 \le y \le 2$?

Solution: - Let $f(x, y) = 2x + 3y$ .In this case $f_x(x, y) = 2$ and $f_y(x, y) = 3$.

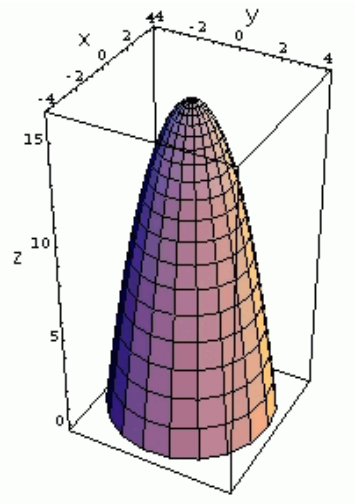
Now by applying the above formula, the surface area S of the region is given by

$$S = \int_{-1}^{2}\int_{0}^{2} \sqrt{14}\, dy\, dx = \int_{0}^{2}\int_{-1}^{2} \sqrt{14}\, dx\, dy = 6\sqrt{14}$$

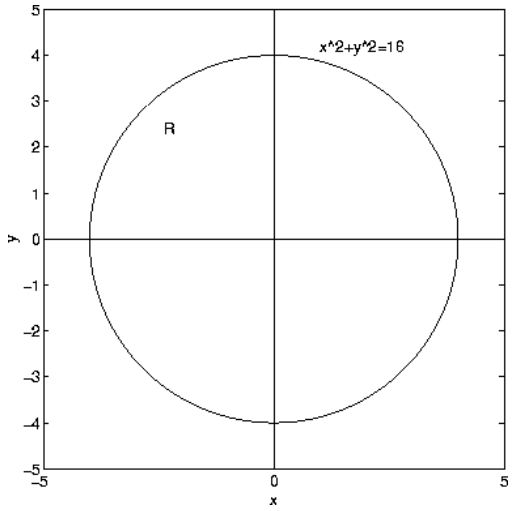
Since, the region of integration R is a rectangle and the integrand is continuous, the value of the integral is independent of the order of integration. Thus the surface area of the region is

$$S = \int_{-1}^{2}\int_{0}^{2} \sqrt{14}\, dy\, dx = \int_{0}^{2}\int_{-1}^{2} \sqrt{14}\, dx\, dy = 6\sqrt{14} \ .$$

Example 2 Find the surface area of the part of the paraboloid $z = 16 - x^2 - y^2$ that lies above the $xy - plane$ (see the the graph given below).



The region R is the disk $0 \le x^2 + y^2 \le 16$ (disk of radius 4 centered at the origin in $xy - plane$).



Solution: - Let $f(x, y) = 16 - x^2 - y^2$. In this case $f_x(x, y) = 2x$ and $f_y(x, y) = -2y$.

Hence, the surface area S is given by

$$S = \iint_{\mathbf{R}} \sqrt{1 + 4x^2 + 4y^2}\ dA$$

Since R is a disk, it is convenient to convert the above integral into polar coordinates. The disk **R** satisfies $0 \le r \le 4$ *and* $0 \le \theta \le 2\pi$. In addition,

$$4x^2 + 4y^2 = 4\left(x^2 + y^2\right) = 4\,.\,r^2\left(\cos^2\theta + \cos^2\theta\right) = 4\,.\,r^2$$

The surface area is given by the integral

$$S = \int_0^{2\pi}\int_0^4 \sqrt{1+4r^2}\ r\ dr\ \ d\theta = \int_0^4\int_0^{2\pi} \sqrt{1+4r^2}\ r\ d\theta\ \ dr$$

Both iterated integrals above can be computed in a straightforward manner.

Thus is

$$S = \int_0^{2\pi}\int_0^4 \sqrt{1+4r^2}\ r\ dr\ \ d\theta = \int_0^4\int_0^{2\pi} \sqrt{1+4r^2}\ r\ d\theta\ \ dr = \frac{\pi}{6}\left(65^{3/2}-1\right)$$

QUICK CHECK ACTIVITY 4.1.7

Find the surface area of:

a) the plane $z = 2 - x - y$ above the rectangle $0 \le x \le 2$ and $0 \le y \le 3$

b) the cylinder $z = 9 - y^2$ above the triangle bounded by $y = x$, $y = -x$, and $y = 3$

c) the surface $z = 16 - x^2 - y^2$ above the circle $x^2 + y^2 \le 9$

Group Activity 4.1.3

1. Find the surface area of the plane region which is

a. The portion of the graph of $f(x,y) = \frac{2}{3}x^{3/2}$ that lies over $R: 0 \le x \le 3,\ 0 \le y \le 2$.

b. The portion of the plane in the first octant.

c. The portion of the paraboloid $z = 9 - x^2 - y^2$ above the *xy – plane* .

d. The portion of the sphere $x^2 + y^2 + z^2 = 4$ that is inside the cylinder $x^2 + y^2 = 1$.

4.2 Triple Integrals

Definition:

Let **D** be the solid region between the graphs of two continuous functions F_1 *and* F_2 on a vertically or horizontally simple region **R** in the $xy-plane$. If f is continuous on D then a unique number $\iiint_D f(x,y,z)dV$ is called the **triple integral** of f on D.

Theorem 4.2.1 (Fubini's Theorem for Triple integrals):

If f is continuous on the rectangular box $B=[a,b]\times[c,d]\times[r,s]$, then

$$\iiint_B f(x,y,z)dV=\int_r^s\int_c^d\int_a^b f(x,y,z)dxdydz$$

For a box-like region, the integral is independent of the order of integration, assuming f(x,y,z) is continuous. Hence, there are total of 6 ways to order the integrations. For example we can integrate with respect to x, then z, then y. In this case we have

$$\int\int\int_R f(x,y,z)dV=\int_c^d\int_r^s\int_a^b f(x,y,z)dxdzdy$$

Consider the following example:

$$\int_0^1\int_1^2\int_1^2(x^2+yz)dzdydx$$

The inner integral is

$$\int_1^2(x^2+yz)dz.$$

Integrating with respect to z, treating x and y as constants, we obtain

$$x^2+\frac{3}{2}y$$

Note that z has completely disappeared from the expression.

$$\int_1^2 (x^2 + 3y/2)dy = x^2 y + 3y^2/4|_{y=1}^{y=2} = x^2 + 9/4$$

Note that y has disappeared from the expression on the right. The outer integral is with respect to x. Thus we have

$$\int_0^1 (x^2 + 9/4)dx = 31/4$$

Students can verify that the same answer is obtained if the order of integration is changed.

Theorem 4.2.2:-

Let D be the solid region between the graphs of two continuous functions F_1 and F_2 on a vertically or horizontally simple region R in the $xy-plane$, and let f be continuous on D. Then

$$\iiint_D f(x,y,z)dV = \iint_R \left(\int_{F_1(x,y)}^{f_2(x,y)} f(x,y,z)dz \right) dA$$

Proof: - Exercise

- If **R** is the vertically simple region between the graphs of g_1 *and* g_2 on $[a,b]$, then

$$\iiint_D f(x,y,z)dV = \int_a^b \left[\int_{g_1(x)}^{g_2(x)} \left(\int_{f_1(x,y)}^{f_2(x,y)} f(x,y,z)dy \right] dx \right.$$

$$= \int_a^b \left[\int_{g_1(x)}^{g_2(x)} \int_{g_1(x)}^{g_2(x)} f(x,y,z)dzdydx \right. \ldots (1)$$

- If R is the horizontally simpler region between the graphs of h_1 *and* h_2 on $[c,d]$, then

$$\iiint_D f(x,y,z)dV = \int_c^d \left[\int_{h_1(y)}^{h_2(y)} \left(\int_{f_1(x,y)}^{f_2(x,y)} f(x,y,z)dz \right) dx \right] dy$$

$$= \int_c^d \left[\int_{h_1(y)}^{h_2(y)} \int_{f_1(x,y)}^{f_2(x,y)} f(x,y,z)dzdxdy \right. \ldots (2)$$

Example 1:

Let R be the rectangular region in the $xy-plane$ bounded by the lines $x=2,\ \ x=\frac{5}{2},$ $y=0,$ *and* $y=\pi$. And, let D be the solid region between the graphs of $z=0$ *and* $z=2$. Find

$$\iiint_D z\,x\sin(xy)\ dv.$$

Solution: -

$$\iiint_D z\,x\sin(xy)du=\int_0^2\int_0^{\pi}\int_2^{\frac{5}{2}} z\,x\sin(xy)dx\,dy\,dz$$

$$=\int_2^{\frac{5}{2}}\int_0^{\pi}\int_0^{2} z\,x\sin(xy)dz\,dy\,dx$$

$$=\int_2^{\frac{5}{2}}\int_0^{\pi}\frac{z^2}{2}x\sin(xy)\ \Big|_0^2 dy\,dx$$

$$=2\int_2^{\frac{5}{2}}\int_0^{\pi} x\sin(xy)dy\,dx$$

$$=2\int_2^{\frac{5}{2}}\int_0^{\pi x} x\sin u\,.\frac{du}{x}dx\ ,\ \ where\ u=xy\ \Rightarrow dy=\frac{du}{x}$$

$$=2\int_2^{\frac{5}{2}}\int_0^{\pi x}\sin u\,.du\ \ dx\quad =-2\int_2^{\frac{5}{2}}\cos u\Big|_0^{\pi x}\,dx$$

$$=-2\int_2^{\frac{5}{2}}(\cos\pi x-1)\,dx\ \ =-2\left[\frac{\sin\pi x}{2}-x\right]_2^{\frac{5}{2}}$$

$$=-2\left[\left(\frac{1}{2}-\frac{5}{2}\right)-(0-2)\right]=4$$

Therefore

$$\iiint_D z\,x\sin(xy)dv=4$$

Example 2

Evaluate $\iiint_D 2xyz\ dV$, where D is the solid region that is bounded by the parabolic cylinder $x = 2 - \frac{1}{2}y^2$ and the planes z = 0, y = x and y = 0

Solution: -

By using the given information

$$y = 2 - \frac{1}{2}y^2 \Leftrightarrow y^2 + 2y - 4 = 0$$

$$\Leftrightarrow y = \frac{-2 \pm \sqrt{4+16}}{2} = \frac{-2 \pm 2\sqrt{5}}{2} = -1 \pm \sqrt{5},$$

$$-\sqrt{2} \le x \le \sqrt{2} \text{ and } 0 \le z \le x + \frac{1}{2}y^2$$

$$\therefore \iiint_D 2xyz\ dv = \int_{-\sqrt{2}}^{\sqrt{2}} \int_{-1-\sqrt{5}}^{-1+\sqrt{5}} \int_0^{x+\frac{1}{2}y^2} 2xyz\ dz\,dy\ dx$$

$$= \int_{-\sqrt{2}}^{\sqrt{2}} \int_{-1-\sqrt{5}}^{-1+\sqrt{5}} xy\left[z^2\right]_0^{x+\frac{1}{2}y^2} dy\ dx$$

$$= \int_{-\sqrt{2}}^{\sqrt{2}} \int_{-1-\sqrt{5}}^{-1+\sqrt{5}} xy\left[\left(x + \frac{1}{2}y^2\right)^2\right] dy\ dx$$

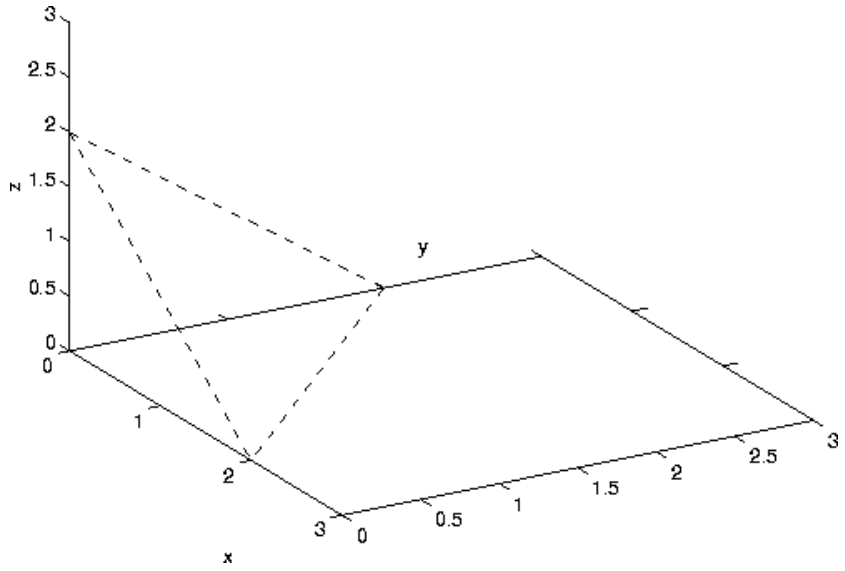
$$= \int_{-\sqrt{2}}^{\sqrt{2}} \int_{-1-\sqrt{5}}^{-1+\sqrt{5}} \left[x^3\ y + 2\ x^2\ y^3 + \frac{1}{4}xy^5\right] dy\ dx$$

$$= \int_{-\sqrt{2}}^{\sqrt{2}} \int_{-1-\sqrt{5}}^{-1+\sqrt{5}} \left[x^3\ y + 2\ x^2\ y^3 + \frac{1}{4}xy^5\right] dy\ dx \text{ (}\textit{Evaluate it}\text{)}$$

Example 3. Evaluate the triple integral

$$\int\int\int_R (x + 2y)dV$$

where R is the tetrahedral region bounded by the planes x = 0, y =0, z = 0 and x + y + z = 2 (see figure below).



There are several ways to compute the integral. We can rewrite the equation of the plane x + y + z = 2 as z = 2 – x - y. Note that $0 \le z \le 2 - x - y$. Hence, we have

$$\int\int\int_R (x+2y)dV = \int\int_D \left[\int_0^{2-x-y} (x+2y)dz\right] dA$$

The inner integral is (remember x and y are constants in this integration)

$$\int_0^{2-x-y} (x+2y)dz = (x+2y)z\Big|_{z=0}^{z=2-x-y} = (x+2y)(2-x-y)$$

The projection of the region R onto the xy-plane is the triangle R shown in the figure above: Hence, we are left with the double integral

$$\int\int_D (x+2y)(2-x-y)dA = \int_0^2 \int_0^{2-x} (x+2y)(2-x-y)dydx$$

We can also evaluate the double integral by integrating with respect to x first, and then with respect to y. In this case

$$\int\int_D (x+2y)(2-x-y)dA = \int_0^2 \int_0^{2-y} (x+2y)(2-x-y)dxdy$$

Students can evaluate and check that the double integral equals 2/3. Would you please determine the final answer?

Quick Check Activity 4.2.1

1. Evaluate the iterated integrals

a. $\int_0^{n3}\int_0^1\int_0^y (z^2+1)e^{y^2}\,dxdzdy$

b. $\int_0^{\pi/2}\int_0^{\frac{\pi}{2}}\int_0^{\sin z} x^2\sin y dxdydz.$

c. $\int_0^{\pi/2}\int_{\sin z}^0\int_0^{2yz}\left(\frac{x}{y}\right)dxdy$

d. $\int_0^a\int_0^b\int_0^c x^2+y^2+z^2 dxdydz$

2. Evaluate the integral at example (2) by doing the integration in the order $dy\ dx\ dz$.

$$\iiint_D (2xyz)du=\int_0^2\int_0^{\sqrt{4-2z}}\int_0^x 2xyzdydxdz$$

Definition

Let D be the solid region between the graphs of two continuous functions F_1 and F_2 on a region R in the $xy-plane$. Then the volume V of D is defined by

$$V=\iiint_D dV \quad \Leftrightarrow V=\iint_{\mathbf{R}}\left[\int_{F_1(x,y)}^{F_2(x,y)} dz\right]dA$$

$$=\iint\left[F_2(x,y)-F_1(x,y)\right]dA$$

Example 1: - Find the volume of the solid **D** in the first octant bounded by

$$y=2x^2 \ \ and \ \ y+4z=8$$

Solution: - By using the given information, the solid region **D** is

$$\mathbf{D}=0\le x\le 2,\quad 0\le y\le 2x^2 \ \ and \ \ 2-\frac{1}{2}x^2\le z\le 0.$$

Therefore the volume of the solid is

$$V=\int_0^2\int_0^{2x^2}\int_{2-\frac{1}{2}x^2}^0 dz\ dy\ dx$$

$$\Leftrightarrow V=\int_0^2\int_0^{2x^2}\left[\frac{1}{2}x^2-2\right]dy\ dx$$

$$V = \int_0^2 \left[\frac{1}{2}x^2 - 2\right] 2x^2 \, dx$$

$$\Leftrightarrow V = \int_0^2 \left[x^4 - 4x^2\right] \, dx$$

$$V = \left[\frac{x^5}{5} - \frac{4}{3}x^3\right]_0^2 = \frac{32}{5} - \frac{32}{3} = -\frac{160-96}{15} = -\frac{64}{15}$$

But since volume of the solid is positive, thus

$$V = \frac{64}{15} \; square \; units \; .$$

Example 2: Verify the formula $V = \frac{4}{3}\pi a^3$ for the volume of a sphere of radius *a*.

Solution:

$$V = \iiint_V 1 \, dV = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$$= \left(\int_0^a r^2 \, dr\right)\left(\int_0^{\pi} \sin\theta \, d\theta\right)\left(\int_0^{2\pi} 1 \, d\phi\right)$$

$$= \left[\frac{r^3}{3}\right]_0^a \left[-\cos\theta\right]_0^{\pi} \left[\phi\right]_0^{2\pi} = \left(\frac{a^3}{3} - 0\right)(+1+1)(2\pi - 0)$$

Therefore

$$V = \frac{4}{3}\pi a^3$$

Quick Check Activity 4.2.2

1. Find the volume of the solid in the first octant bounded by $y^2 + 64z4 = 4$ and the plane y = x
2. Consider the solid in the first octant cut from the cylindrical solid $y^2 + z^2 + x^2 \le 1$ by the planes $y = x \; and \; x = 0$. Evaluate $\iiint_D zdu.$
3. Find the volume of the solid region below the surface $f(x, y) = e^x . \cos y$ for $f(x, y)$ in the region **R:** $0 \le x \le 1, \; 0 \le y \le \frac{\pi}{2}$

4.2.1 Triple Integrals in Cylindrical Coordinates

Just as certain double integrals are easier to evaluate by means of polar coordinates than by rectangular coordinates, certain triple integrals are easier to evaluate by coordinates other than rectangular coordinates. In this section we introduce two new types of coordinates: Cylindrical and Spherical coordinates.

Cylindrical Coordinates: - Let (x, y, z) be the rectangular coordinates of a point **P** in $xyz-space$. If (r,θ) is a polar coordinate for the point (x, y) in the $xy-plane$, then we call (r,θ,z) a cylindrical coordinate for **P**.

Given the rectangular coordinates (x, y, z) of a point **P**, we can determine a set of cylindrical coordinate for **P** with the aid of the formulas

$$x^2+y^2=r^2 \quad and \quad \tan\theta=\frac{y}{x} \quad (for \quad x\neq 0)$$

Conversely, from any set (r,θ,z) of cylindrical coordinates of a point p we can determine the rectangular coordinate (x, y, z) of **P** by the formulae

$$x=r\cos\theta \quad and \quad y=r\sin\theta$$

Theorem 4.2.3: - Let D be the solid region between the graphs of $F_1(r,\theta)$ and $F_2(r,\theta)$ on R, where R is the plane region between the polar graphs of $h_1(\theta)$ and $h_2(\theta)$ on $\alpha\le\theta\le\beta$ is continuous on D. Then

$$\iiint_D f(x,y,z)\,dv = \int_\alpha^B\int_{h_1(\theta)}^{h_2(\theta)}\int_{F_1(r,\theta)}^{F_2(r,\theta)} f(r,\theta,z)\,r\,dz\,d\theta\,dr$$

Proof: - Exercise

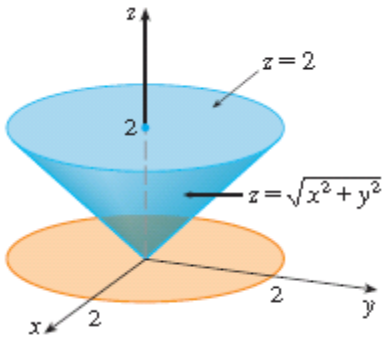
Examples: -

1. Evaluate $\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{\sqrt{x^2+y^2}}^{2}\left(x^2+y^2\right)dzdydx$

Solution: The iterated integral is a triple integral over the solid region

$$D=\left\{(x,y,z)\middle| -2\le x\le 2, -\sqrt{4-x^2}\le y\le\sqrt{4-x^2}, \sqrt{x^2+y^2}\le z\le 2\right\}$$

and the projection of D onto xy-plane is the disk $x^2+y^2\le 4$. The lower surface of D is the cone $z=\sqrt{x^2+y^2}$ and its upper surface is the plane $z=2$ (see figure below).



This region has much simpler description in cylindrical coordinates.

$D=\{(r,\theta,z)|0\le\theta\le 2\pi,0\le r\le 2,r\le z\le 2\}$

Therefore we have

$$\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{\sqrt{x^2+y^2}}^{2}(x^2+y^2)dzdydx=\iiint_D(x^2+y^2)dV$$

$$=\int_0^{2\pi}\int_0^2\int_r^2 r^2 r\,dz\,dr\,d\theta$$

$$=\int_0^{2\pi}d\theta\int_0^2 r^3(2-r)dr$$

$$=2\pi\left[\frac{r^4}{2}-\frac{r^5}{5}\right]_0^2=\frac{16}{3}\pi$$

2. Let D be the solid region bounded by the cylinder $x^2 + y^2 = 1$ and the planes $z=0$ *and* $z=4$. *Evaluate* $\iiint_D (x^2+y^2)dv$

Solution: - Since $x^2+y^2=1$, then $r=0$, $r=1$, $\theta=0$ *and* $\theta=2\pi$,

$$\iiint_D (x^2+y^2)dv=\int_0^4\int_0^1\int_0^{2\pi} r^2.\,r\,d\theta\,dr\,dz$$

$$=\int_0^4\int_0^1\left[r^3\,\theta\,dr\,dz\right]_{\theta=0}^{\theta=2\pi}=2\pi\int_0^4\int_0^1 r^3\,\theta\,dr\,dz$$

$$=2\pi\int_0^4\left[\frac{r^4}{4}\right]_{r=0}^{1}dz=\frac{\pi}{2}\int_0^4 dz=2\pi$$

Thus $\iiint_D (x^2+y^2)dv=2\pi$

3. Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{a^2-x^2-y^2} x^2 dzdydx, a>0.$

Solution: - Since $z=a^2-x^2-y^2 \Rightarrow z=a^2-r^2 \quad (as \quad x^2+y^2=r^2)$ and

$$y=0 \quad to \quad y=\sqrt{a^2-x^2} \quad and \quad y^2+x^2=a^2$$

$$\Rightarrow r=a \,\&\, r=0 \rightarrow a$$

$$\Rightarrow \theta \in \left[0, \frac{\pi}{2}\right]$$

$$\therefore \int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{a^2-x^2-y^2} x^2 dzdydx = \int_0^{\pi/2} \int_0^a \int_0^{a^2-r^2} (r\cos\theta)^2 r dzdrd\theta$$

$$= \int_0^{\pi/2} \int_0^a \int_0^{a^2-r^2} r^3 cez\,\theta dzdrd\theta.$$

Quick Check Activity 4.2.3

1. Evaluate the iterated integral $\int_0^{2\pi} \int_1^2 \int_0^5 e^z r dzdrd\theta = 3\pi(e^5-1)$
2. Find the volume of the solid D that is bounded above by hemisphere $z=\sqrt{25-x^2-y^2},$ below by $xy-plane$, and laterally by the cylinder $x^2+y^2=9$.
3. Show that the volume of a cylinder with radius r_0 and height h is given by $V=\pi\ r_0^{\,2} h$

4.2.2 Triple Integrals in Spherical Coordinates

Spherical coordinate system is a simplified evaluation of triple integrals over Solid regions bounded by surfaces such as **Spheres** and **Cones**.

Let (x,y,z) and (r,θ,z) be, respectively sets of rectangular and cylindrical coordinates for a point **P** in space, with $r \geq 0$.

Let

$\rho =$ The length of the line segment **PO**

$\phi =$ The angle **PO** makes with the positive z-axis, with $0 \leq \phi \leq \pi$.

$\theta =$ The angle makes with the positive x-axis.

The point **P** is specified by the three quantities $\rho,\ \phi\ and\ \theta$, and we call the triple (ρ, ϕ, θ) set of spherical coordinates for **P**.

From trigonometry we find that

$$r = \rho\sin\phi,\ z = \rho\cos\phi$$

These equations, along with the polar coordinate formulas

$$x = r\cos\theta \qquad and \qquad y = r\sin\theta$$

yield the following formulas.

$$x = \rho\ \sin\phi\ \cos\theta$$

$$y = \rho\ \sin\phi\ \sin\theta$$

$$z = \rho\ \cos\theta$$

Theorem 4.2.4: -

Let α and β be real numbers with $\alpha \le \beta \le \alpha + 2\ \pi$ α. Let h_1, h_2, F_1 and F_2 be continuous functions with $0 \le h_1 \le h_2 \le \pi$ and $0 \le F_1 \le F_2$ Let D be the solid region consisting of all points in space whose spherical coordinates (ρ,ϕ,θ) satisfy

$$\alpha \le \theta \le \beta$$

$$h_1(\theta) \le \phi \le h_2(\theta)$$

$$F_1(\phi,\theta) \le \rho \le F_2(\phi,\theta)$$

If f is continuous on D, then

$$\iiint_D f(x,y,z)dV = \int_\alpha^\beta \int_{h_1(\theta}^{h_2(\theta)} \int_{F_2(\phi,\ \theta)}^{F_2(\phi,\ \theta)} f(\rho,\theta,\phi)\rho^2 \sin\phi\, d\theta\ d\rho\ d\theta$$

Proof: - Exercise

Remark: - In the above Theorem, if $f(x,y,z) = 1$, then

$$\iiint_D dV = \int_\alpha^\beta \int_{h_1(\theta}^{h_2(\theta)} \int_{F_2(\phi,\ \theta)}^{F_2(\phi,\ \theta)} \rho^2 \sin\phi\ \ d\theta\ d\rho\ d\phi$$

is the volume of the solid.

Example 1:

The density of an object is equal to the reciprocal of the distance from the origin. Find the mass and the average density inside the sphere $r = a$. By using spherical polar coordinates**.**

Solution:

Density:

$$\rho = \frac{1}{r}$$

Mass:

$$m = \iiint_V \rho \, dV = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} \frac{1}{r} r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$$= \left(\int_0^a r \, dr\right)\left(\int_0^{\pi} \sin\theta \, d\theta\right)\left(\int_0^{2\pi} 1 \, d\phi\right)$$

$$= \left[\frac{r^2}{2}\right]_0^a \left[-\cos\theta\right]_0^{\pi} \left[\phi\right]_0^{2\pi} = \left(\frac{a^2}{2} - 0\right)(+1+1)(2\pi - 0)$$

Therefore,

$$m = \underline{\underline{2\pi a^2}}$$

Average density =

$$\bar{\rho} = \frac{\text{mass}}{\text{volume}} = \frac{m}{V} = \frac{2\pi a^2}{\frac{4}{3}\pi a^3} = \frac{3}{2a}$$

Note that the mass is finite even though the density is infinite at the origin!

Group Activity 4.2.1

1. Evaluate the iterated integral

   a. $\int_0^{2\pi}\int_0^{\pi/2}\int_0^{\sin\phi} \rho^3 \sin\phi \;\; d\rho \;\; d\phi \;\; d\theta$

   b. $\int_0^{\pi}\int_0^{\pi/2}\int_0^{\sin\phi} \rho^2 \sin\phi \;\; d\rho \;\; d\phi \;\; d\theta$

   c. $\int_0^{\frac{\pi}{4}}\int_0^{\pi}\int_0^{2\cos\phi} \rho^2 \sin\phi \;\; d\rho \, d\theta \, d\phi$

   d. $\int_0^{\frac{\pi}{4}}\int_0^{2\pi}\int_0^{\sec\phi} \rho^3 \sin\phi\cos\phi \, d\rho \, d\theta \, d\phi$

2. Evaluate the following triple integrals by using Spherical coordinates.

   a. $\int_0^1\int_0^{\sqrt{1-x^2}}\int_{\sqrt{x^2-y^2}}^{\sqrt{2-x^2-y^2}} dz\ dy\ dx$

   b. $\int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} \frac{1}{x^2+y^2+z^2} dz\ dy\ dx$

3. Let D be the solid region between the spheres $\rho=1$ *and* $\rho=2$. Evaluate $\iiint_D z^2 du$.

4. Show that the volume of a sphere of radius r_0 is $\frac{4}{3}\pi r_0^{\,3}$.

4.3 CHANGE OF VARIABLES IN MULTIPLE INTEGRALS

Change of variables in a double integral:

A change of variables is sometimes useful in evaluating double integrals. We have already seen one example of this: conversion to polar coordinates. The new variables r and θ are related to x and y by the equations $x = r\cos\theta$ $y = r\sin\theta$

More generally, we consider a change of variables that is given by a transformation T from the uv-plane to the xy-plane where x and y are related to u and v by the equations

$$x = g(u,v) \qquad y = h(u,v)$$

If T is a one to one transformation, then it has an inverse transformation T^{-1} from the uv-plane to the xy-plane.

Definition: The jacobian of the transformation T given by $x = g(u,v)$ and $y = h(u,v)$ is

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$$

The Jacobian of the transformation from Cartesian to plane polar coordinates is

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} x_r & y_r \\ x_\theta & y_\theta \end{vmatrix} = r$$

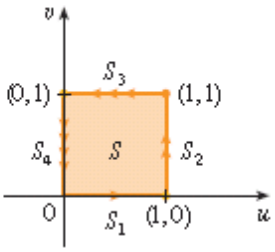
The element of area is therefore $dA = dx\,dy = r\,dr\,d\theta$

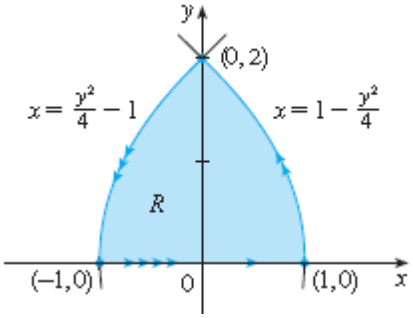
Theorem 4.3.1: Suppose that T is a transformation whose jacobian is non-zero and that maps a region S in the uv-plane on to a region R in the *xy*-plane. Suppose that f is continous on R and that R and S are vertically or horizontally simple regions. Suppose also that T is one-to-one, except perhaps on the boundary of S.Then

$$\iint_R f(x,y)dA = \iint_S f\left(x(u,v), y(u,v)\right)\left|\frac{\partial(x,y)}{\partial(u,v)}\right|dudv$$

Example 1: Use the change of variables $x = u^2 - v^2, y = 2v$ to evaluate the integral $\iint_R ydA$, where R is the region bounded by the x-axis and the parabolas $y^2 = 4 - 4x$ and $y^2 = 4 - 4x, y \geq 0.$

Solution: The region R is pictured in fig given below.





First we need to compute the jacobian:

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 2u & -2v \\ 2v & 2u \end{vmatrix} = 4u^2 + 4v^2 \geq 0$$

Therefore by theorem above

$$\iint_R ydA = \iint_S 2uv\left|\frac{\partial(x,y)}{\partial(u,v)}\right|dudv = \int_0^1\int_0^1 2uv\left(4u^2 + 4v^2\right)dudv$$

$$= 8\int_0^1\int_0^1 \left(u^3v + uv^3\right)dudv = 8\int_0^1 \left[\frac{u^4v}{4} + \frac{u^2v^3}{2}\right]_0^1 dv$$

$$= 8\int_0^1 \left(\frac{v}{4} + \frac{v^3}{2}\right)dv = 8\left[\frac{v^2}{8} + \frac{v^4}{8}\right]_0^1 = 2$$

Note the above example is not difficult because we are given a suitable change of variables. If we are not supplied with the transformation, then the first step is to find a suitable change of variables. If is difficult to integrate we take the suggested form of the transformation.

Example 2: Evaluate the integral $\iint_R e^{x+y}e^{x-y}dA$ where R is the trapezoidal region with

Vertices $(1,0),(2,0),(0,-2)$ and $(0,-1)$.

Solution: Since it is not easy to integrate $e^{x+y}e^{x-y}$, we make change of variables suggested by form of the function

i) $u = x+y$, $\quad v = x-y$

ii) $x = \frac{1}{2}(u+v)$, $\quad y = \frac{1}{2}(u-v)$

The jacobian of T is

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{-1}{2} \end{vmatrix} = -\frac{1}{2}$$

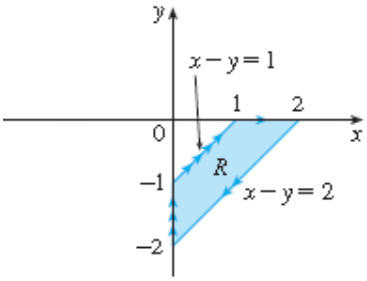
To find the region S in the *uv*-plane corresponding to R, we note that the sides of R lie on the lines

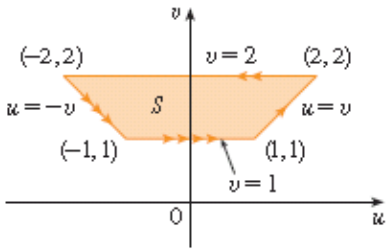
$$y = 0 \quad x-y = 2 \quad x = 0 \quad x-y = 1$$

and from either equations (i) or (ii), the image lines in the uv-plane are

$$u = v \qquad v = 2 \qquad u = -v \qquad v = 1$$

Thus the region S is the trapezoidal region with vertices $(1,1),(2,2),(-2,2)$ and $(-1,1)$ shown in fig below





Since $S = \{(u,v) | 1 \le v \le 2, -v \le u \le v\}$

Therefore,

$$\iint_R e^{x+y}e^{x-y}dA = \int_1^2\int_{-v}^{v} e^{uv}\left|\frac{\partial(x,y)}{\partial(u,v)}\right|dvdu$$

$$= -\frac{1}{2}\int_1^2\int_{-v}^{v} e^{uv}dudv = -\frac{1}{2}\int_1^2 (e-e^{-1})vdv = \frac{3}{4}(e-e^{-1})$$

Change of Variables in Triple integrals

There is a similar change of variables formula for triple integrals. Let T be a transformation that maps a region S in uvw-space in to a region R in xyz-space by a formula

$$x = g(u,v,w) \qquad y = h(u,v,w) \qquad z = k(u,v,w)$$

The jacobian of T is the following 3×3 determinant

$$\frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

With similar hypothesis to those in double integral, we have the following formula for triple integrals.

$$\iiint_R f(x,y,z)dV = \iiint_S f(x(u,v,w), y(u,v,w), z(u,v,w)) \left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right| du dv dw$$

The concepts for double integrals (surfaces) extend naturally to triple integrals (volumes).

The element of volume, in terms of the Cartesian coordinate system (x, y, z) and another orthogonal coordinate system (u, v, w), is

$$dV = dx\,dy\,dz = \frac{\partial(x,y,z)}{\partial(u,v,w)}\,du\,dv\,dw .$$

And

$$\iiint_V f(x,y,z)\,dV = \int_{w_1}^{w_2} \int_{v_1(w)}^{v_2(w)} \int_{u_1(v,w)}^{u_2(v,w)} f\left(x(u,v,w), y(u,v,w), z(u,v,w)\right) \frac{\partial(x,y,z)}{\partial(u,v,w)}\,du\,dv\,dw$$

The most common choices for non-Cartesian coordinate systems in R^3 are:

Cylindrical Polar Coordinates:

$$x = r\cos\phi$$
$$y = r\sin\phi$$
$$z = z$$

for which the differential volume is

$$dV = \frac{\partial(x,y,z)}{\partial(r,\phi,z)}\, dr\, d\phi\, dz = r\, dr\, d\phi\, dz$$

Spherical Polar Coordinates:

$$x = r\sin\theta\cos\phi$$
$$y = r\sin\theta\sin\phi$$
$$z = r\cos\theta$$

for which the differential volume is

$$dV = \frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}\, dr\, d\theta\, d\phi = r^2\sin\theta\, dr\, d\theta\, d\phi$$

Problem 2: Use a suitable three-dimensional change of variables to integrate the function x^2+y^2 over the solid region between the upper sheet of the hyperboloid $4x^2+9y^2+49=25z^2$ and the plane z=3.

Additional Problems

1. Use a suitable change of variables to evaluate $\iint_R x\,dA$, where *R* is the region in the first quadrant bounded by the lines *y=2x*, *x=2y*, and the hyperbolae *xy*=1 and *xy*=4.
2. In parametrizing the solid region inside a hyperboloid of one sheet, we started with a parametrization of the hyperboloid and introduced r, with limits of 0 and 1, as a factor of the first two coordinates, in order to parametrize the solid region.
3. Verify that we would obtain the same answer for both the volume and the integral of z^2 if we made r a factor only of the first coordinate.

4.4 Summary

- Suppose that $f(x,y)$ is a non negative continuous function of two variables x *and* y on a rectangular region in the $xy-$plane. The double integral

$\iint_R f(x,y)\ dA$, where $dA = dx\,dy$ *or* $dA = dy\,dx$

represents the volume under the plane region R.

- If the rectangle R is given by $R:\ a \le x \le b\ \wedge\ c \le y \le d$, then

$$\iint_R f(x,y)\ dA = \int_a^b \int_c^d f(x,y)dy\,dx = \int_c^d \int_a^b f(x,y)dx\,dy$$

- Suppose that the region **R** is defined by $G_1(x) \le y \le G_2(x)$ with $a \le x \le b$. Here **R** is called a ***vertically simple region***. The double integral on **R** is given by

$$\iint_{\mathbf{R}} f(x,y)\ dA = \int_a^b \int_{G_1(x)}^{G_2(x)} f(x,y)\ dy\ dx$$

- If **R** is a **horizontally simple region** - that is, if the region is defined by $c \le y \le d$ and $H_1(y) \le x \le H_2(y)$, then

$$\iint_{\mathbf{R}} f(x,y)\,dA = \int_c^d \int_{H_1(y)}^{H_2(y)} f(x,y)\,dx\,dy$$

- In case **R is a simple region**, $\iint_{\mathbf{R}} f(x,y)dA$ can be evaluated as either

$$\int_a^b \int_{g_1(x)}^{y_2(x)} f(x,y)dydx \quad or \quad \int_c^d \int_{h_1(x)}^{h_2(x)} f(x,y)dxdy\ .$$

- If the region **R** is of the form $g_1(r) \le \theta \le g_2(r)$ with $a \le r \le b$, then the double integral of the function $f(x,y)$ in polar coordinates is given by the iterated integral

$$\iint_{\mathbf{R}} f(x,y)\ dA = \int_a^b \int_{g_1(r)}^{g_2(r)} f(r\cos\theta + r\sin\theta)\ r\ d\theta\ dr$$

Since every point in a plane has both Cartesian and polar coordinates. Suppose a point **P** in the plane has polar coordinates (r, θ) and Cartesian coordinates (x, y). Then from the definition of sine and the cosine we deduce that

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta$$

$$x^2 + y^2 = r^2 \quad \text{and} \quad \tan\theta = \frac{y}{x},\ x \neq 0$$

- If f is non-negative function on **R**, the volume V of the region between the graph of f and R is given by

$$V = \int_{\alpha}^{\beta}\int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta, r\sin\theta) r\,dr\,d\theta.$$

If $f(x, y) = 1$, the area A of the region R is given by

$$A = \int_{\alpha}^{\beta}\int_{h_1(\theta)}^{h_2(\theta)} r\,dr\,d\theta.$$

- Let R be a vertically or horizontally simple region, and let f have continuous partial derivatives on R. The **surface area** S of is defined by the graph of f on R

$$S = \iint_R \sqrt{[f_x(x, y)]^2 + [f_y(x, y)]^2 + 1}\ \ dA$$

- Let **D** be the solid region between the graphs of two continuous functions F_1 *and* F_2 on a vertically or horizontally simple region **R** in the $xy - plane$. If f is continuous on D then a unique number $\iiint_D f(x, y, z)dV$ is called the **triple integral** of f on D.

- Let D be the solid region between the graphs of two continuous functions F_1 and F_2 on a vertically or horizontally simple region R in the $xy - plane$, and let f be continuous on D. Then

$$\iiint_D f(x, y, z)dV = \iint_R \left(\int_{F_1(x,y)}^{f_2(x,y)} f(x, y, z)dz\right)dA$$

❖ If **R** is the vertically simple region between the graphs of g_1 *and* g_2 on $[a, b]$, then

$$\iiint_D f(x,y,z)dV = \int_a^b [\int_{g_1(x)}^{g_2(x)} (\int_{f_1(x,y)}^{f_2(x,y)} f(x,y,z)dy]dx$$

❖ If R is the horizontally simpler region between the graphs of h_1 *and* h_2 on $[c, d]$, then

$$\iiint_D f(x,y,z)dV = \int_c^d [\int_{h_1(y)}^{h_2(y)} (\int_{f_1(x,y)}^{f_2(x,y)} f(x,y,z)dz)dx]dy$$

$$= \int_c^d [\int_{h_1(y)}^{h_2(y)} \int_{f_1(x,y)}^{f_2(x,y)} f(x,y,z)dzdx]dy$$

❖ Let D be the solid region between the graphs of two continuous functions F_1 and F_2 on a region R in the $xy-plane$. Then the volume V of D is defined by

$$V = \iiint_D dV$$

$$\Leftrightarrow V = \iint_{\mathbf{R}} \left[\int_{F_1(x,y)}^{F_2(x,y)} dz\right] dA$$

$$= \iint [F_2(x,y) - F_1(x,y)]\, dA$$

❖ Let (x, y, z) be the rectangular coordinates of a point **P** in $xyz-space$. If (r,θ) is a polar coordinate for the point (x, y) in the $xy-plane$, then we call (r,θ,z) a cylindrical coordinate for **P**. Given the rectangular coordinates (x, y, z) of a point **P**, we can determine a set of cylindrical coordinate for **P** with the aid of the formulas

$$x^2 + y^2 = r^2 \quad and \quad \tan\theta = \frac{y}{x} \quad (for \quad x \neq 0)$$

Conversely, from any set (r,θ,z) of cylindrical coordinates of a point p we can determine the rectangular coordinate (x, y, z) of **P** by the formula

$$x = r\cos\theta \quad and \quad y = r\sin\theta$$

❖ Let D be the solid region between the graphs of $F_1(r,\theta)$ and $F_2(r,\theta)$ on R, where R is the plane region between the polar graphs of $h_1(\theta)$ and $h_2(\theta)$ on $\alpha \le \theta \le \beta$ is continuous on D. Then

$$\iiint_D f(x,y,z)dv = \int_\alpha^B \int_{h_1(\theta)}^{h_2(\theta)} \int_{F_1(r,\theta)}^{F_2(r,\theta)} f(r,\theta,z)\, r \; dz \; d\theta \; d$$

❖ Let α and β be real numbers with $\alpha \le \beta \le \alpha + 2\ \pi$ α. Let h_1, h_2, F_1 and F_2 be continuous functions with $0 \le h_1 \le h_2 \le \pi$ and $0 \le F_1 \le F_2$ Let D be the solid region consisting of all points in space whose spherical coordinates (ρ,ϕ,θ) satisfy

$$\alpha \le \theta \le \beta$$

$$h_1(\theta) \le \phi \le h_2(\theta)$$

$$F_1(\phi,\theta) \le \rho \le F_2(\phi,\theta)$$

If f is continuous on D, then

$$\iiint_D f(x,y,z)dV = \int_\alpha^\beta \int_{h_1(\theta}^{h_2(\theta)} \int_{F_2(\phi,\ \theta)}^{F_2(\phi,\ \theta)} f(\rho,\theta,\phi)\rho^2 \sin\phi \, d\theta \; d\rho \; d\theta$$

4.5 Review Exercise

1. Compute the double integral

a. $\iint_R \left(4x + 9x^2y^2\right) dA, \ \ where \ \ R = \{(x,y): 0 \le x \le 3 \ , \ 1 \le y \le 2\}$

b. $\iint_R 2\, e^{4x+2y}\ dA, \ \ where \ \ R = \{(x,y): 0 \le x \le 1 \ , \ 0 \le y \le 1\}$

c. $\iint_R 2\, e^{-x^2-y^2}\ dA, \ \ where \ \ R = \{(x,y): 1 \le x^2 + y^2 \le 4\}$

d. $\iint_R 2\ xy\ dA, \ \ where \ \ R \ is \ the \ region \ bounded \ by \ \ y = x, \ y = 2 - x \ \ and \ \ y = 0$

e. $\int_{-1}^{1} \int_{x^2}^{2x} (2xy - 1) dy \ dx$

f. $\int_{0}^{1} \int_{2x}^{2} \left(3xy^2 + 4\right) dy \ dx$

g. $\iint_R xy\ dA$, *where* R *is the region bounded by* $r = 2\cos\theta$

h. $\iint_R \sin(x^2 + y^2)\ dA$, *where* R *is the region bounded by* $x^2 + y^2 = 4$

i. $\iint_R 4\,xy\ dA$, *where* R *is the region bounded by* $y = x^2 - 4$ *and* $y = \ln x$

j. $\iint_R 6\,x^2 y\ dA$, *where* R *is the region bounded by* $y = x^2 - 1$ *and* $y = \cos x$

2. Find the volume of the solid region

a. Bounded by

i. $z = 1 - x^2,\ z = 0,\ \ y = 0,\ y = 1$

ii. $z = 4 - x^2 - y^2,\ ,\ z = 0,\ \ x = 0,\ x + y = 1$

iii. $x + 2y + z = 8$ *and the coordinate planes*.

iv. $x + 5y + 7z = 1$ *and the coordinate planes*.

v. $z = \sqrt{y^2 + x^2}$ *and* $z = 4$

vi. $z = \sqrt{y^2 + x^2}$ *and* $x = 2$

b. Between

i. $z = x^2 + y^2,\ \ z = 8 - x^2 - y^2$

ii. $z = \sqrt{y^2 + x^2}$ *and* $z^2 + x^2 + y^2 = 4$

c.

i. Under $e^{\sqrt{x^2+y^2}}$ and inside $x^2 + y^2 = 4$

ii. Under $z = 6 - x^2 - y^2$ *inside* $x^2 + y^2 = 1$

iii. Under $z = x$ *inside* $r = \cos\theta$

3. Change the order of integration

a. $\int_0^2 \int_0^{x^2} f(x, y)\ dy\,dx$

b. $\int_0^2 \int_{x^2}^4 f(x, y)\ dy\,dx$

4. Convert to polar coordinates and evaluate the integral

a. $\int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} 2x \, dy \, dx$ b. $\int_0^2 \int_0^{\sqrt{4-x^2}} 2\sqrt{x^2+y^2} \, dy \, dx$

5. By using a double integral find the area of the solid region

a. Bounded by $y = x^2$, $y = 2 - x$ *and* $y = 0$

b. One leaf of $r = \sin 4\theta$

6. Evaluate or estimate the surface area The portion of

a. $z = 2x + 4y$ *between* $y = x,\ y = 2$ *and* $x = 0$

b. $z = x^2 + 6y$ *between* $y = x^2$ *and* $y = 4$

c. $z = xy$ *inside* $y^2 + x^2 = 8,\ y = 4$, *and first octant*

d. $z = \sin(x^2 + y^2)$ *inside* $y^2 + x^2 = \pi$

e. $z = \sqrt{x^2 + y^2}$ *below* $z = 4$

f. $z = x + 2y + 3z = 6$ *in the first octant*

7. Set up the triple integral $\iiint_D f(x,y,z)\, dV$ in an appropriate coordinate system. If $f(x,y,z)$ is given, evaluate the integral

a. $f(x,y,z) = z(x+y)$,

where $D = \{(x,y,z) : 0 \le x \le 2,\ -1 \le y \le 1,\ -1 \le z \le 1\}$

b. $f(x,y,z) = 2xy\, e^{yz})$,

where $D = \{(x,y,z) : 0 \le x \le 2,\ -1 \le y \le 1,\ -1 \le z \le 1\}$

c. $f(x,y,z) = \sqrt{x^2 + y^2 + z^2}$, where D is above $z = \sqrt{x^2 + y^2}$ and below $4 = x^2 + y^2 + z^2$

d. D is the region below $z = \sqrt{4 - x^2 - y^2}$, above $z = 0$ and inside $x^2 + y^2 = 1$

e. D is the region below $z = \sqrt{4 - x^2 - y^2}$, above $z = 0$

8. Evaluate the integral after changing coordinate systems

a. $\int_0^1 \int_x^{\sqrt{2-x^2}} \int_0^{\sqrt{x^2+y^2}} e^z \; dz\, dy\, dx$

b. $\int_0^{\sqrt{2}} \int_x^{\sqrt{4-y^2}} \int_0^2 4z \; dz\, dx\, dy$

c. $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}} \sqrt{x^2+y^2+z^2} \; dz\, dy\, dx$

d. $\int_{-2}^2 \int_0^{\sqrt{4-y^2}} \int_0^{\sqrt{4-x^2-y^2}} dz\, dy\, dx$

9. Write the given equation in

a. Cylindrical coordinates

b. Spherical coordinates

i. $y = 3$

ii. $x^2 + y^2 = 9$

iii. $x^2 + y^2 + z^2 = 4$

iv. $y = x$

v. $z = \sqrt{x^2 + y^2}$

vi. $z = 4$

5. References

1. *Ellis, R***;** Calculus with Analytic Geometry, Third Edition.

2. *Etigen, Sallas & Hille's*; Calculus of One and Several Variables, Eighth Edition.

3 *Anton, H.*: Calculus, Six Editions.

4. *Anton, H.; Bivens, I.; Davis, S.(2012): Calculus, Early Transcendentals, 10th edition, John Wiley & Sons, INC, USA.*

4. *Smith, R.T & Mintor, R. B.*, Calculus, Second Edition.